# *ZOO 201*

# Animal Diversity: Invertebrate

## Lecture 151 to 271

## Finalterm Merge PPT

# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

Nervous System

Lecture 151

# Nervous System

- Crustacean nervous systems show trends similar to those in annelids and arachnids.
- Primitively, the ventral nervous system is ladderlike.
- Higher crustaceans show a tendency toward centralization and cephalization.

# Nervous System

- Crayfish have supraesophageal and subesophageal ganglia that receive sensory input from receptors in the head and control the head appendages.
- The ventral nerves and segmental ganglia fuse, and giant neurons in the ventral nerve cord function in escape responses.

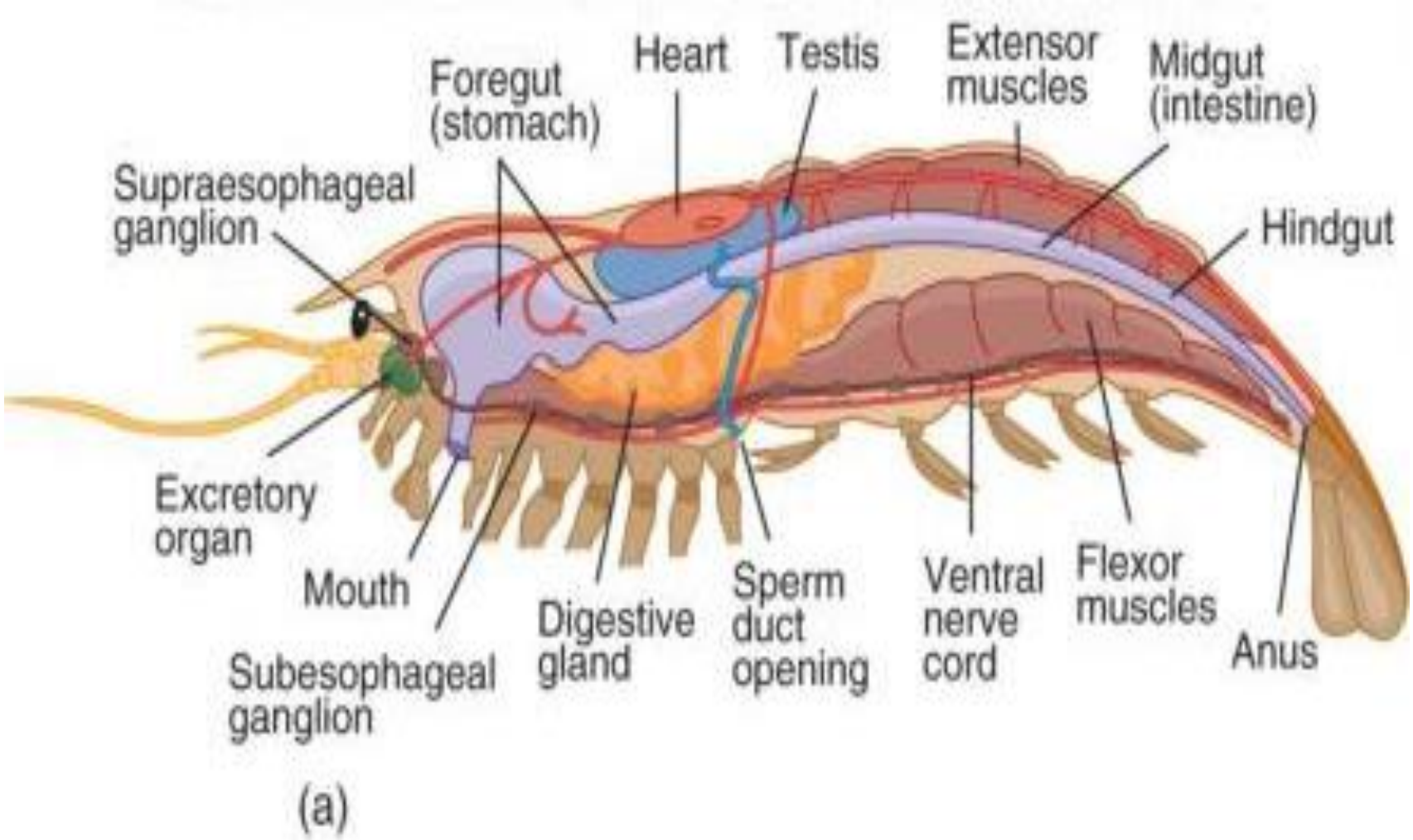
# Nervous System

- When nerve impulses are conducted posteriorly along giant nerve fibers of a crayfish, powerful abdominal flexor muscles of the abdomen contract alternately with weaker extensor muscles, causing the abdomen to flex (the propulsive stroke) and then extend (the recovery stroke).

# Nervous System

- The telson and uropods form a paddle like “tail” that propels the crayfish posteriorly.

# Nervous System



# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

Sensory structures: Statocyst

Lecture 152

# Sensory Structures: Statocyst

- In addition to antennae, the sensory structures of crayfish include compound eyes, simple eyes, statocysts, chemoreceptors, proprioceptors, and tactile setae.
- Chemical receptors are widely distributed over the appendages and the head.

# Sensory Structures: Statocyst

- Many of the setae covering the mouthparts and antennae are chemoreceptors used in sampling food and detecting pheromones.
- A single pair of statocysts is at the bases of the first antennae.

# Sensory Structures: Statocyst

- A statocyst is a pit like invagination of the exoskeleton that contains setae and a group of cemented sand grains called a **statolith**.
- Crayfish movements move the statolith and displace setae.

# Sensory Structures: Statocyst

- Statocysts provide information regarding movement, orientation with respect to the pull of gravity, and vibrations of the substrate.
- Because the statocyst is cuticular, it is replaced with each molt.
- Sand is incorporated into the statocyst when the crustacean is buried in sand.

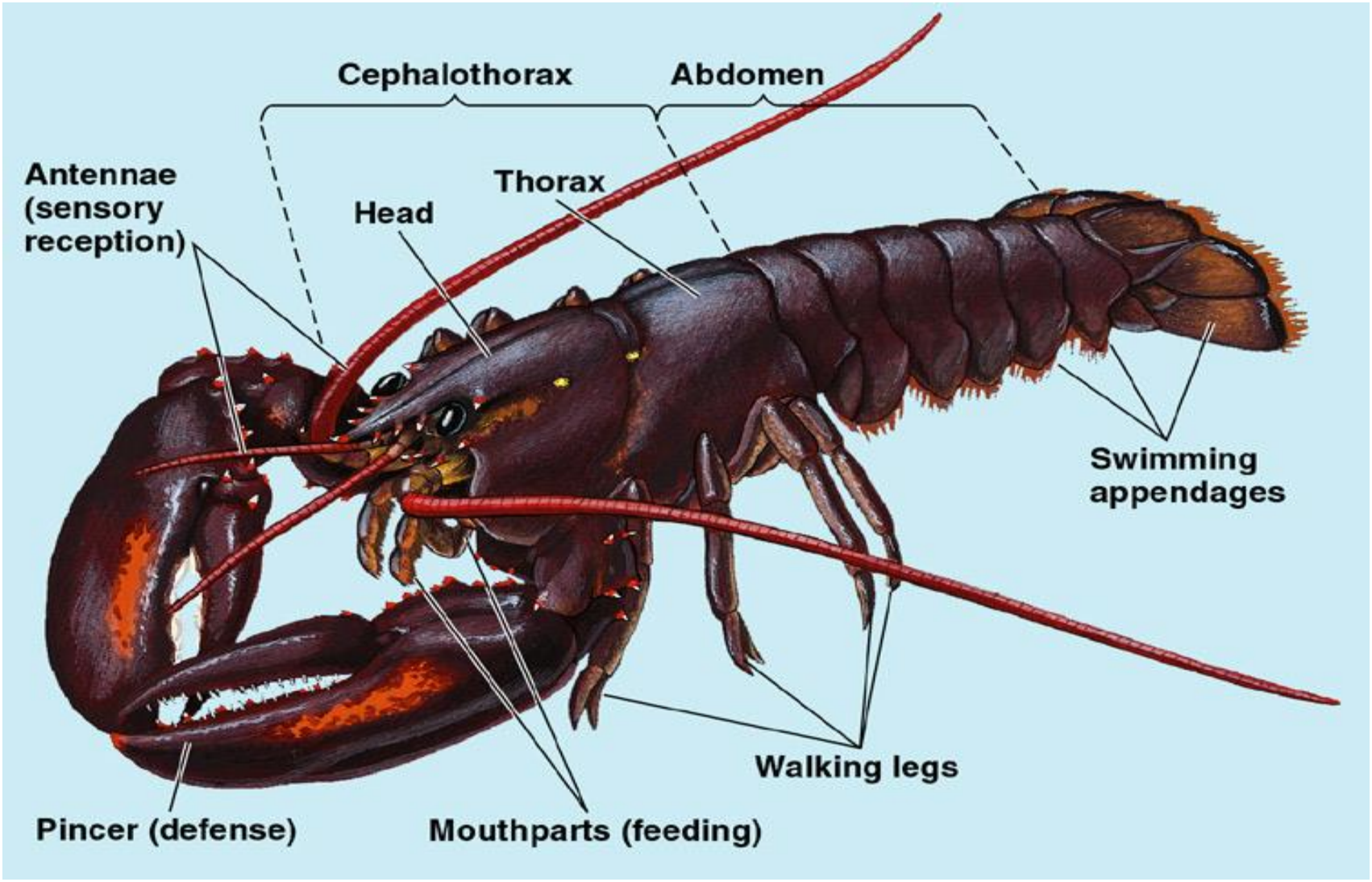
# Sensory Structures: Statocyst

- Other receptors involved with equilibrium, balance, and position senses are tactile receptors on the appendages and at joints.
- When a crustacean is crawling or resting, stretch receptors at the joints are stimulated.

# Sensory Structures: Statocyst

- Crustaceans detect tilting from changing patterns of stimulation.
- These widely distributed receptors are important to crustaceans that lack statocysts.

# Sensory Structures: Statocyst



# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

Compound eyes and ocelli

Lecture 153

# Compound Eyes and Ocelli

- Crayfish have compound eyes mounted on movable eyestalks.
- The lens system consists of 25 to 14,000 individual receptors called **ommatidia**.
- Compound eyes also occur in insects.

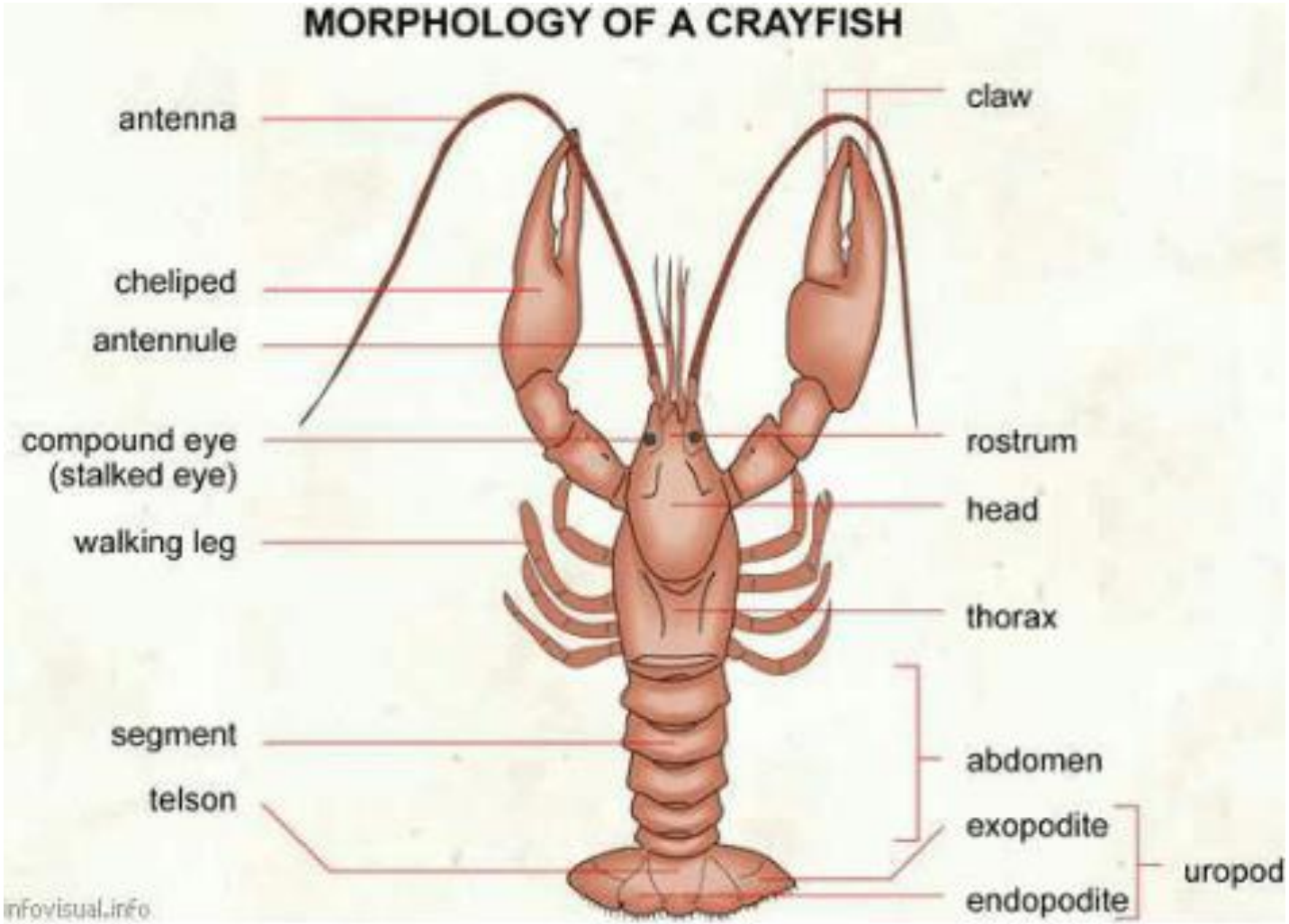
# Compound Eyes and Ocelli

- Larval crustaceans have a single, median photoreceptor consisting of a few sensilla.
- These simple eyes, called **ocelli**, allow larval crustaceans to orient toward or away from the light, but do not form images.

# Compound Eyes and Ocelli

- Many larvae are planktonic and use their ocelli to orient toward surface waters.

# Compound Eyes and Ocelli



# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

Androgenic glands

Lecture 154

# Androgenic Glands

- Androgenic glands in the cephalothorax of males mediate another endocrine function.
- Females possess rudiments of these glands during development, but the glands never mature.

# Androgenic Glands

- Normally, androgenic hormone(s) promotes the development of testes and male characteristics, such as gonopods.

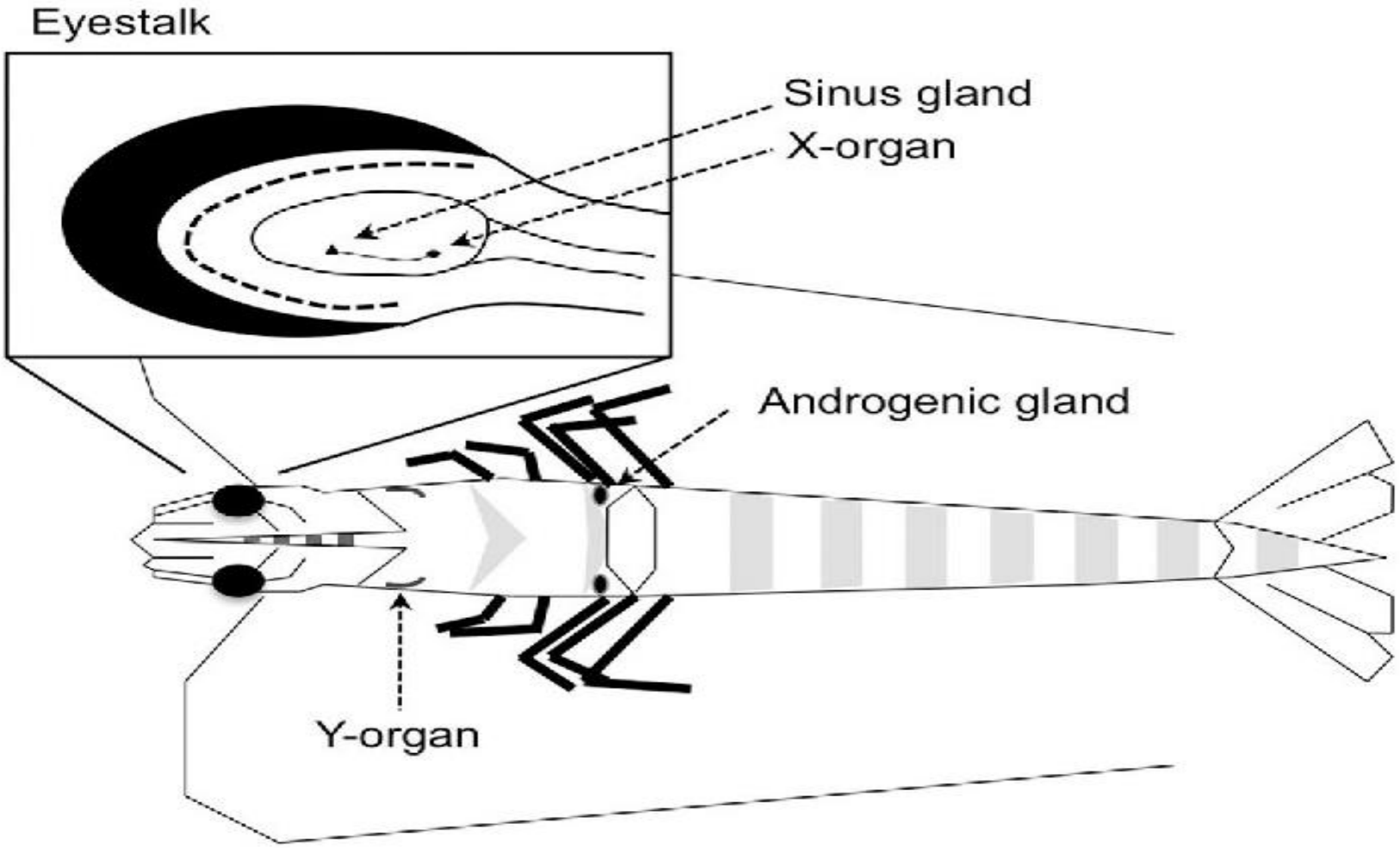
# Androgenic Glands

- The removal of androgenic glands from males results in the development of female sex characteristics, and if androgenic glands are experimentally implanted into a female, she develops testes and gonopods.

# Androgenic Glands

- Hormones probably regulate many other crustacean functions.
- Some that have been investigated include the development of female brooding structures in response to ovarian hormones, the seasonal regulation of ovarian functions, and the regulation of heart rate and body color changes by eyestalk hormones.

# Androgenic Glands



# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

Excretory organs: Antennal Glands/ Maxillary gGands

Lecture 155

# Excretory Organs: Antennal Glands/Maxillary Glands

- The excretory organs of crayfish are called **antennal glands** (green glands) because they are at the bases of the second antennae and are green in living crayfish.
- In other crustaceans, they are called **maxillary glands** because they are at the bases of the second maxillae.

# Excretory Organs: Antennal Glands/Maxillary Glands

- They are structurally similar to the coxal glands of arachnids and presumably had a common evolutionary origin.
- Excretory products form by the filtration of blood. Ions, sugars, and amino acids are reabsorbed in the tubule before the diluted urine is excreted.

# Excretory Organs: Antennal Glands/Maxillary Glands

- As with most aquatic animals, ammonia is the primary excretory product.
- However, crayfish do not rely solely on the antennal glands to excrete ammonia.
- Ammonia also diffuses across thin parts of the exoskeleton.

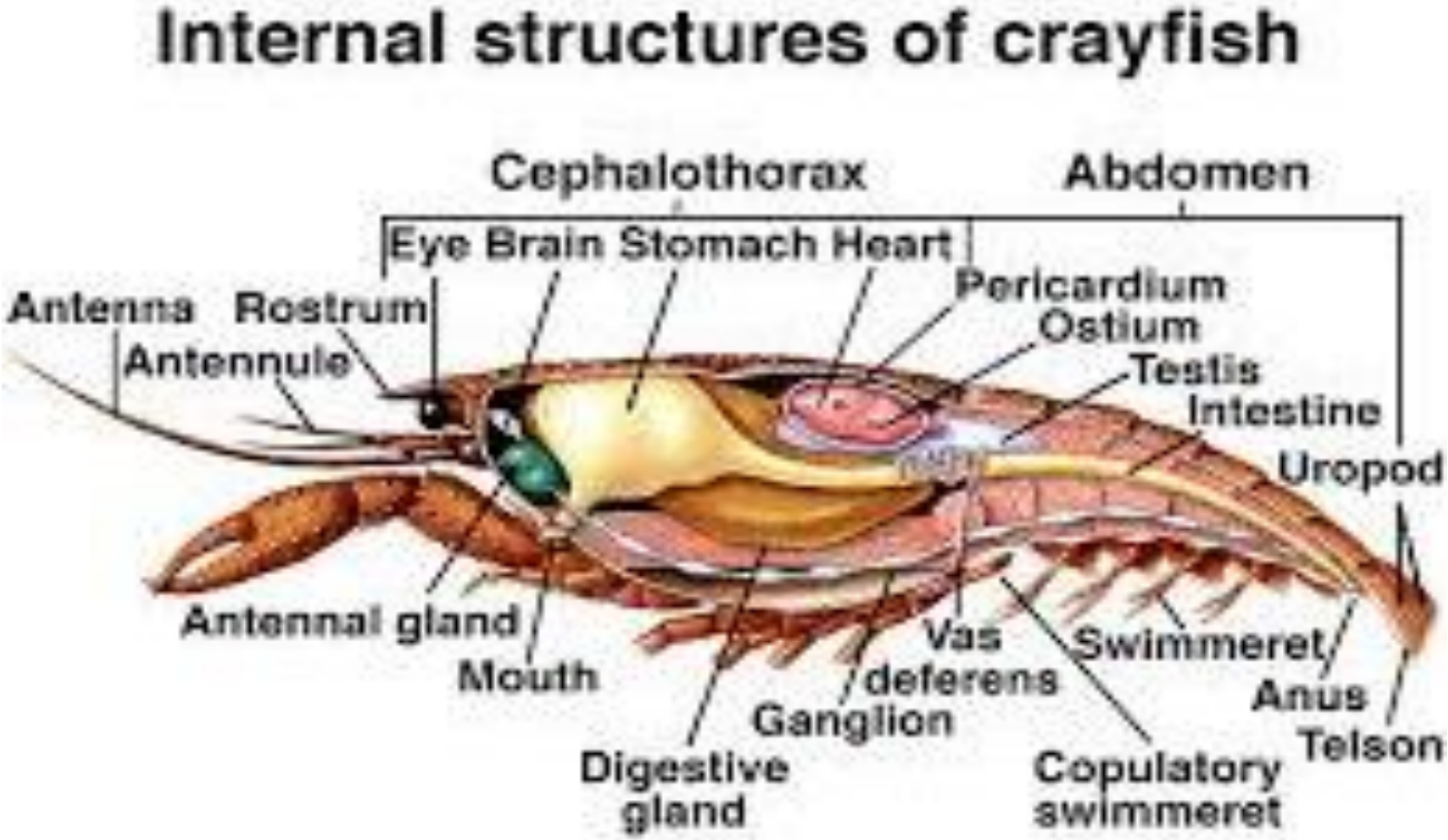
# Excretory Organs: Antennal Glands/Maxillary Glands

- Even though it is toxic, ammonia is water soluble, and water rapidly dilutes it.
- All freshwater crustaceans face a continual influx of freshwater and loss of ions.

# Excretory Organs: Antennal Glands/Maxillary Glands

- Thus, the elimination of excess water and the reabsorption of ions become extremely important functions.
- Gill surfaces are also important in ammonia excretion and water and ion regulation (osmoregulation).

# Excretory Organs: Antennal Glands/Maxillary Glands



# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

### Reproduction and Development

Lecture 156

# Reproduction and Development

- Crayfish, and all other crustaceans except the barnacles, are dioecious.
- Gonads are in the dorsal portion of the thorax, and gonoducts open at the base of the third (females) or fifth (males) periopods.
- Mating occurs just after a female has molted.

# Reproduction and Development

- The male turns the female onto her back and deposits non flagellated sperm near the openings of the female's gonoducts.
- Fertilization occurs after copulation, as the eggs are shed.
- The eggs are sticky and securely fasten to the female's pleopods.

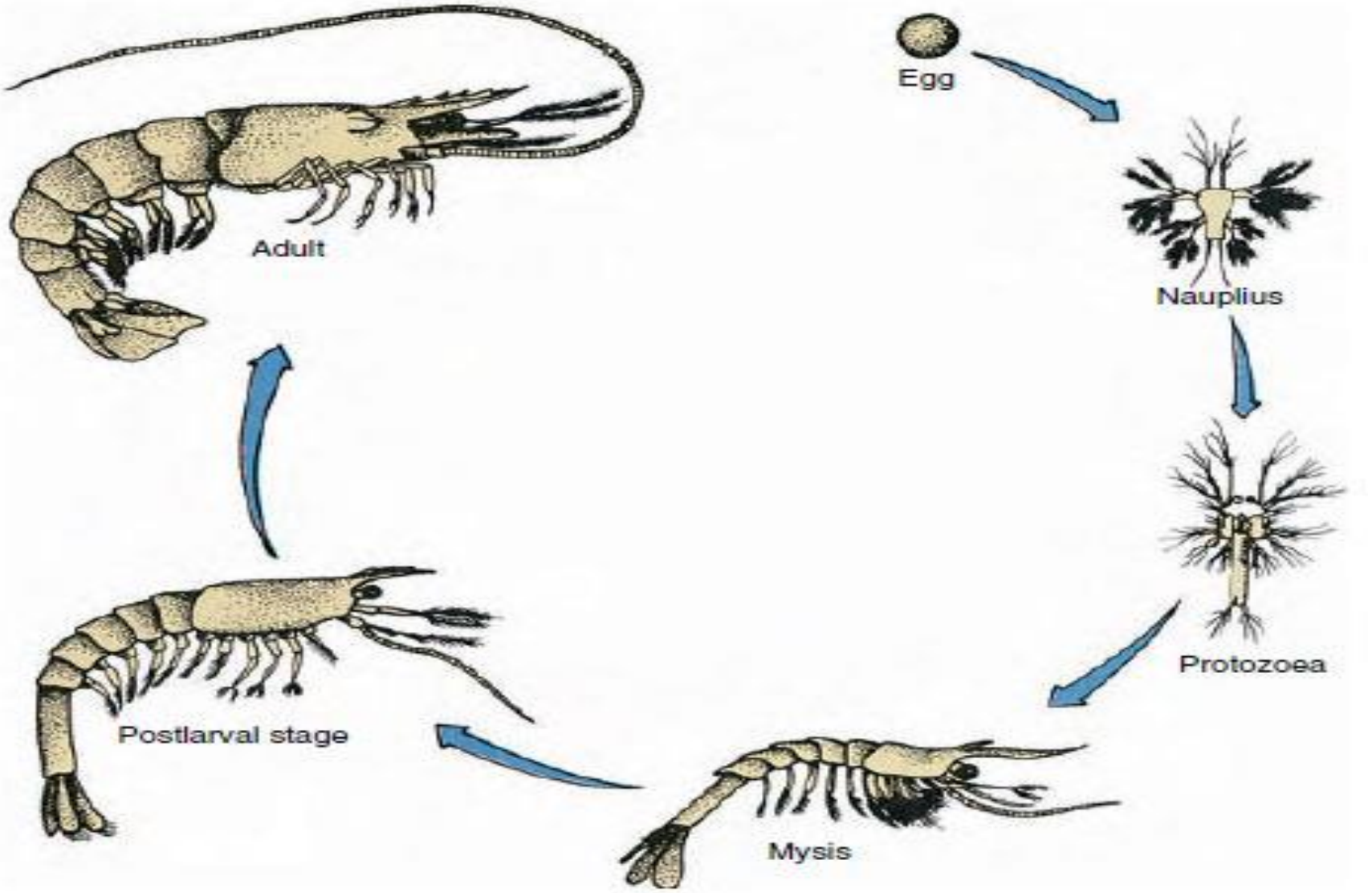
# Reproduction and Development

- Fanning movements of the pleopods over the eggs keep the eggs aerated.
- The development of crayfish embryos is direct, with young hatching as miniature adults.
- Many other crustaceans have a planktonic, free-swimming larva called a **nauplius**.

# Reproduction and Development

- In some, the nauplius develops into a miniature adult.
- Crabs and their relatives have a second larval stage called a **zoea**.
- When all adult features are present, except sexual maturity, the immature is called the **post larva**.

# Reproduction and Development



# Reproduction and Development

- Two other orders of malacostracans have members familiar to most humans.
- Members of the order Isopoda include “pillbugs.”
- Isopods are dorsoventrally flattened, may be either aquatic or terrestrial, and scavenge decaying plant and animal material.

# Reproduction and Development

- Some have become modified for clinging to and feeding on other animals.
- Terrestrial isopods live under rocks and logs and in leaf litter.
- Members of the order Amphipoda have a laterally compressed body that gives them a shrimplike appearance.

# Reproduction and Development

- Amphipods move by crawling or swimming on their sides along the substrate.
- Some species are modified for burrowing, climbing, or jumping.
- Amphipods are scavengers, and a few species are parasites.

END

# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

### Class Branchiopoda

Lecture 157

# Class Branchiopoda

- Members of the class Branchiopoda (Gr. *branchio,* gill *podos,* foot) primarily live in freshwater.
- All branchiopods possess flattened, leaflike appendages used in respiration, filter feeding, and locomotion.
- Fairy shrimp and brine shrimp comprise the order Anostraca.

# Class Branchiopoda

- Fairy shrimp usually live in temporary ponds that spring thaws and rains form.
- Eggs are brooded, and when the female dies, and the temporary pond begins to dry, the embryos become dormant in a resistant capsule.

# Class Branchiopoda

- Embryos lay on the forest floor until the pond fills again the following spring, at which time they hatch into nauplius larvae.
- Animals, wind, or water currents may carry the embryos to other locations.
- Their short and uncertain life cycle is an adaptation to living in ponds that dry up.

# Class Branchiopoda

- The vulnerability of these slowly swimming and defenseless crustaceans probably explains why they live primarily in temporary ponds, a habitat that contains few larger predators.
- Brine shrimp also form resistant embryos.
- They live in salt lakes and ponds (e.g., the Great Salt Lake in Utah).

# Class Branchiopoda

- Members of the order Cladocera are called **water fleas.**
- A large carapace covers their bodies, and they swim by repeatedly thrusting their second antennae downward to create a jerky, upward locomotion.

# Class Branchiopoda

- Females reproduce parthenogenetically (without fertilization) in spring and summer, and can rapidly populate a pond or lake.
- Eggs are brooded in an egg case beneath the carapace.
- At the next molt, the egg case is released and either floats or sinks to the bottom of the pond or lake.

# Class Branchiopoda

- In response to decreasing temperature, changing photoperiod, or decreasing food supply, females produce eggs that develop parthenogenetically into males.
- Sexual reproduction produces resistant “winter eggs” that overwinter and hatch in the spring.

# Class Branchiopoda

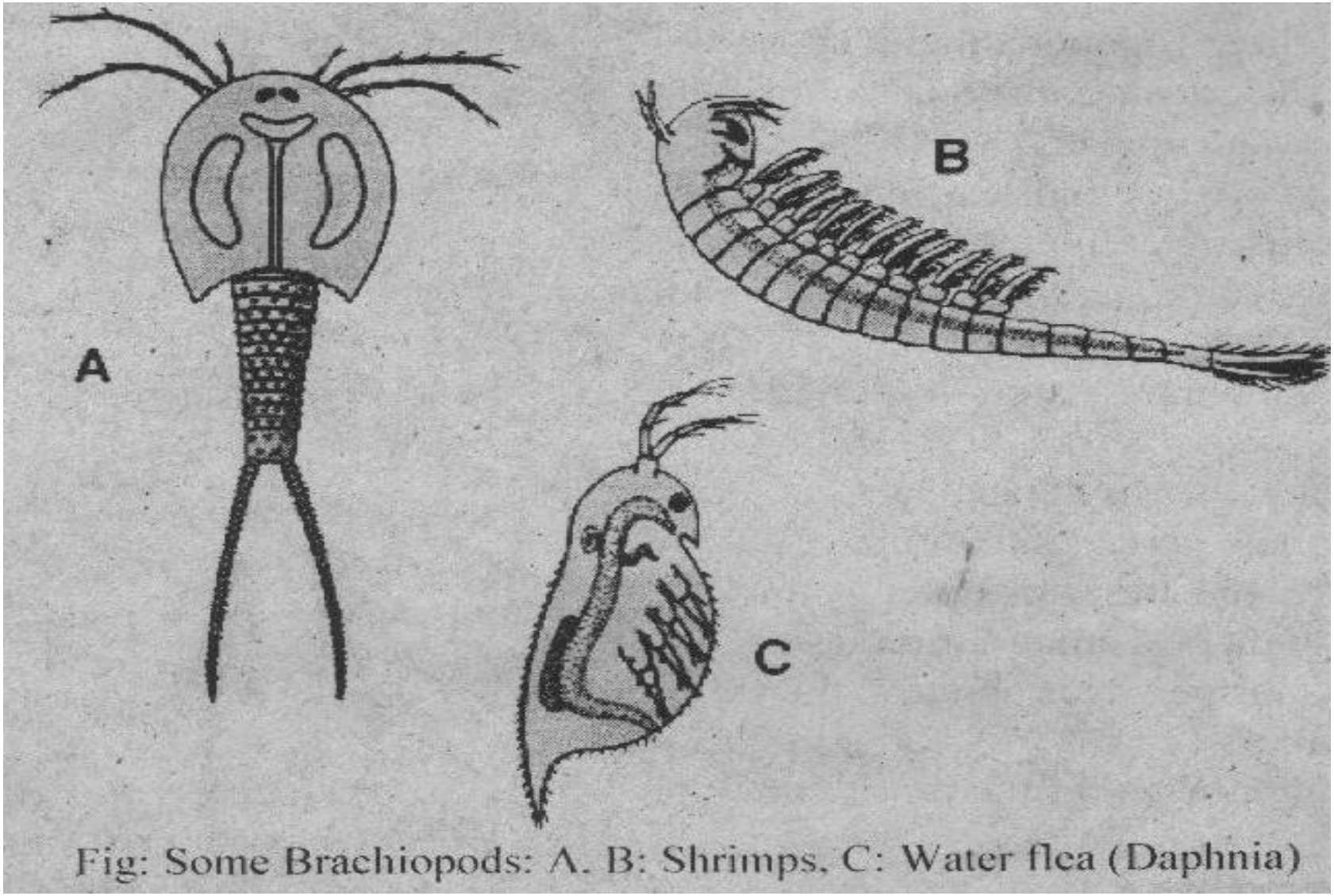



Fig: Some Brachiopods: A. B: Shrimps, C: Water flea (Daphnia)

# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

### Class Copepoda

Lecture 158

# Class Copepoda

- Members of the class **Copepoda** (Gr. *kope,* oar *podos,* foot) include some of the most abundant crustaceans.
- There are both marine and freshwater species.
- Copepods have a cylindrical body and a median ocellus that develops in the nauplius stage and persists into the adult stage.

# Class Copepoda

- The first antennae (and the thoracic appendages in some) are modified for swimming, and the abdomen is free of appendages.
- Most copepods are planktonic and use their second maxillae for filter feeding.

# Class Copepoda

- Their importance in marine food webs was noted in the “Evolutionary Perspective”.
- A few copepods live on the substrate, a few are predatory, and others are commensals or parasites of marine invertebrates, fishes, or marine mammals.

# Class Copepoda

## Class Copepoda

Copepods



# Animal Diversity: Invertebrates

## Sub-Phylum Crustacea

### Class Cirripedia

Lecture 159

# Class Cirripedia

- Members of the class Cirripedia, the barnacles, are sessile and highly modified as adults.
- They are exclusively marine and include about one thousand species.
- Most barnacles are monoecious.

# Class Cirripedia

- The planktonic nauplius of barnacles is followed by a planktonic larval stage, called a **cypris** larva, which has a bivalved carapace.
- Cypris larvae attach to the substrate by their first antennae and metamorphose to adults.
- In the process of metamorphosis, the abdomen is reduced, and the gut tract becomes U-shaped.

# Class Cirripedia

- Thoracic appendages are modified for filtering and moving food into the mouth.
- Calcareous plates cover the larval carapace in the adult stage.
- Barnacles attach to a variety of substrates, including rock outcroppings, ship bottoms, whales, and other animals.

# Class Cirripedia

- Some barnacles attach to their substrate by a stalk.
- Others are non stalked and are called acorn barnacles.
- Barnacles that colonize ship bottoms reduce both ship speed and fuel efficiency.
- Much time, effort, and money have been devoted to research on keeping ships free of barnacles.

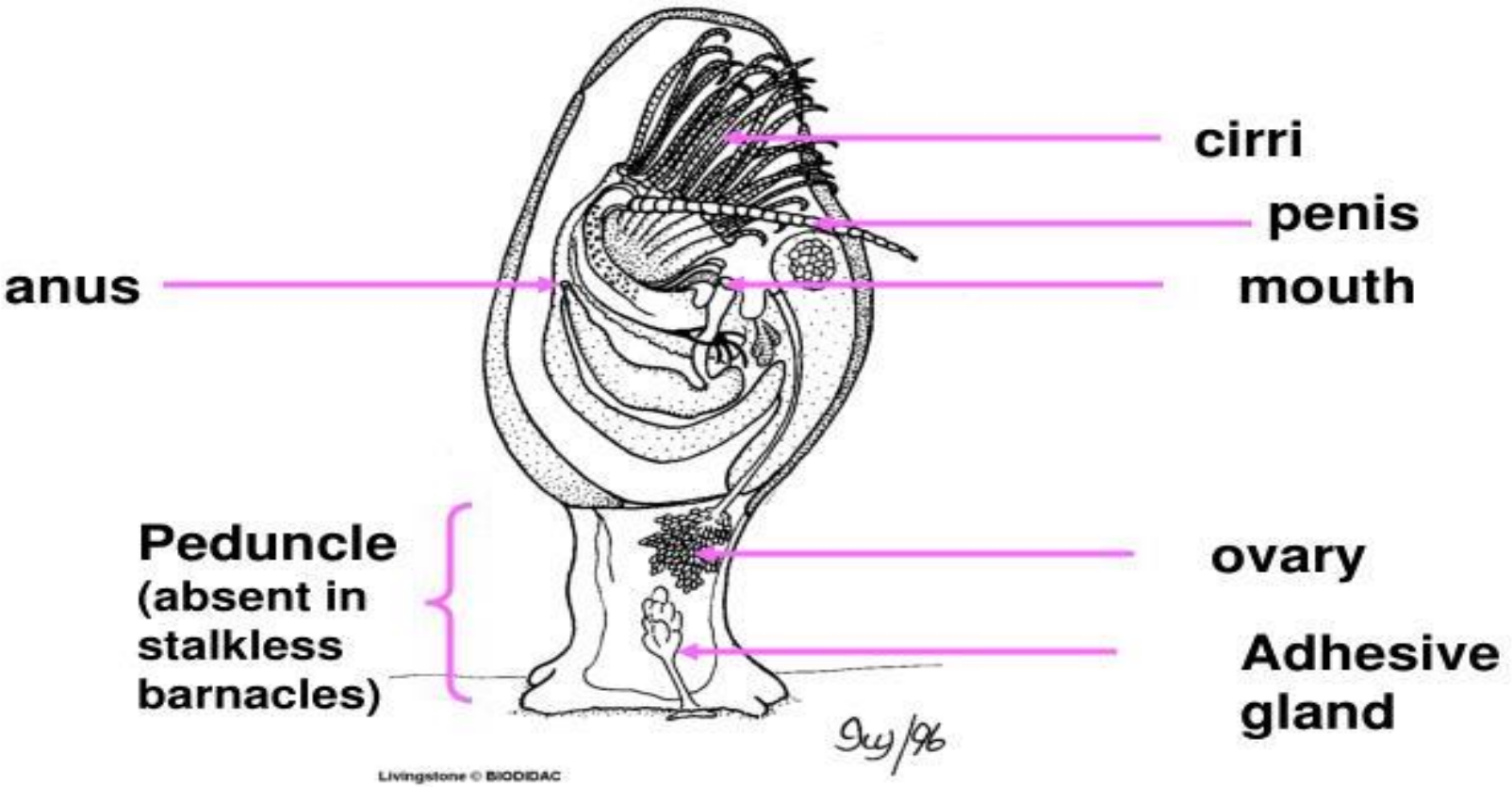
# Class Cirripedia

- Some barnacles have become highly modified parasites.
- **The** evolution of parasitism in barnacles is probably a logical consequence of living attached to other animals.

# Class Cirripedia

## Class Cirripedia

- modified body form



# Animal Diversity: Invertebrates

## Sub-Phylum Chelicerata

### Class Pycnogonida

Lecture 160

# Class Pycnogonida

- Members of the class Pycnogonida are the sea spiders.
- All are marine and are most common in cold waters.
- Pycnogonids live on the ocean floor and frequently feed on cnidarian polyps and ectoprocts.

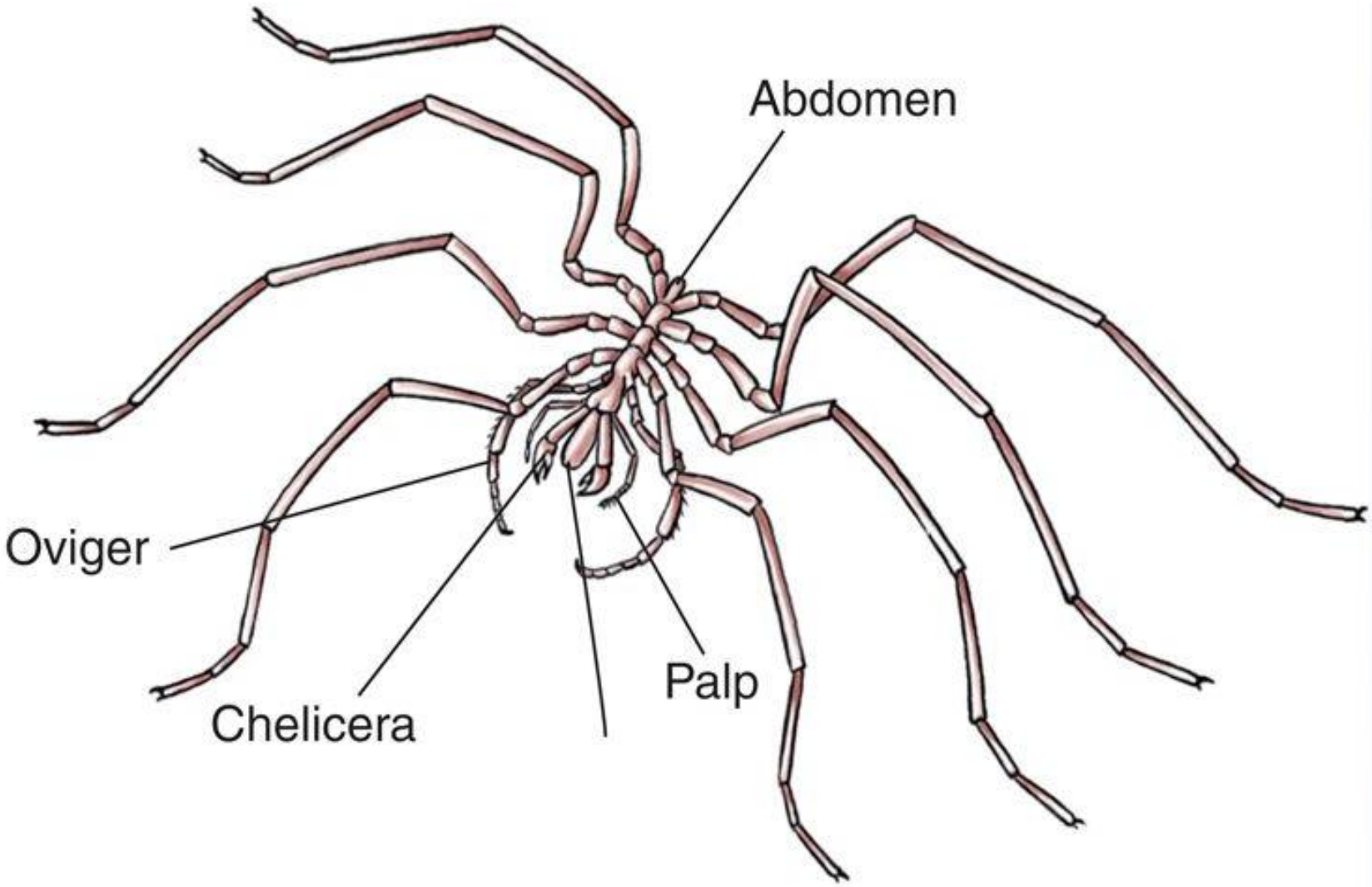
# Class Pycnogonida

- Some sea spiders feed by sucking prey tissues through a proboscis.
- Others tear at prey with their chelicerae.
- Pycnogonids are dioecious.
- Gonads are U-shaped, and branches of the gonads extend into each leg.

# Class Pycnogonida

- Gonopores are on one of the pairs of legs.
- As the female releases eggs, the male fertilizes them, and the fertilized eggs are cemented into spherical masses and attached to a pair of elongate appendages of the male, called **ovigers**, where they are brooded until hatching.

# Class Pycnogonida



# Animal Diversity: Invertebrates

## Phylum Arthropoda

Further phylogenetic considerations

Lecture 161

# Further Phylogenetic Considerations

- The arthropods have been very successful.
- This is evidenced by the diverse body forms and lifestyles of copepods, crabs, lobsters, crayfish, and barnacles, which demonstrate adaptive radiation.
- Few aquatic environments are without some crustaceans.

# Further Phylogenetic Considerations

- The subphylum Chelicerata is a very important group of animals from an evolutionary standpoint, even though they are less numerous in terms of numbers of species and individuals than are many of the crustacean groups.

# Further Phylogenetic Considerations

- **Their arthropod** exoskeleton and the evolution of excretory and respiratory systems that minimize water loss resulted in ancestral members of this subphylum becoming some of the first terrestrial animals.

# Further Phylogenetic Considerations

- Chelicerates, however, are not the only terrestrial arthropods.
- In terms of numbers of species and numbers of individuals, chelicerates are dwarfed in terrestrial environments by the fourth arthropod lineage—the insects and their relative.

# Further Phylogenetic Considerations

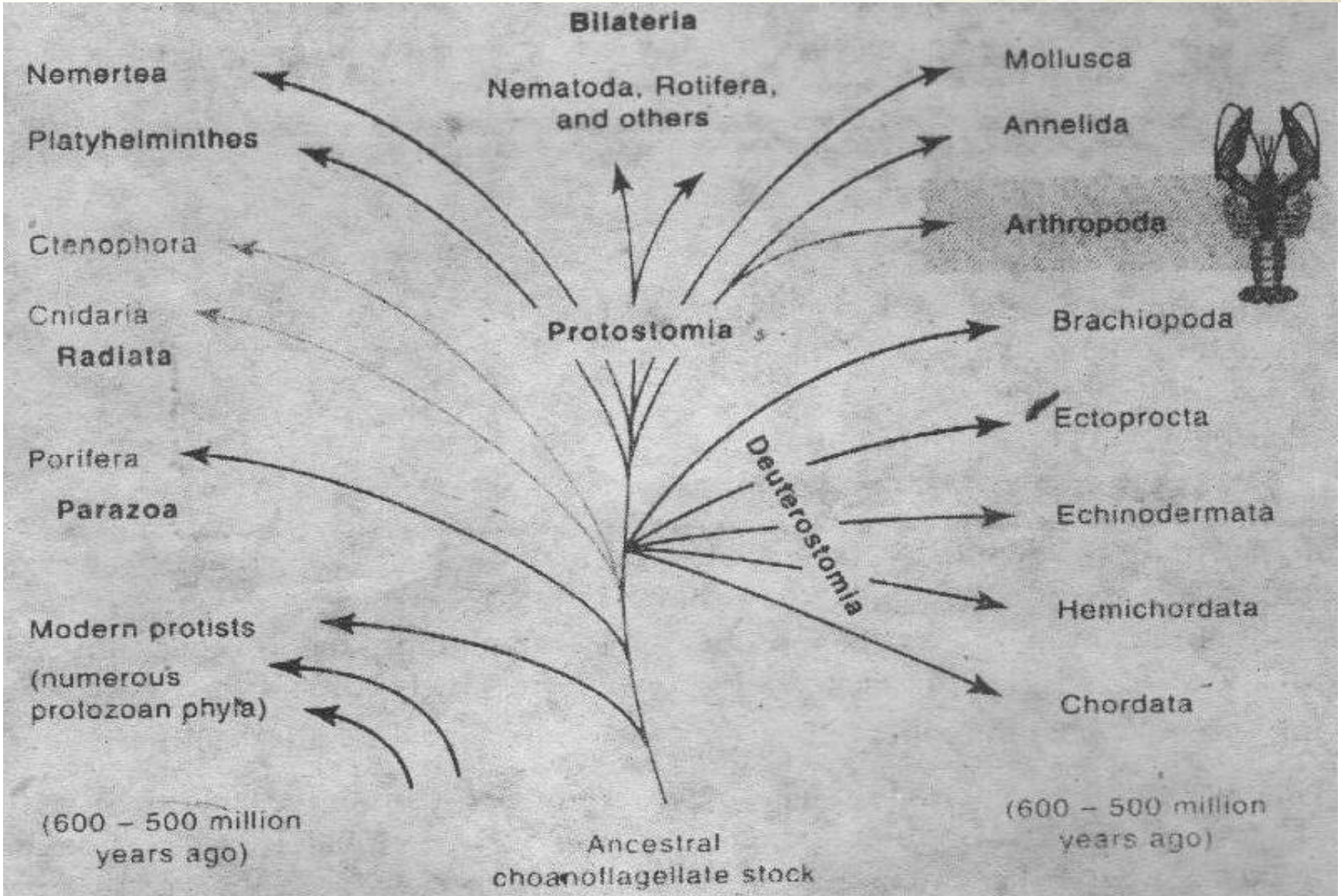



Fig: Evolutionary Relationships of the Arthropods.

# The Body Wall and Nematocysts

- Cells of epidermis and gastrodermis differentiate into a number of cell types for:
  - protection
  - food gathering
  - coordination
  - movement
  - digestion
  - absorption
- Between the epidermis and gastrodermis is a jellylike layer called **mesoglea**.

# The Body Wall and Nematocysts

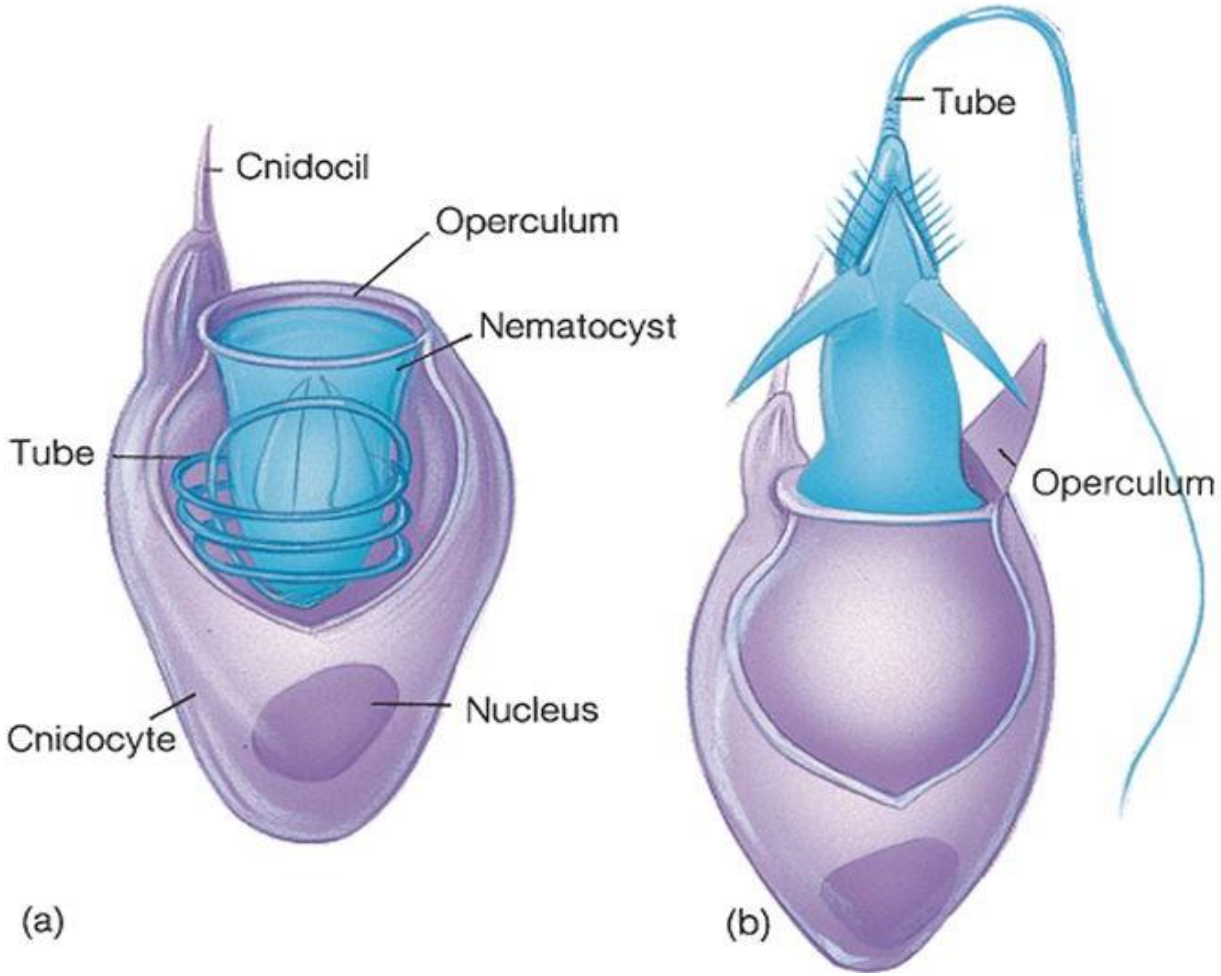
- Cells are present in the middle layer of some cnidarians, but they have their origin in either the epidermis or the gastrodermis.
- One kind of cell is characteristics of the phylum.

# The Body Wall and Nematocysts

- Epidermal and/or gastrodermal cells called **cnidocytes** produce structures called **nematocysts**, which are used for attachment, defense and feeding.
- A **nematocyst** is a fluid-filled, intracellular capsule enclosing a coiled hollow tube.

# The Body Wall and Nematocysts

**Cnidocyte Structure and Nematocyst Discharge**



# The Body Wall and Nematocysts

- A lid like operculum caps the capsule at one end.
- The cnidocyte has a modified cilium, called a **cnidocil**.
- Stimulation of the cnidocil forces open the operculum, discharging the coiled tube.
- Zoologist have described nearly 30 kinds of nematocysts.

# The Body Wall and Nematocysts

- Nematocysts used in food gathering and defense may discharge a long tube armed with spines that penetrates the prey.
- The spines have hollow tips that deliver paralyzing toxins.

# The Body Wall and Nematocysts

- Other nematocysts contain unarmed tubes that wrap around prey or a substrate.
- Still other nematocysts have sticky secretions that help the animal anchor itself.
- Six or more kinds of nematocysts may be present in one individual.

# Animal Diversity: Invertebrates

## Sub-Phylum Myriapoda

### Class Diplopoda

Lecture 162

# Class Diplopoda

- The class Diplopoda (Gr. *diploos,* two fold *podus,* foot) contains the millipedes.
- Ancestors of this group appeared on land during the Devonian period and were among the first terrestrial animals.

# Class Diplopoda

- Millipedes have 11 to 100 trunk segments derived from an embryological and evolutionary fusion of primitive metameres.
- An obvious result of this fusion is the occurrence of two pairs of appendages on each apparent trunk segment.
- Each segment is actually the fusion of two segments.

# Class Diplopoda

- Fusion is also reflected internally by two ganglia, two pairs of ostia, and two pairs of tracheal trunks per apparent segment.
- Most millipedes are round in cross section, although some are more flattened.
- Millipedes are worldwide in distribution and are nearly always found in or under leaf litter, humus, or decaying logs.

# Class Diplopoda

- Their epicuticle does not contain much wax; therefore, their choice of habitat is important to prevent desiccation.
- Their many legs, simultaneously pushing against the substrate, help millipedes bulldoze through the habitat.
- Millipedes feed on decaying plant matter using their mandibles in a chewing or scraping fashion.

# Class Diplopoda

- A few millipedes have mouthparts modified for sucking plant juices.
- Millipedes roll into a ball when faced with desiccation or when disturbed.
- Many also possess repugnatorial glands that pro-duce hydrogen cyanide, which repels other animals.

# Class Diplopoda

- Hydrogen cyanide is not synthesized and stored as hydrogen cyanide because it is caustic and would destroy millipede tissues.
- Instead, a precursor compound and an enzyme mix as they are released from separate glandular compartments.

# Class Diplopoda

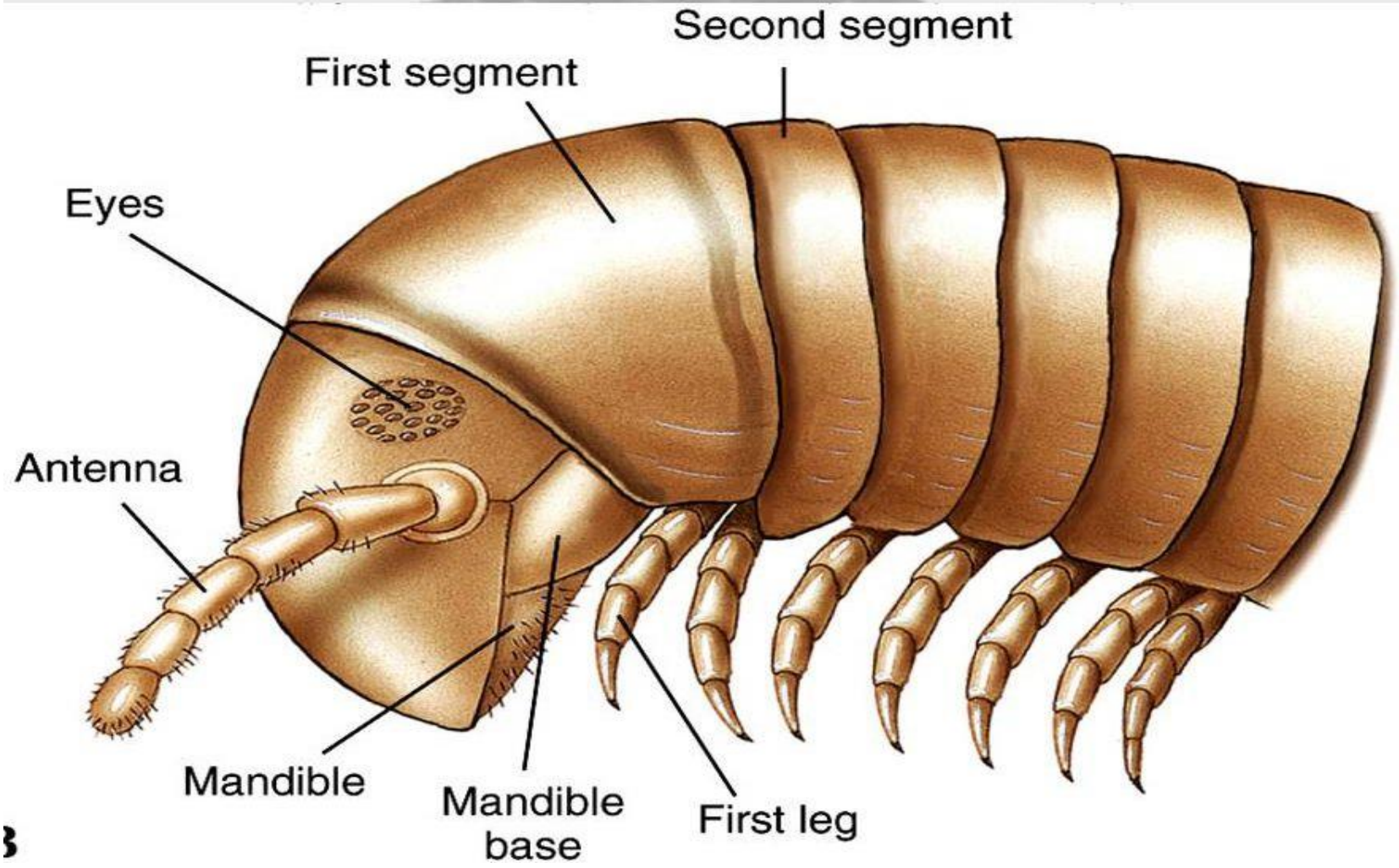
- Repellants increase the likelihood that the millipede will be dropped unharmed and decrease the chances that the same predator will try to feed on another millipede.
- Male millipedes transfer sperm to female millipedes with modified trunk appendages, called gonopods, or in spermatophores.

# Class Diplopoda

- Eggs are fertilized as they are laid and hatch in several weeks.
- Immatures acquire more legs and segments with each molt until they reach adulthood.

# Class Diplopoda

## SP Myriapoda, Class Diplopoda



# Animal Diversity: Invertebrates

## Sub-Phylum Myriapoda

### Class Chilopoda

Lecture 163

# Class Chilopoda

- Members of the class Chilopoda (Gr. *cheilos,* lip and *podus,* foot) are the centipedes.
- Most centipedes are nocturnal and scurry about the surfaces of logs, rocks, or other forest-floor debris.
- Like millipedes, most centipedes lack a waxy epicuticle and therefore require moist habitats.

# Class Chilopoda

- Their bodies are flattened in cross section, and they have a single pair of long legs on each of their 15 or more trunk segments.
- The last pair of legs is usually modified into long sensory appendages.
- Centipedes are fast-moving predators.

# Class Chilopoda

- Food usually consists of small arthropods, earthworms, and snails; however, some centipedes feed on frogs and rodents.
- Poison claws (modified first-trunk appendages called maxillipeds) kill or immobilize prey.
- Maxillipeds, along with mouth appendages, hold the prey as mandibles chew and ingest the food.

# Class Chilopoda

- Most centipede venom is essentially harmless to humans, although many centipedes have bites that are comparable to wasp stings; a few human deaths have been reported from large, tropical species.
- Centipede reproduction may involve courtship displays in which the male lays down a silk web using glands at the posterior tip of the body.

# Class Chilopoda

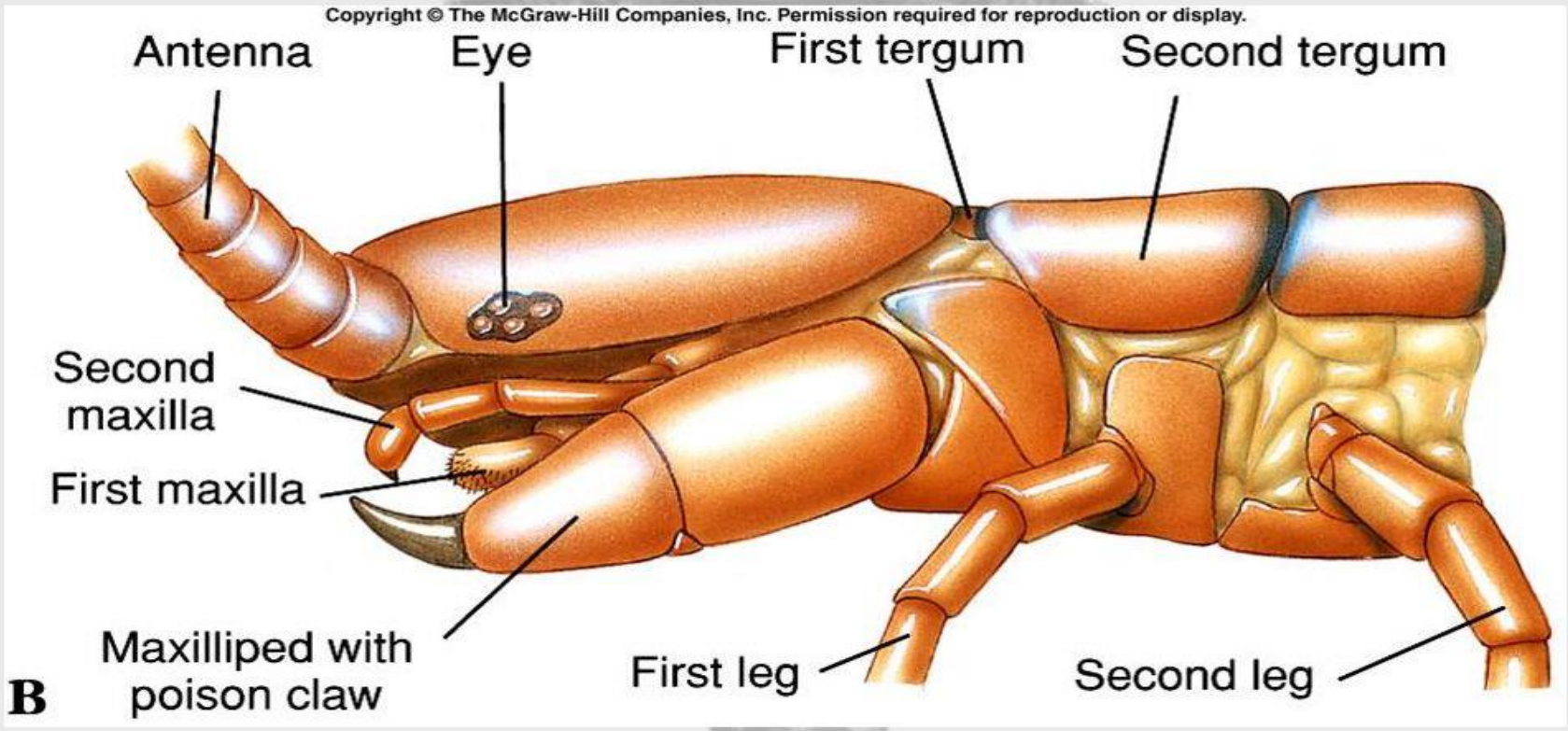
- He places a spermatophore in the web, which the female picks up and introduces into her genital opening.
- Eggs are fertilized as they are laid.
- A female may brood and guard eggs by wrapping her body around the eggs, or they may be deposited in the soil.

# Class Chilopoda

- Young are similar to adults except that they have fewer legs and segments.
- Legs and segments are added with each molt.

# Class Chilopoda

## SP Myriapoda, Class Chilopoda




## Centipedes

# Animal Diversity: Invertebrates

## Phylum Euarthropoda

### Class Hexapoda: General features and external structures

Lecture 164

# General Features and External Structures

- Members of the class Hexapoda (Gr. *hexa,* six *and podus,* feet) are, in terms of numbers of species and individuals, the most successful land animals.
- Order-level classification varies, depending on the authority consulted.

# General Features and External Structures

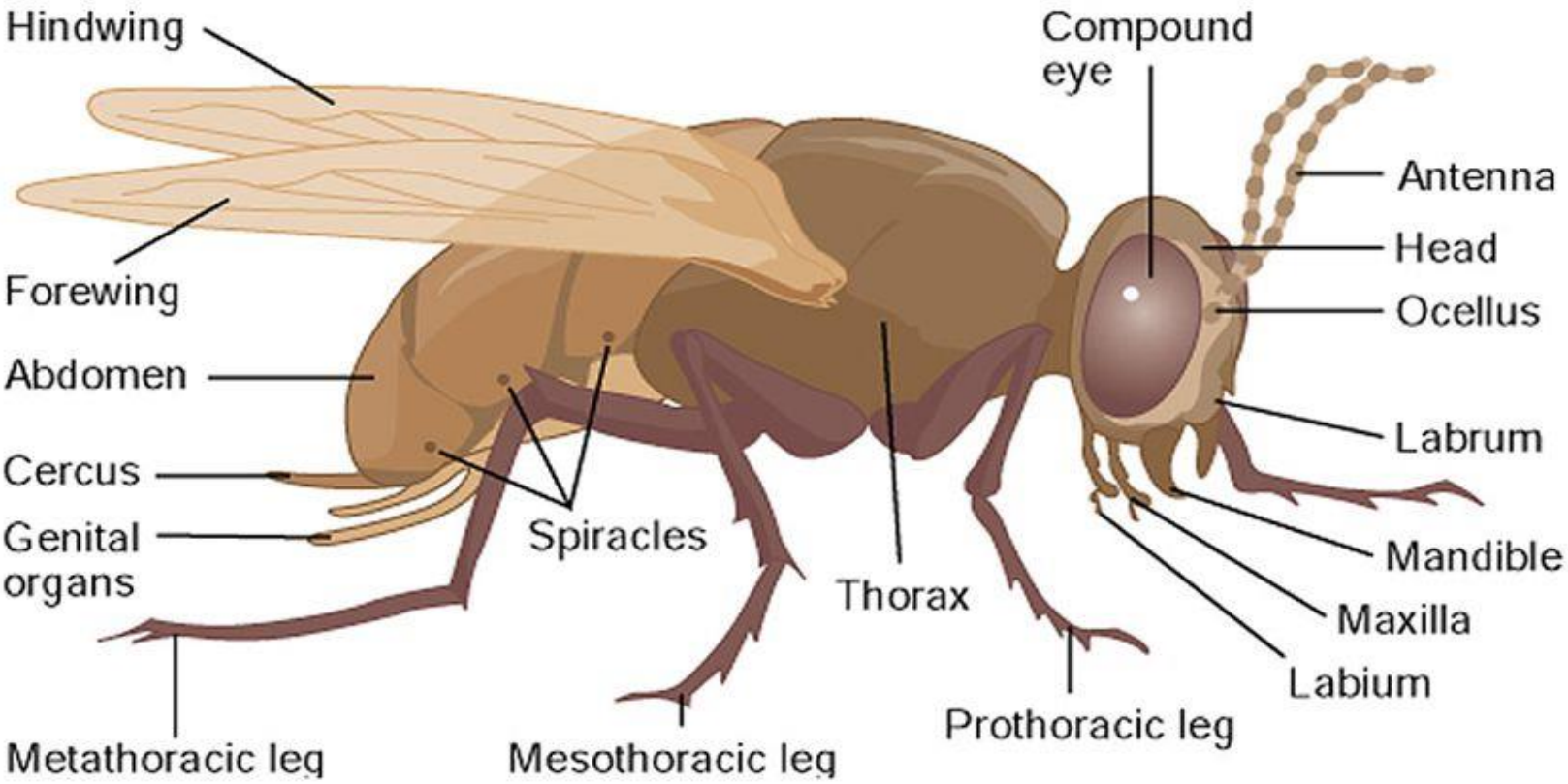
- In spite of obvious diversity, common features make insects easy to recognize.
- Many insects have wings and one pair of antennae, and virtually all adults have three pairs of legs.
- The body of an insect is divided into three tagmata: **head**, **thorax**, and **abdomen**.

# General Features and External Structures

- The head bears a single pair of antennae, mouthparts, compound eyes, and zero, two, or three ocelli.
- The thorax consists of three segments.
- They are, from anterior to posterior, the **prothorax,** the **mesothorax**, and the **metathorax**.

# General Features and External Structures

**External Structure of a Generalized Insect**



# Class Hexapoda: General Features and External Structures

- One pair of legs attaches along the ventral margin of each thoracic segment, and a pair of wings, when present, attaches at the dorsolateral margin of the mesothorax and metathorax.
- Wings have thickened, hollow veins for increased strength.

# Class Hexapoda: General Features and External Structures

- The thorax also contains two pairs of spiracles, which are openings to the tracheal system.
- Most insects have 10 or 11 abdominal segments, each of which has a lateral fold in the exoskeleton that allows the abdomen to expand when the insect has gorged itself or when it is full of mature eggs.

# Class Hexapoda: General Features and External Structures

- Each abdominal segment has a pair of spiracles.
- Also present are genital structures used during copulation and egg deposition, and sensory structures called cerci.
- Gills are present on abdominal segments of certain immature aquatic insects.

# Animal Diversity: Invertebrates

## Sub-Phylum Myriapoda

Classes Pauropoda and Symphyla

Lecture 165

# Classes Pauropoda and Symphyla

- Members of the class Pauropoda (Gr. *pauros,* small and *podus,* foot) are soft-bodied animals with 11 segments.
- These animals live in forest-floor litter, where they feed on fungi, humus, and other decaying organic matter.

# Classes Pauropoda and Symphyla

- Their very small size and thin, moist exoskeleton allow gas exchange across the body surface and diffusion of nutrients and wastes in the body cavity.

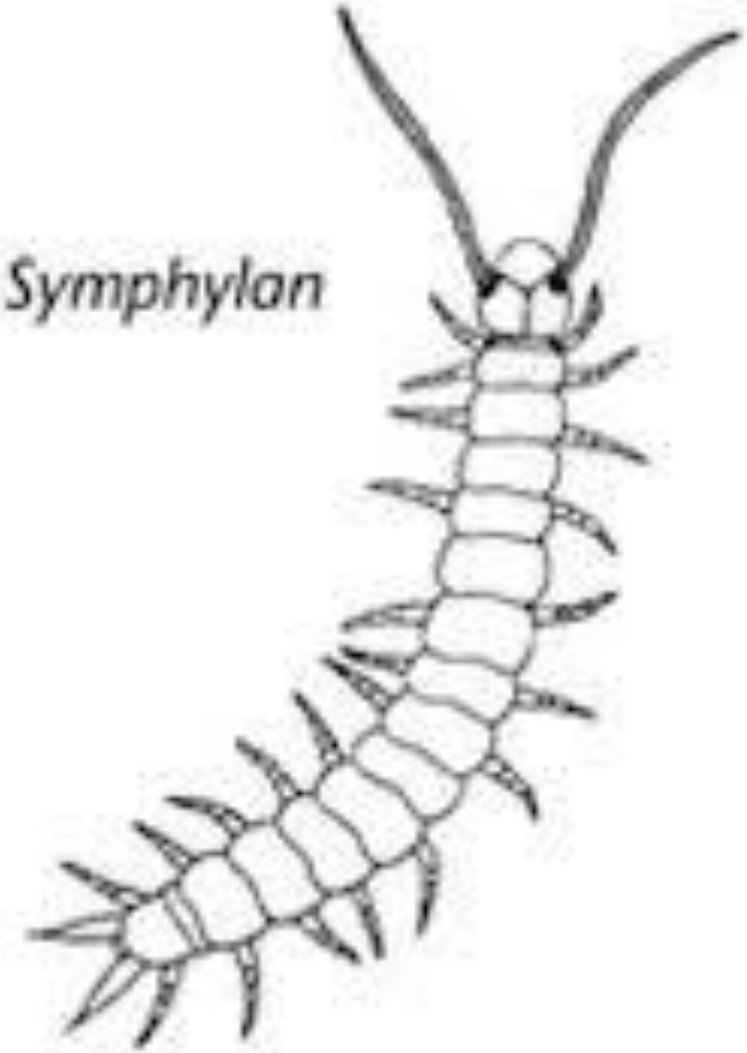
# Classes Pauropoda and Symphyla

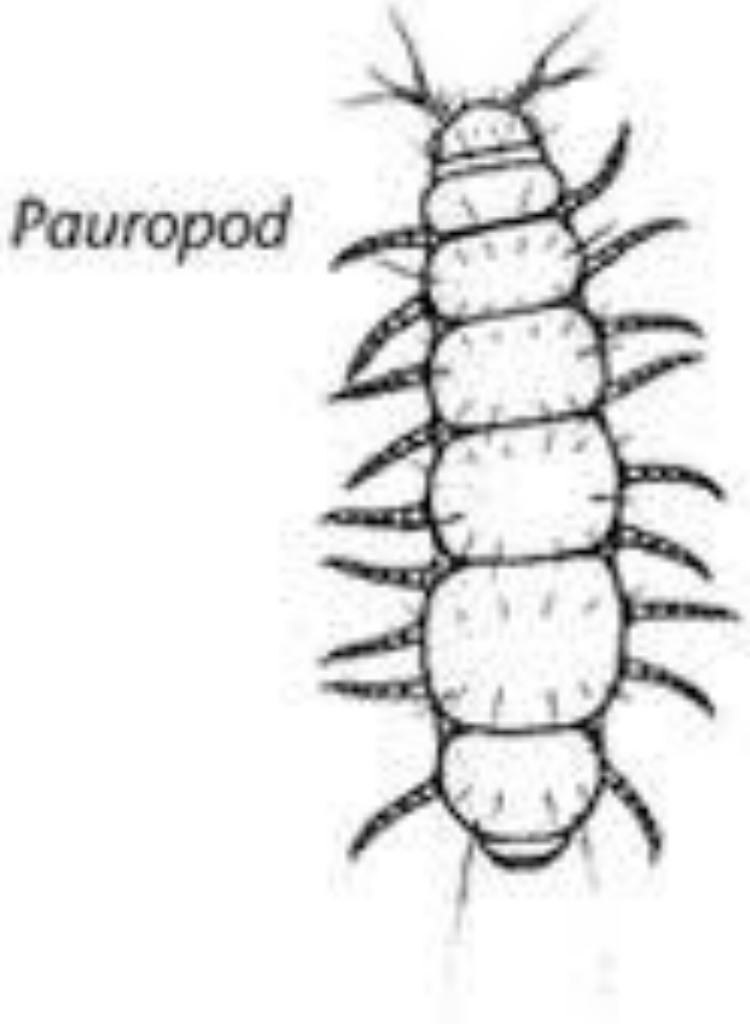
- Members of the class Symphyla (sim-filah) (Gr. *sym,* same *phyllos,* leaf) are small arthropods (2 to 10 mm in length) that occupy soil and leaf mold, superficially resemble centipedes, and are often called **garden centipedes**.

# Classes Pauropoda and Symphyla

- They lack eyes and have 12 leg-bearing trunk segments.
- The posterior segment may have one pair of spinnerets or long, sensory bristles.
- Symphylans normally feed on decaying vegetation; however, some species are pests of vegetables and flowers.

# Classes Pauropoda and Symphyla



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Insect Flight

Lecture 166

# Insect Flight

- Insects move in diverse ways.
- **From an evolutionary perspective**, however, flight is the most important form of insect locomotion.
- Insects were the first animals to fly.

# Insect Flight

- One of the most popular hypotheses on the origin of flight states that wings may have evolved from rigid, lateral outgrowths of the thorax that probably protected the legs or spiracles.
- Later, these fixed lobes could have been used in gliding from the top of tall plants to the forest floor.

# Insect Flight

- The ability of the wing to flap, tilt, and fold back over the body probably came later.

- Another requirement for flight was the evolution of limited thermoregulatory abilities.

# Insect Flight

- Thermoregulation is the ability to maintain body temperatures at a level different from environmental temperatures.
- Relatively high body temperatures, perhaps 25° C or greater, are needed for flight muscles to contract rapidly enough for flight.

# Insect Flight






# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Synchronous flight

Lecture 167

# Synchronous Flight

- Some insects use a **direct or synchronous flight mechanism**, in which muscles acting on the bases of the wings contract to produce a downward thrust, and muscles attaching dorsally and ventrally on the exoskeleton contract to produce an upward thrust.

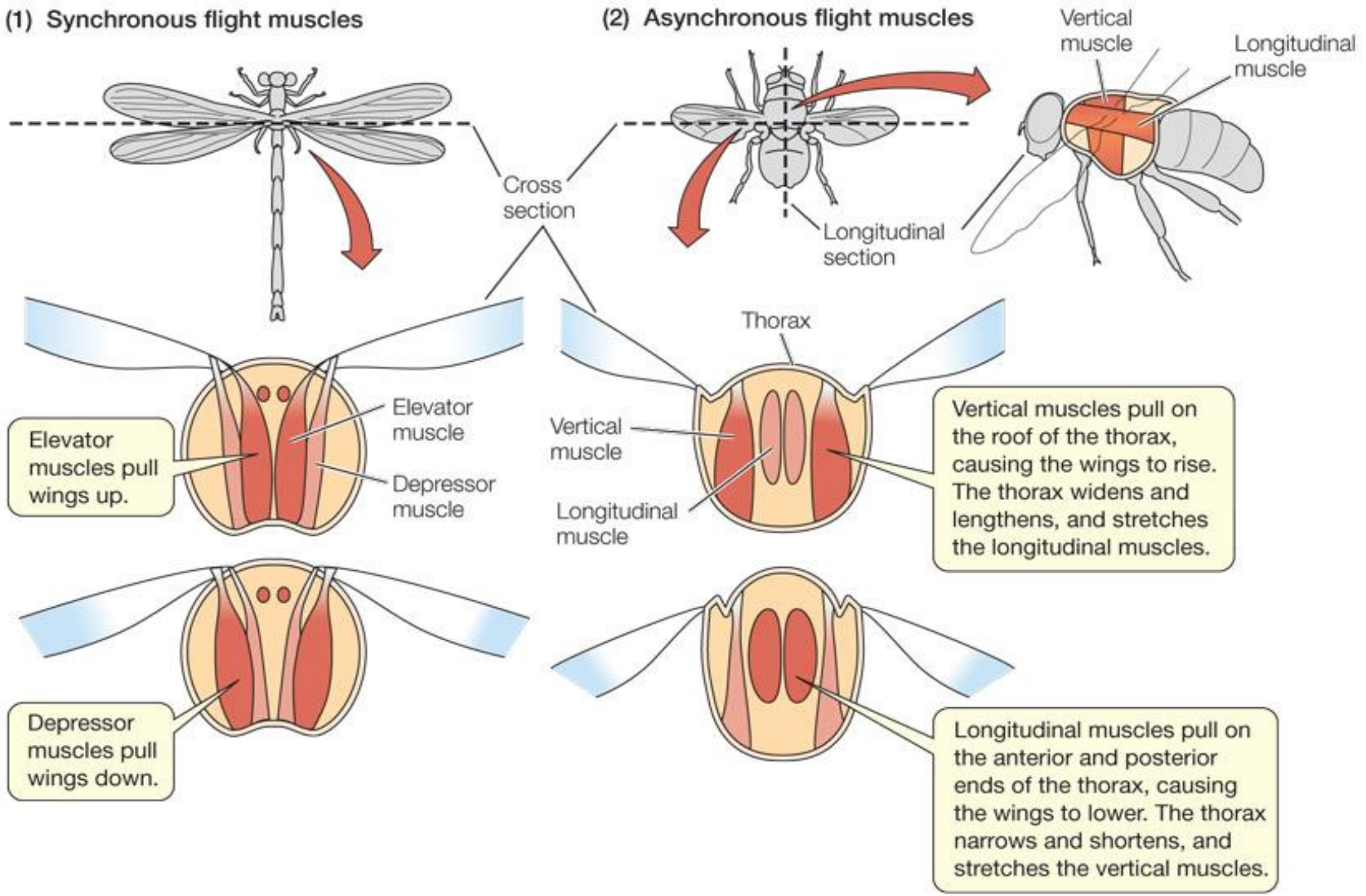
# Synchronous Flight

- The synchrony of direct flight mechanisms depends on the nerve impulse to the flight muscles that must precede each wing beat.
- Butterflies, dragonflies, and grasshoppers are examples of insects with a synchronous flight mechanism.

# Synchronous Flight

- **Synchronous** muscle is the most common form of muscle found within an **insect's** body.
- It also occurs in **flight** muscle within **insects** that beat their wings less than 100 times per second.

# Synchronous Flight



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Asynchronous Flight

Lecture 168

# Asynchronous Flight

- Other insects use an **indirect or asynchronous flight mechanism.**
- Muscles act to change the shape of the exoskeleton for both upward and downward wing strokes.
- Dorsoventral muscles pulling the dorsal exoskeleton (tergum) downward produce the upward wing thrust.

# Asynchronous Flight

- The downward thrust occurs when longitudinal muscles contract and cause the exoskeleton to arch upward.
- The resilient properties of the exoskeleton enhance the power and velocity of these strokes.
- During a wing beat, the thorax is deformed, storing energy in the exoskeleton.

# Asynchronous Flight

- At a critical point midway into the down stroke, stored energy reaches a maximum, and at the same time, resistance to wing movement suddenly decreases.
- The wing then “clicks’’ through the rest of the cycle, using energy stored in the exoskeleton.

# Asynchronous Flight

- Asynchrony of this flight mechanism arises from the lack of one-to-one correspondence between nerve impulses and wing beats.
- A single nerve impulse can result in approximately 50 cycles of the wing, and frequencies of 1,000 cycles per second (cps) have been recorded in some midges.

# Asynchronous Flight

- The asynchrony between wing beat and nerve impulses is dependent on flight muscles being stretched during the “click’’ of the thorax.
- The stretching of longitudinal flight muscles during the upward beat of the wing initiates the subsequent contraction of these muscles.

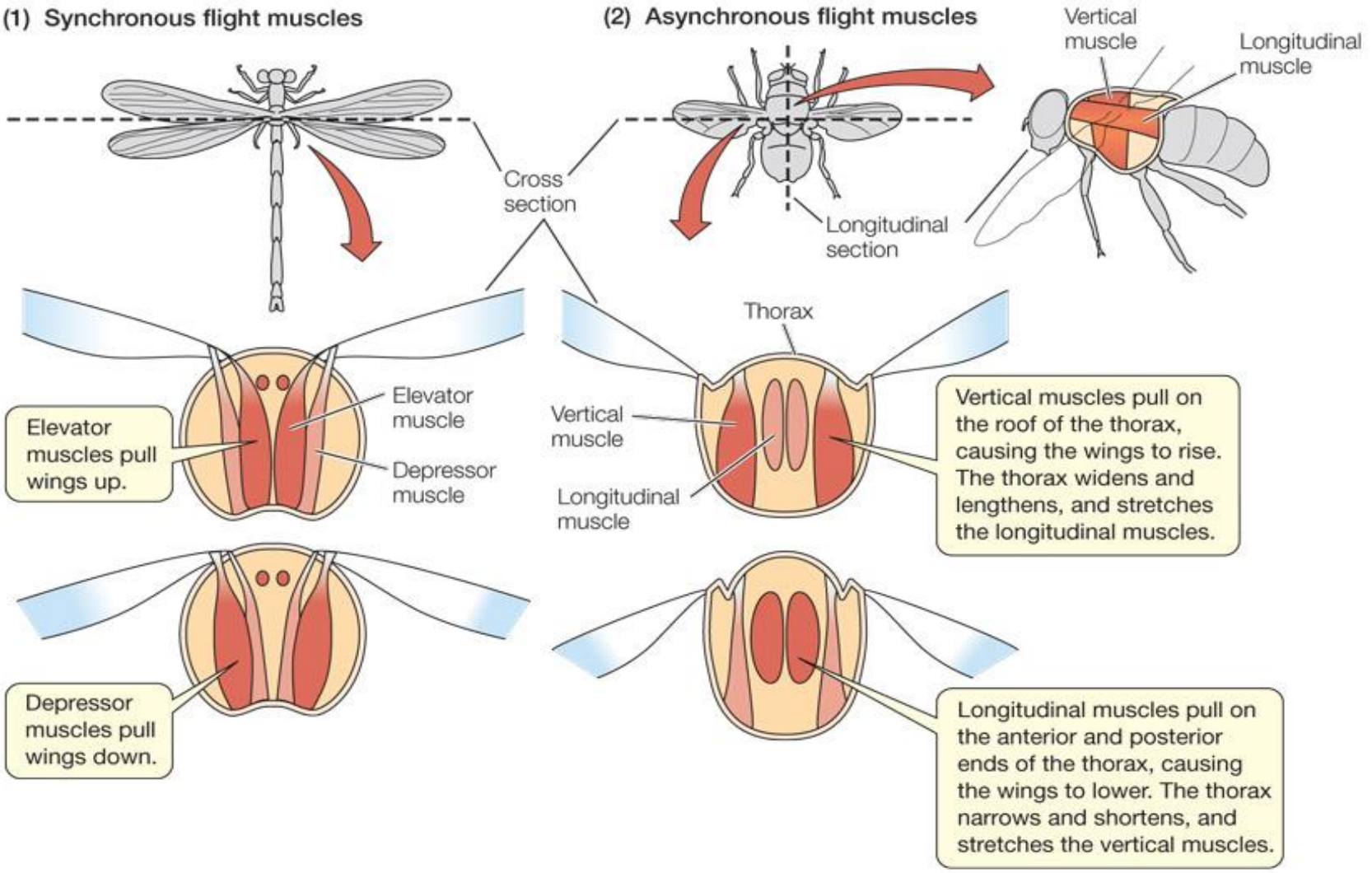
# Asynchronous Flight

- Similarly, stretching during the downward beat of the wing initiates subsequent contraction of dorsoventral flight muscles.
- Indirect flight muscles are frequently called **fibrillar flight muscles.**
- **Flies and wasps are examples of insects** with an asynchronous flight mechanism.

# Asynchronous Flight

- Simple flapping of wings is not enough for flight.
- The tilt of the wing must be controlled to provide lift and horizontal propulsion.
- In most insects, muscles that control wing tilting attach to sclerotized plates at the base of the wing.

# Asynchronous Flight



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Other forms of Locomotion

Lecture 169

# Other Forms of Locomotion

- Insects walk, run, jump, or swim across the ground or other substrates.
- When they walk, insects have three or more legs on the ground at all times, creating a very stable stance.
- When they run, fewer than three legs may be in contact with the ground.

# Other Forms of Locomotion

- A fleeing cockroach (order Blattaria) reaches speeds of about 5 km/hour, although it seems much faster when trying to catch one.
- The apparent speed is the result of their small size and ability to quickly change directions.

# Other Forms of Locomotion

- Jumping insects, such as grasshoppers (order Orthoptera), usually have long, metathoracic legs in which leg musculature is enlarged to generate large, propulsive forces.
- Energy for a flea's (order Siphonaptera) jump is stored as elastic energy of the exoskeleton.
- Muscles that flex the legs distort the exoskeleton.

# Other Forms of Locomotion

- A catch mechanism holds the legs in this “cocked’’ position until special muscles release the catches and allow the stored energy to quickly extend the legs.

# Other Forms of Locomotion

- This action hurls the flea for distances that exceed 100 times its body length.
- A comparable distance for a human long jumper would be the length of two football fields.

END

# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Gaseous Exchange: Tracheal system

Lecture 170

# Gaseous Exchange: Tracheal System

- Gas exchange with air requires a large surface area for the diffusion of gases.
- In terrestrial environments, these surfaces are also avenues for water loss.

# Gaseous Exchange: Tracheal System

- Respiratory water loss in insects, as in some arachnids, is reduced through the invagination of respiratory surfaces to form highly branched systems of chitin-lined tubes, called **tracheae.**

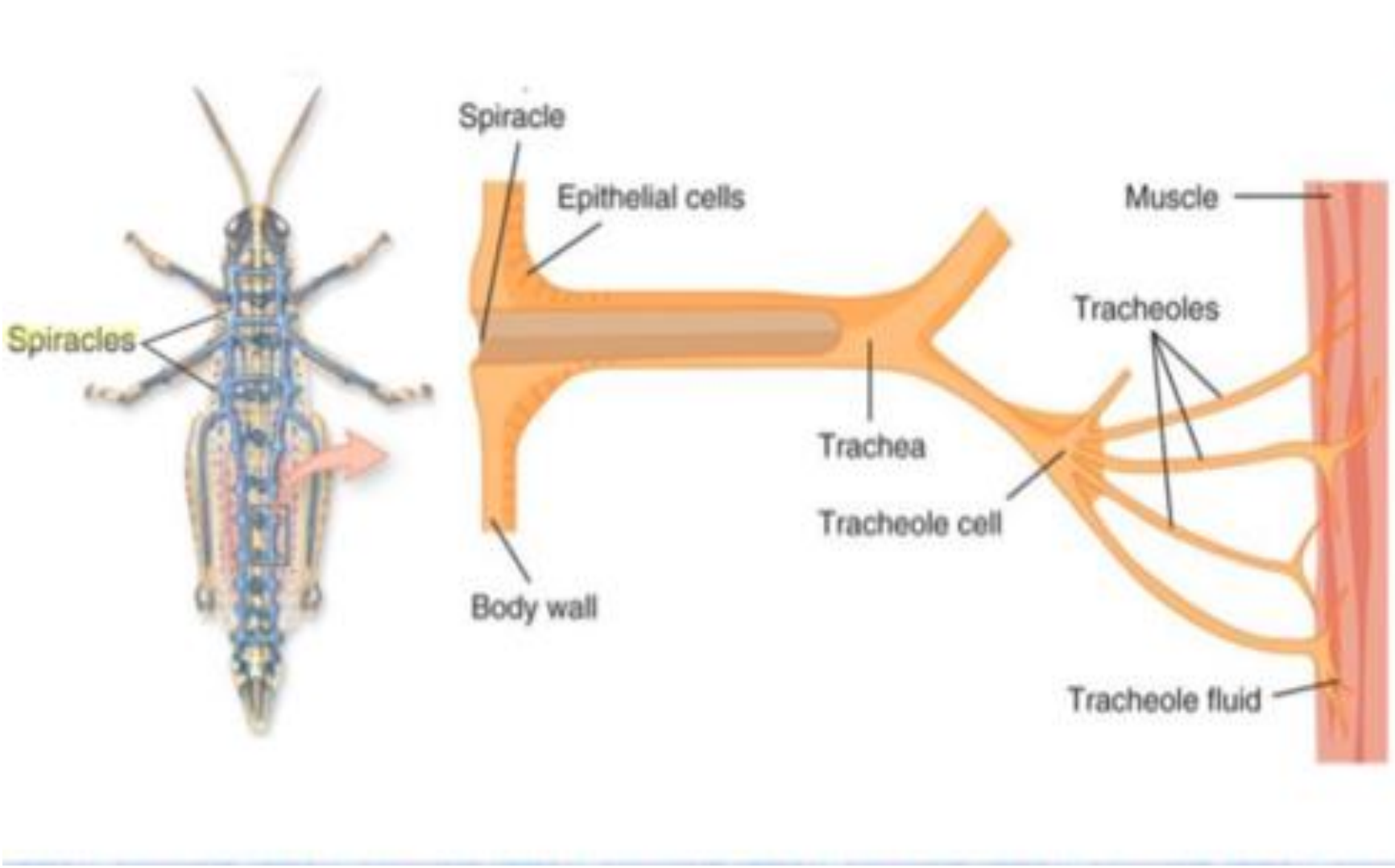
# Gaseous Exchange: Tracheal System

- Tracheae open to the outside of the body through spiracles, which usually have some kind of closure device to prevent excessive water loss.
- Spiracles lead to tracheal trunks that branch, eventually giving rise to smaller branches, the tracheoles.

# Gaseous Exchange: Tracheal System

- Tracheoles end intracellularly and are especially abundant in metabolically active tissues, such as flight muscles.
- No cells are more than 2 or 3 μm from a tracheole.

# Gaseous Exchange: Tracheal System



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

What is Ventilating Mechanism in Insects?

Lecture 171

# What is Ventilating Mechanism in Insects?

- Most insects have ventilating mechanisms that move air into and out of the tracheal system.
- For example, contracting flight muscles alternatively compress and expand the larger tracheal trunks and thereby ventilate the tracheae.

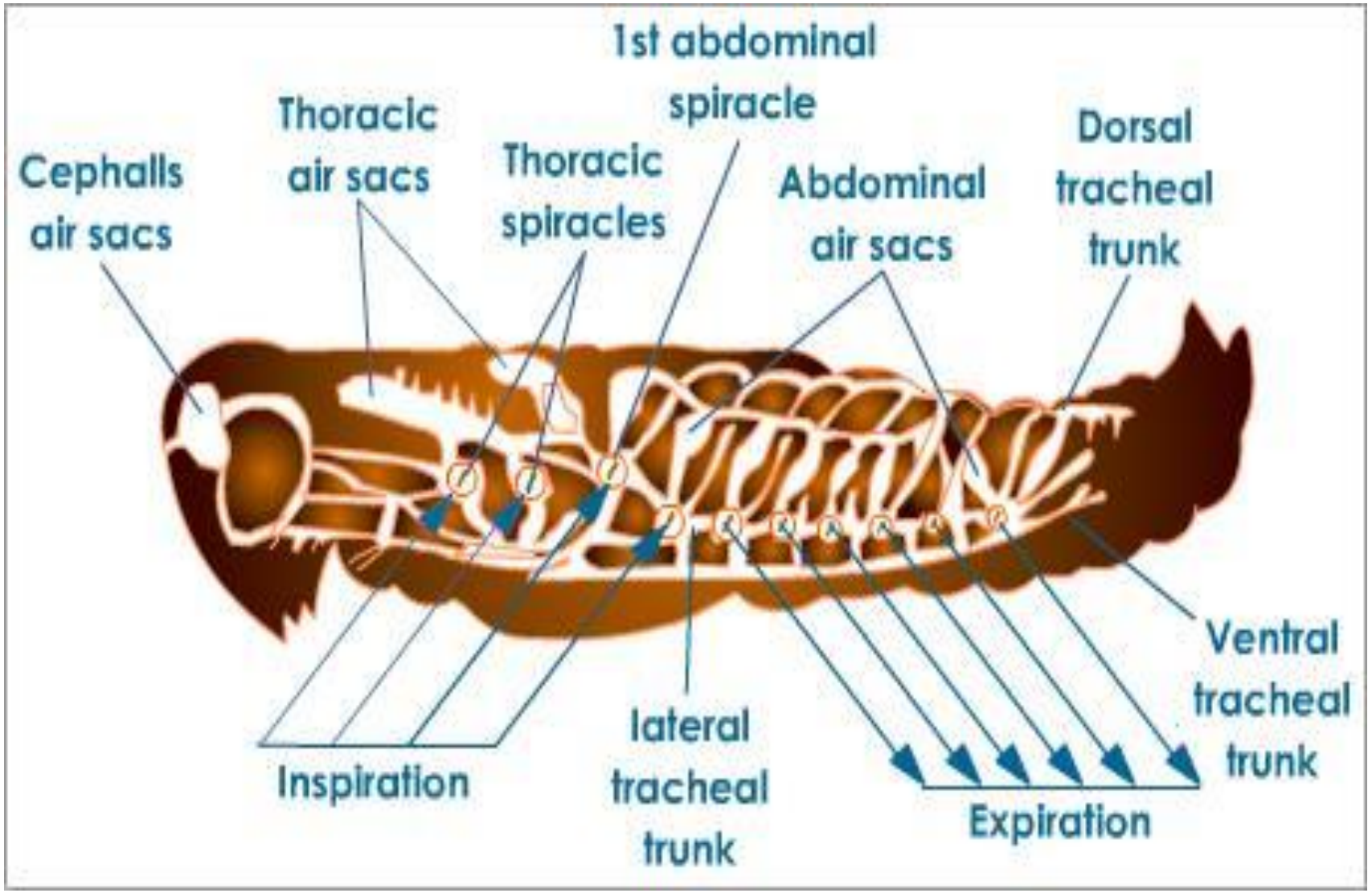
# What is Ventilating Mechanism in Insects?

- In some inspects, carbon dioxide that metabolically active cells produce is sequestered in the hemocoel as bicarbonate ions ($HCO_3$).
- As oxygen diffuses from the tracheae to the body tissues, and is not replaced by carbon dioxide, a vacuum is created that draws more air into the spiracles.

# What is Ventilating Mechanism in Insects?

- This process is called passive suction.
- Periodically, the sequestered bicarbonate ions are converted back into carbon dioxide, which escapes through the tracheal system.
- Other insects contract abdominal muscles in a pumplike fashion to move air into and out of their tracheal systems.

# What is Ventilating Mechanism in Insects?



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Nutrition and Digestive System, Circulation in Insects

Lecture 172

# Nutrition and Digestive System, Circulation in Insects

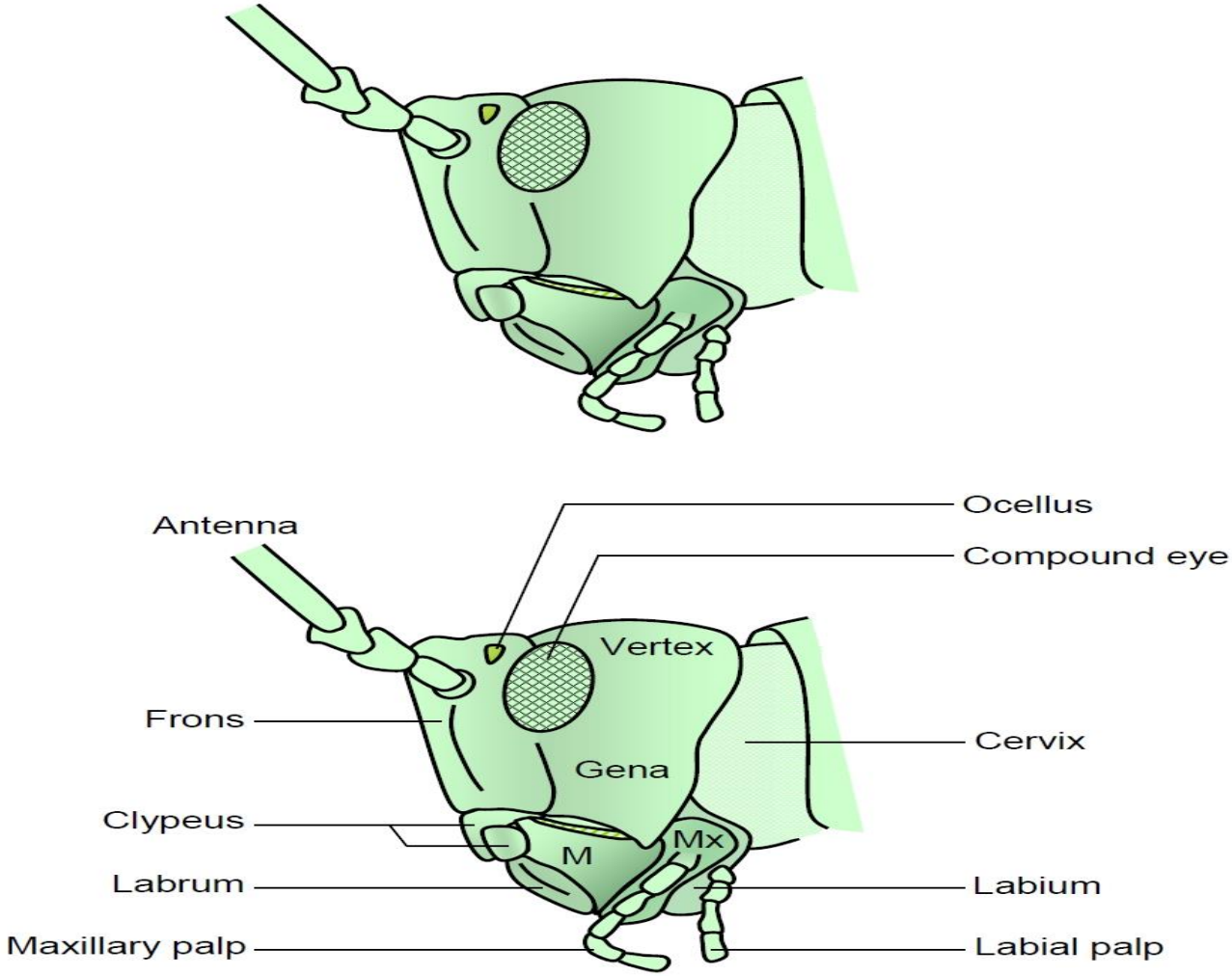
- The diversity of insect feeding habits parallels the diversity of insects themselves.
- An upper, liplike structure is called the **labrum**.
- It is sensory, and unlike the remaining mouthparts, is not derived from segmental, paired appendages.

# Nutrition and Digestive System, Circulation in Insects

- Mandibles are sclerotized, chewing mouthparts.
- The maxillae often have cutting surfaces and bear a sensory palp.
- The **labium is a** sensory lower lip.
- All of these aid in food handling.

# Nutrition and Digestive System, Circulation in Insects

Biting Mouthparts of the Grasshopper

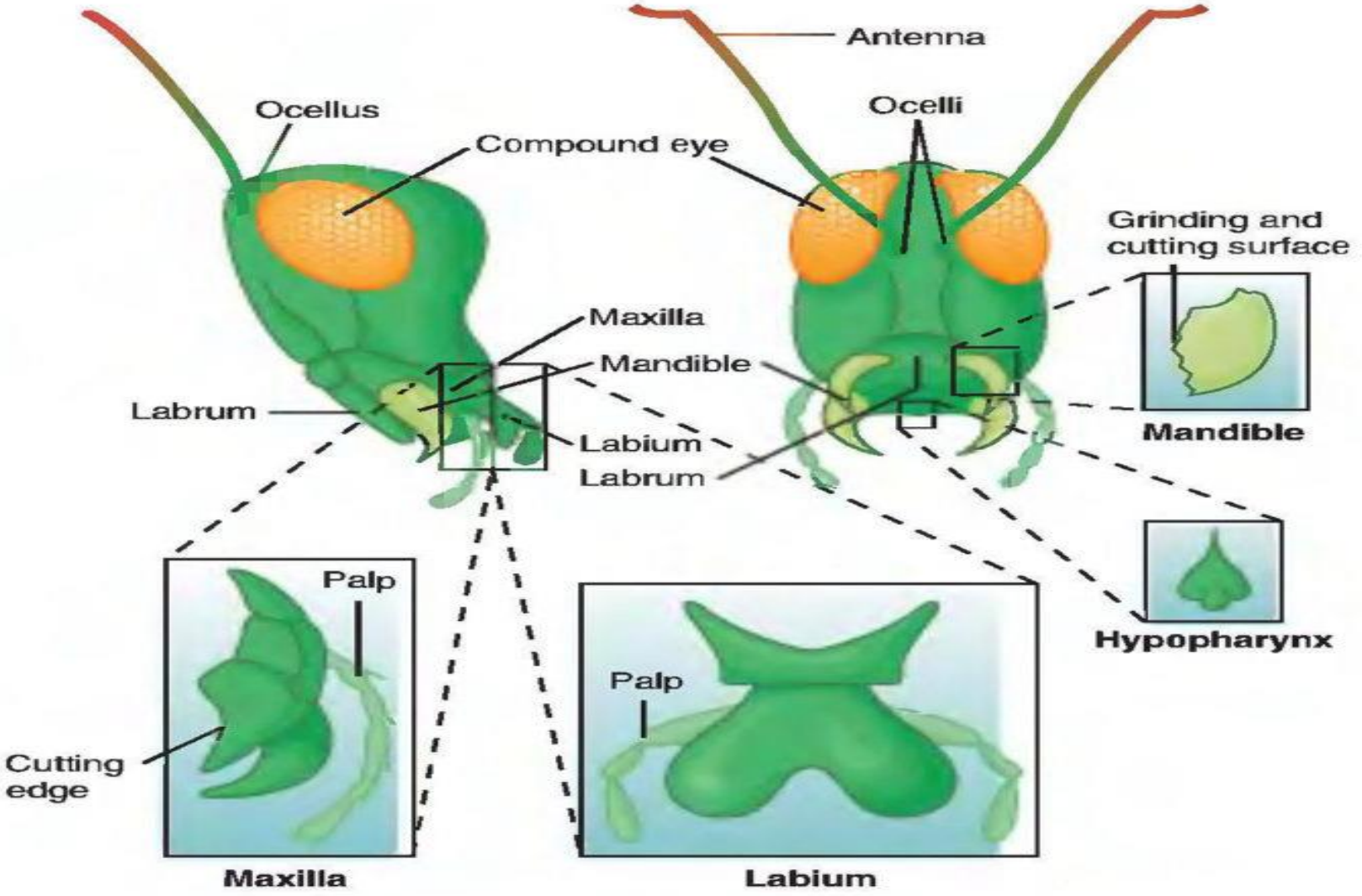



M: mandible; Mx: maxilla

# Nutrition and Digestive System, Circulation in Insects

- Variations on this plan are specializations for sucking or siphoning plant or animal fluids.
- The digestive tract, as in all arthropods, consists of a foregut, a midgut, and a hindgut.
- Enlargements for storage and diverticula that secrete digestive enzymes are common.

# Nutrition and Digestive System, Circulation in Insects



# Nutrition and Digestive System, Circulation in Insects

- The **circulatory** system of insects is similar to that described for other arthropods, although the blood vessels are less well developed.
- Blood distributes nutrients, hormones, and wastes, and amoeboid blood cells participate in body defense and repair mechanisms.
- Blood is not important in gas transport.

END

# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Thermoregulation: Shivering Thermogenesis

Lecture 173

# Thermoregulation: Shivering Thermogenesis

- **Thermoregulation** is a requirement for flying insects.
- Virtually all insects warm themselves by basking in the sun or resting on warm surfaces.
- Because they use external heat sources in temperature regulation, insects are generally considered ectotherms.

# Thermoregulation: Shivering Thermogenesis

- Other insects (e.g., some moths, alpine bumblebees, and beetles) can generate heat by rapid contraction of flight muscles, a process called **shivering thermogenesis.**
- Metabolic heat generated in this way can raise the temperature of thoracic muscles from near 0 to 30° C.

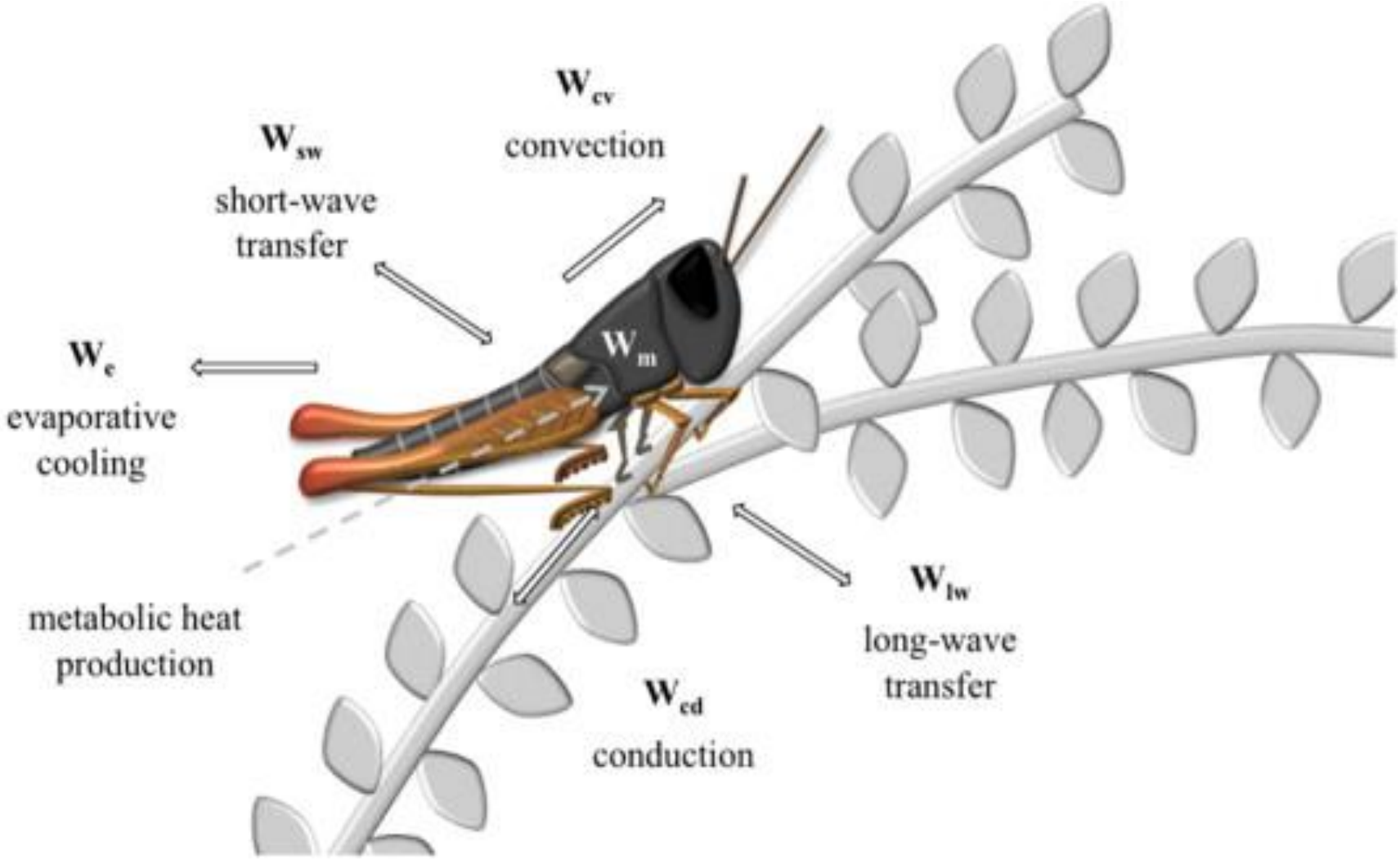
# Thermoregulation: Shivering Thermogenesis

- Because some insects rely to a limited extent on metabolic heat sources, they have a variable body temperature and are sometimes called heterotherms.
- Insects are also able to cool themselves by seeking cool, moist habitats.

# Thermoregulation: Shivering Thermogenesis

- Honeybees can cool a hive by beating their wings at the entrance of the hive, thus circulating cooler outside air through the hive.

# Thermoregulation: Shivering Thermogenesis



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

What are ectotherms and Heterotherms?

Lecture 174

# What are Ectotherms and Heterotherms?

- **Insect thermoregulation** is the process whereby insects maintain body temperatures within certain boundaries.

# What are Ectotherms and Heterotherms?

- While many insects are ectotherms (animals in which their heat source is primarily from the environment), others are **endotherms** (animals which can **produce** heat internally by biochemical processes).

# What are Ectotherms and Heterotherms?

- These endothermic insects are better described as regional heterotherms because they are not uniformly endothermic.
- The body temperatures of aquatic ectotherms are usually very close to those of the water.

# What are Ectotherms and Heterotherms?

- Ectotherms do not require as much food as warm-blooded animals (endotherms) of the same size, but most cannot deal as well with cold surroundings.

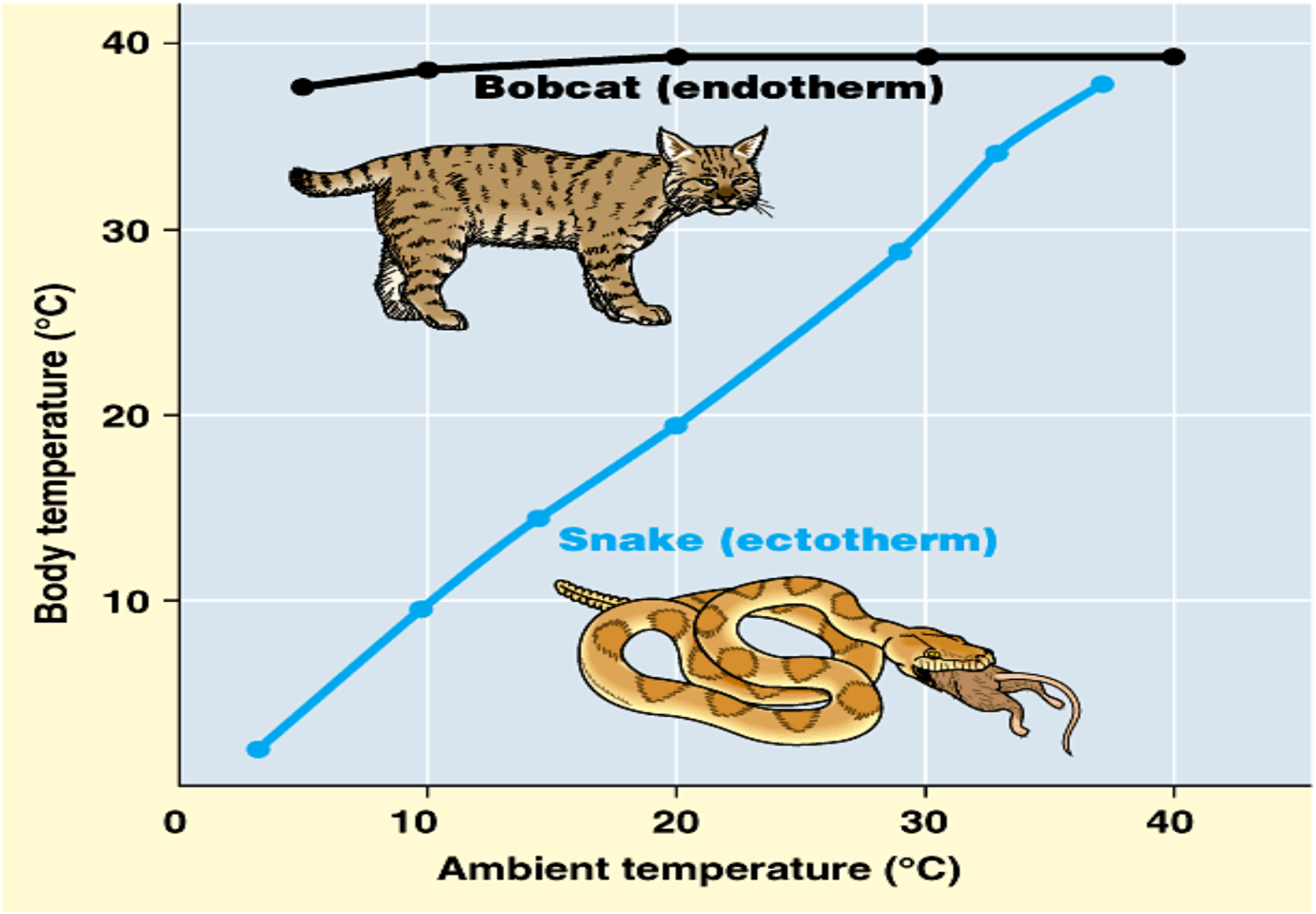
# What are Ectotherms and Heterotherms?

- Heterothermy (“hetero”, different) occurs in **animals which are able to switch from endothermy to ectothermy.**

# What are Ectotherms and Heterotherms?

- This usually happens in small birds and mammals with high metabolic rates (quite active and with very high energetic requirements), which **decrease their body temperature during inactivity periods**.
- These inactivity periods usually are yearly or daily.

# What are Ectotherms and Heterotherms?



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Behavioral Adaptations to Regulate Temperature

Lecture 175

# Behavioral Adaptations to Regulate Temperature

- Many animals regulate their body temperature through behavior, such as seeking sun or shade or huddling together for warmth.
- **Endotherms** can alter metabolic heat production to maintain body temperature using both shivering and non-shivering thermogenesis.

# Behavioral Adaptations to Regulate Temperature

- **Vasoconstriction** shrinking and **vasodilation** expansion of blood vessels to the skin can alter an organism's exchange of heat with the environment.

# Behavioral Adaptations to Regulate Temperature

- A **countercurrent heat exchanger** is an arrangement of blood vessels in which heat flows from warmer to cooler blood, usually reducing heat loss.
- Some animals use body insulation and evaporative mechanisms, such as sweating and panting, in body temperature regulation.

# Behavioral Adaptations to Regulate Temperature

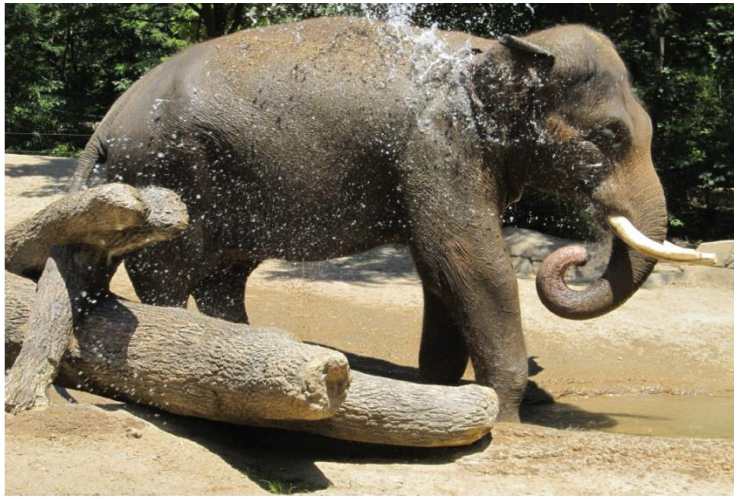
- Heterothermic **insects** ha ve adapted to make use of the excess **heat** produced by flight muscles to increase their thoracic **temperature** pre-flight.

# Behavioral Adaptations to Regulate Temperature

- The other described way of thermoregulation is called **behavioral** thermor egulation because body **temperature** is controlled by **behavioral** means, such as basking in the sun.

# Behavioral Adaptations to Regulate Temperature









# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Nervous System of Insects

Lecture 176

# Nervous System of Insects

- The nervous system of insects is similar to the pattern described for annelids and other arthropods.
- The supraesophageal ganglion is associated with sensory structures of the head.

# Nervous System of Insects

- Connectives join the supraesophageal ganglion to the subesophageal ganglion, which innervates the mouthparts and salivary glands and has a general excitatory influence on other body parts.

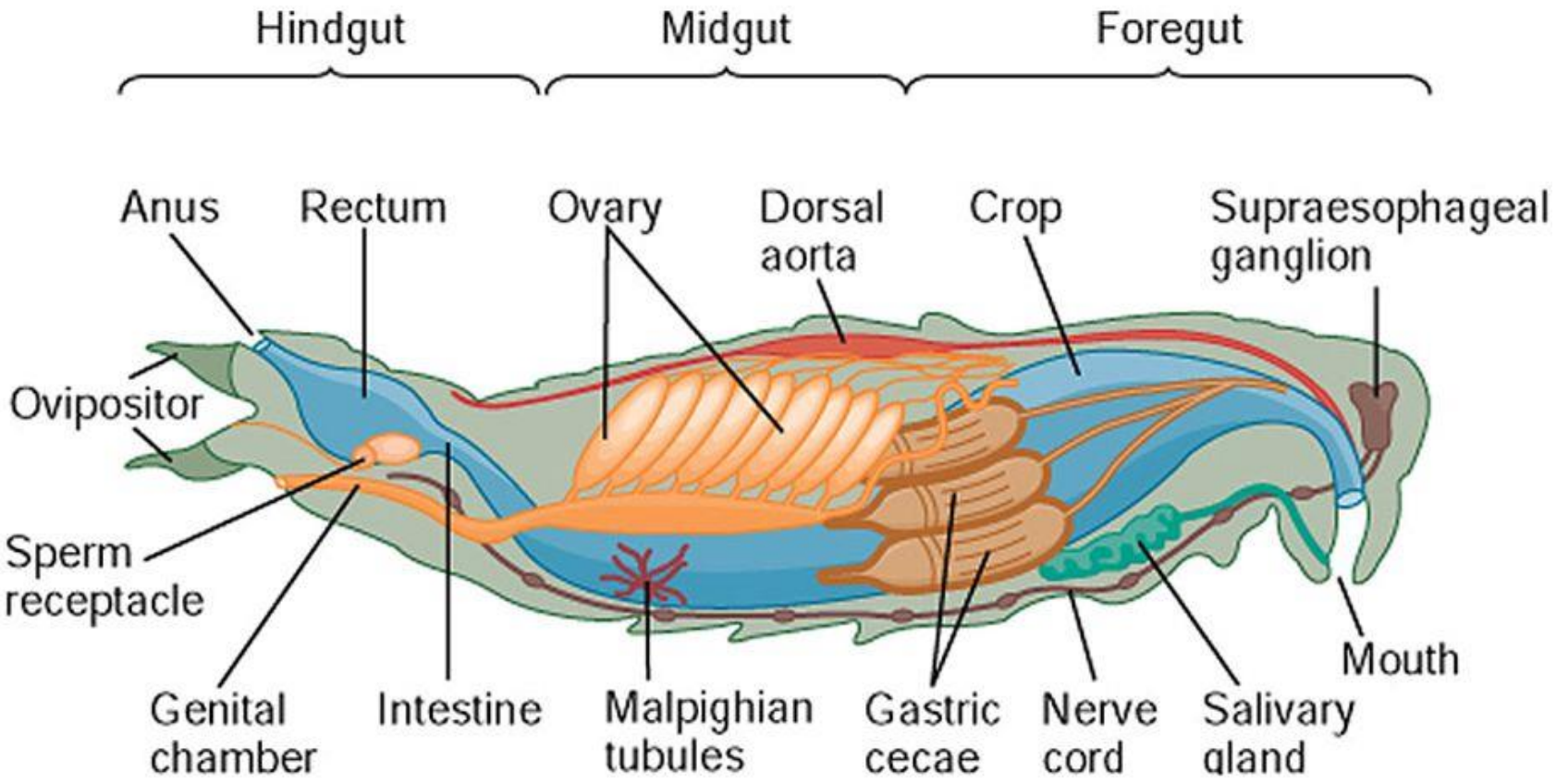
# Nervous System of Insects

- Segmental ganglia of the thorax and abdomen fuse to various degrees in different taxa.
- Insects also possess a well-developed visceral nervous system that innervates the gut, reproductive organs, and heart.

# Nervous System of Insects

**Internal Structure of a Generalized Insect**

**Fig. 15.8**



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Learning and Memory of Insects

Lecture 177

# Learning and Memory of Insects

- Research has demonstrated that insects are capable of some learning and have a memory.

# Learning and Memory of Insects

- For example, bees (order Hymenoptera) instinctively recognize flowerlike objects by their shape and ability to absorb ultraviolet light, which makes the center of the flower appear dark.

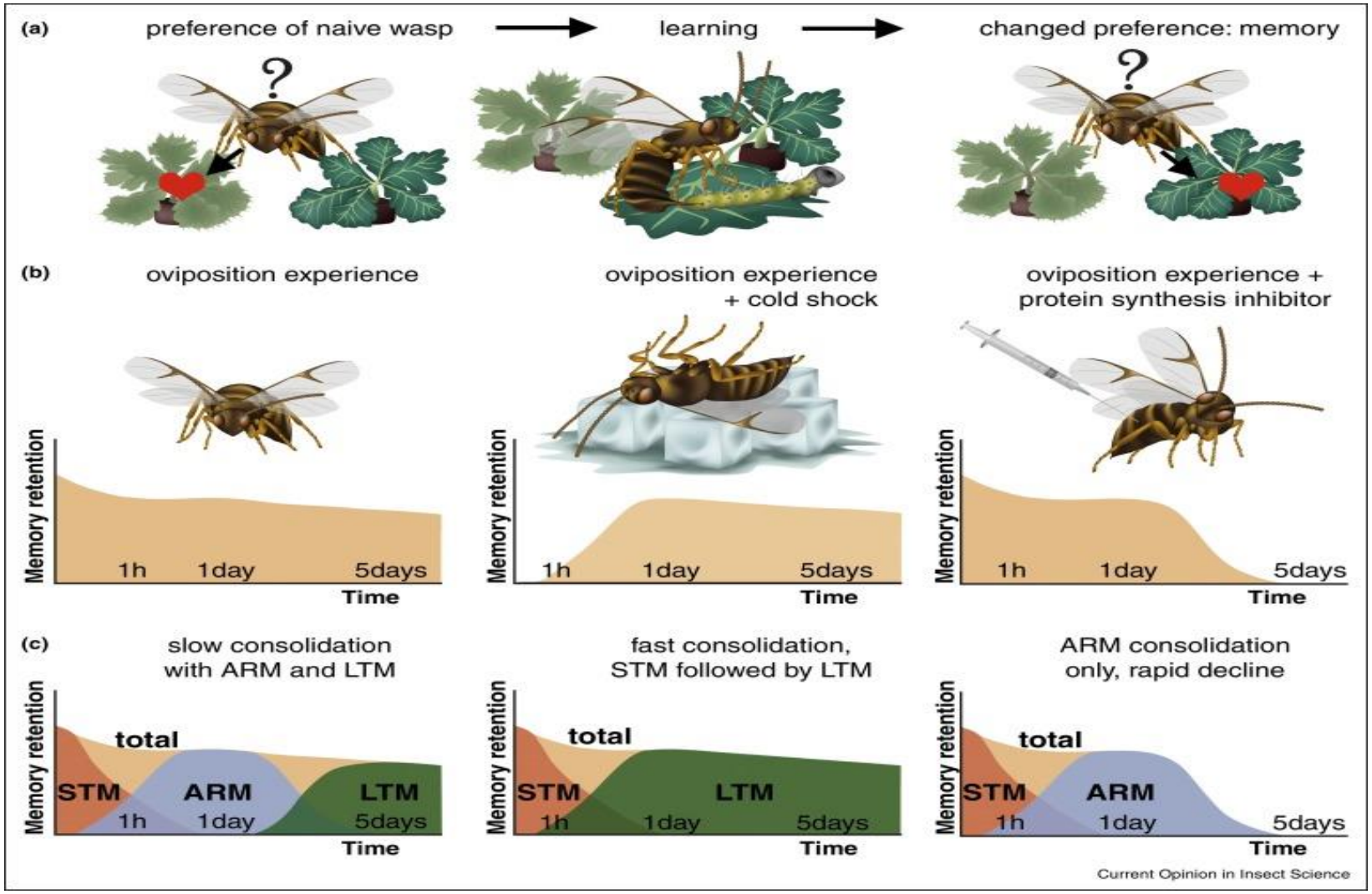
# Learning and Memory of Insects

- If a bee is rewarded with nectar and pollen, it learns the odor of the flower.
- Bees that feed once at artificially scented feeders choose that odor in 90% of subsequent feeding trials.

# Learning and Memory of Insects

- Odor is a very reliable cue for bees because it is more constant than color and shape.
- Wind, rain, and herbivores may damage the latter.

# Learning and Memory of Insects



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Sense Organs

Lecture 178

# Sense Organs

- Sense organs of insects are similar to those found in other arthropods, although they are usually specialized for functioning on land.
- Mechanoreceptors perceive physical displacement of the body or of body parts.

# Sense Organs

- Setae are distributed over the mouthparts, antennae, and legs.
- Touch, air movements, and vibrations of the substrate can displace setae.
- Stretch receptors at the joints, on other parts of the cuticle, and on muscles monitor posture and position.

# Sense Organs

- All sense organs (receptors) act as transducers, converting light energy, chemical energy, or mechanical energy from the environment into electrical energy of nerve impulses in sensory neurons.

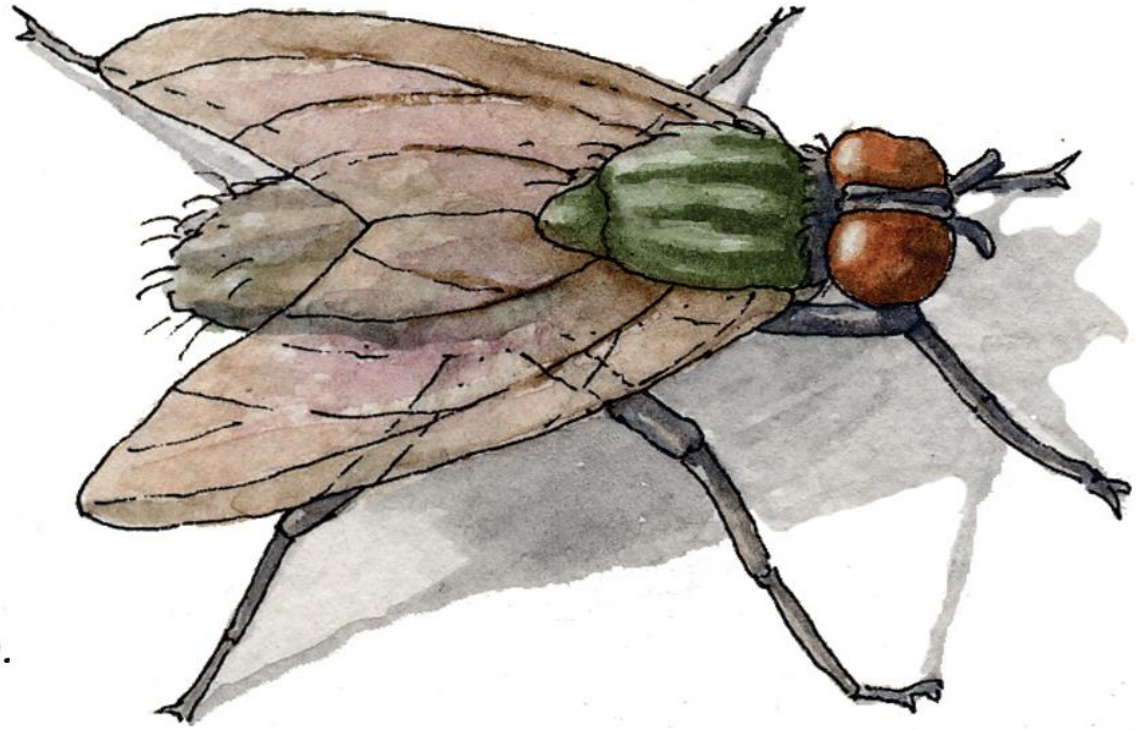
# Sense Organs

- Signals generated by insect sensory receptors travel to the brain or ventral nerve cord where they stimulate appropriate behavioral responses: finding resources (e.g. food, mate, etc.), avoiding danger, or reacting to changes in the environment.

# Sense Organs

- All sensory receptors are derived from embryonic ectoderm and are integral parts of the insect’s exoskeleton.

# Sense Organs



*A fly's feet have claws for grabbing and hairs that grip smooth, wet areas with surface tension.*



*They even have tiny hairs that are sensory organs to taste everywhere they land!*

# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Hearing: Johnston's organs

Lecture 178

# Hearing: Johnston's Organs

- **Hearing** is a mechanoreceptive sense in which airborne pressure waves displace certain receptors.
- All insects can respond to pressure waves with generally distributed setae; others have specialized receptors.

# Hearing: Johnston's Organs

- For example, **Johnston's organs are** in the base of the antennae of most insects, including mosquitoes and midges (order Diptera).
- Long setae that vibrate when certain frequencies of sound strike them cover the antennae of these insects.

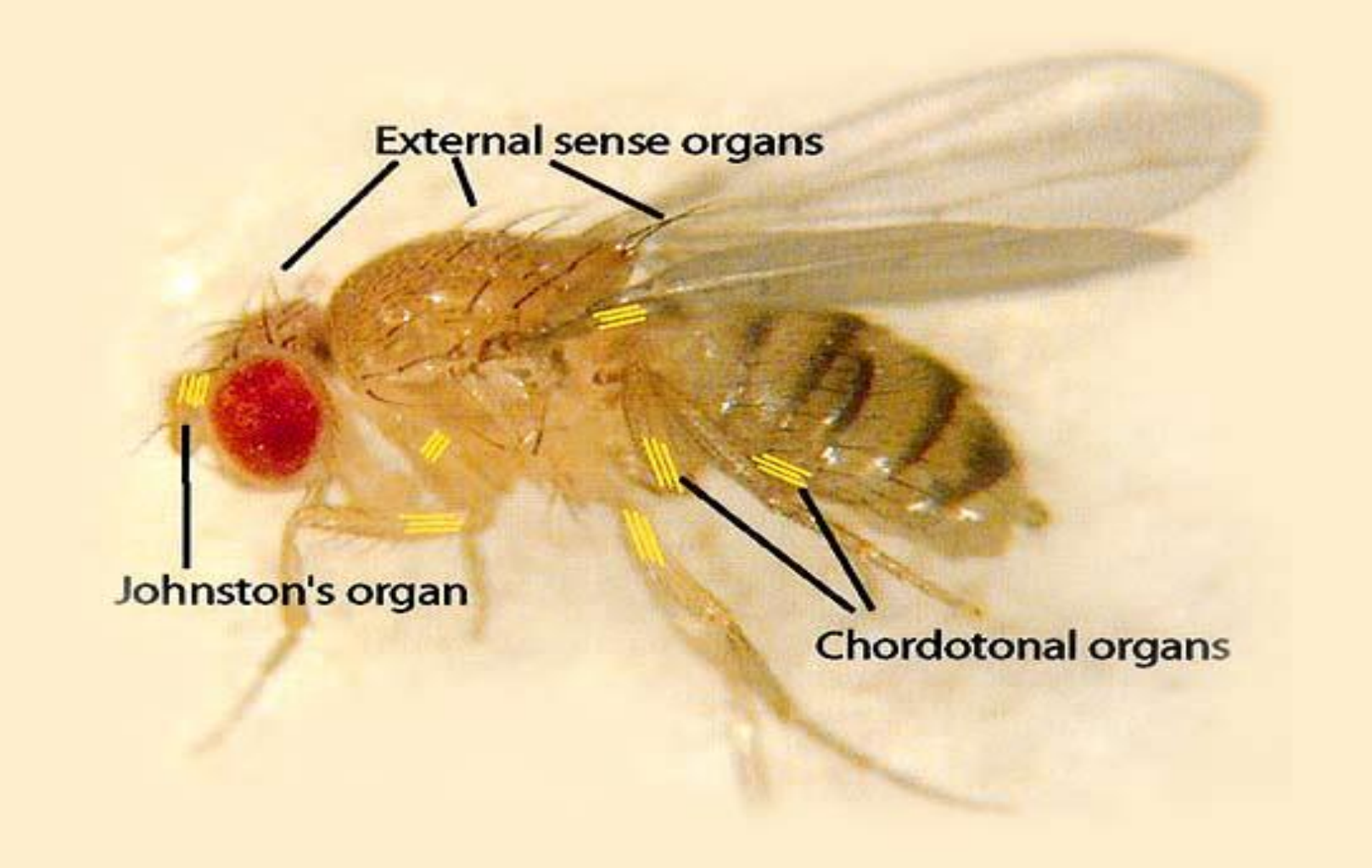
# Hearing: Johnston's Organs

- Vibrating setae move the antenna in its socket, stimulating sensory cells.
- Sound waves in the frequency range of 500 to 550 cycles per second (cps) attract and elicit mating behavior in male mosquitoes (*Aedes aegypti* ).

# Hearing: Johnston's Organs

- These waves are in the range of sounds that the wings of females produce.

# Hearing: Johnston's Organs



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Hearing: Johnston's organs

Lecture 179

# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Tympanic Organs

Lecture 180

# Tympanic Organs

- **Tympanal (tympanic) organs are in the legs of crickets and katydids (order** Orthoptera), in the abdomen of grasshoppers (order Orthoptera) and some moths (order Lepidoptera), and in the thorax of other moths.

# Tympanic Organs

- **Tympanal** organs consist of a thin, cuticular membrane covering a large air sac.
- The air sac acts as a resonating chamber.
- Just under the membrane are sensory cells that detect pressure waves.

# Tympanic Organs

- Grasshopper tympanal organs can detect sounds in the range of 1,000 to 50,000 cps.
- The human ear can detect sounds between 20 and 20,000 cps.
- Bilateral placement of tympanal organs allows insects to discriminate the direction and origin of a sound.

# Tympanic Organs

## Tympanal Organ



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Chemoreception

Lecture 181

# Chemoreception

- Insects use chemoreception in feeding, selection of egg laying sites, mate location, and sometimes, social organization.

# Chemoreception

- Chemoreceptors are usually abundant on the mouthparts, antennae, legs, and ovipositors, and take the form of hairs, pegs, pits, and plates that have one or more pores leading to internal nerve endings.
- Chemicals diffuse through these pores and bind to and excite nerve endings.

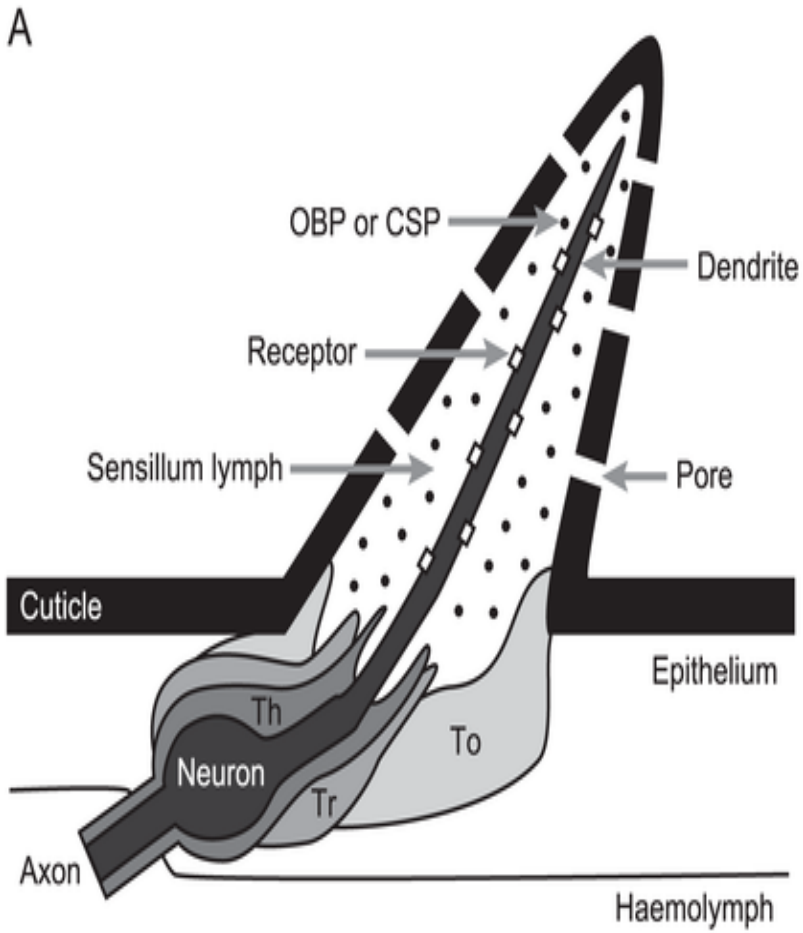
# Chemoreception

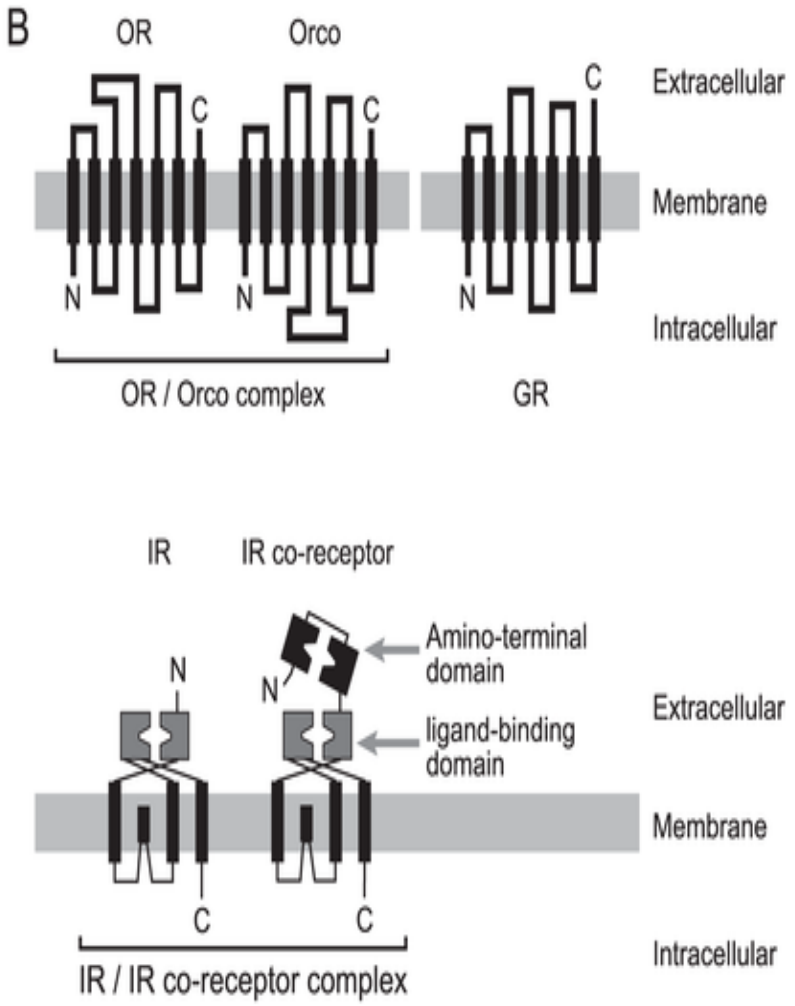
- Insect chemoreception is mediated by a large and diverse super family of seven transmembrane domain receptors.
- These receptors were first identified in *Drosophila*, but have since been found in other insects, including mosquitoes and moths.

# Chemoreception

- Expression and functional analysis of these receptors have been used to identify receptor ligands and to map receptors to functional classes of neurons.
- Many receptors detect general odorants or tastants, whereas some detect pheromones.

# Chemoreception



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Structure and Function of Compound Eyes

Lecture 182

# Structure and Function of Compound Eyes

- All insects are capable of detecting light and may use light in orientation, navigation, feeding, or other functions.
- **Compound eyes are well developed in most adult insects.**

# Structure and Function of Compound Eyes

- They are similar in structure and function to those of other arthropods, although zoologists debate their possible homology (common ancestry) with those of crustaceans, horseshoe crabs, and trilobites.

# Structure and Function of Compound Eyes

- Compound eyes consist of a few to 28,000 receptors, called **ommatidia, that fuse into a multifaceted eye.**
- **The outer** surface of each ommatidium is a lens and is one facet of the eye.
- Below the lens is a crystalline cone. The lens and the crystalline cone are light-gathering structures.

# Structure and Function of Compound Eyes

- Certain cells of an ommatidium, called retinula cells, have a special light-collecting area, called the rhabdom.
- The rhabdom converts light energy into nerve impulses.

# Structure and Function of Compound Eyes

- Pigment cells surround the crystalline cone, and sometimes the rhabdom, and prevent the light that strikes one rhabdom from reflecting into an adjacent ommatidium.
- Although many insects form an image of sorts, the concept of an image has no real significance for most species.

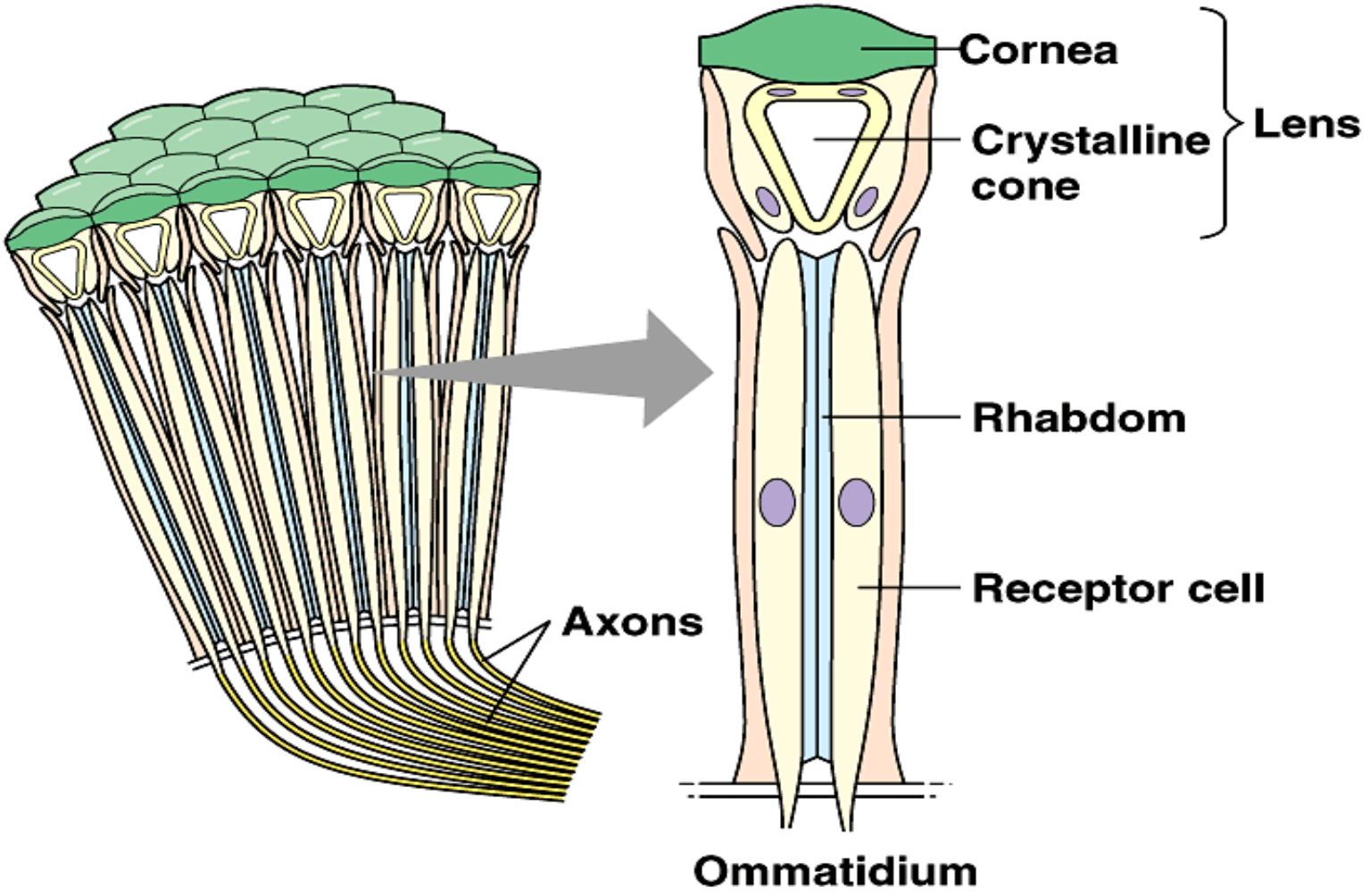
# Structure and Function of Compound Eyes

- The compound eye is better suited for detecting movement.
- Movement of a point of light less than 0.1° can be detected as light successively strikes adjacent ommatidia.
- For this reason, bees are attracted to flowers blowing in the wind, and predatory insects select moving prey.

# Structure and Function of Compound Eyes

- Compound eyes detect wavelengths of light that the human eye cannot detect, especially in the ultraviolet end of the spectrum.
- In some insects, compound eyes also detect polarized light, which may be used for navigation and orientation.

# Structure and Function of Compound Eyes



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Structure and Function of Ocelli in Insects

Lecture 183

# Structure and Function of Ocelli in Insects

- Ocelli consist of 500 to 1,000 receptor cells beneath a single cuticular lens.
- Ocelli are sensitive to changes in light intensity and may be important in the regulation of daily rhythms.

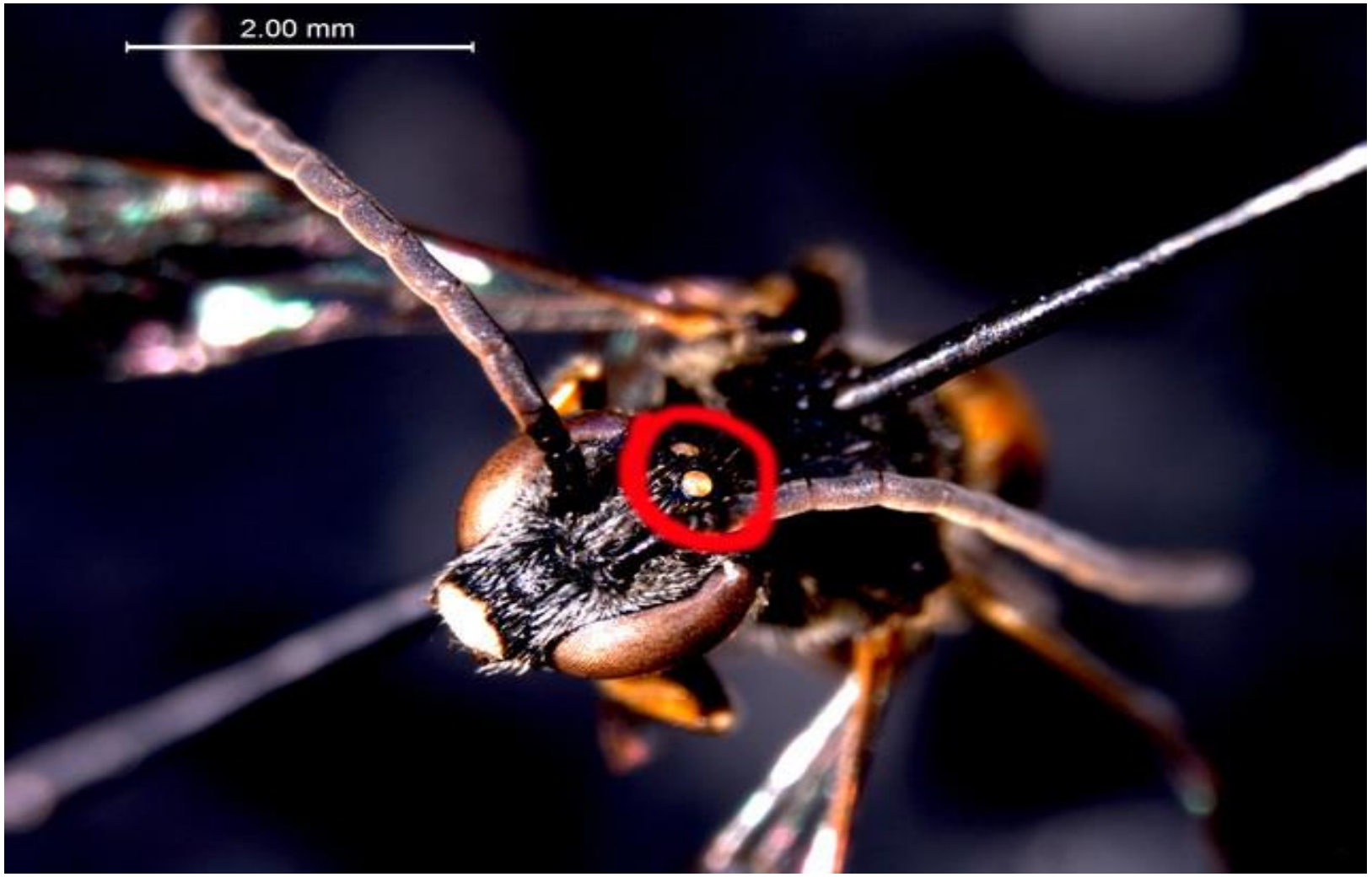
# Structure and Function of Ocelli in Insects

- Ocelli (singular Ocellus) are simple photo-receptors (light detecting organs).
- They consist of a single **lens** and several **sensory** cells.
- Unlike compound eyes, ocelli do not form a complex image of the environment but are used to detect movement.

# Structure and Function of Ocelli in Insects

- One suggestion which has been made a number of times is that ocelli are concerned in some way with flight, and insects which are good at flying tend to have well developed ocelli.

# Structure and Function of Ocelli in Insects



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Excretion

Lecture 184

# Excretion

- The primary insect excretory structures are the Malpighian tubules and the rectum.
- Malpighian tubules end blindly in the hemocoel and open to the gut tract at the junction of the midgut and the hindgut.

# Excretion

- Microvilli cover the inner surface of their cells.
- Various ions are actively transported into the tubules, and water passively follows.
- Uric acid is secreted into the tubules and then into the gut, as are amino acids and ions.

# Excretion

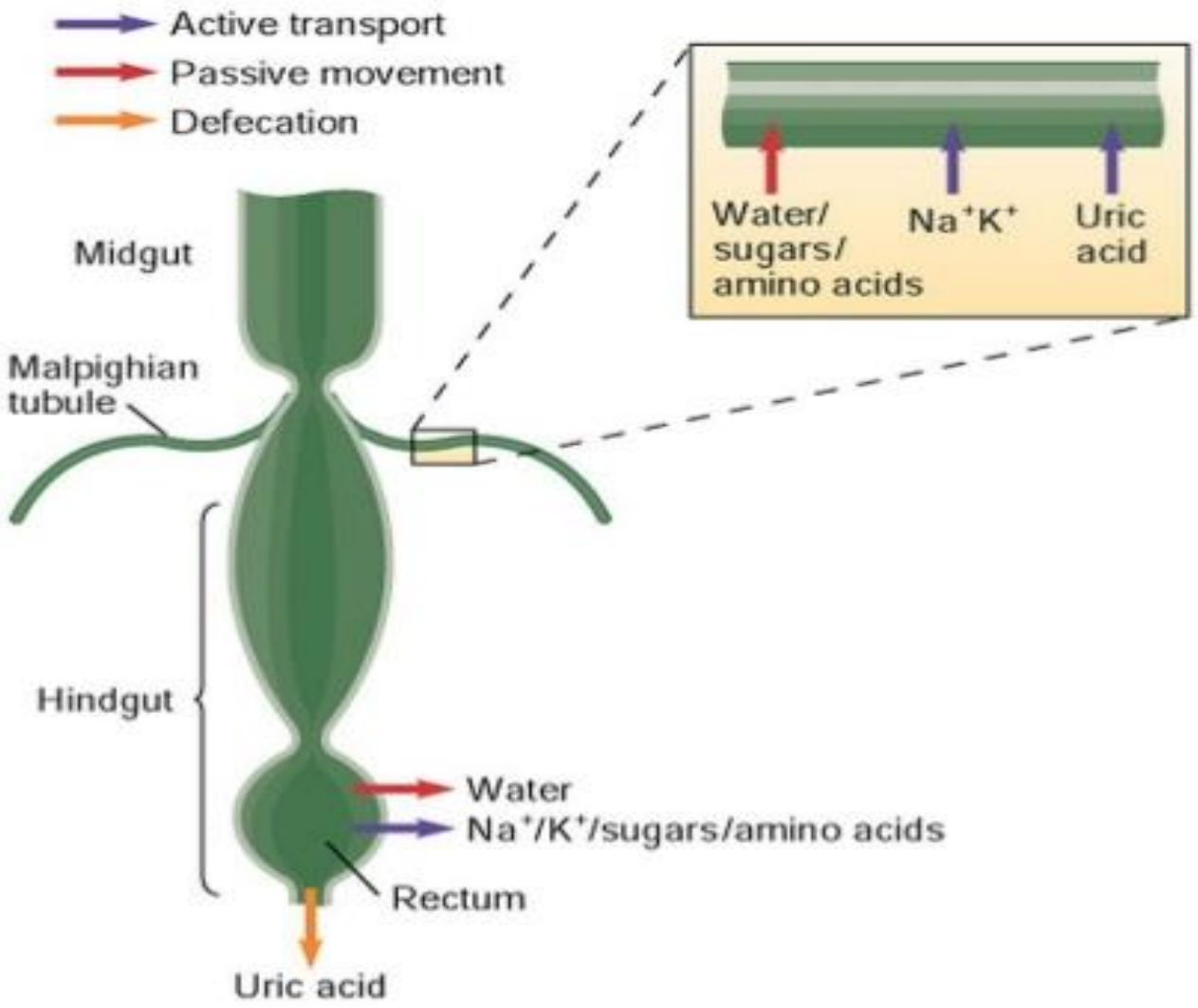
- In the rectum, water, certain ions, and other materials are reabsorbed, and the uric acid is eliminated.
- Excretion of uric acid is advantageous for terrestrial animals because it minimizes water loss.
- There is, however, an evolutionary trade-off.

# Excretion

- The conversion of primary nitrogenous wastes (ammonia) to uric acid is energetically costly.
- Nearly half of the food energy a terrestrial insect consumes may be used to process metabolic wastes.
- In aquatic insects, ammonia simply diffuses out of the body into the surrounding water.

# Excretion

**Insect Excretion**



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Chemical Regulation

Lecture 185

# Chemical Regulation

- The endocrine system controls many physiological functions of insects, such as cuticular sclerotization, osmoregulation, egg maturation, cellular metabolism, gut peristalsis, and heart rate.

# Chemical Regulation

- As in all arthropods, ecdysis is under neuroendocrine control.
- In insects, the subesophageal ganglion and two endocrine glands the corpora allata and the prothoracic glands control these activities.

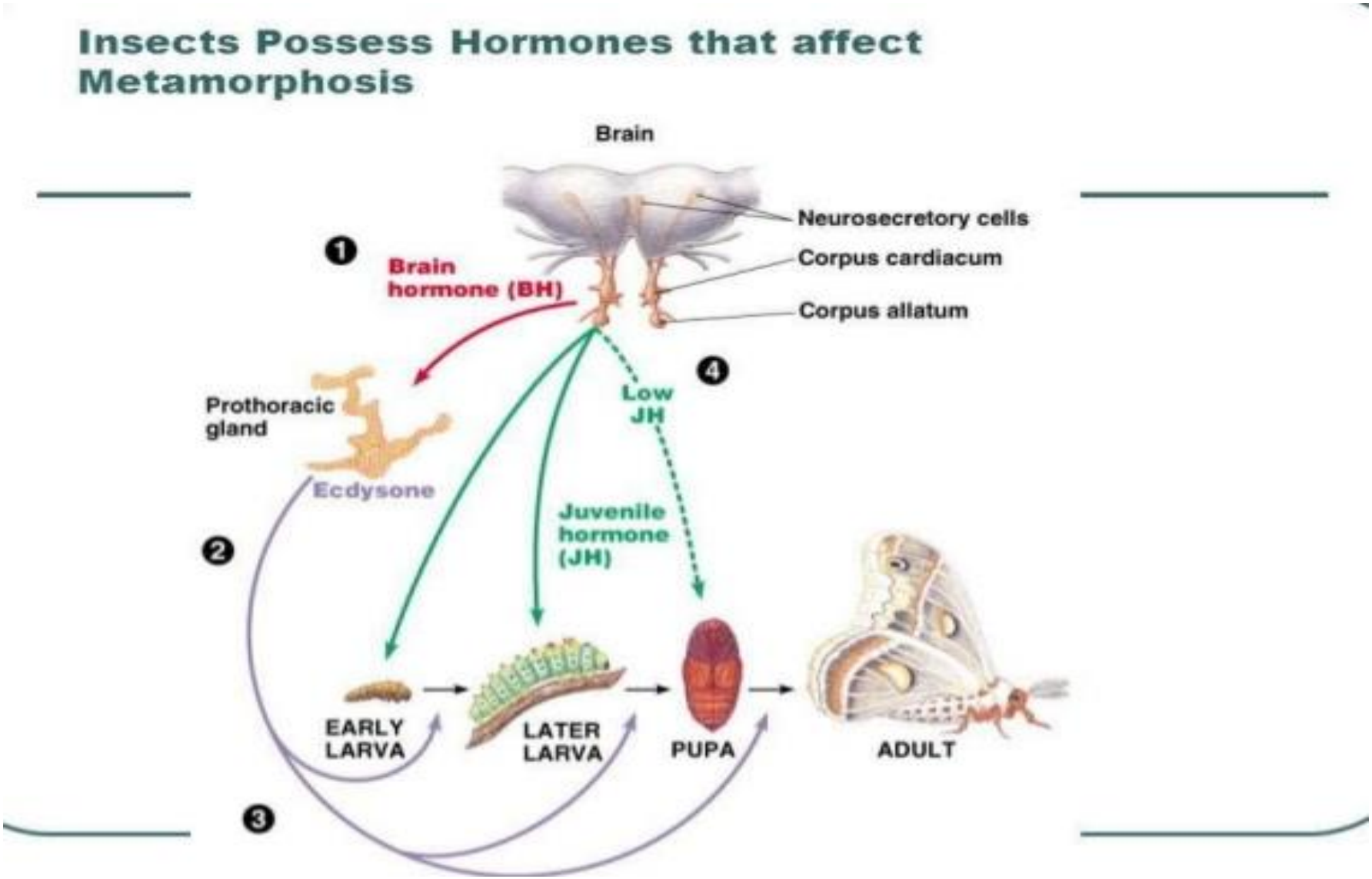
# Chemical Regulation

- There are two main classes of hormones in **insects**:
- 1) the true hormones produced by epithelial glands and belonging to the ecdysteroids or juvenile hormones.
- 2) the neuropeptide hormones produced by neurosecretory cells.

# Chemical Regulation

- An **insect growth regulator (IGR)** is a substance (chemical) that inhibits the life cycle of an insect.
- IGRs are typically used as insecticides to control populations of harmful insect pests such as cockroaches and flea s.

# Chemical Regulation



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Control of Ecdysis by Neuroendocrine System

Lecture 186

# Control of Ecdysis in Neuroendocrine System

- Neurosecretory cells of the subesophageal ganglion manufacture ecdysiotropin.
- This hormone travels in neurosecretory cells to a structure called the corpora cardiaca.

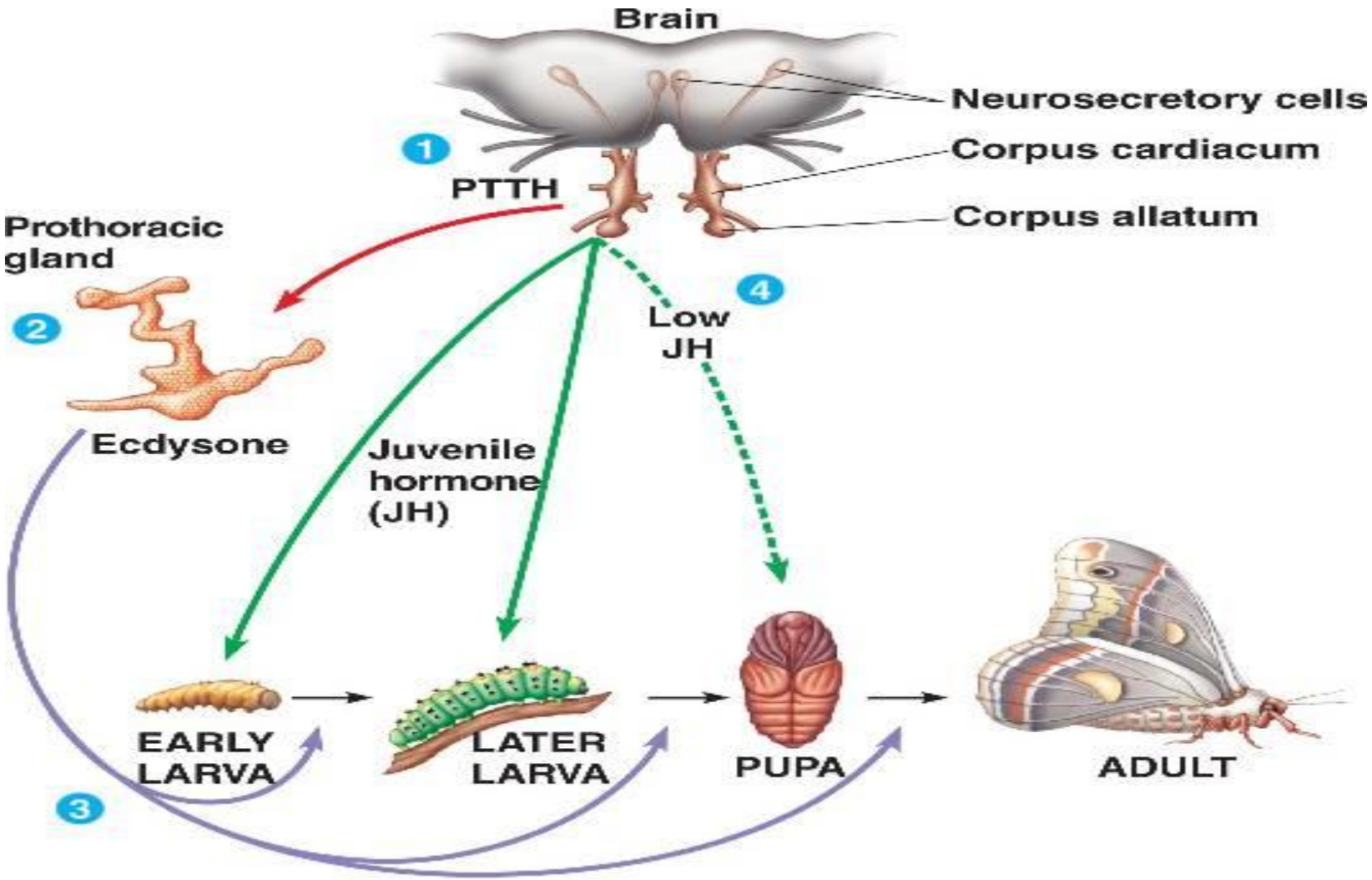
# Control of Ecdysis in Neuroendocrine System

- The corpora cardiaca then releases thoracotropic hormone, which stimulates the prothoracic gland to secrete ecdysone.
- Ecdysone initiates the reabsorption of the inner portions of the procuticle and the formation of the new exoskeleton.

# Control of Ecdysis in Neuroendocrine System

- Other hormones are also involved in ecdysis.
- The recycling of materials absorbed from the procuticle, changes in metabolic rates, and pigment deposition are a few of probably many functions that hormones control.

# Control of Ecdysis in Neuroendocrine System



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Role of Juvenile Hormone in Metamorphosis

Lecture 187

# Role of Juvenile Hormone in Metamorphosis

- In immature stages, the corpora allata produces and releases small amounts of juvenile hormone.
- The amount of juvenile hormone circulating in the hemocoel determines the nature of the next molt.

# Role of Juvenile Hormone in Metamorphosis

- Large concentrations of juvenile hormone result in a molt to a second immature stage, intermediate concentrations result in a molt to a third immature stage, and low concentrations result in a molt to the adult stage.

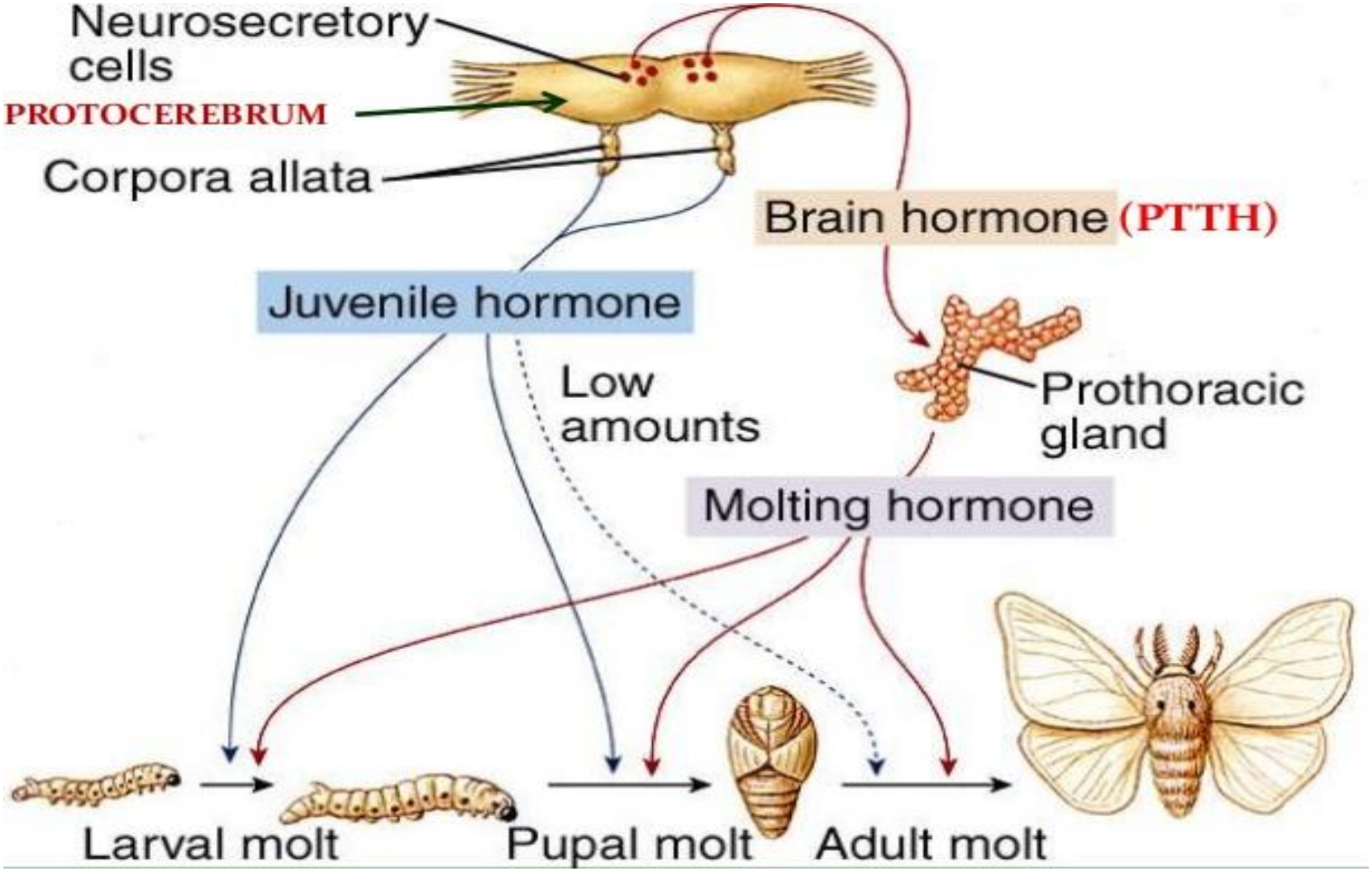
# Role of Juvenile Hormone in Metamorphosis

- Decreases in the level of circulating juvenile hormone also lead to the degeneration of the prothoracic gland so that, in most insects, molts cease once adulthood is reached.

# Role of Juvenile Hormone in Metamorphosis

- Interestingly, after the final molt, the level of juvenile hormone increases again, but now it promotes the development of accessory sexual organs, yolk synthesis, and the egg maturation.

# Role of Juvenile Hormone in Metamorphosis



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Pheromones

Lecture 188

# Pheromones

- **Pheromones** are chemicals an animal releases that cause behavioral or physiological changes in another member of the same species.
- Zoologists have described many different insect uses of pheromones.

# Pheromones

- Pheromones are often so specific that the stereoisomer (chemical mirror image) of the pheromone may be ineffective in initiating a response.

# Pheromones

- Wind or water may carry pheromones several kilometers, and a few pheromone molecules falling on a chemoreceptor of another individual may be enough to elicit a response.

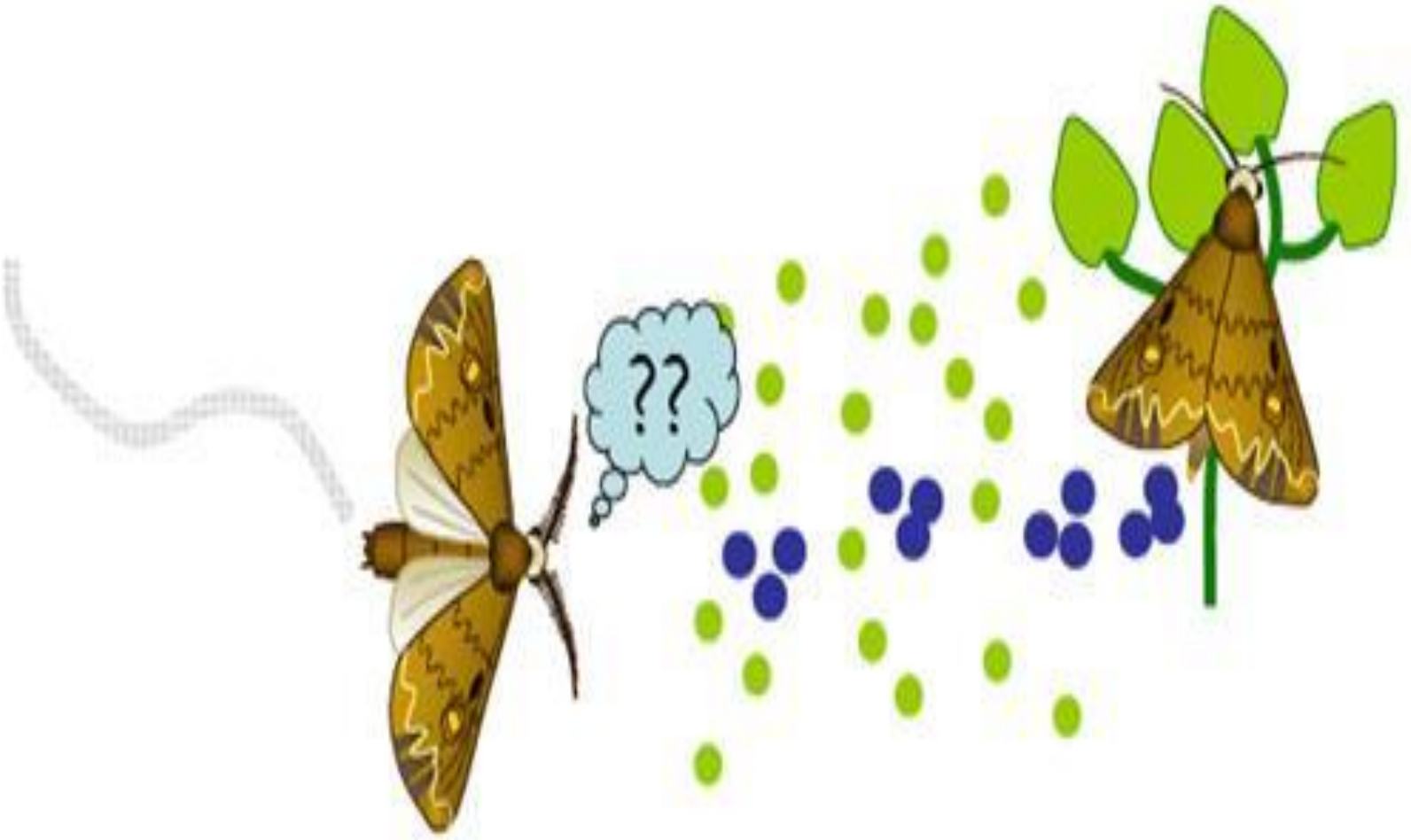
# Pheromones

- Releaser **pheromones** ini tiate immediate behavioral responses in **insects** upon reception, while primer **pheromones** caus e physiological changes in an animal that ultimately result in a behavior response.

# Pheromones

- Thus a male **insect** may be attracted to and attempt to copulate with an inanimate object that has sex **pheromone** on it.

# Pheromones



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Sex Pheromones

Lecture 189

# Sex Pheromones

- Excite or attract members of the opposite sex; accelerate or retard sexual maturation.
- Example: Female moths produce and release pheromones that attract males.

# Sex Pheromones

- Sex pheromones act as a signal to attract potential mates over long distances (e.g. moths).
- Sensitive chemoreceptive sensilla i n insects facilitate the detection of very low concentrations of sex pheromones in the environment.

# Sex Pheromones

- Release of sex pheromones may be governed by factors such as time of day, weather, and the availability of host plants.
- Furthermore, both the immature and adult stages of insects can sequester chemicals from host plants and use them as precursors for sex pheromones.

# Sex Pheromones

- Sex pheromones are highly species specific and, as a rule, a unique blend of the natural pheromone components is needed for attraction.

# Sex Pheromones

Sex Pheromones

- Chemical signalling through pheromones may be an effective means for females to attract males for copulation.



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Sex pheromones

Lecture 190

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# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

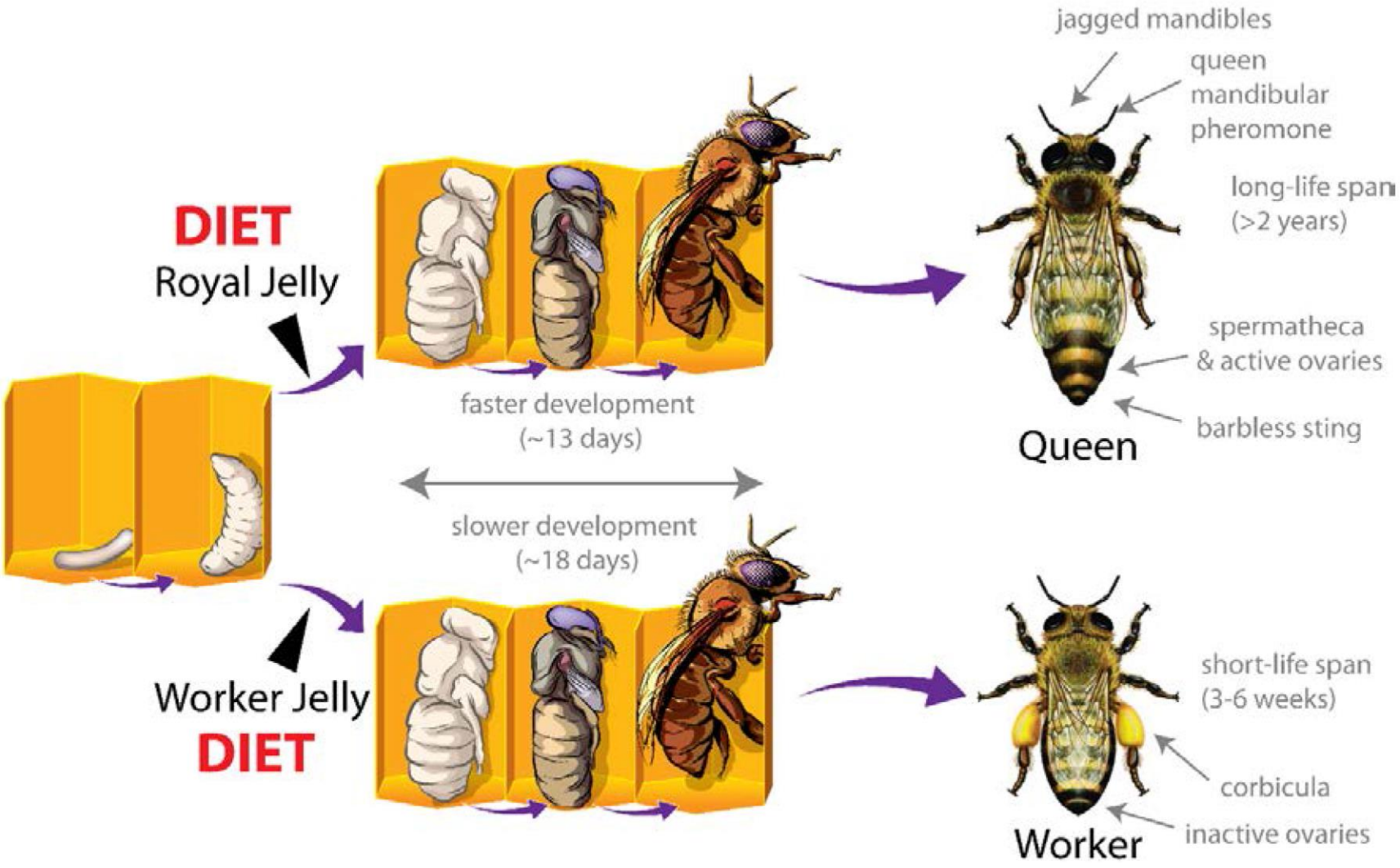
Caste Regulating Pheromones

Lecture 191

# Caste Regulating Pheromones

- Used by social insects to control the development of individuals in a colony.
- Example: The amount of “royal jelly” fed a female bee larva determines whether the larva will become a worker or a queen.

# Caste Regulating Pheromones



# Caste Regulating Pheromones

- Termites, a major group of social insects, exhibit many interesting phenomena related to social physiology, such as mechanisms of caste regulation in a colony.
- In their colonies, there are different types of individuals, i.e., castes, which show distinctive phenotypes specialized in specific colony tasks.

# Caste Regulating Pheromones

- Termite castes comprise reproductives, soldiers and workers, and the caste composition can be altered depending on circumstances.
- For the regulation of caste compositions, interactions among individuals, i.e., social interactions, are thought to be important.

END

# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Aggregation Pheromones

Lecture 192

# Aggregation Pheromones

- Produced to attract individuals to feeding or mating sites.
- Example: Certain bark beetles aggregate on pine trees during an attack on a tree.

# Aggregation Pheromones

- Male-produced sex attractants often are referred to as aggregation pheromones because they typically result in the arrival of both sexes at a calling site.

# Aggregation Pheromones

- Aggregation pheromones have been reported for members of the Coleoptera, Dictyoptera, Hemiptera, Homoptera and Orthoptera and have been identified for hundred of species.

# Aggregation Pheromones

- Aggregation pheromones function in many ways including mate selection, defense against predators, and overcoming host resistance by mass attack.
- A group of individuals (more than a pair) at one location may be referred as aggregation, whether comprised of one sex or both sexes.

# Aggregation Pheromones



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Alarm Pheromones

Lecture 193

# Alarm Pheromones

- Warn other individuals of danger; may cause orientation toward the pheromone source and elicit a subsequent attack or flight from the source.
- Example: A sting from one bee alarms other bees in the area, who are likely to attack.

# Alarm Pheromones

- Social insects, ants, bees, wasps, and the unrelated termites, all show colony defense coordinated by alarm pheromones.
- Honeybees release an alarm pheromone when they sting an intruder such as a bear intent on robbing their nest of its honey.

# Alarm Pheromones

- Many freshwater fish species have an alarm pheromone released from skin glands when these are damaged by a predator's closing its jaws on the victim.
- Other fish detecting the alarm pheromone show escape behavior and greater vigilance.

# Alarm Pheromones

- Alarm pheromones generally consist of low-molecular-weight, highly volatile compounds that easily spread throughout a colony yet evaporate quickly to terminate the aggression when the danger no longer exists.

# Alarm Pheromones



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Trailing Pheromones

Lecture 194

# Trailing Pheromones

- Laid down by foraging insects to help other members of a colony identify the location and quantity of food found by one member of the colony.
- Example: Ants often trail on a pheromone path to and from a food source.

# Trailing Pheromones

- The pheromone trail is reinforced each time an ant travels over it.
- Trail pheromones often serve as a multi purpose chemical secretion that leads members of its own species towards a food source, while representing a territorial mark in the form of an allomone to organisms outside of their species.

# Trailing Pheromones

- Specifically, trail pheromones are often incorporated with secretions of more than one exocrine gland to produce a higher degree of specificity.

# Trailing Pheromones

- Considered one of the primary chemical signaling methods in which many social insects depend on, trail pheromone deposition can be considered one of the main facets to explain the success of social insect communication today.

# Trailing Pheromones



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Reproductive Potential of Insects

Lecture 195

# Reproductive Potential of Insects

- **One of the reasons for insects' success is their high reproductive** potential.
- Reproduction in terrestrial environments, however, has its risks.
- Temperature, moisture, and food supplies vary with the season.

# Reproductive Potential of Insects

- Internal fertilization requires highly evolved copulatory structures, because gametes dry quickly on exposure to air.
- In addition, mechanisms are required to bring males and females together at appropriate times.

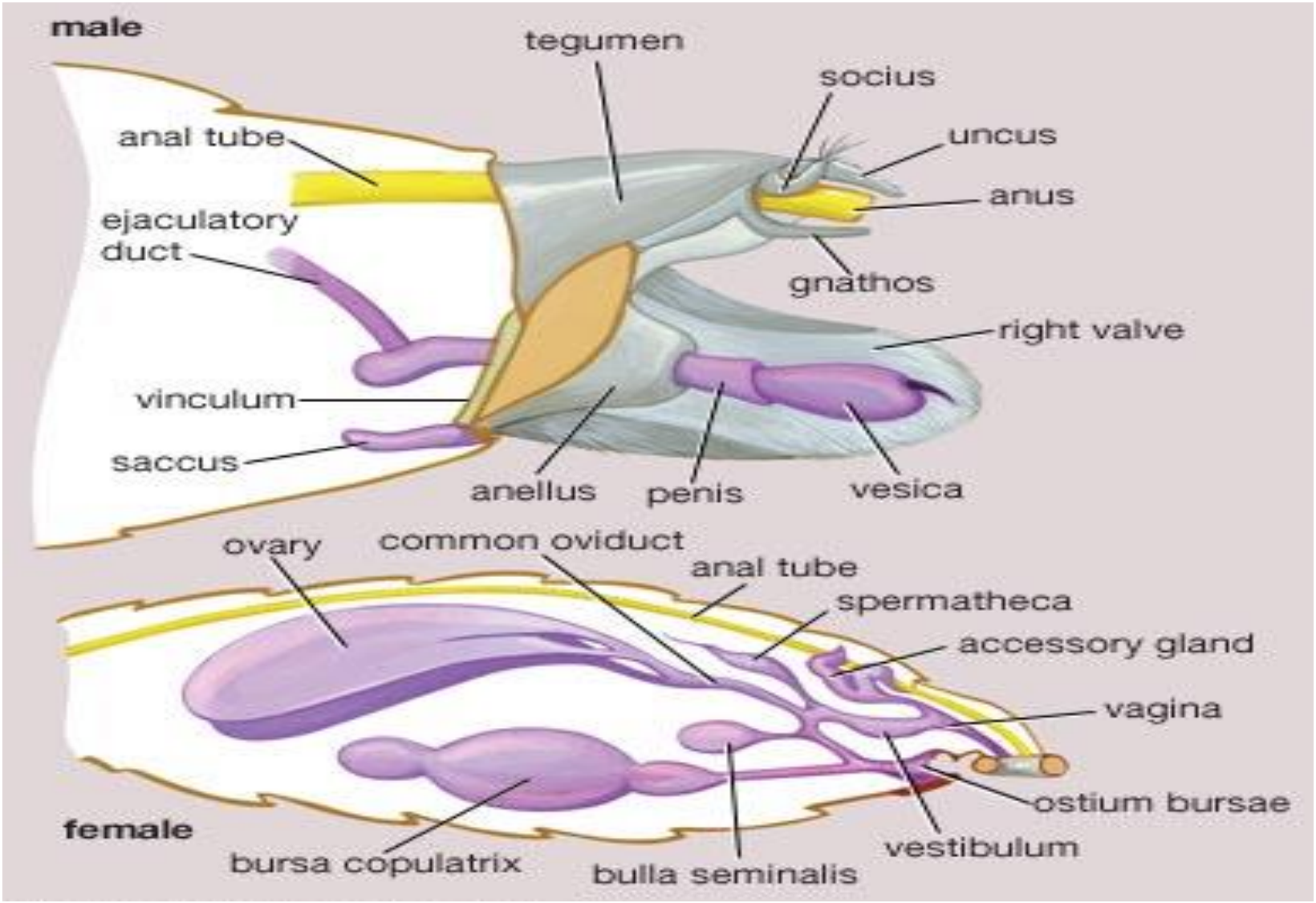
# Reproductive Potential of Insects

- Most insects reproduce o viparously, i.e. by laying eggs.
- The eggs are produced by the female in a pair of ovaries.
- Sperm, produced by the male in one testis or more commonly two, is transmitted to the female during mating by means of external genitalia.

# Reproductive Potential of Insects

- The sperm is stored within the female in one or more spermathecae.
- At the time of fertilization, the eggs travel along oviducts to be fertilized by the sperm and are then expelled from the body ("laid"), in most cases via an ovipositor.

# Reproductive Potential of Insects



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Sexual Maturity of Insects by Internal Factors

Lecture 196

# Sexual Maturity of Insects by Internal Factors

- Complex interactions between internal and external environmental factors regulate sexual maturity.
- Internal regulation includes interactions between endocrine glands (primarily the corpora allata) and reproductive organs.

# Sexual Maturity of Insects by Internal Factors

- In insects, the largest and most obvious endocrine glands are found in the prothorax, just behind the head.

# Sexual Maturity of Insects by Internal Factors

- These prothoracic glands manufacture **ecdy steroids**, a group of closely-related steroid hormones (including ecdysone) that stimulate synthesis of chitin and protein in epidermal cells and trigger a cascade of physiological events that culminates in molting.

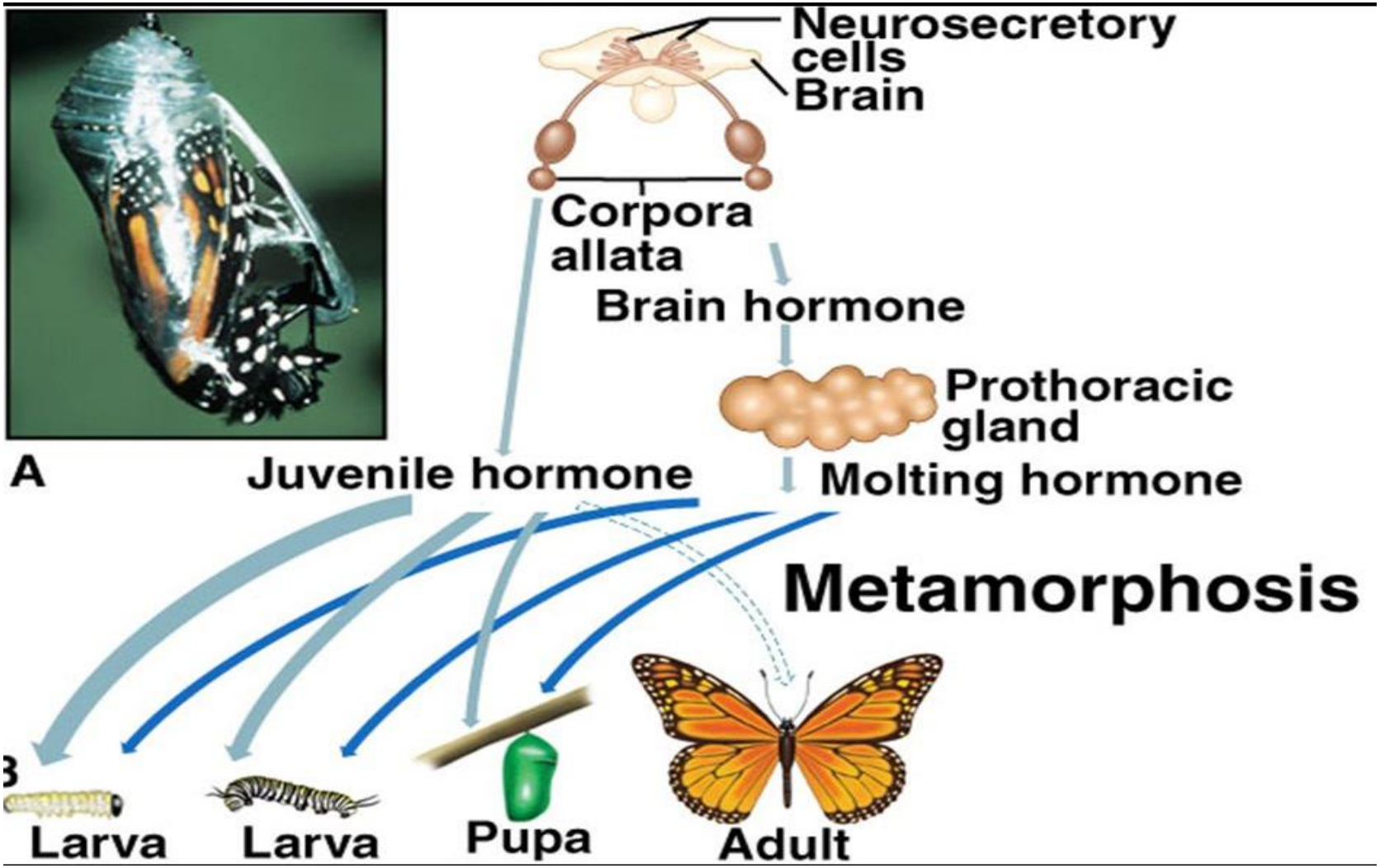
# Sexual Maturity of Insects by Internal Factors

- For this reason, the ecdysteroids are often called “molting hormones”.
- Once an insect reaches the adult stage, its prothoracic glands atrophy (wither away) and it will never molt again.

# Sexual Maturity of Insects by Internal Factors

- The corpora allata, another pair of neurohemal organs, lie just behind the corpora cardiaca.
- They manufacture **juvenile hormone** (JH for short), a compound that inhibits development of adult characteristics during the immature stages and promotes sexual maturity during the adult stage.

# Sexual Maturity of Insects by Internal Factors



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Sexual Maturity of Insects by External Factors

Lecture 197

# Sexual Maturity of Insects by External Factors

- External regulating factors may include the quantity and quality of food.
- For example, the eggs of mosquitoes (order Diptera) do not mature until after the female takes a meal of blood, and the number of eggs produced is proportional to the quantity of blood ingested.

# Sexual Maturity of Insects by External Factors

- Many insects use the photoperiod (the relative length of daylight and darkness in a 24-hour period) for timing reproductive activities because it indicates seasonal changes.
- Population density, temperature, and humidity also influence reproductive activities.

# Sexual Maturity of Insects by External Factors

- Males die shortly after mating, whereas females enter diapause and store sperm which fertilize eggs that mature and are deposited following diapause termination.

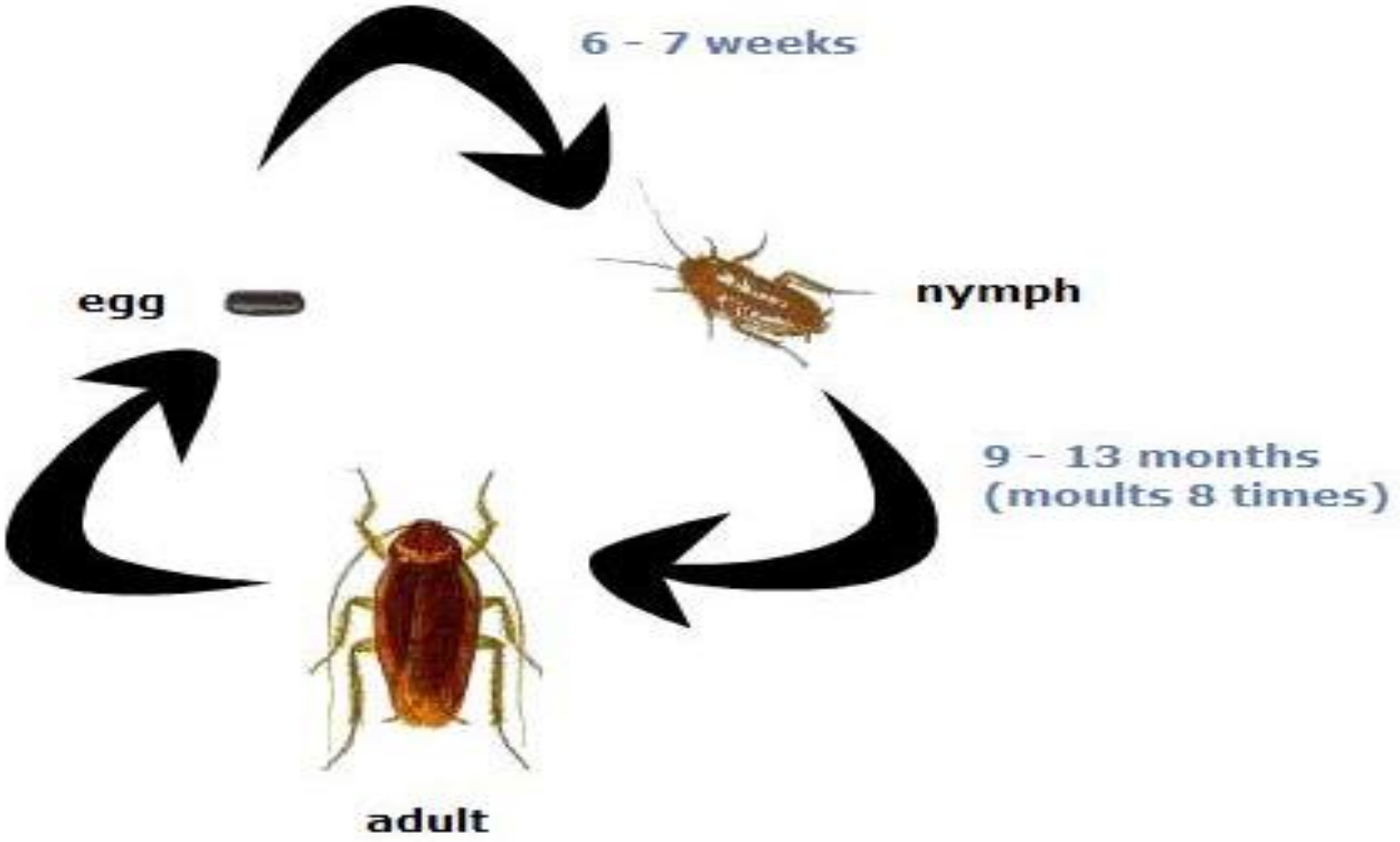
# Sexual Maturity of Insects by External Factors

- Alternatively, females and males may endure the unfavorable period in non receptive and refractory states, respectively, with both mating and oviposition following diapause.

# Sexual Maturity of Insects by External Factors

- While female and male behavioral states in this pattern appear to be controlled by similar environmental (normally photoperiodic) cues acting via the neuroendocrine system, the regulation of male refractoriness is often less stringent.

# Sexual Maturity of Insects by External Factors



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Mating Behavior in Insects

Lecture 198

# Mating Behavior in Insects

- Most insects have complex mating behaviors for locating and recognizing a potential mate, for positioning a mate for copulation, or for pacifying an aggressive mate.

# Mating Behavior in Insects

- Mating behavior may involve pheromones (moths, order Lepidoptera), visual signals (fireflies, order Coleoptera), and auditory signals (cicadas, order Homoptera; and grasshoppers, crickets, and katydids, order Orthoptera).

# Mating Behavior in Insects

- Once other stimuli have brought the male and female together, tactile stimuli from the antennae and other appendages help position the insects for mating.

# Mating Behavior in Insects

- Mating behavior starts by locating mates.
- Male and female have different strategies toward copulation and reproduction.

# Mating Behavior in Insects

- After locating mates is courtship; one male and female have encountered each other, they may proceed directly to copulation or may first engage in various rituals, collectively known as courtship.
- The final ritual is copulation.

# Mating Behavior in Insects



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Insect Development

Lecture 199

# Insect Development

- Insect evolution has resulted in the divergence of immature and adult body forms and habits.
- For insects in the superorder Endopterygota immature stages, called **larval instars,** are a time of growth and accumulation of reserves for the transition to adulthood.

# Insect Development

- The adult stage is associated with reproduction and dispersal.
- In these orders, insects tend to spend a greater part of their lives in juvenile stages.

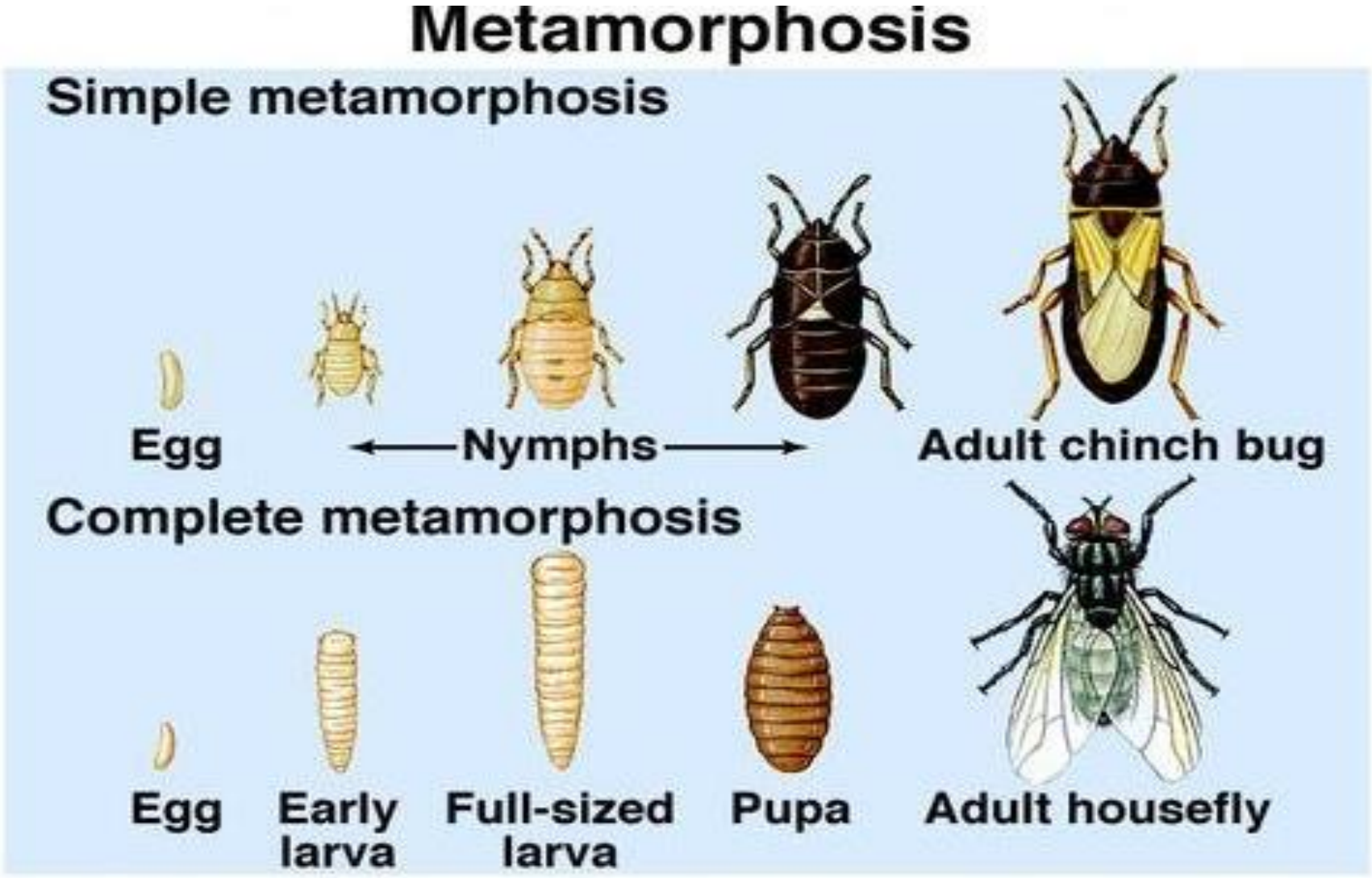
# Insect Development

- The developmental patterns of insects reflect degrees of divergence between immatures and adults and are classified into three (or sometimes four) categories.

# Insect Development

- Changes occur in the insect during the development from **egg** to the adult.
- Complete metamorphosis consists of **four** stages: **egg, larva ,pupa**, and adult.
- Incomplete metamorphosis consists of three stages: **egg,** nymph, and adult.

# Insect Development



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Ametabolous
Metamorphosis

Lecture 200

# Ametabolous Metamorphosis

- In insects that display **ametabolous** (**Gr. *a,* without** and *metabolos,* change) **metamorphosis, the primary differences between** adults and larvae are body size and sexual maturity.
- Both adults and larvae are wingless.

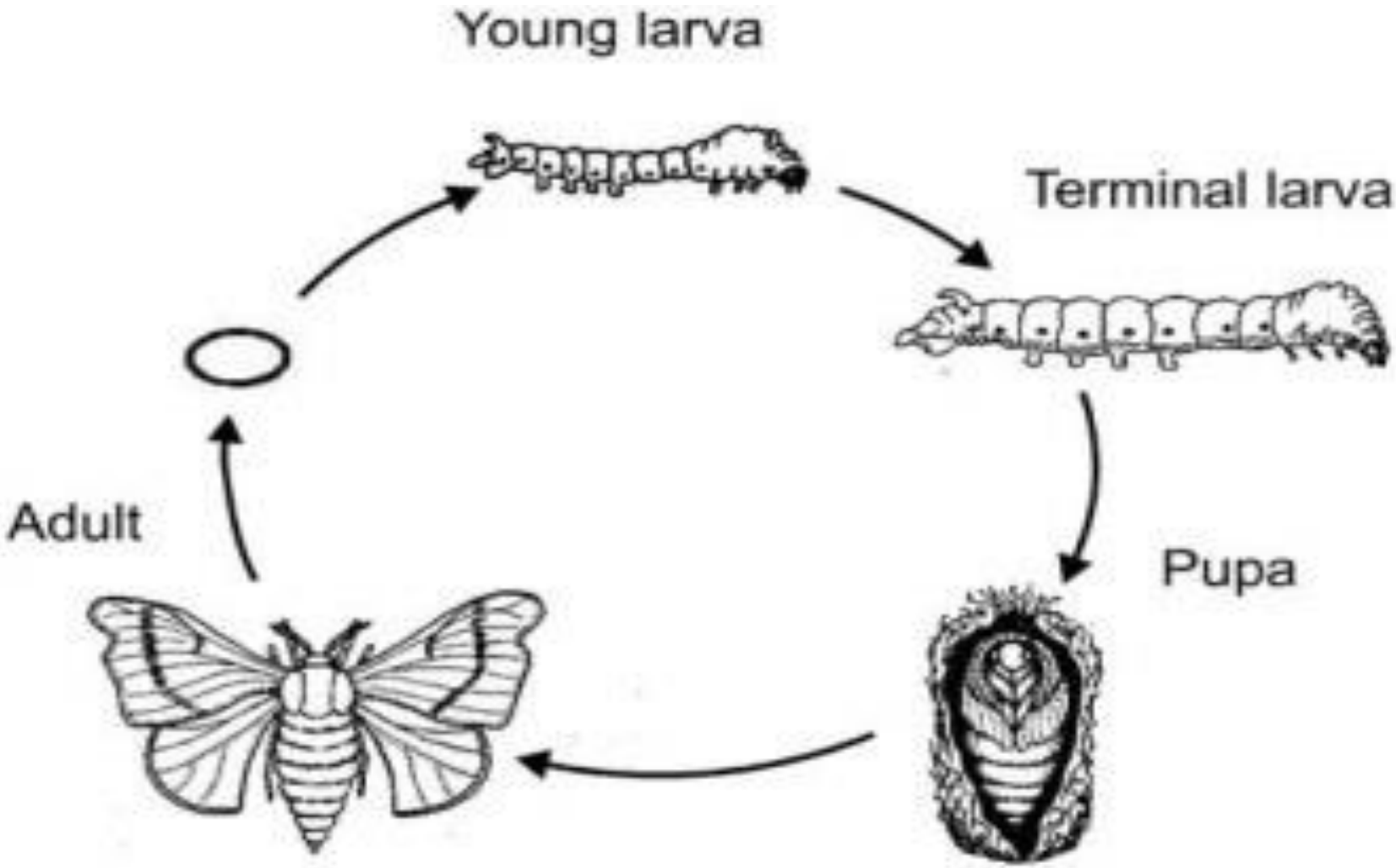
# Ametabolous Metamorphosis

- The number of molts in the ametabolous development of a species varies, and unlike most other insects, molting continues after sexual maturity.
- Silverfish (order Thysanura) have ametabolous metamorphosis.

# Ametabolous Metamorphosis

- **Ametabolous** insect lifecycles are unique in that the adult lays eggs, and immature insects that first hatch from the eggs resemble miniature versions of the adult.
- Immature silverfish molt six to seven times until they reach their adult, sexually mature stage.

# Ametabolous Metamorphosis



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Paurometabolous Metamorphosis

Lecture 201

# Paurometabolous Metamorphosis

- **Paurometabolous (Gr. *pauros,* small) metamorphosis** involves a species-specific number of molts between egg and adult stages, during which immatures gradually take on the adult form.

# Paurometabolous Metamorphosis

- The external wings develop (except in those insects, such as lice, that have secondarily lost wings), adult body size and proportions are attained, and the genitalia develop during this time.
- Immatures are called **nymphs.**

# Paurometabolous Metamorphosis

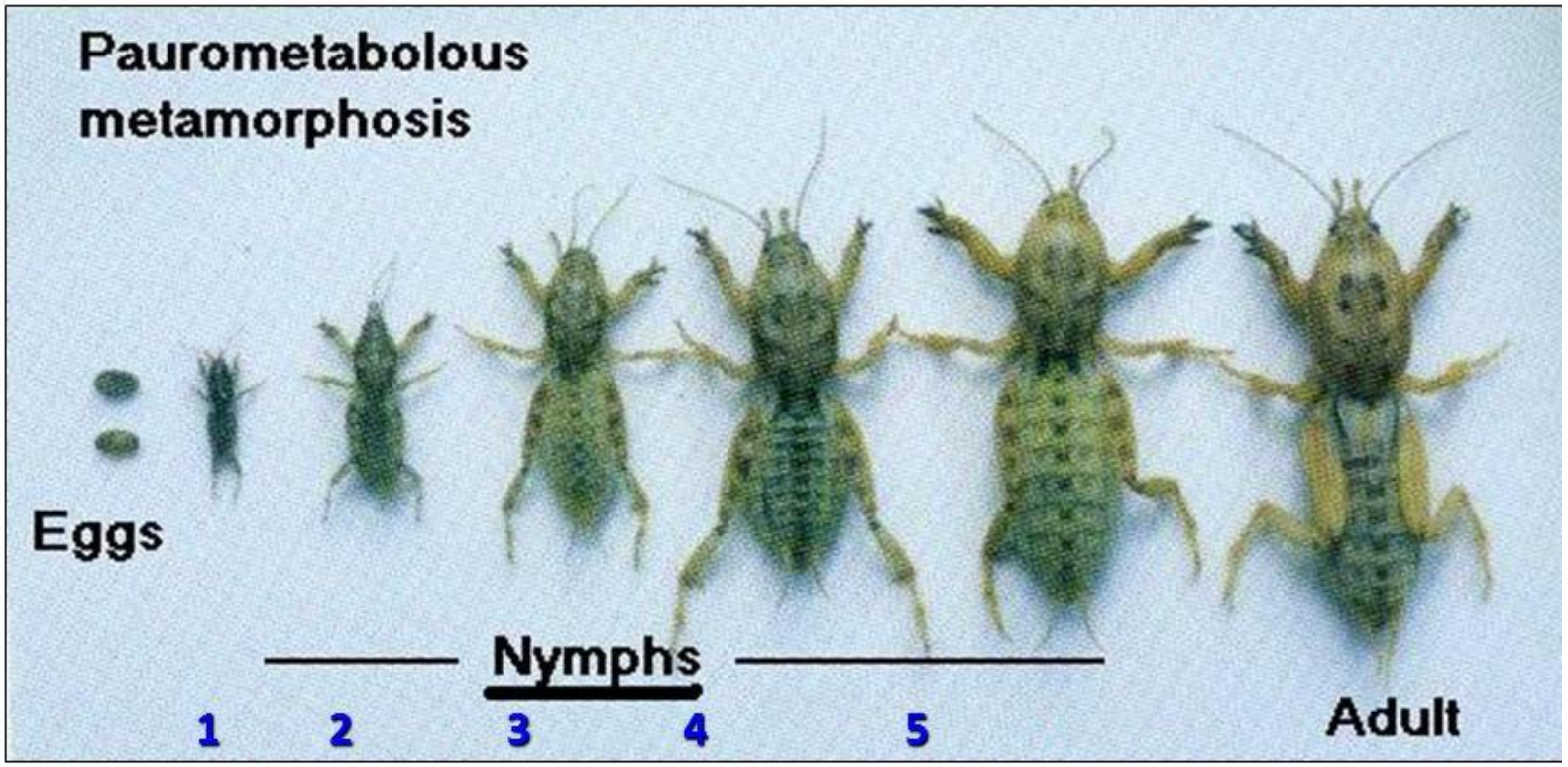
- In this form of development, immature stages (nymphs) resemble small adults and typically have external wing buds.
- They live in the same habitat as adults, typically taking the same food.

# Paurometabolous Metamorphosis

- Grasshoppers (order Orthoptera) and chinch bugs (order Hemiptera) show paurometabolous metamorphosis.

# Paurometabolous Metamorphosis

Paurametabolous – gradual metamorphosis




instars

# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Hemimetabolous
Mmetamorphosis

Lecture 202

# Hemimetabolous Metamorphosis

- Some zoologists use an additional classification for insects that have a series of gradual changes in their development, but whose immature form is much different from the adult form usually due to the presence of gills (e.g., mayflies, order Ephemeroptera; dragonflies, order Odonata).

# Hemimetabolous Metamorphosis

- This kind of development is called **hemimetabolous (Gr. *hemi,* half) metamorphosis**, and the immatures are aquatic and called **naiads (L. *naiad,* water** nymph).

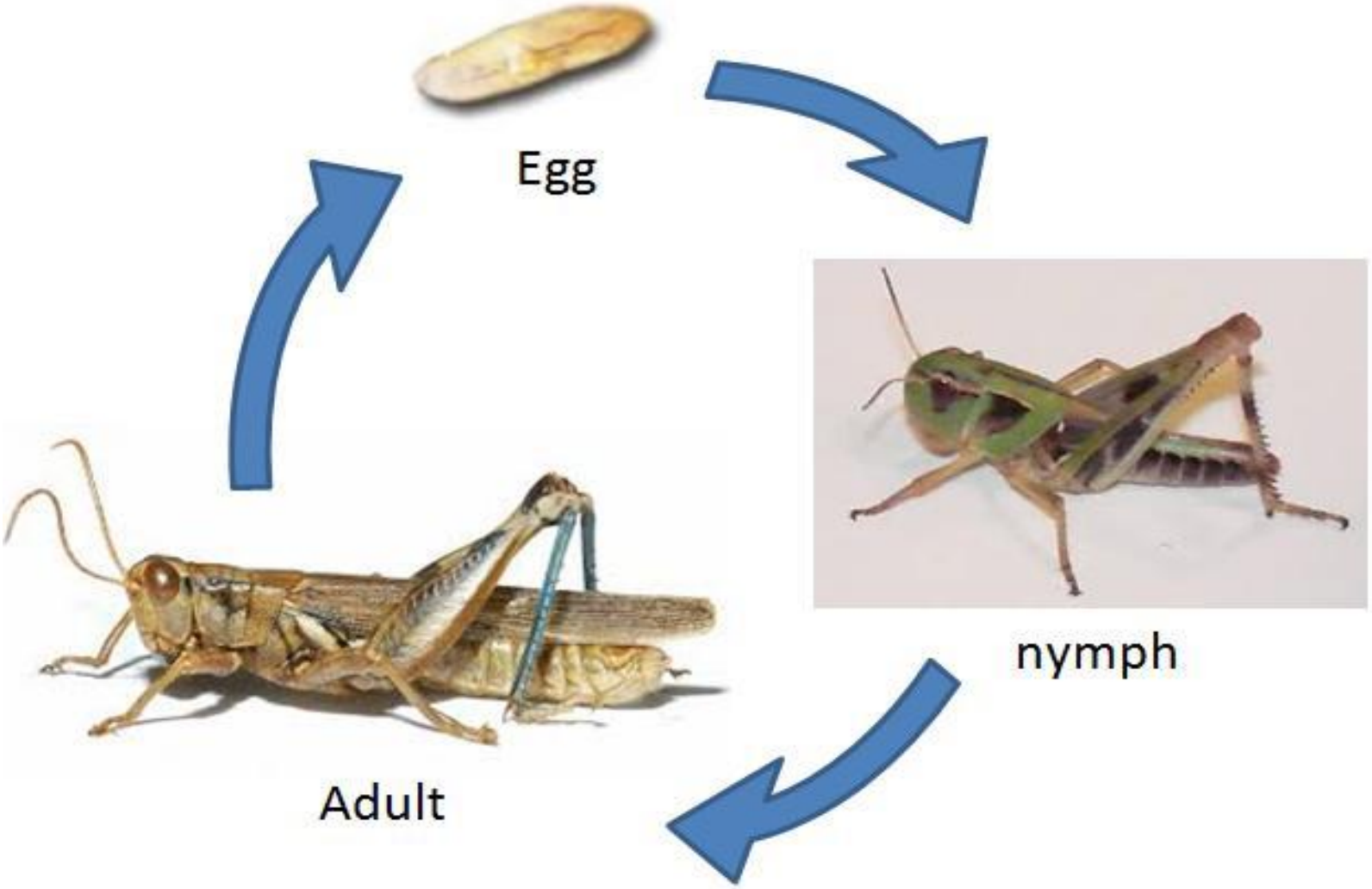
# Hemimetabolous Metamorphosis

- Insects that undergo **hemimetabolous metamorphosis** emerge from eggs into nymphs (called larvae by some entomologists) that are similar in shape to the adults.
- They go through several nymphal stages (called instars) before they undergo a final molt into their adult form.

# Hemimetabolous Metamorphosis

- Also known as incomplete or simple **metamorphosis**.
- The nymph, or immature insect, resembles the adult in form and eating habits, differing in size, body proportions, and color pattern.

# Hemimetabolous Metamorphosis



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Holometabolous Metamorphosis

Lecture 203

# Holometabolous Metamorphosis

- In **holometabolous (Gr. *holos,* whole) metamorphosis, immatures** are called larvae because they are very different from the adult in body form, behavior, and habitat.
- The number of larval instars is species specific, and the last larval molt forms the **pupa.**

# Holometabolous Metamorphosis

- The pupa is a time of apparent inactivity but is actually a time of radical cellular change, during which all characteristics of the adult insect develop.
- A protective case may enclose the pupal stage.

# Holometabolous Metamorphosis

- The last larval instar (e.g., moths, order Lepidoptera) constructs a **cocoon partially or entirely from silk.**
- **The chrysalis (e.g., butterflies, order Lepidoptera) and puparium (e.g.,** flies, order Diptera) are the last larval exoskeletons and are retained through the pupal stage.

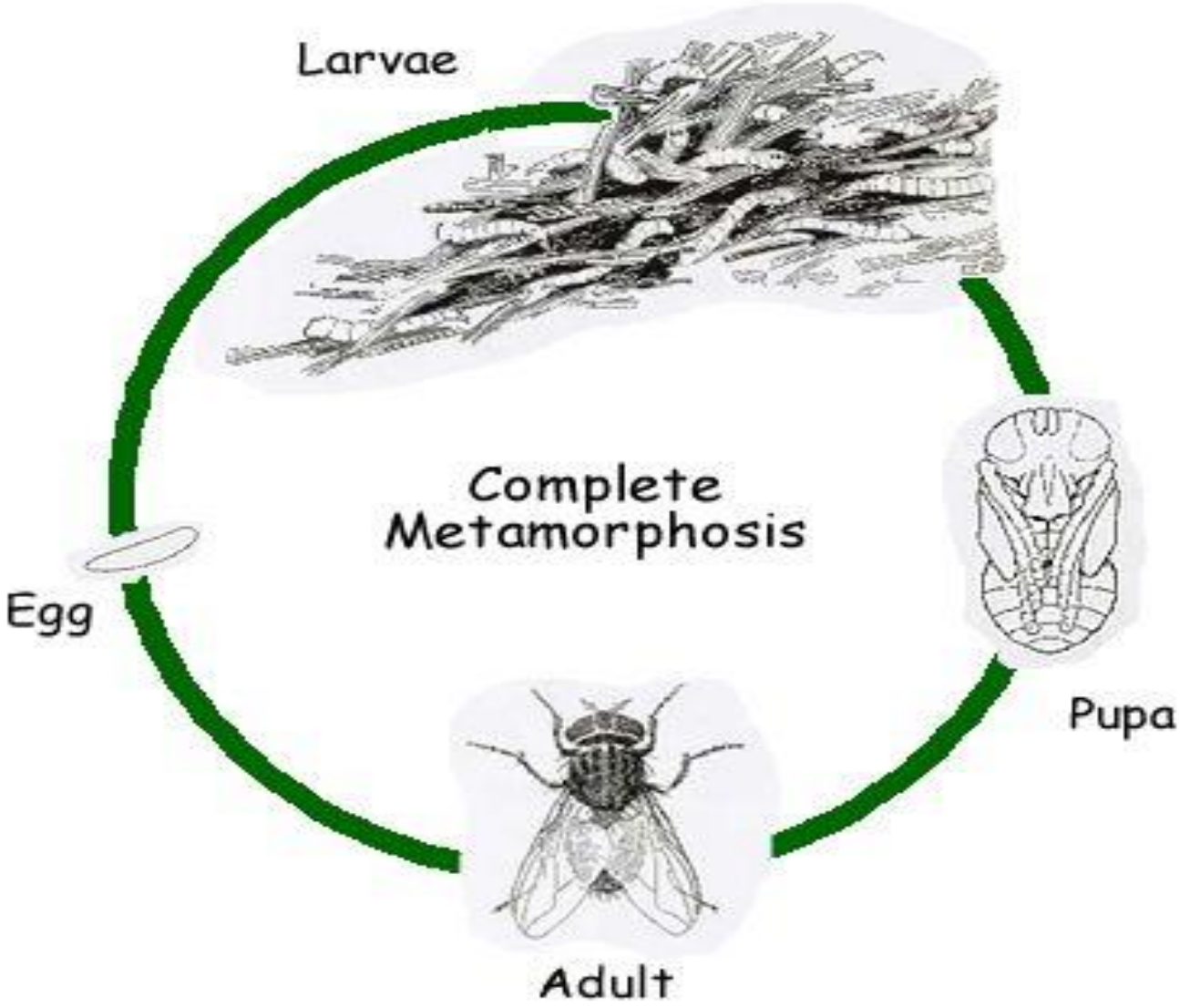
# Holometabolous Metamorphosis

- Other insects (e.g., mosquitoes, order Diptera) have pupae that are unenclosed by a larval exoskeleton, and the pupa may be active.

# Holometabolous Metamorphosis

- The final molt to the adult stage usually occurs within the cocoon, chrysalis, or puparium, and the adult then exits, frequently using its mandibles to open the cocoon or other enclosure.
- This final process is called emergence or eclosion.

# Holometabolous Metamorphosis



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Innate Behavior of Insects

Lecture 204

# Innate Behavior of Insects

- Insects have many complex behavior patterns.
- Most of these are innate (genetically programmed).
- For example, a newly emerged queen in a honeybee hive will search out and try to destroy other queen larvae and pupae in the hive.

# Innate Behavior of Insects

- This behavior is innate because no experiences taught the potential queen that her survival in the hive required the death or dispersal of all other potential queens.

# Innate Behavior of Insects

- Similarly, no experience taught her how queen-rearing cells differ from the cells containing worker larvae and pupae.
- Some insects are capable of learning and remembering, and these abilities play important roles in insect behavior.

# Innate Behavior of Insects



Spider spinning a web



Bird building a nest



Caterpillar making a cocoon



Dolphin leaping from the water

# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Social Insects

Lecture 205

# Social Insects

- Social behavior has evolved in many insects and is particularly evident in those insects that live in colonies.
- Usually, different members of the colony are specialized, often structurally as well as behaviorally, for performing different tasks.

# Social Insects

- Social behavior is most highly evolved in the bees, wasps, and ants (order Hymenoptera) and in termites (order Isoptera).
- Each kind of individual in an insect colony is called a **caste.**
- **Often, three or** four castes are present in a colony.

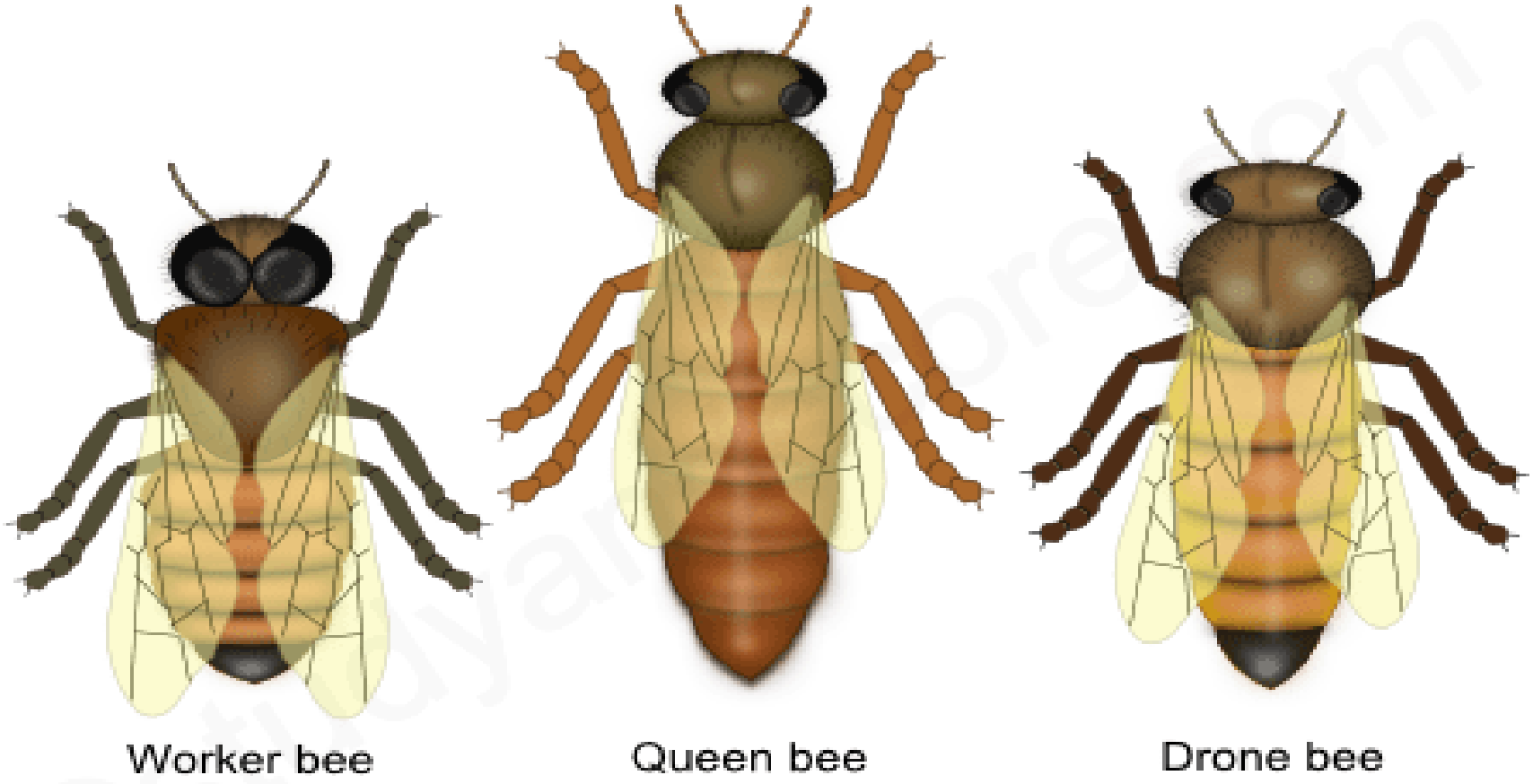
# Social Insects

- Reproductive females are referred to as queens.
- Workers may be sterile males and females (termites) or sterile females (order Hymenoptera), and they support, protect, and maintain the colony.

# Social Insects

- Their reproductive organs are often degenerate.
- Reproductive males inseminate the queen (s) and are called kings or drones.
- Soldiers are usually sterile and may possess large mandibles to defend the colony.

# Social Insects




POLYMORPHISM (CASTE SYSTEM) IN HONEY BEES

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# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Caste System in Honey Bees

Lecture 206

# Caste System in Honey Bees

- Honeybees (order Hymenoptera) have three of these castes in their colonies.
- A single queen lays all the eggs.
- Workers are female, and they construct the comb out of wax that they produce.

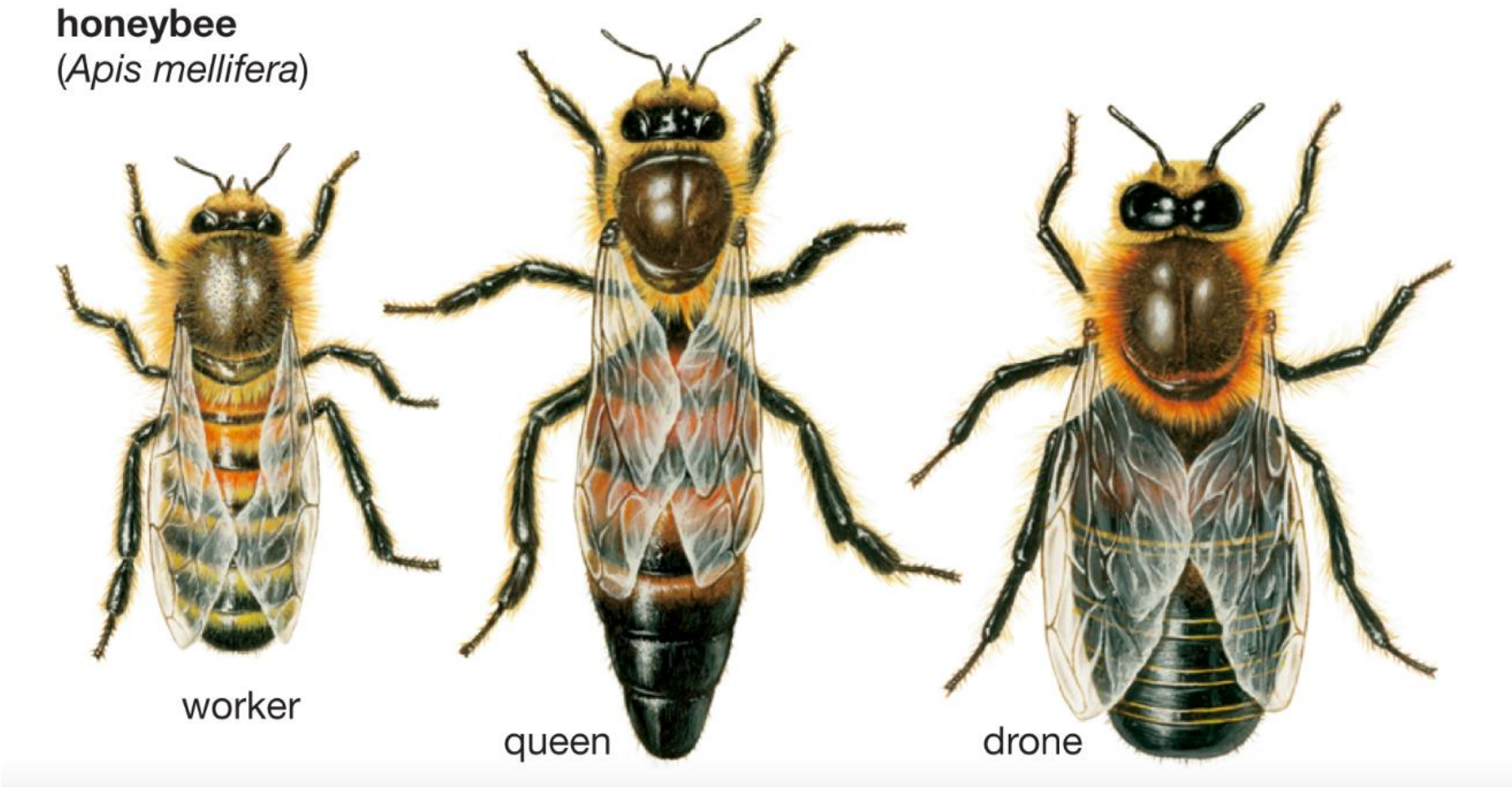
# Caste System in Honey Bees

- They also gather nectar and pollen, feed the queen and drones, care for the larvae, and guard and clean the hive.
- These tasks are divided among workers according to age.
- Younger workers take care of jobs around the hive, and older workers forage  for nectar and pollen.

# Caste System in Honey Bees

- Except for those that overwinter, workers live for about one month.
- Drones develop from unfertilized eggs, do not work, and are fed by workers until they leave the hive to attempt mating with a queen.

# Caste System in Honey Bees



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Control of Honey Bee Caste System

Lecture 207

# Control of Honey Bee Caste System

- A **pheromone** that the queen releases controls the honeybee caste system.
- Workers lick and groom the queen and other workers.
- In so doing, they pick up and pass to other workers a caste regulating pheromone.

# Control of Honey Bee Caste System

- This pheromone inhibits the workers from rearing new queens.
- As the queen ages, or if she dies, the amount of caste-regulating pheromone in the hive decreases.

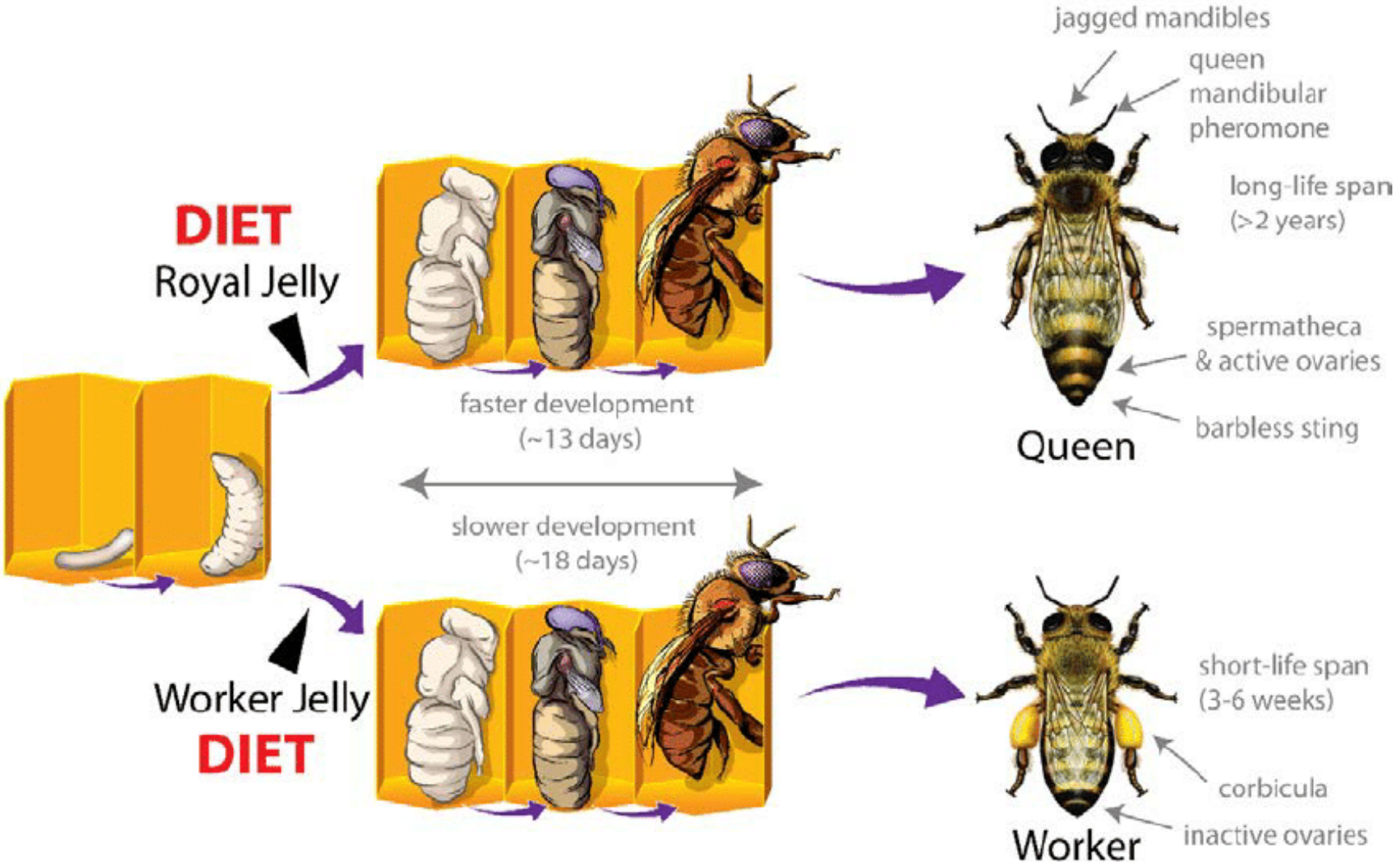
# Control of Honey Bee Caste System

- As the pheromone decreases, workers begin to feed the food for queens (“royal jelly”) to several female larvae developing in the hive.
- This food contains chemicals that promote the development of queen characteristics.

# Control of Honey Bee Caste System

- The larvae that receive royal jelly develop into queens, and as they emerge, the new queens begin to eliminate each other until only one remains.
- The queen that remains goes on a mating flight and returns to the colony, where she lives for several years.

# Control of Honey Bee Caste System



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Role of Insects in Pollination

Lecture 208

# Role of Insects in Pollination

- Only about 0.5% of insect species adversely affect human health and welfare.
- Many others have provided valuable services and commercially valuable products, such as wax, honey, and silk, for thousands of years.

# Role of Insects in Pollination

- Insects are responsible for the pollination of approximately 65% of all plant species.
- Insects and flowering plants have coevolutionary relationships that directly benefit humans.

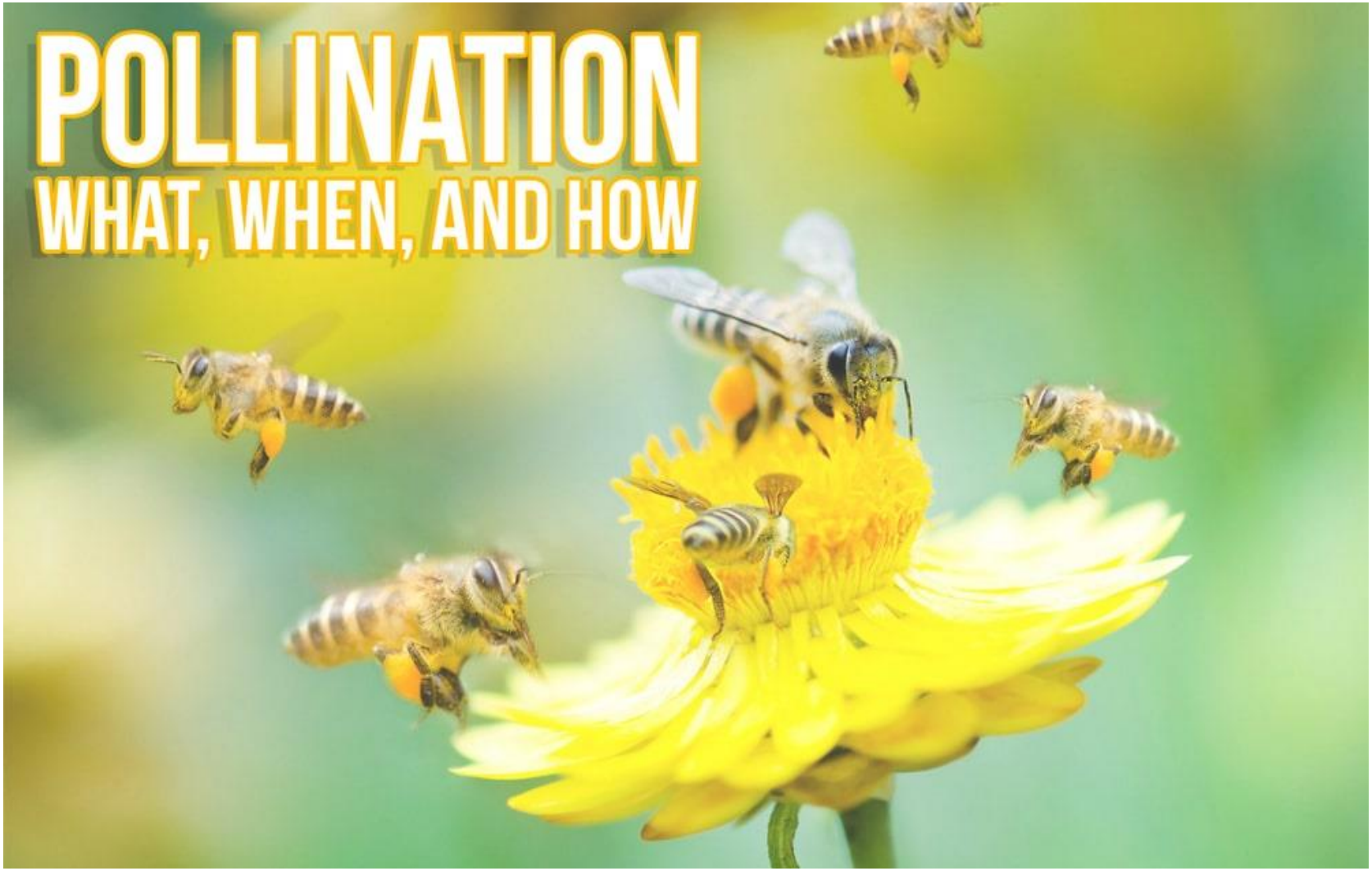
# Role of Insects in Pollination

- The annual value of insect-pollinated crops is estimated at $19 billion per year in the United States.

# Role of Insects in Pollination

- **Insect pollination** is crucial to most gardens and is as simple as **insects** like bees, butterflies and wasps flying from flower to flower in order to collect nectar.
- In the process, pollen collects on their bodies and rubs off on other flowers that they visit.

# Role of Insects in Pollination



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Biological Control by Insects

Lecture 209

# Biological Control by Insects

- Insects are also agents of biological control.
- The classic example of one insect regulating another is the vedalia (lady bird) beetles' control of cottony-cushion scale.
- The scale insect, *Icerya purchasi,* was introduced into California in the 1860s.

# Biological Control by Insects

- Within 20 years, the citrus industry in California was virtually destroyed.
- The vedalia beetle (*Vedalia cardinalis* ) was brought to the United States in 1888 and 1889 and cultured on trees infested with scale.

# Biological Control by Insects

- In just a few years, the scale was under control, and the citrus industry began to recover.

# Biological Control by Insects



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Importance of soil Dwelling Insects

Lecture 210

# Importance of Soil Dwelling Insects

- Many other insects are also beneficial.
- Soil-dwelling insects play important roles in aeration, drainage, and turnover of soil, and they promote decay processes.
- Other insects serve important roles in food webs.

# Importance of Soil Dwelling Insects

- Insects are used in teaching and research, and have contributed to advances in genetics, population ecology, and physiology.
- Insects have also given endless hours of pleasure to those who collect them and enjoy their beauty.

# Importance of Soil Dwelling Insects

- Soil-dwelling insects are hidden in the soil and hard to detect until the sudden appearance of damage creates the demand for rapid, curative treatment.

# Importance of Soil Dwelling Insects



# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

Insects as Parasites and Vectors of Disease

Lecture 211

# Insects as Parasites and Vectors of Disease

- Some insects, however, are parasites and vectors of disease.
- Parasitic insects include head, body, and pubic lice (order Anoplura); bedbugs (order Hemiptera); and fleas (order Siphonaptera).

# Insects as Parasites and Vectors of Disease

- **Parasites** transmitted by **insects** often circulate in the blood of the host, with the **parasite** residing in and damaging organs or other parts of the body.

# Insects as Parasites and Vectors of Disease

- Other insects transmit disease-causing microorganisms, nematodes, and flatworms.
- Insect-transmitted diseases, such as malaria, yellow fever, bubonic plague, encephalitis, leishmaniasis, and typhus, have changed the course of history.

# Insects as Parasites and Vectors of Disease

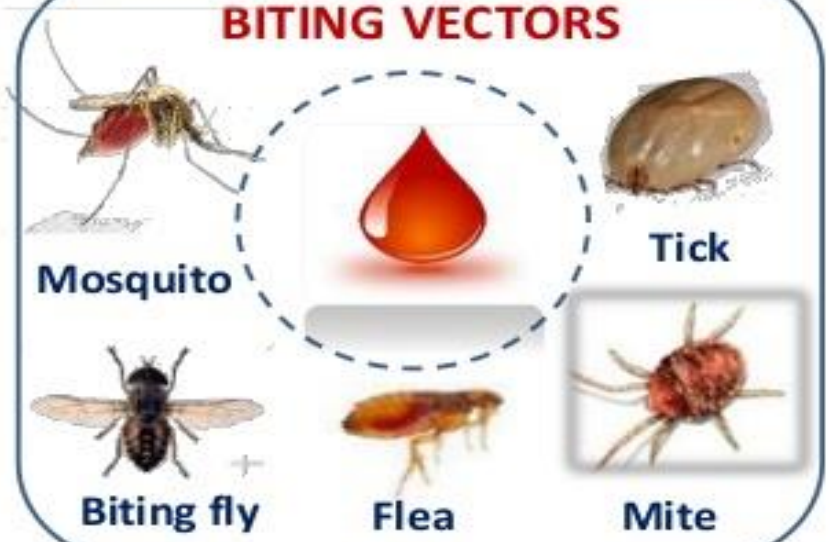
- Triatomine bugs are the vectors for Chagas disease.
- An **insect** that transmits a disease is known as a **vector**, and the disease is referred to as a **vector-borne disease**.

# Insects as Parasites and Vectors of Disease

## What are vectors?

**VECTORS** are common **INSECTS** that carry & transmit diseases causing bacteria, viruses & parasites

### BITING VECTORS




### CARRIER VECTORS




**Transmit more than 250 food borne diseases** leading to

- Nausea, vomiting and diarrhoea
- Malaise, headache, fever
- Deaths: Globally- **19 Lakhs/year**
  South-East Asia- **10 Lakhs/year**

# Animal Diversity: Invertebrates

## Sub-Phylum Hexapoda

### Role of Insects as Pests

Lecture 212

# Role of Insects as Pests

- Other insects are pests of domestic animals and plants.
- Some reduce the health of domestic animals and the quality of animal products.
- Insects feed on crops and transmit plant diseases, such as Dutch elm disease, potato virus, and asters yellow.

# Role of Insects as Pests

- Insect pests destroy crops in the field through their biting, chewing, boring, sucking and defoliation activities.
- Spots of injuries by insects may predispose crops to disease attack.

# Role of Insects as Pests

- They reduce the quality of produce in the field as well as in the store e.g. Potato tuber moth (PTM) on Irish Potatoes.
- They render vegetables and fruits unattractive and unmarketable e.g. damage on crucifer leaves by diamond back moth (Plutella xylostella).

# Role of Insects as Pests

**4. Potato tuber moth:**

***Phthorimoea operculella***

**(Gelechiidae: Lepidoptera)**







# Role of Insects as Pests

- Annual lost revenue from insect damage to crops or insect-transmitted diseases in the United States is approximately $5 billion.

END

# Animal Diversity: Invertebrates

## Phylum Arthropoda

### Further Phylogenetic Consideration

Lecture 213

# Further Phylogenetic Considerations

- A fundamental question regarding arthropod evolution concerns whether or not the arthropod taxa represent fundamentally different evolutionary lineages.
- Many zoologists believe that the living arthropods should be divided into three separate phyla: Chelicerata, Crustacea, and Uniramia.

# Further Phylogenetic Considerations

- **A polyphyletic** origin of these groups implies convergent evolution of remarkably similar arthropodan features in all three (or at least two of three) phyla.
- Although evidence exists for dual origins of tracheae, mandibles, and compound eyes, many zoologists believe that convergence in all other arthropod traits is unlikely.

# Further Phylogenetic Considerations

- Members of all four subphyla are present in the fossil record from the early Paleozoic era, and there currently are no known fossils of arthropods from Precambrian times.
- The fossil record, therefore, is of little help in discovering the evolutionary relationships among the arthropod subphyla.

# Further Phylogenetic Considerations

- Zoologists must rely on comparative anatomy, comparative embryology, and molecular studies to investigate these relationships.
- Central to the questions surrounding arthropod phylogeny are two important issues.
- One issue is whether or not the biramous limbs of crustaceans and trilobites are homologous.

# Further Phylogenetic Considerations

- Homology of these appendages would imply that the trilobites were closely related to the crustaceans; therefore, many zoologists view the trilobites as an important ancestral group.

# Further Phylogenetic Considerations

- It is possible to envision crustaceans, and possibly the arachnids, arising from the trilobites, but envisioning a similar origin for the myriapods and insects is more difficult.

# Further Phylogenetic Considerations

- A second important issue is whether or not the mandibles of uniramians and crustaceans are homologous.
- Superficially, the mandibles of members of these groups are structurally similar and have similar functions.

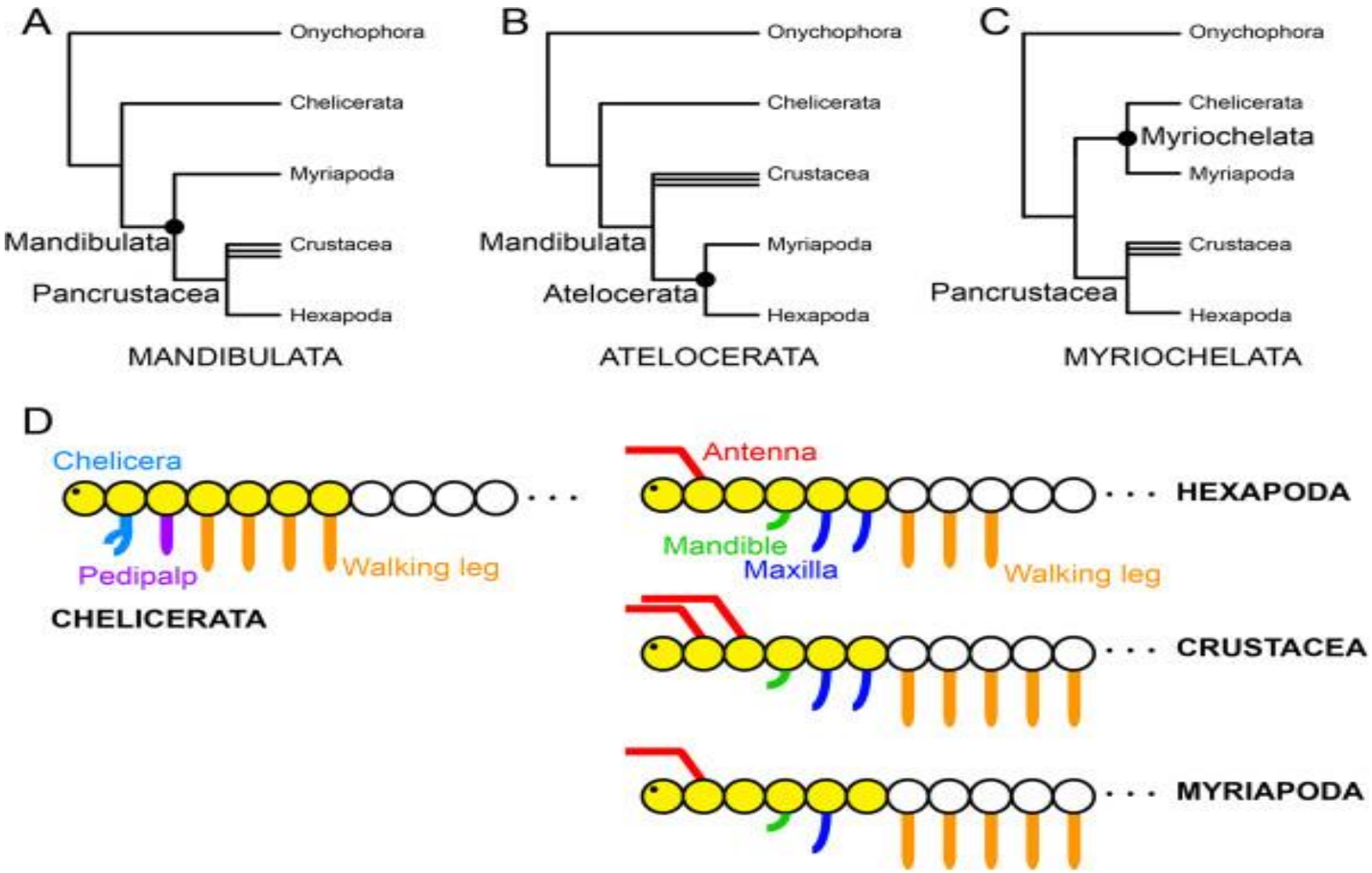
# Further Phylogenetic Considerations

- Muscle arrangements and methods of articulation, however, are different enough that many zoologists doubt that these appendages are homologous.
- Discussions of arthropod phylogeny also center around the origins and possible homologies of arthropod compound eyes, tracheal systems, and Malpighian tubules.

# Further Phylogenetic Considerations

- Questions regarding the evolutionary relationships within the arthropods are difficult to answer.
- These questions will probably remain unanswered until new fossils are discovered or data from molecular studies provide more information on ancestral arthropods.

# Further Phylogenetic Considerations



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Echinoderms: Relationship to other Animals

Lecture 214

# Echinoderms: Relationship to Other Animals

- **Most zoologists believe that echinoderms share a common ancestry** with hemichordates and chordates because of the deuterostome characteristics that they share;
- An anus that develops in the region of the blastopore,

# Echinoderms: Relationship to Other Animals

- A coelom that forms from outpockets of the embryonic gut tract (vertebrate chordates are an exception),
- Radial, indeterminate cleavage.

# Echinoderms: Relationship to Other Animals

- Unfortunately, no known fossils document a common ancestor for these phyla or demonstrate how the deuterostome lineage was derived from ancestral diploblastic or triploblastic stocks.

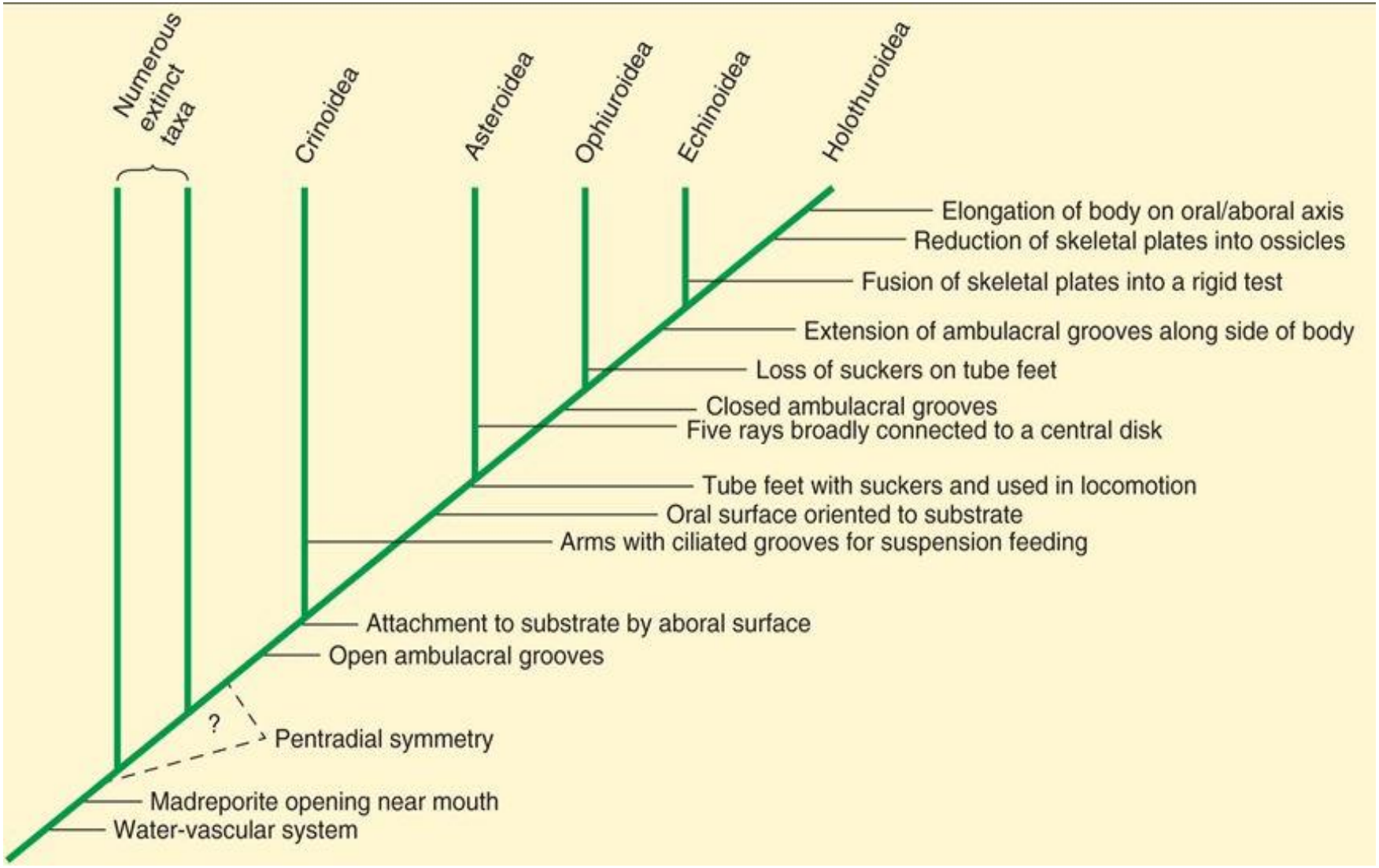
# Echinoderms: Relationship to Other Animals

- **Although echinoderm adults are radially symmetrical,** most zoologists believe that echinoderms evolved from bilaterally symmetrical ancestors.

# Echinoderms: Relationship to Other Animals

- Evidence for this relationship includes bilaterally symmetrical echinoderm larval stages and extinct forms that were not radially symmetrical.

# Echinoderms: Relationship to Other Animals



# Echinoderm: General Characteristics

## Phylum Echinodermata

Echinoderm: General Characteristic s

Lecture 215

# Echinoderm: General Characteristics

- Phylum Echinodermata (Gr. *echinos,* spiny *derma,* skin *ata,* to bear) attached to their substrates and probably lived as filter feeders a feature found in only one class of modern echinoderms.

# Echinoderm: General Characteristics

- Characteristics of the phylum Echinodermata include:
- 1. Calcareous endoskeleton in the form of ossicles that arise from mesodermal tissue
- 2. Adults with pentaradial symmetry and larvae with bilateral symmetry

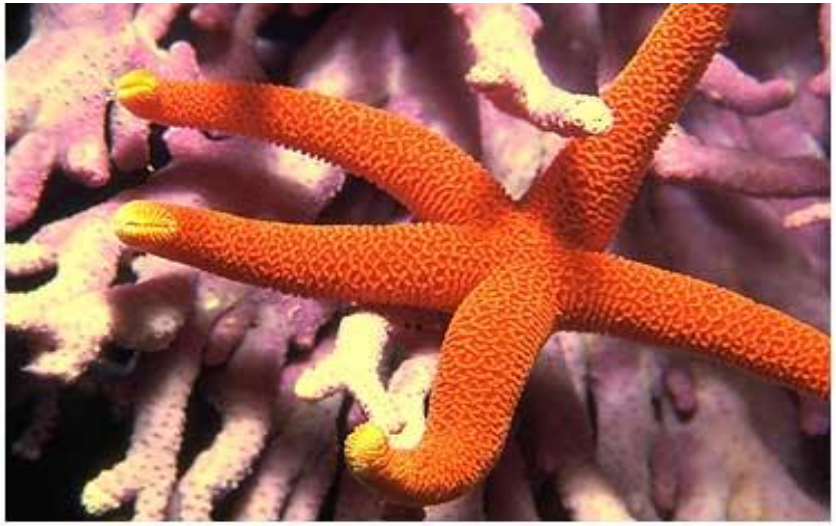
# Echinoderm: General Characteristics

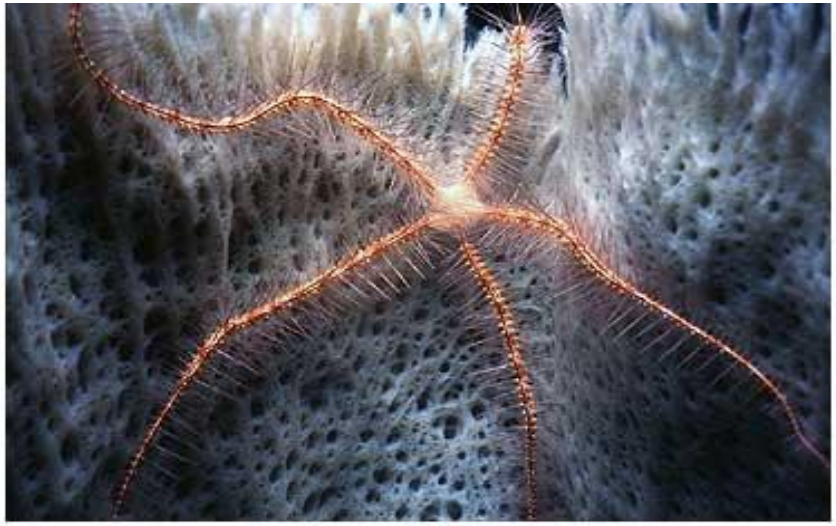
- 3. Water-vascular system composed of water-filled canals used in locomotion, attachment, and/or feeding
- 4. Complete digestive tract that may be secondarily reduced
- 5. Hemal system derived from coelomic cavities
- 6. Nervous system consisting of a nerve net, nerve ring, and radial

# Echinoderm: General Characteristics

- The approximately seven thousand species of living echinoderms are exclusively marine and occur at all depths in all oceans.
- Radial symmetry is adaptive for sedentary or slowly moving animals because it allows a uniform distribution of sensory, feeding, and other structures around the animal.

# Echinoderm: General Characteristics









# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Echinoderm Skeleton

Lecture 216

# Echinoderm Skeleton

- The echinoderm skeleton consists of a series of calcium carbonate plates called **ossicles.**
- These plates are derived from mesoderm, held in place by connective tissues, and covered by an epidermal layer.

# Echinoderm Skeleton

- If the epidermal layer is abraded away, the skeleton may be exposed in some body regions.
- The skeleton is frequently modified into fixed or articulated spines that project from the body surface.

# Echinoderm Skeleton

- Some echinoderms have very tightly interlocking plates, like sea urchins.
- In sea urchins and sand dollars, these interlocking plates form an enclosure, almost like a shell, called a **test**.
- Sea urchins also have spines protruding from their tests, in life.

# Echinoderm Skeleton

- Others have more movable plates, like sea stars or starfish.
- Sea cucumbers, the worm-like echinoderms, have an endoskeleton that is nearly microscopic.
- They may not have as strong a defense system, but they are much more flexible and movable.

# Echinoderm Skeleton












# Animal Diversity: Invertebrates

## Phylum Echinodermata

Echinoderm skeleton and Pentaradial Body form of Echinoderms

Lecture 217

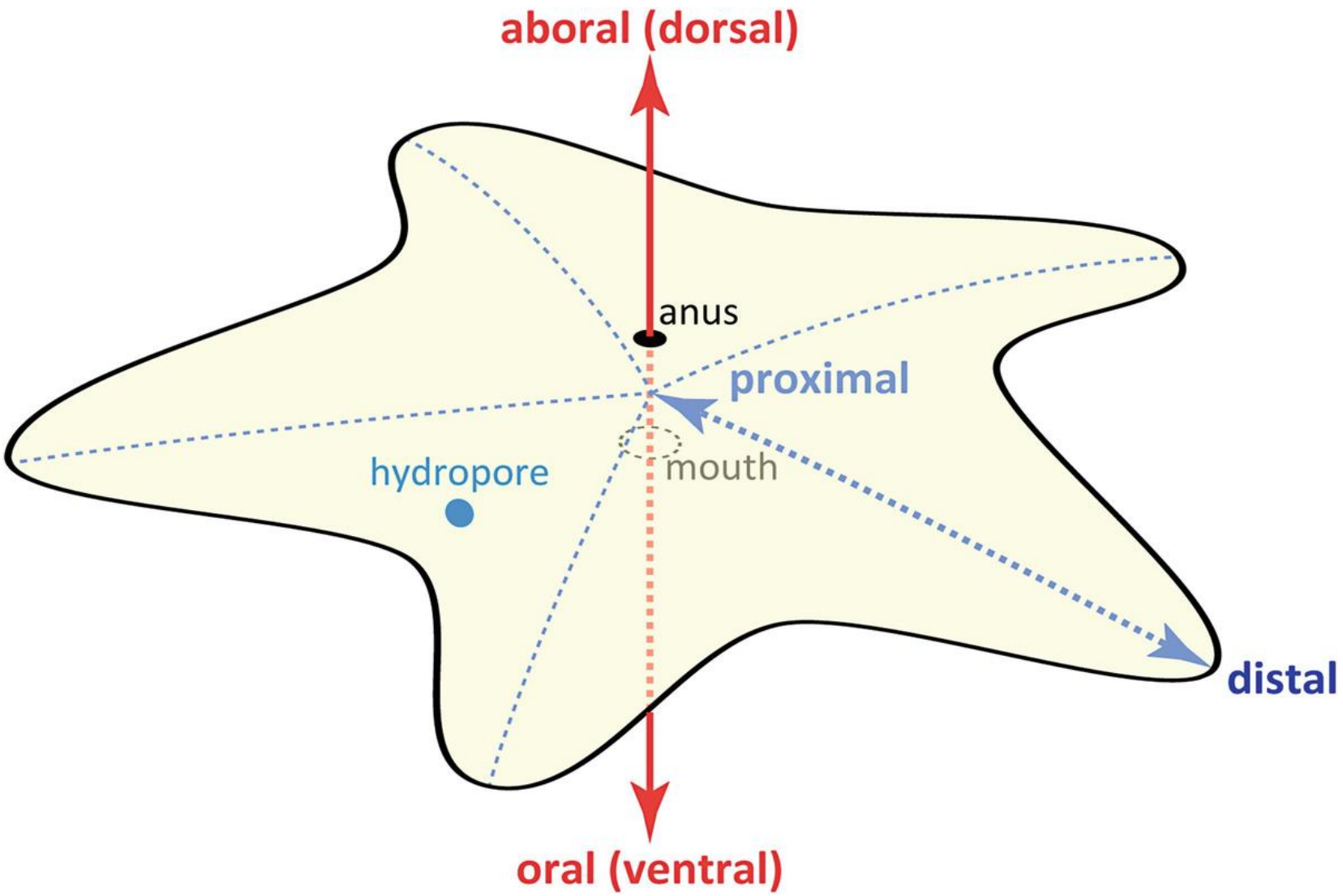
# Echinoderm Skeleton & Pentaradial Body Form of Echinoderm

- **The evolution of the skeleton may be responsible for the** pentaradial body form of echinoderms.
- The joints between two skeletal plates represent a weak point in the skeleton.

# Echinoderm Skeleton & Pentaradial Body Form of Echinoderm

- Pentaradial symmetry is a form of radial symmetry where body parts are organized around an **oral** or **aboral** axis in sets of five.
- Pentaradial symmetry allows sedentary animals to have sensory, feeding, and other structures evenly apportioned around the body's axis.

# Echinoderm Skeleton & Pentaradial Body Form of Echinoderm



# Echinoderm Skeleton & Pentaradial Body Form of Echinoderm

- By not having weak joints directly opposite one another, the skeleton is made stronger than if the joints were arranged opposite each other.

# Echinoderm Skeleton & Pentaradial Body Form of Echinoderm

END

- The arrangement of the echinoderm body in this five set pattern may be adaptive as a result of their presence of skeletal ossicles.
- A skeleton is stronger if joints are not directly across from one another, as would be the case if the skeleton was organized in even sets. (i.e. 2,4,6,etc.)

# Animal Diversity: Invertebrates

## Phylum Echinodermata

Water Vascular System of Echinoderms

Lecture 218

# Water Vascular System of Echinoderms

- The **water-vascular system of echinoderms is a series of** water-filled canals, and their extensions are called tube feet.
- It originates embryologically as a modification of the coelom and is ciliated internally.

# Water Vascular System of Echinoderms

- The water-vascular system includes a ring canal that surrounds the mouth.
- The ring canal usually opens to the outside or to the body cavity through a stone canal and a sieve like plate, called the madreporite.

# Water Vascular System of Echinoderms

- **Tiedemann** bodies are swellings often associated with the ring canal.
- They are believed to be sites for the production of phagocytic cells, called **coelomocytes.**

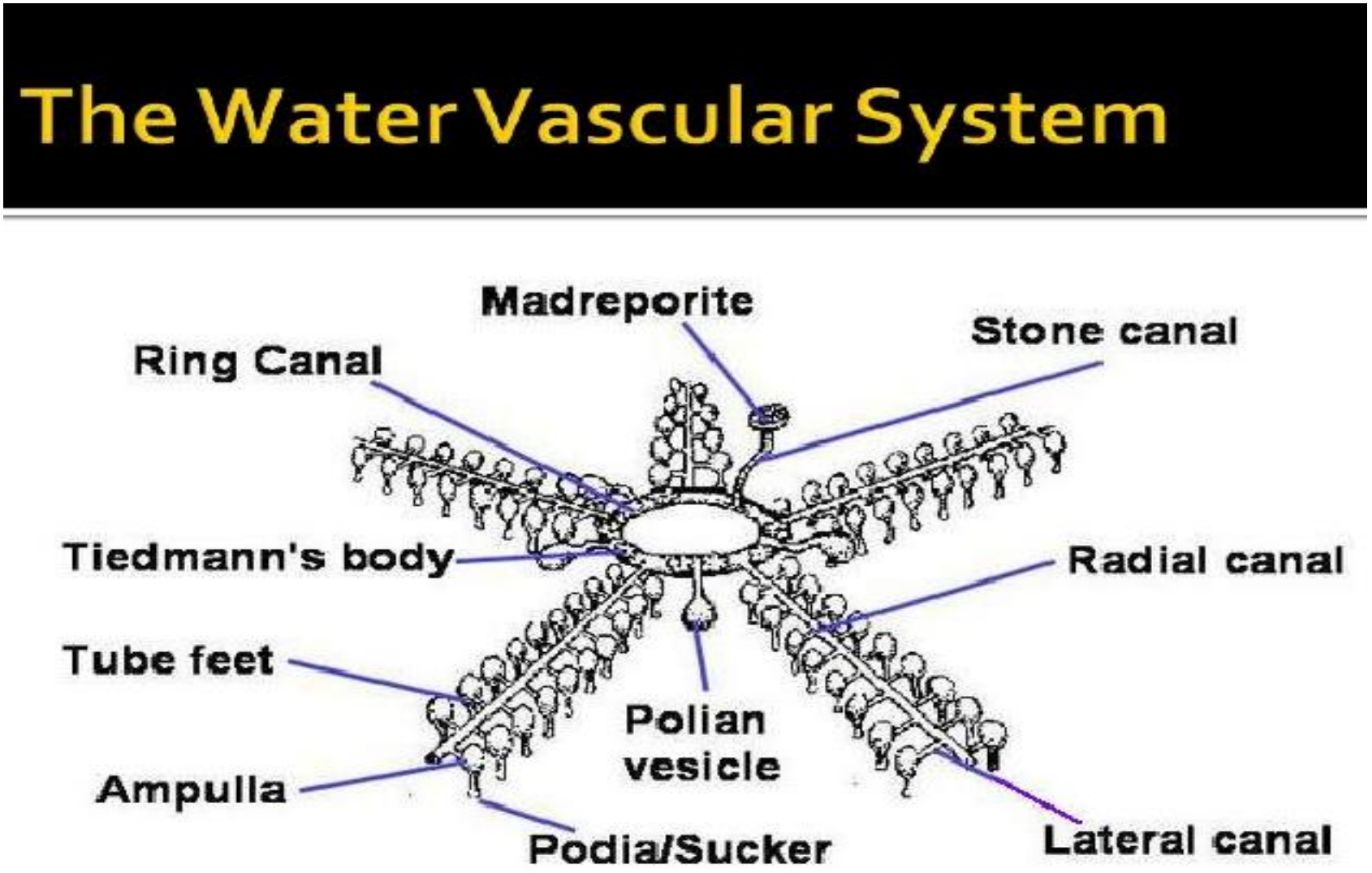
# Water Vascular System of Echinoderms

- Polian vesicles are sacs that are also associated with the ring canal.
- Five (or a multiple of five) radial canals branch from the ring canal.
- Radial canals are associated with arms of star-shaped echinoderms.

# Water Vascular System of Echinoderms

- In other echinoderms, they may be associated with the body wall and arch toward the aboral pole.
- Many lateral canals branch off each radial canal and end at the tube feet.

# Water Vascular System of Echinoderms



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Structure and Function of Tube Feet

Lecture 219

# Structure and Function of Tube Feet

- **Tube feet are extensions of the canal system and usually** emerge through openings in skeletal ossicles.
- Internally, tube feet usually terminate in a bulblike, muscular ampulla.

# Structure and Function of Tube Feet

- When an ampulla contracts, it forces water into the tube foot, which then extends.
- Valves prevent the backflow of water from the tube foot into the lateral canal.
- A tube foot often has a suction cup at its distal end.

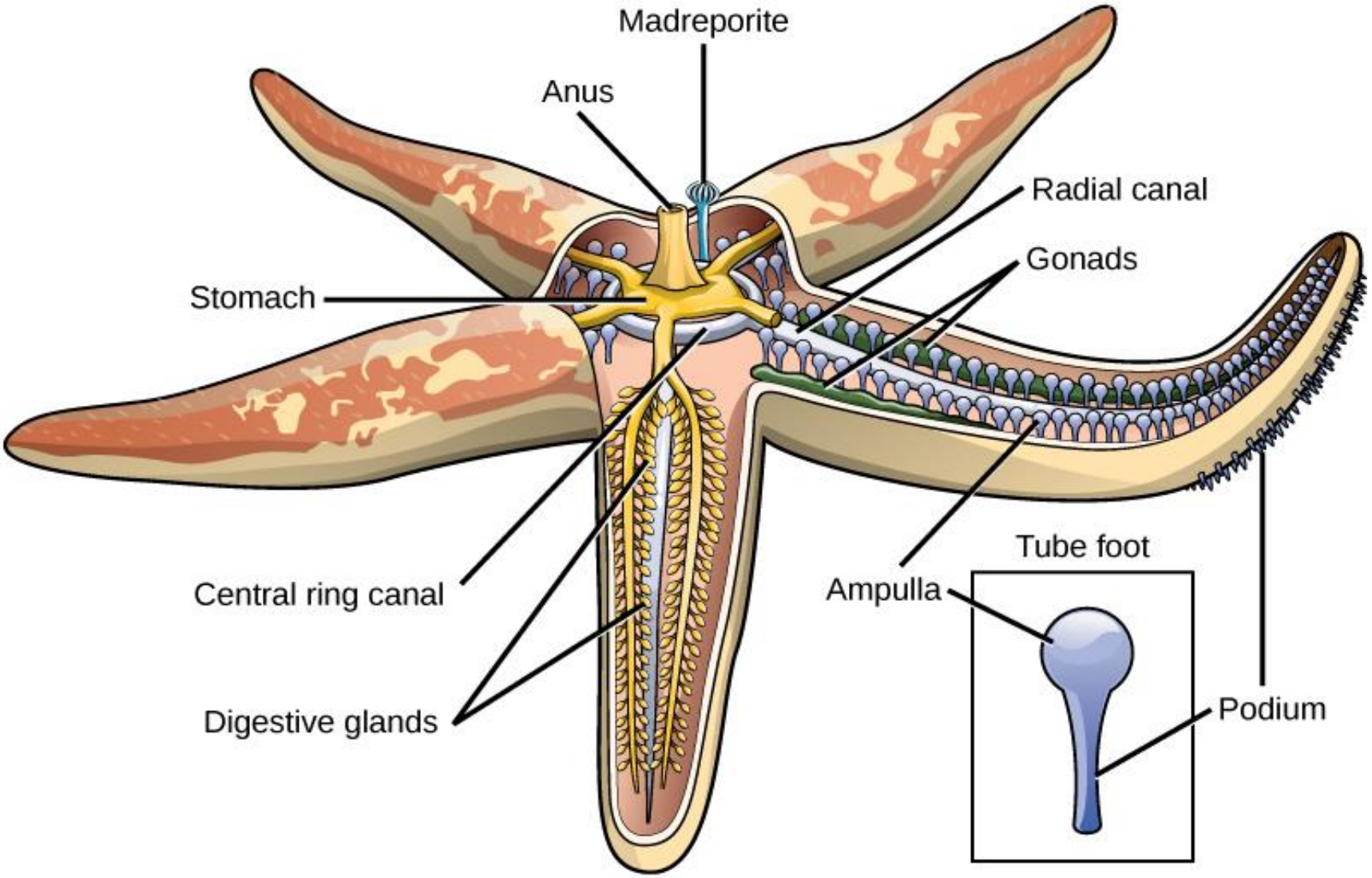
# Structure and Function of Tube Feet

- When the foot extends and contacts solid substrate, muscles of the suction cup contract and create a vacuum.
- In some taxa, tube feet have a pointed or blunt distal end.

# Structure and Function of Tube Feet

- These echinoderms may extend their tube feet into a soft substrate to secure contact during locomotion or to sift sediment during feeding.

# Structure and Function of Tube Feet



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Functions of Water Vascular System

Lecture 220

# Functions of Water Vascular System

- The madreporite may serve as an inlet to replace water lost from the water-vascular system and may help equalize pressure differences between the water-vascular system and the outside.

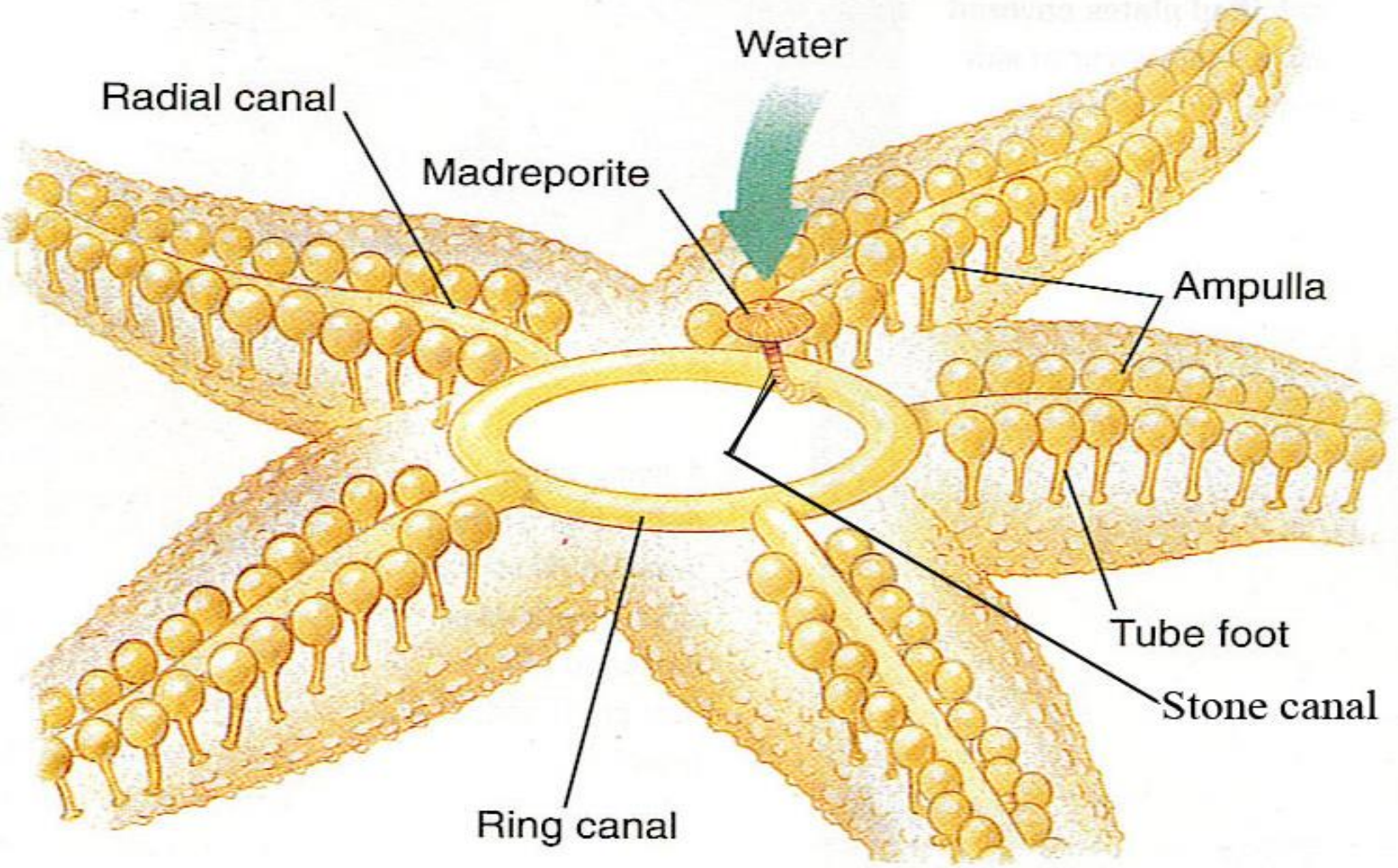
# Functions of Water Vascular System

- Polian vesicles function in fluid storage for the water-vascular system.
- The water-vascular system has other functions in addition to locomotion.
- The original function of water-vascular systems was probably feeding, not locomotion.

# Functions of Water Vascular System

- In addition, the soft membranes of the water-vascular system permit diffusion of respiratory gases and nitrogenous wastes across the body wall.
- Other terms sometimes used to refer to the water vascular system are "ambulacral system" and "aquiferous system".

# Functions of Water Vascular System



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Hemal System

Lecture 221

# Hemal System

- A **hemal system consists of strands of tissue that encircle an** echinoderm near the ring canal of the water-vascular system and run into each arm near the radial canals.

# Hemal System

- The hemal system has been likened to a vestigial circulatory system; however, its function is largely unknown.
- It may aid in the transport of large molecules, hormones, or coelomocytes, which are cells that engulf and transport waste particles within the body.

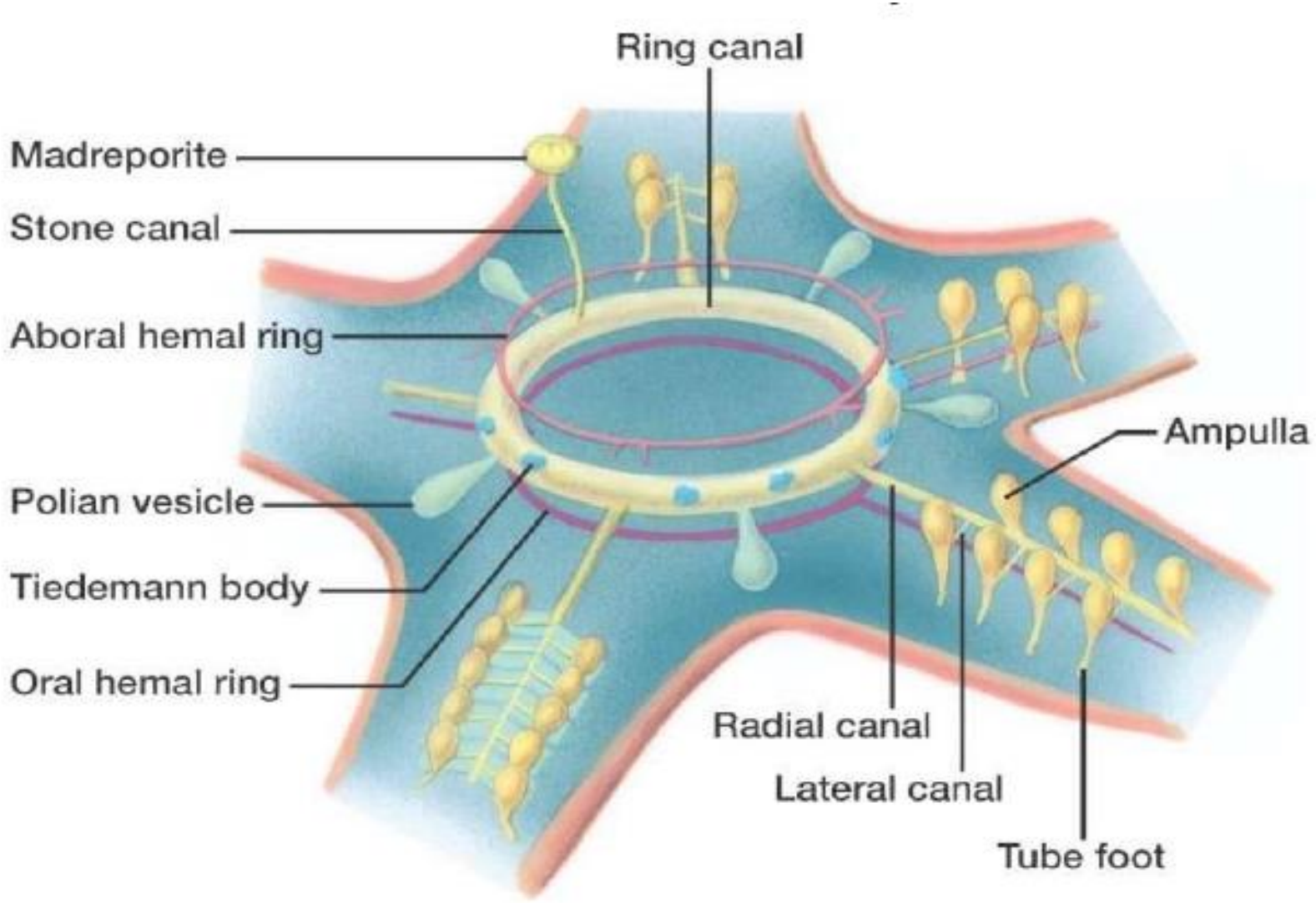
# Hemal System

- Circulation in cushion star occurs in three places, the perivisceral coelom, the water vasular system, and the hemal system.
- The perivisceral coelem is the area inside the body but outside of the major organs.

# Hemal System

- The water vascular system is a circulatory system which forces water into and out of the tube feet through contracting muscles.
- The hemal system is a series of chanels throughout the body that transport and distribute food materials.

# Hemal System



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Class Asteroidea: GeneralFfeatures

Lecture 222

# Class Asteroidea: General Features

- The sea stars make up the class Asteroidea (Gr. *aster,* star *oeides,* in the form of) and include about 1,500 species.
- They often live on hard substrates in marine environments, although some species also live in sandy or muddy substrates.

# Class Asteroidea: General Features

- Sea stars may be brightly colored with red, orange, blue, or gray.
- Asterias is an orange sea star common along the Atlantic coast of North America and is frequently studied in introductory zoology laboratories.

# Class Asteroidea: General Features

- Sea stars usually have five arms that radiate from a central disk.
- The oral opening, or mouth, is in the middle of one side of the central disk.

# Class Asteroidea: General Features

- It is normally oriented downward, and movable oral spines surround it.
- Movable and fixed spines project from the skeleton and roughen the aboral surface.

# Class Asteroidea: General Features

## Class Asteroidea – sea stars








# Animal Diversity: Invertebrates

## Phylum Echinodermata

Dermal Branchiae

Lecture 223

# Dermal Branchiae

- Thin folds of the body wall, called **dermal branchiae, extend between ossicles** and **function** in gas exchange.

# Dermal Branchiae

- **Papulae, known as, Dermal branchiae** or **skin** gills, are projections of the coelom of **Asteroidea** that serve in respiration and waste removal.

# Dermal Branchiae

- Papulae are soft, covered externally with the epidermis, and lined internally with peritoneum.
- They extend through the mesodermal ossicles and are protected from microscopic larvae by pedicellariae.

# Dermal Branchiae

- In a living specimen, the **dermal** gills can be extended or withdraw.

# Dermal Branchiae

Dermal Branchiae

- Soft bumps on the body that absorb oxygen from the water



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Pedicellariae

Lecture 224

# Pedicellariae

- In some sea stars, the aboral surface has numerous pincherlike structures called **pedicellariae**, which clean the body surface of debris and have protective functions.

# Pedicellariae

- Pedicellariae may be attached on a movable spine, or they may be immovably fused to skeletal ossicles.
- A **pedicellaria** (**plural**: **pedicellariae**) is a small wrench- or claw-shaped appendage with movable jaws, called valves.

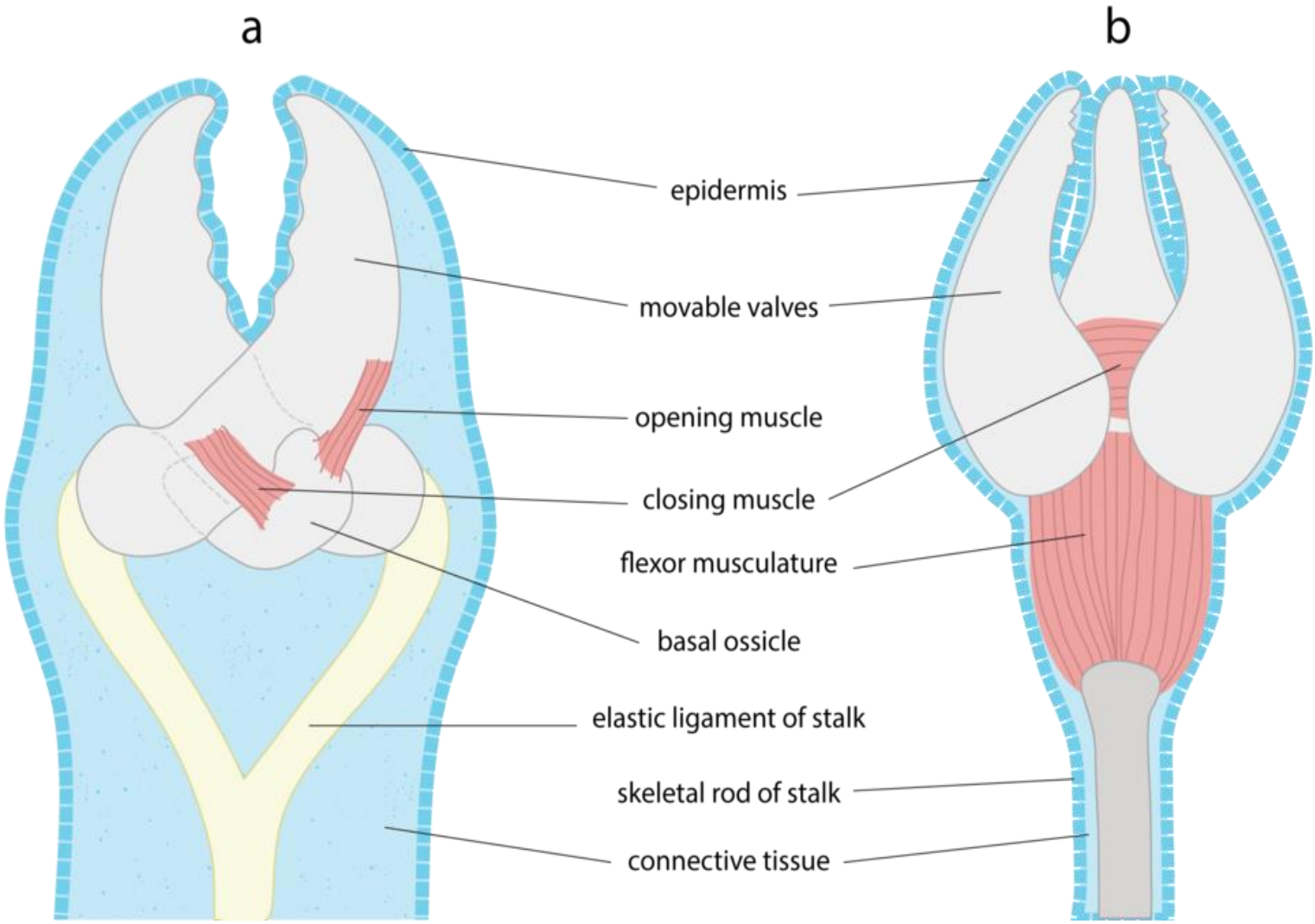
# Pedicellariae

- Each pedicellaria is an effector organ with its own set of muscles, neuropils, and sensory receptors and is therefore capable of reflex responses to the environment.

# Pedicellariae

- Pedicellariae are poorly understood but in some taxa, they are thought to keep the body surface clear of algae, encrusting organisms, and other debris in conjunction with the ciliated epidermis present in all echinoderms.

# Pedicellariae



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Ambulacral Groove

Lecture 225

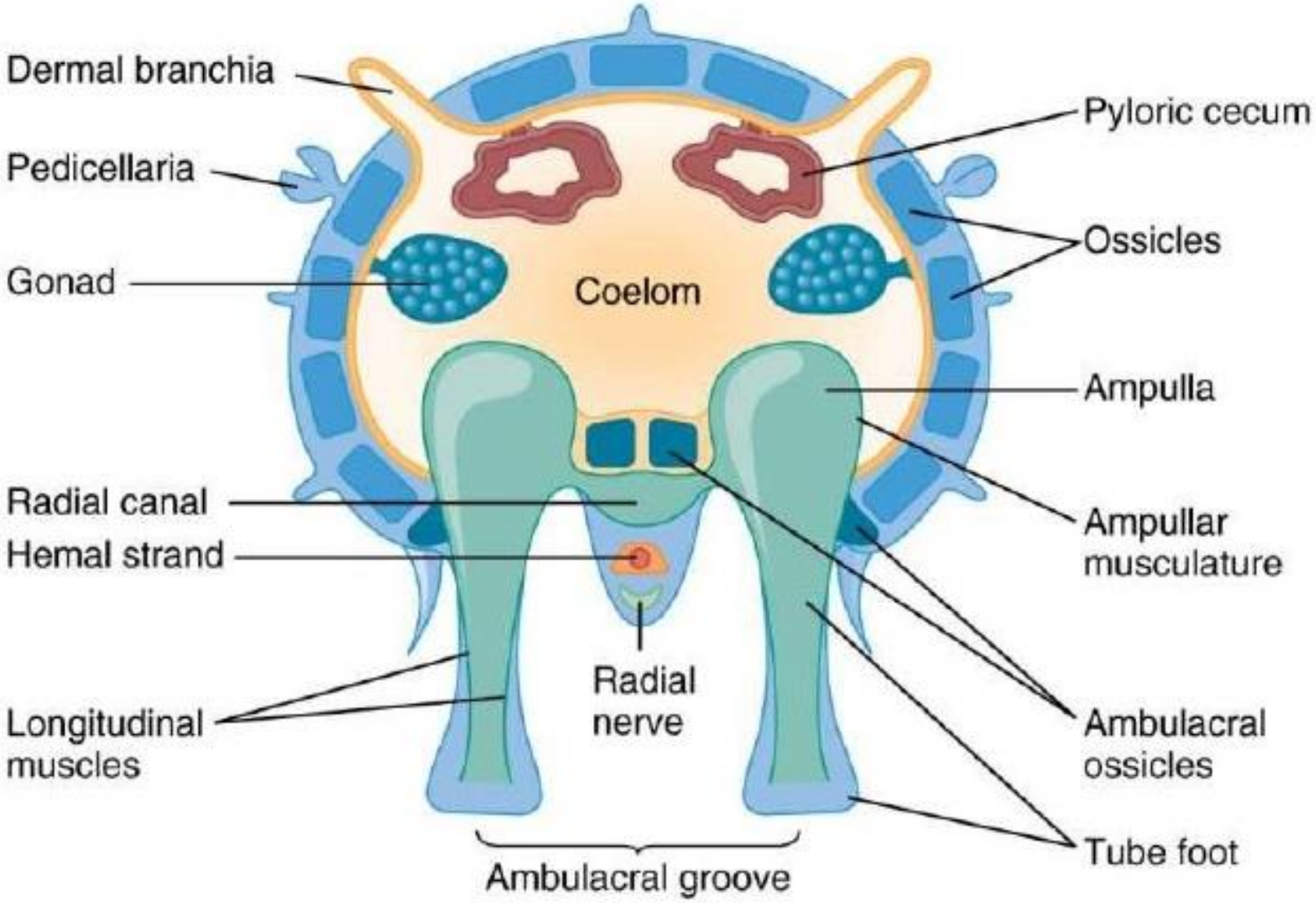
# Ambulacral Groove

- A series of ossicles in the arm form an **ambulacral groove** that runs the length of the oral surface of each arm.
- The ambulacral groove houses the radial canal, and paired rows of tube feet protrude through the body wall on either side of the ambulacral groove.

# Ambulacral Groove

- For example, **sea stars** or "**star fish**" have an ambulacral groove on their oral side (underside).
- This ambulacral groove extends from the mouth to the end of each ray or arm.

# Ambulacral Groove



# Ambulacral Groove

- Each groove of each arm in turn has four rows of hollow tube feet that can be extended or withdrawn.

# Ambulacral Groove

- Opposite the ambulacral groove is an ambulacral ridge on the aboral side of each ray, known as an ambulacrum.
- These
have interambulacra
bet--ween them.

END

# Animal Diversity: Invertebrates

## Phylum Echinodermata

Locomotion in Star fish

Lecture 226

# Locomotion in Star Fish

- Tube feet of sea stars move in a stepping motion.
- Alternate extension, attachment, and contraction of tube feet move sea stars across their substrate.

# Locomotion in Star Fish

- The nervous system coordinates the tube feet so that all feet move the sea star in the same direction; however, the tube feet do not move in unison.

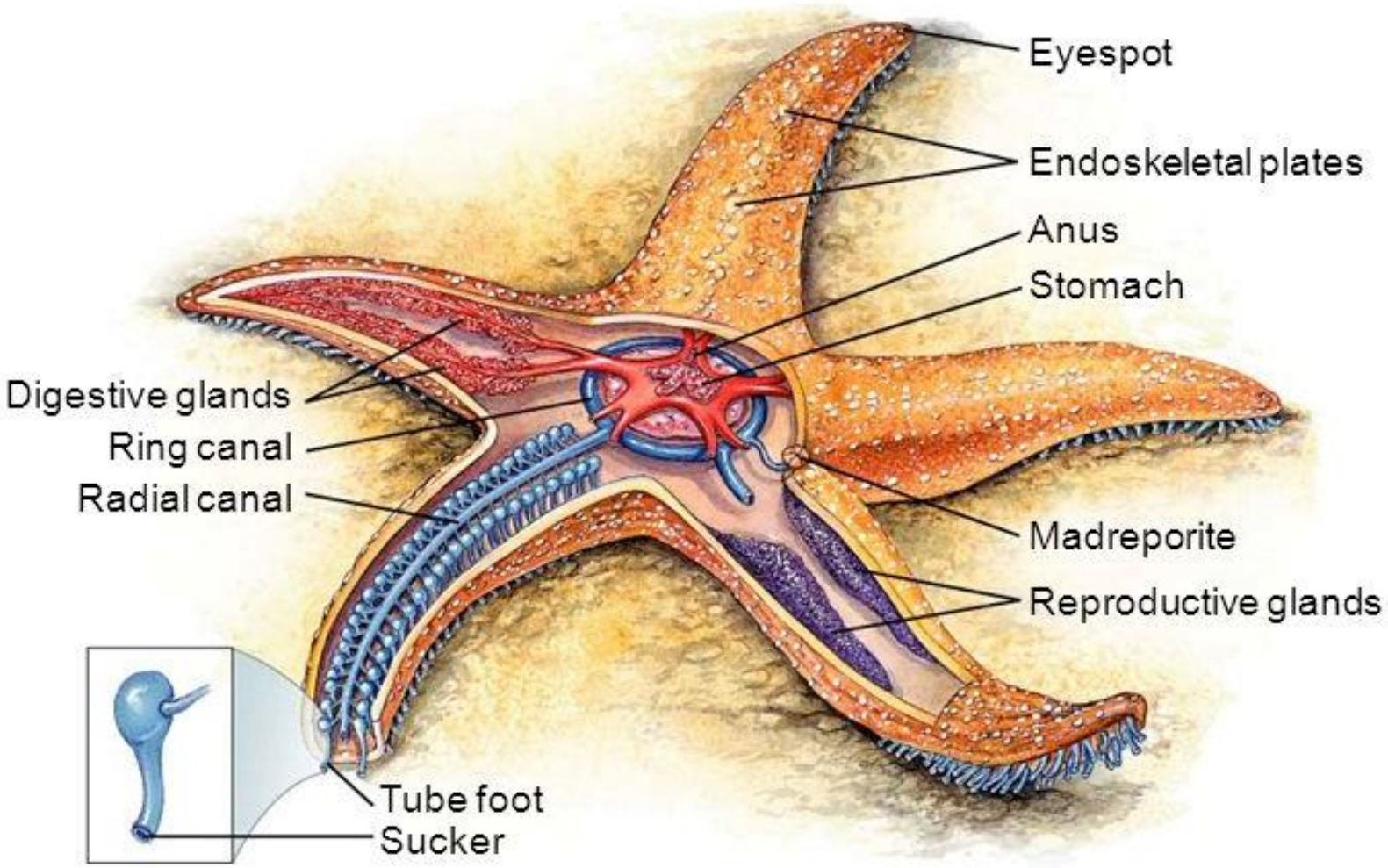
# Locomotion in Star Fish

- The suction disks of tube feet are effective attachment structures, allowing sea stars to maintain their position, or move from place to place, in spite of strong wave action.

# Locomotion in Star Fish

- Sea stars move using a water vascular system.
- Water comes into the system via the madreporite.
- It is then circulated from the stone canal to the ring canal and into the radial canals.
- The radial canals carry water to the ampullae and provide suction to the tube feet.

# Locomotion in Star Fish



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Feeding and Digestion in Star Fish

Lecture 227

# Feeding and Digestion in Star Fish

- Sea stars feed on snails, bivalves, crustaceans, polychaetes, corals, detritus, and a variety of other food items.
- The mouth opens to a short esophagus and then to a large stomach that fills most of the coelom of the central disk.

# Feeding and Digestion in Star Fish

- The stomach is divided into two regions.
- The larger, oral stomach, sometimes called the **cardiac stomach**, receives ingested food.
- It joins the smaller, aboral stomach, sometimes called the **pyloric stomach**.

# Feeding and Digestion in Star Fish

- The aboral (pyloric) stomach gives rise to ducts that connect to secretory and absorptive structures called pyloric cecae.
- Two pyloric cecae extend into each arm.
- A short intestine leads to rectal cecae (uncertain functions) and to a nearly nonfunctional anus, which opens on the aboral surface of the central disk.

# Feeding and Digestion in Star Fish

- Some sea stars ingest whole prey, which are digested extracellularly within the stomach.
- Undigested material is expelled through the mouth.
- Many sea stars feed on bivalves by forcing the valves apart.

# Feeding and Digestion in Star Fish

- When a sea star feeds on a bivalve, it wraps itself around the bivalve's ventral margin.
- Tube feet attach to the outside of the shell, and the body-wall musculature forces the valves apart.
- This is possible because the sea star changes tube feet when the muscles of engaged tube feet begin to tire.

# Feeding and Digestion in Star Fish

- When the valves are opened about 0.1 mm, increased coelomic pressure everts the oral (cardiac) portion of the sea star's stomach into the bivalve shell.
- Digestive enzymes are released, and partial digestion occurs in the bivalve shell.

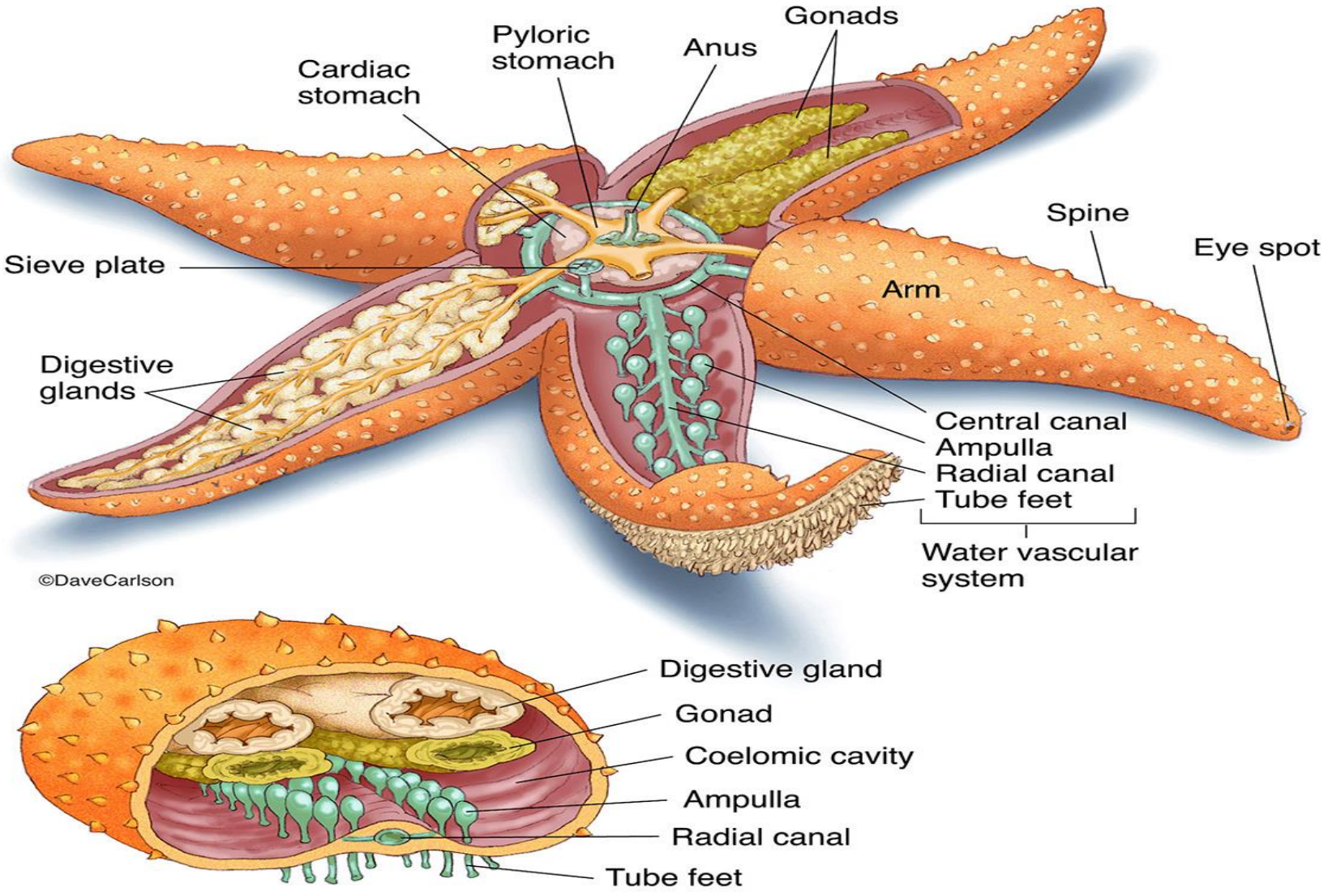
# Feeding and Digestion in Star Fish

- This digestion further weakens the bivalve's adductor muscles, and the shell eventually opens completely.
- Partially digested tissues are taken into the aboral (pyloric) portion of the stomach, and into the pyloric cecae for further digestion and absorption.

# Feeding and Digestion in Star Fish

- After feeding and initial digestion, the sea star retracts the stomach, using stomach retractor muscles.

# Feeding and Digestion in Star Fish



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Gaseous Exchange in Star Fish

Lecture 228

# Gaseous Exchange in Star Fish

- Gases, nutrients, and metabolic wastes are transported in the coelom by diffusion and by the action of ciliated cells lining the body cavity.

# Gaseous Exchange in Star Fish

- Gas exchange and excretion of metabolic wastes (principally ammonia) occur by diffusion across dermal branchiae, tube feet, and other membranous structures.

# Gaseous Exchange in Star Fish

- Respiration or gas exchange occurs in two locations on a starfish.
- The first location is through the thin skin on the ends of the tube feet.
- Gas exchange also occurs in the dermal gills or papulae.
- These gills are lined with cilia that move to produce a current.

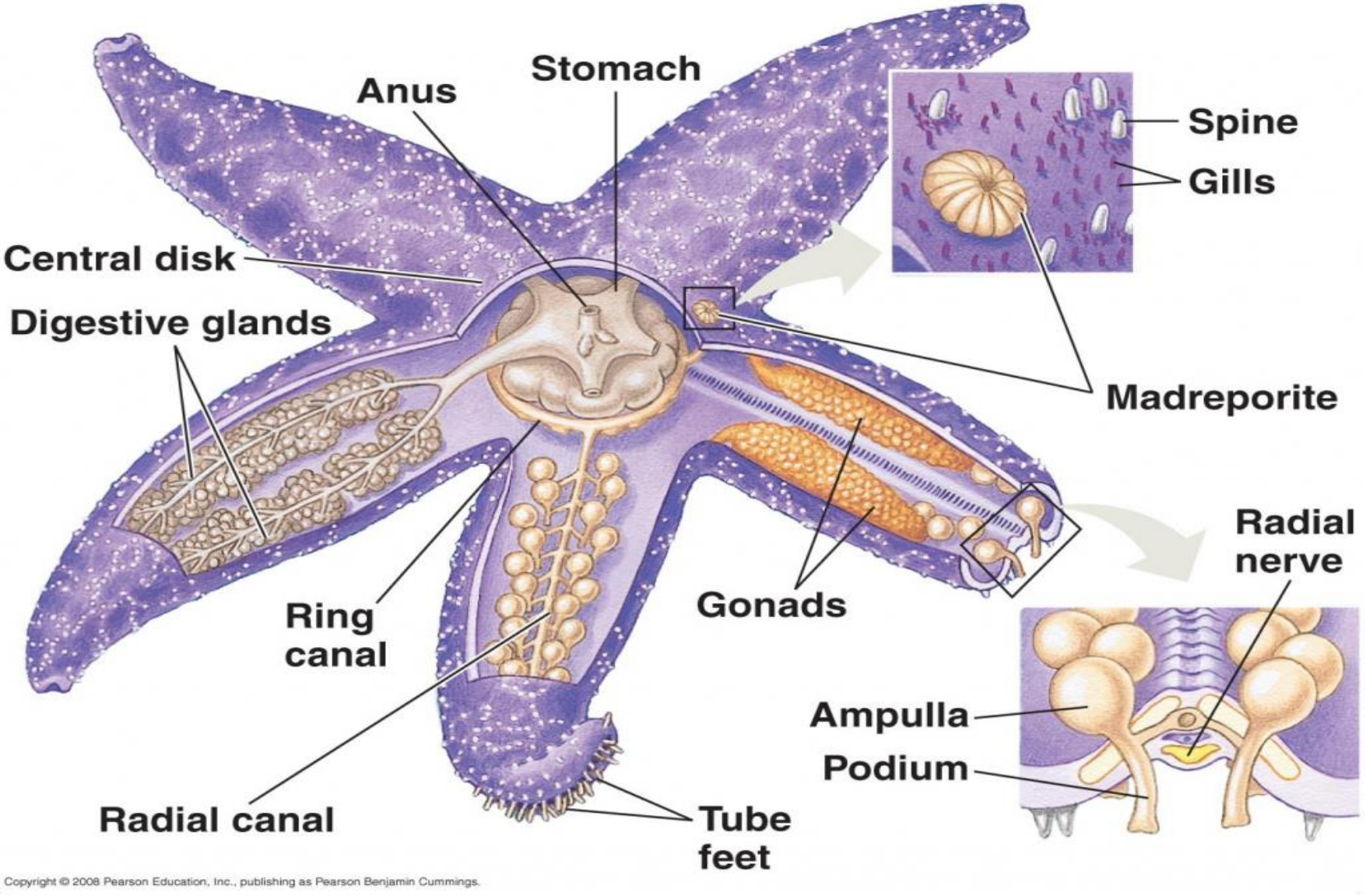
# Gaseous Exchange in Star Fish

- The current brings fresh water so the starfish can absorb oxygen and freshens the surrounding water where the starfish may have just excreted carbon dioxide.

# Gaseous Exchange in Star Fish

- A sea star's hemal system consists of strands of tissue that encircle the mouth near the ring canal, extend aborally near the stone canal, and run into the arms near radial canals.

# Gaseous Exchange in Star Fish




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# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Nervous System in Star Fish

Lecture 229

# Nervous System in Star Fish

- The nervous system of sea stars consists of a nerve ring that encircles the mouth and radial nerves that extend into each arm.

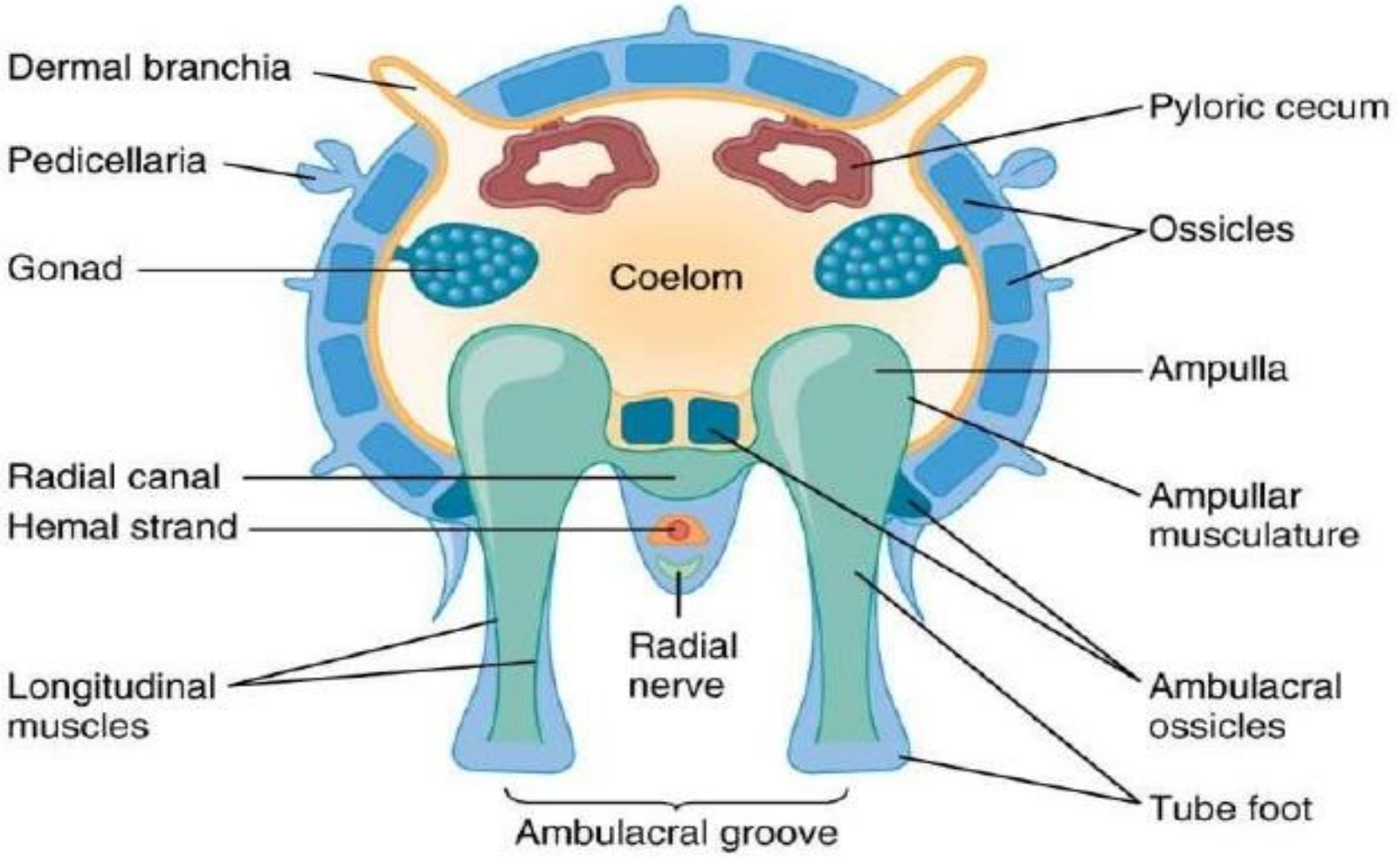
# Nervous System in Star Fish

- Radial nerves lie within the ambulacral groove, just oral to the radial canal of the water-vascular system and the radial strands of the hemal system.

# Nervous System in Star Fish

- Radial nerves coordinate the functions of tube feet.
- Other nervous elements are in the form of a nerve net associated with the body wall.
- The nervous system of the starfish is very simple, there is no brain and there are not even any ganglia to coordinate movement.

# Nervous System in Star Fish



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Sensory Functions in Star Fish

Lecture 230

# Sensory Functions in Star Fish

- Most sensory receptors are distributed over the surface of the body and tube feet.
- Sea stars respond to light, chemicals, and various mechanical stimuli.

# Sensory Functions in Star Fish

- They often have specialized photoreceptors at the tips of their arms.
- These are actually tube feet that lack suction cups but have a pigment spot surrounding a group of ocelli.

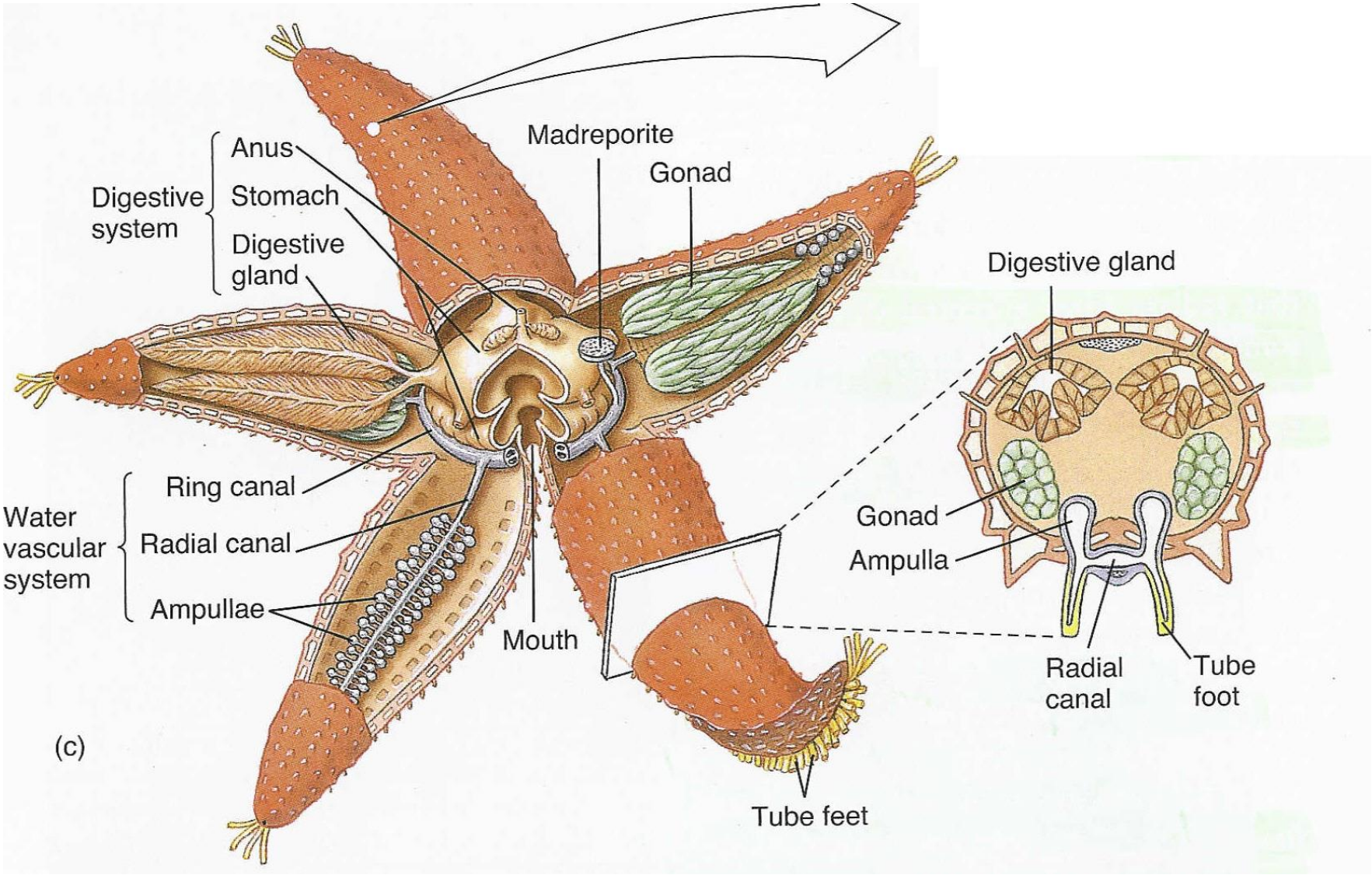
# Sensory Functions in Star Fish

- The ring nerves and radial nerves have **sensory** and motor components and coordinate the **starfish's** balance and directional systems.

# Sensory Functions in Star Fish

- The **sensory** component receives input from the **sensory** organs while the motor nerves control the tube feet and musculature.
- The **starfish** does not have the capacity to plan its actions.

# Sensory Functions in Star Fish



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Regeneration in Sea Star

Lecture 231

# Regeneration in Sea Star

- Sea stars are well known for their powers of regeneration.
- They can regenerate any part of a broken arm.
- In a few species, an entire sea star can be regenerated from a broken arm if the arm contains a portion of the central disk.

# Regeneration in Sea Star

- Regeneration is a slow process, taking up to a year for complete regeneration.
- Asexual reproduction involves division of the central disk, followed by regeneration of each half.

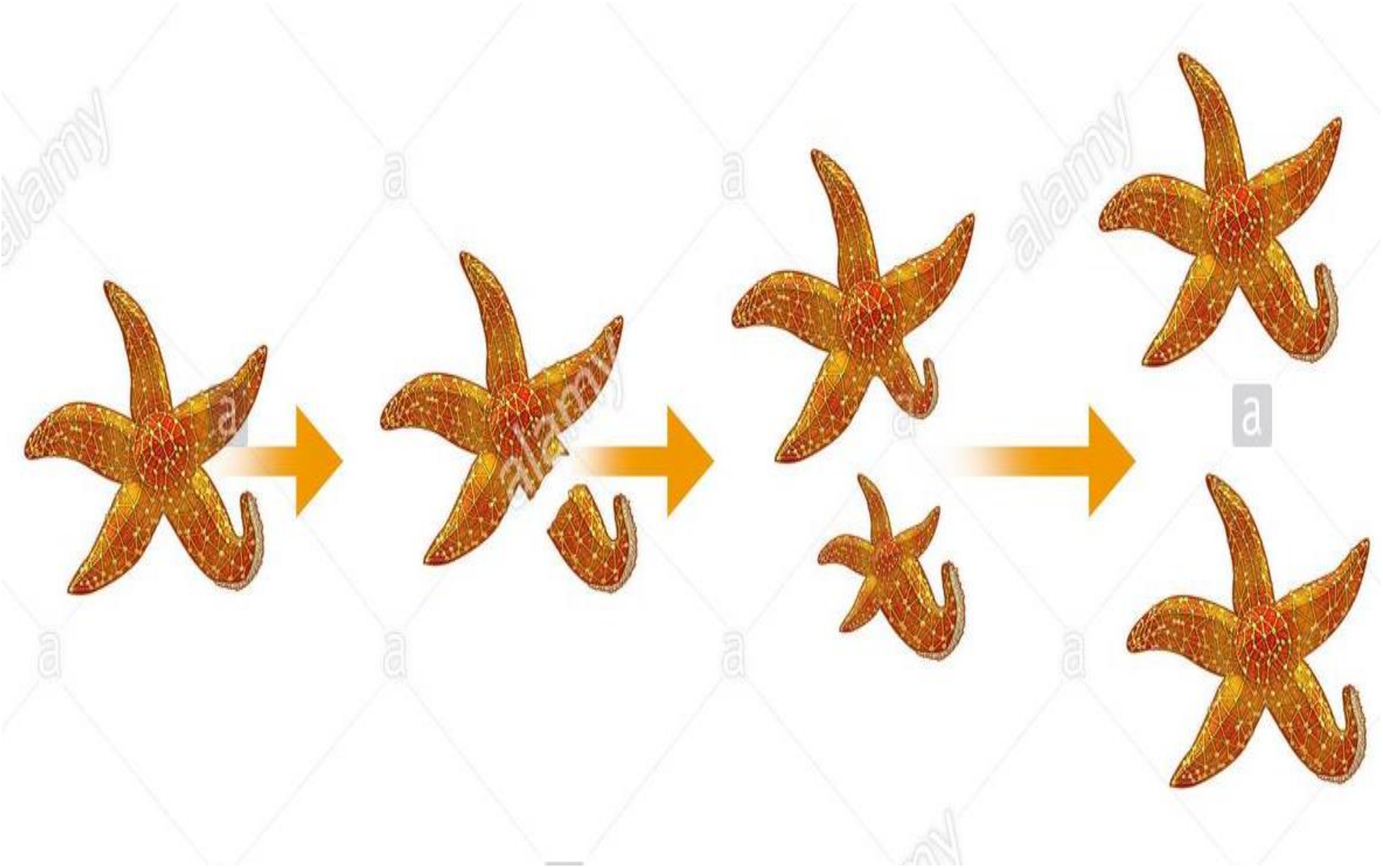
# Regeneration in Sea Star

- A sea star that has the ability to regenerate amputated limbs must first undergo a repair phase to heal the exposed wound.
- Once the wound is healed, the sea star can begin to generate new cells, which in turn, sparks new growth.

# Regeneration in Sea Star

- Cell proliferation that results in the growth of the new limb occurs in the final phase.
- If the final phase is interrupted, the resulting new limb may be deformed.

# Regeneration in Sea Star



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Reproduction in sea star

Lecture 232

# Reproduction in Sea Star

- Sea stars are dioecious, but sexes are indistinguishable externally.
- Two gonads are present in each arm, and these enlarge to nearly fill an arm during the reproductive periods.
- Gonopores open between the bases of each arm.

# Reproduction in Sea Star

- In sexual reproduction, fertilization occurs in the water with males and females releasing sperm and eggs into the environment.
- The fertilized embryos, which are free-swimming animals, become part of the zooplankton in most species.
- Eventually the larvae undergo metamorphosis, settle to the bottom and grow into adults.

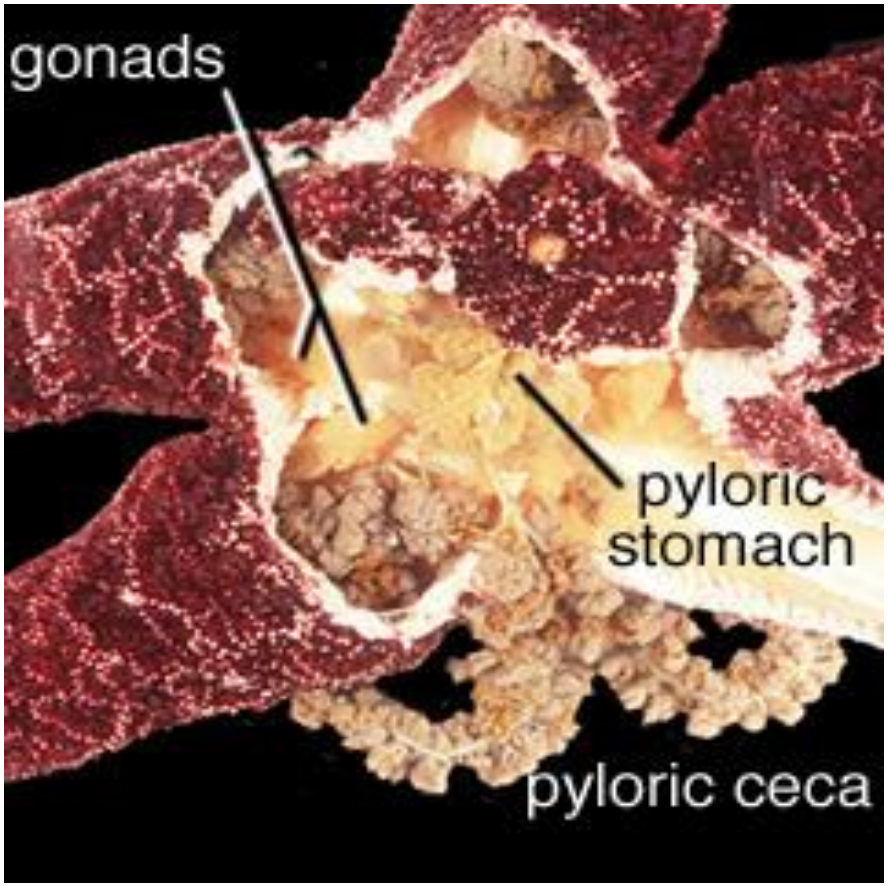
# Reproduction in Sea Star

- The embryology of echinoderms has been studied extensively because of the relative ease of inducing spawning and maintaining embryos in the laboratory.
- External fertilization is the rule.

# Reproduction in Sea Star

- Because gametes cannot survive long in the ocean, maturation of gametes and spawning must be coordinated if fertilization is to take place.

# Reproduction in Sea Star






# Animal Diversity: Invertebrates

## Phylum Echinodermata

Role of Photoperiod and Pheromones in Spawning

Lecture 233

# Role of Photoperiod and Pheromones in Spawning

- The photoperiod (the relative length of light and dark in a 24-hour period) and temperature are environmental factors used to coordinate sexual activity.

# Role of Photoperiod and Pheromones in Spawning

- Gametogenesis and gonadal growth in the west coast sea star *Pisaster ochraceus* normally begins in the fall and leads to large gonads full of gametes in the spring, when spawning occurs.

# Role of Photoperiod and Pheromones in Spawning

- In addition, gamete release by one individual is accompanied by the release of spawning pheromones, which induce other sea stars in the area to spawn, increasing the likelihood of fertilization.

# Role of Photoperiod and Pheromones in Spawning

- The first individual of a species to spawn may release a pheromone that serves to attract other starfish to aggregate and to release their gametes synchronously.
- In other species, a male and female may come together and form a pair.

END

# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Development in Sea Star

Lecture 234

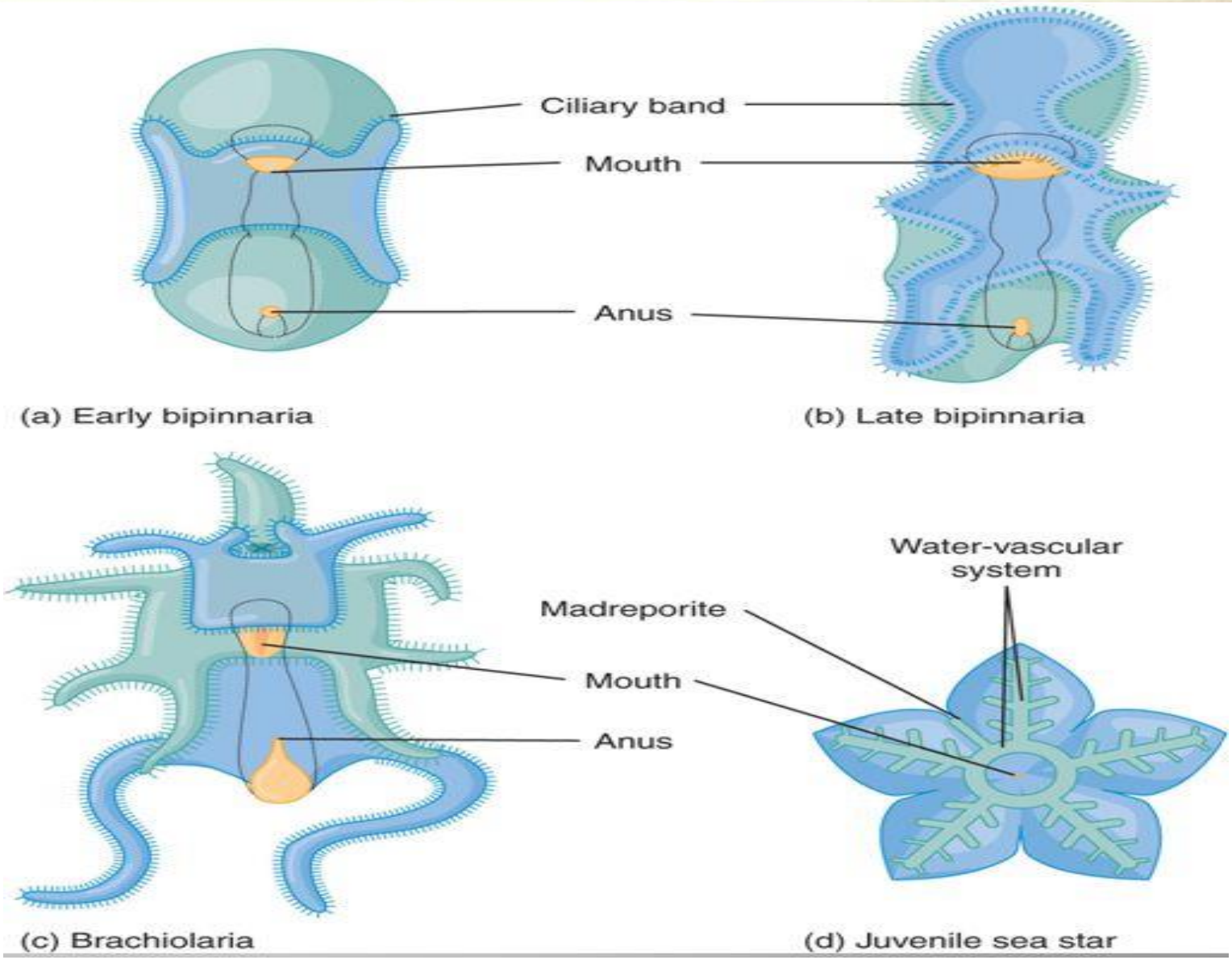
# Development in Sea Star

- Embryos are planktonic, and cilia are used in swimming.
- After gastrulation, bands of cilia differentiate, and a bilaterally symmetrical larva, called a **bipinnaria** larva, forms.
- The larva usually feeds on planktonic protists.

# Development in Sea Star

- The development of larval arms results in a **brachiolaria** larva, which settles to the substrate, attaches, and metamorphoses into a juvenile sea star.

# Development in Sea Star



# Development in Sea Star

- Later embryonic stages are ciliated and swim and feed in the plankton.
- In a few species, embryos develop from yolk stored in the egg during gamete formation.

# Development in Sea Star

- Oocytes averaged 137 µm in diameter, the smallest reported in the genus.
- Fertilized oocytes developed into feeding bipinnaria larvae that fed in the plankton for at least 4 weeks, and in some cases up to 33 weeks, before metamorphosis.

END

# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Class Ophiuroidea: General Features

Lecture 235

# Class Ophiuroidea: General Features

- The class Ophiuroidea (Gr. *ophis,* snake *oura,* tail *oeides,* in the form of) includes the basket stars and the brittle stars.
- With over two thousand species, this is the most diverse group of echinoderms.

# Class Ophiuroidea: General Features

- Ophiuroids, however, are often overlooked because of their small size and their tendency to occupy crevices in rocks and coral or to cling to algae.

# Class Ophiuroidea: General Features

- The arms of ophiuroids are long and, unlike those of asteroids, are sharply set off from the central disk, giving the central disk a pentagonal shape.
- Neither dermal branchiae nor pedicellariae are present in ophiuroids.

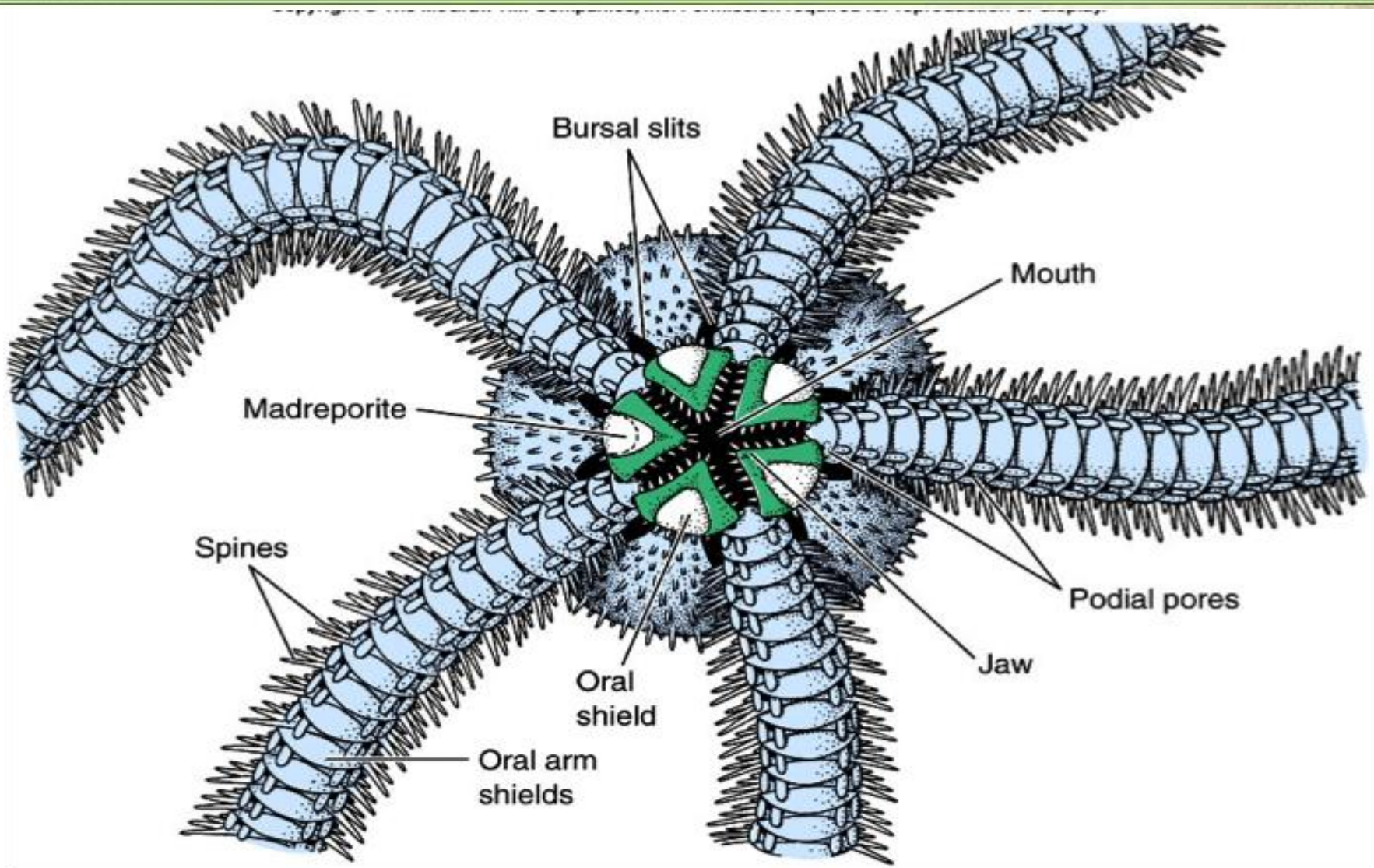
# Class Ophiuroidea: General Features

- The tube feet of ophiuroids lack suction disks and ampullae, and the contraction of muscles associated with the base of a tube foot extends the tube foot.
- Unlike the sea stars, the madreporite of ophiuroids is on the oral surface.

# Class Ophiuroidea: General Features

- **Ophiuroidea** closely related to starfish.
- They crawl across the sea floor using their flexible arms for locomotion.
- The ophiuroids generally have five long, slender, whip-like arms which may reach up to 60 cm (24 in) in length on the largest specimens.

# Class Ophiuroidea: General Features



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Difference Between Brittle Star and Basket Star

Lecture 236

# Difference Between Brittle Star and Basket Star

- Brittle stars have unbranched arms, and most have a central disk that ranges in size from 1 to 3 cm.
- Basket stars have arms that branch repeatedly.

# Difference Between Brittle Star and Basket Star

- The central disk of brittle stars and basket stars is very obvious, since the arms attach to the disc, rather than joining to each other at the base like they do in true sea stars.
- Brittle stars usually have 5, but may have up to 10 arms.

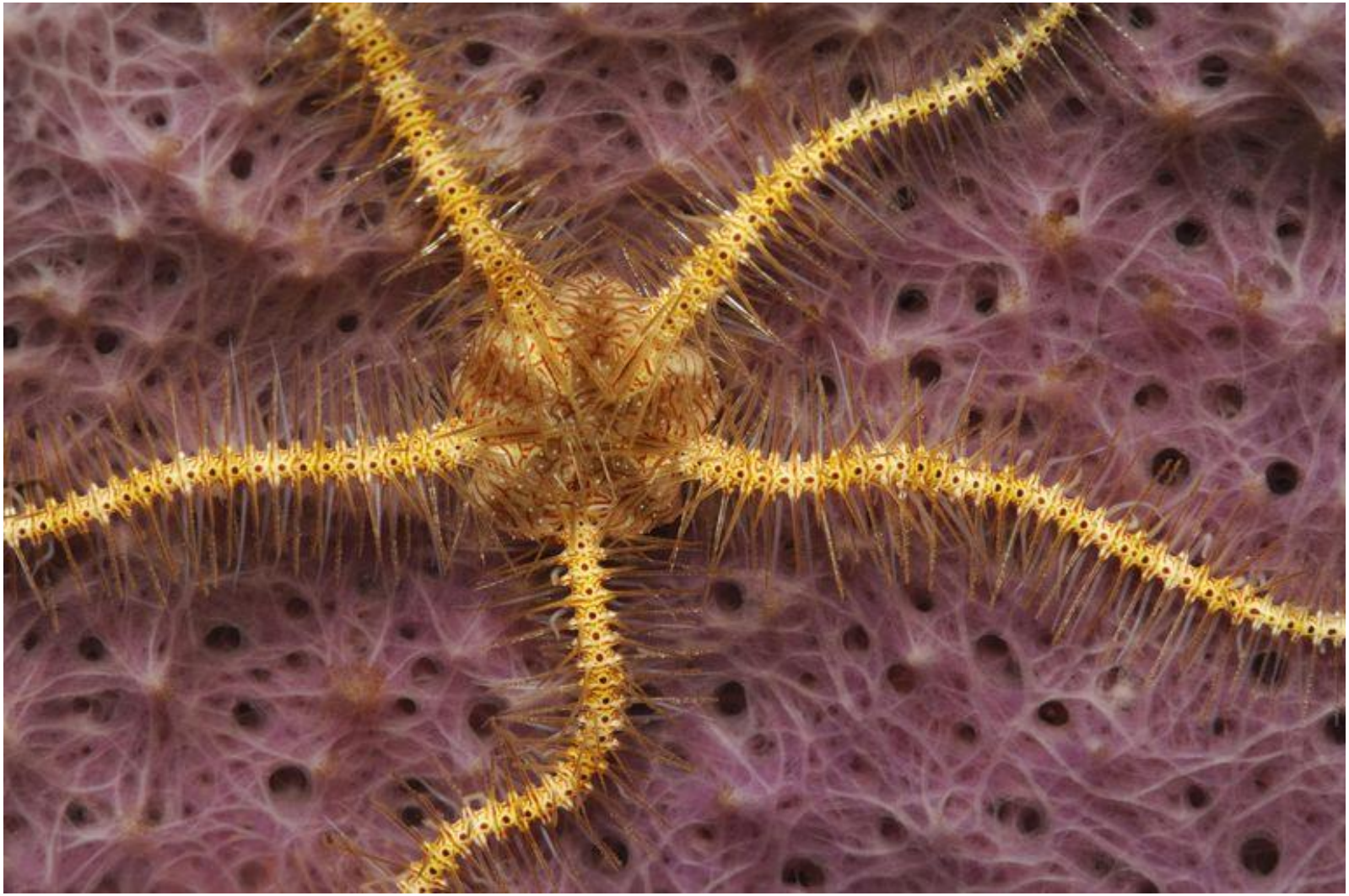
# Difference Between Brittle Star and Basket Star

- Basket stars have 5 arms that branch into many slender, highly mobile arms.
- The arms are covered with calcite plates or thick skin.

# Difference Between Brittle Star and Basket Star

- The central disk of brittle stars and basket stars is usually relatively small, under one inch, and the whole organism itself may be under an inch in size.
- The arms of some species can be quite long, though, with some basket stars measuring over 3 feet across when their arms are extended.

# Difference Between Brittle Star and Basket Star



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Difference Between Sea Star and Brittle Star

Lecture 237

# Difference Between Sea Star and Brittle Star

- **Sea stars** are usually star-shaped or **pentagonal.**
- They have a set of **calcareous** plates embedded into the fleshy tissue of their body wall.

# Difference Between Sea Star and Brittle Star

- The mouth is at the centre of the undersurface.
- When feeding, the sea star is able move its sack-like stomach out through the mouth and partly digests the food outside its body.

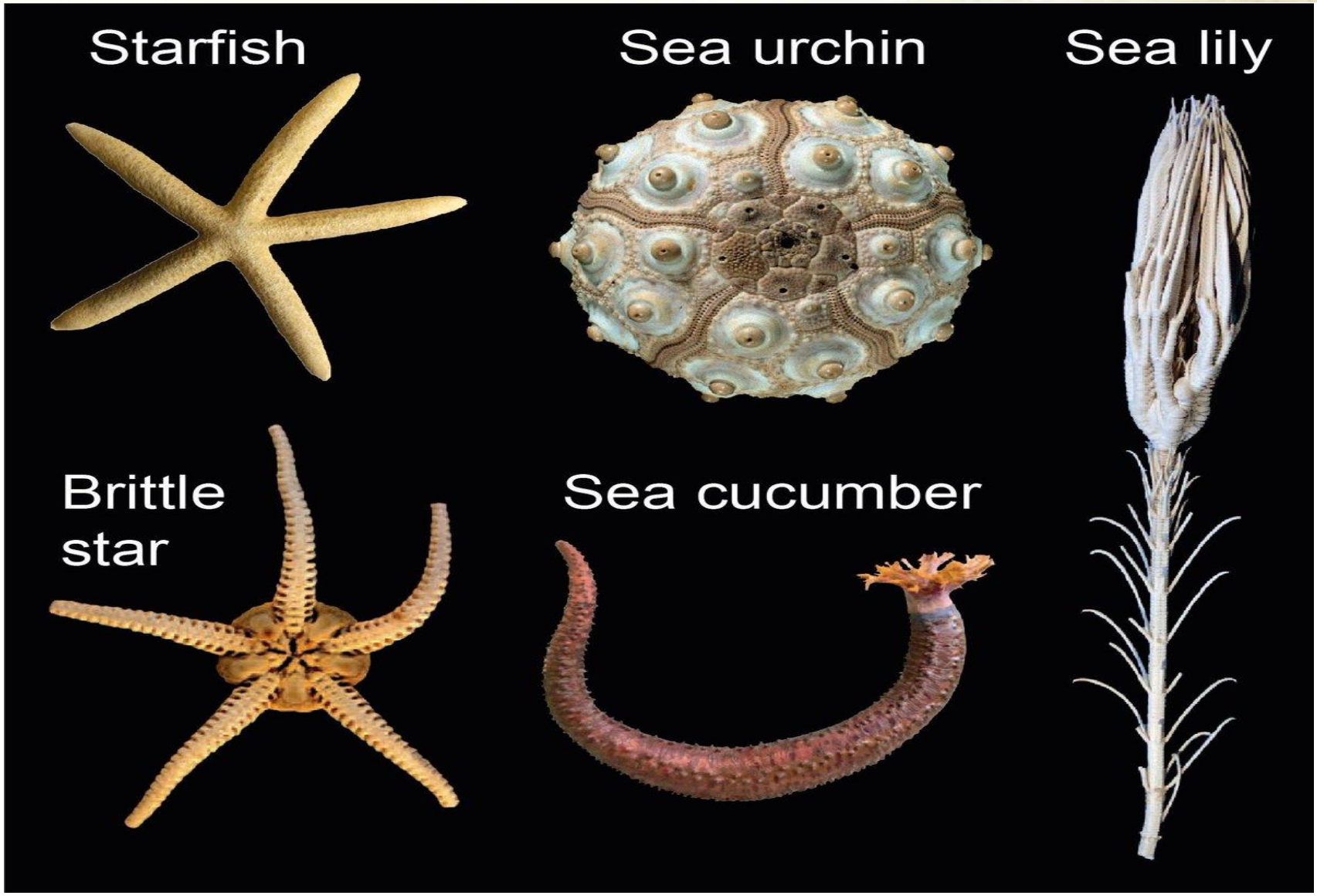
# Difference Between Sea Star and Brittle Star

- **Brittle stars** have a distinct central disc and five radiating slender, highly flexible arms.
- The arms are solid and not hollow like the sea stars.

# Difference Between Sea Star and Brittle Star

- Brittle stars move by a sinuous flexing of the arms rather than movement of tube feet.
- They feed on small organic particles of food.

# Difference Between Sea Star and Brittle Star



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Locomotion in Ophiuroids

Lecture 238

# Locomotion in Ophiuroids

- The water-vascular system of ophiuroids is not used for locomotion.
- Instead, the skeleton is modified to permit a unique form of grasping and movement.
- Superficial ossicles, which originate on the aboral surface, cover the lateral and oral surfaces of each arm.

# Locomotion in Ophiuroids

- The ambulacral groove containing the radial nerve, hemal strand, and radial canal is thus said to be “closed.”
- Ambulacral ossicles are in the arm, forming a central supportive axis.

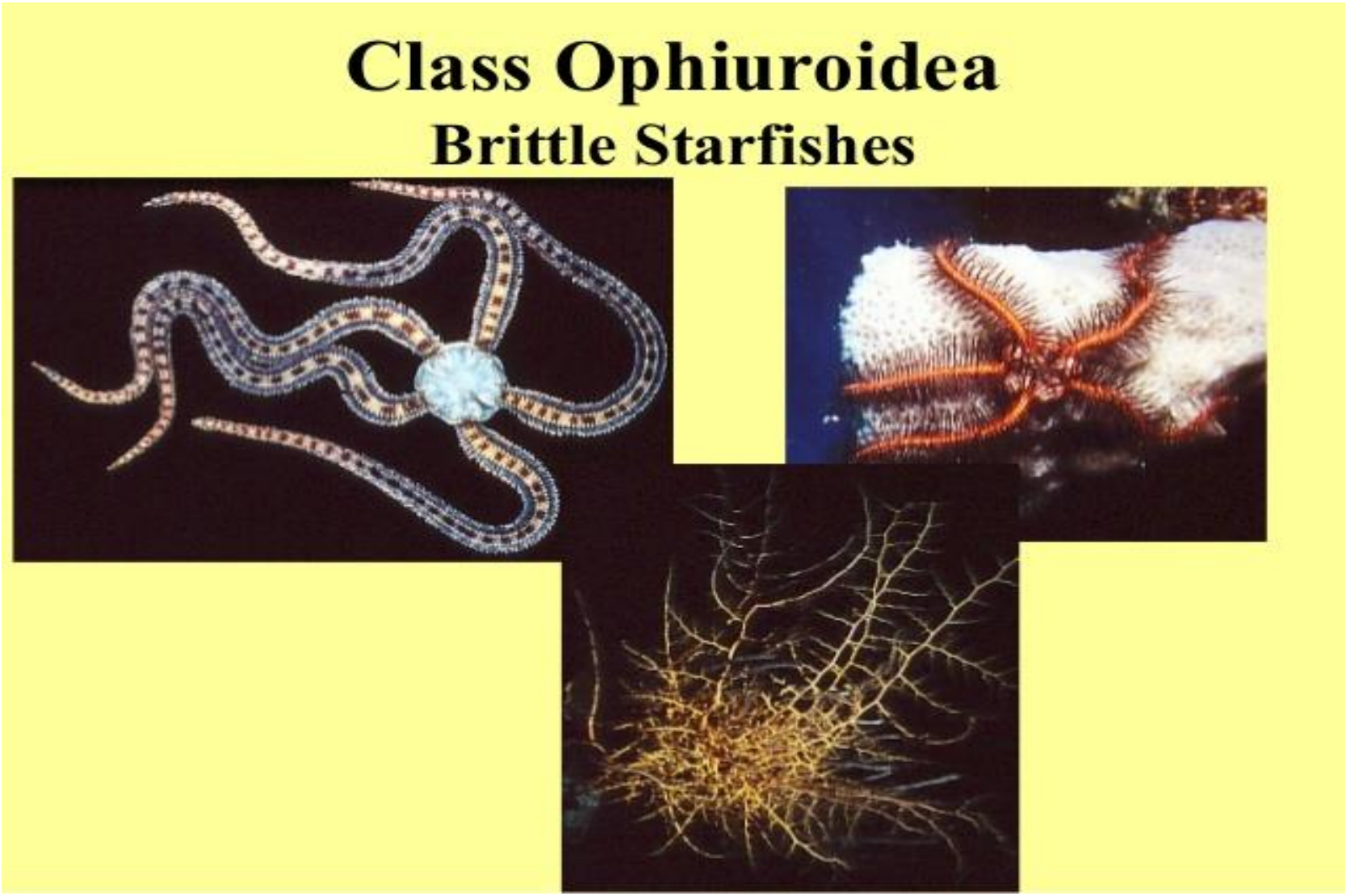
# Locomotion in Ophiuroids

- Successive ambulacral ossicles articulate with one another and are acted upon by relatively large muscles to produce snakelike movements (hence the derivation of the class name) that allow the arms to curl around a stalk of algae or to hook into a coral crevice.

# Locomotion in Ophiuroids

- During locomotion, the central disk is held above the substrate, and two arms pull the animal along, while other arms extend forward and/or trail behind the animal.

# Locomotion in Ophiuroids



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Feeding and Digestion in Ophiuroids

Lecture 239

# Feeding and Digestion in Ophiuroids

- Ophiuroids are predators and scavengers.
- They use their arms and tube feet in sweeping motions to collect prey and particulate matter, which are then transferred to the mouth.

# Feeding and Digestion in Ophiuroids

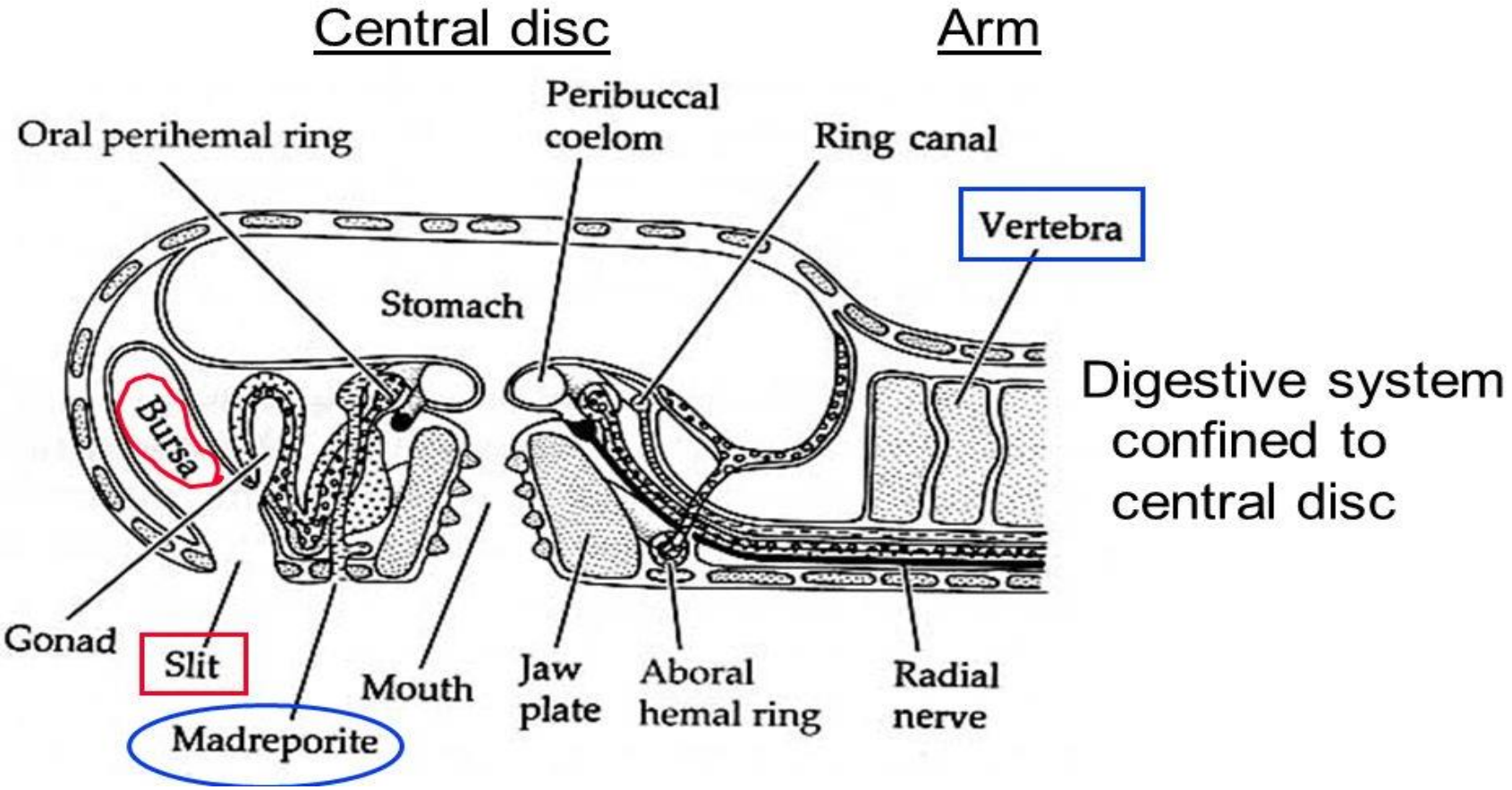
- Some ophiuroids are filter feeders that wave their arms and trap plankton on mucus covered tube feet.
- Trapped plankton is passed from tube foot to tube foot along the length of an arm until it reaches the mouth.

# Feeding and Digestion in Ophiuroids

- The mouth of ophiuroids is in the center of the central disk, and five triangular jaws form a chewing apparatus.
- The mouth leads to a saclike stomach.
- There is no intestine, and no part of the digestive tract extends into the arms.

# Feeding and Digestion in Ophiuroids

## Ophiuroid Cross-section




Digestive system confined to central disc

# Animal Diversity: Invertebrates

## Phylum Echinodermata

Gaseous exchange in Ophiuroids

Lecture 240

# Gaseous Exchange in Ophiuroids

- The coelom of ophiuroids is reduced and is mainly confined to the central disk, but it still serves as the primary means for the distribution of nutrients, wastes, and gases.

# Gaseous Exchange in Ophiuroids

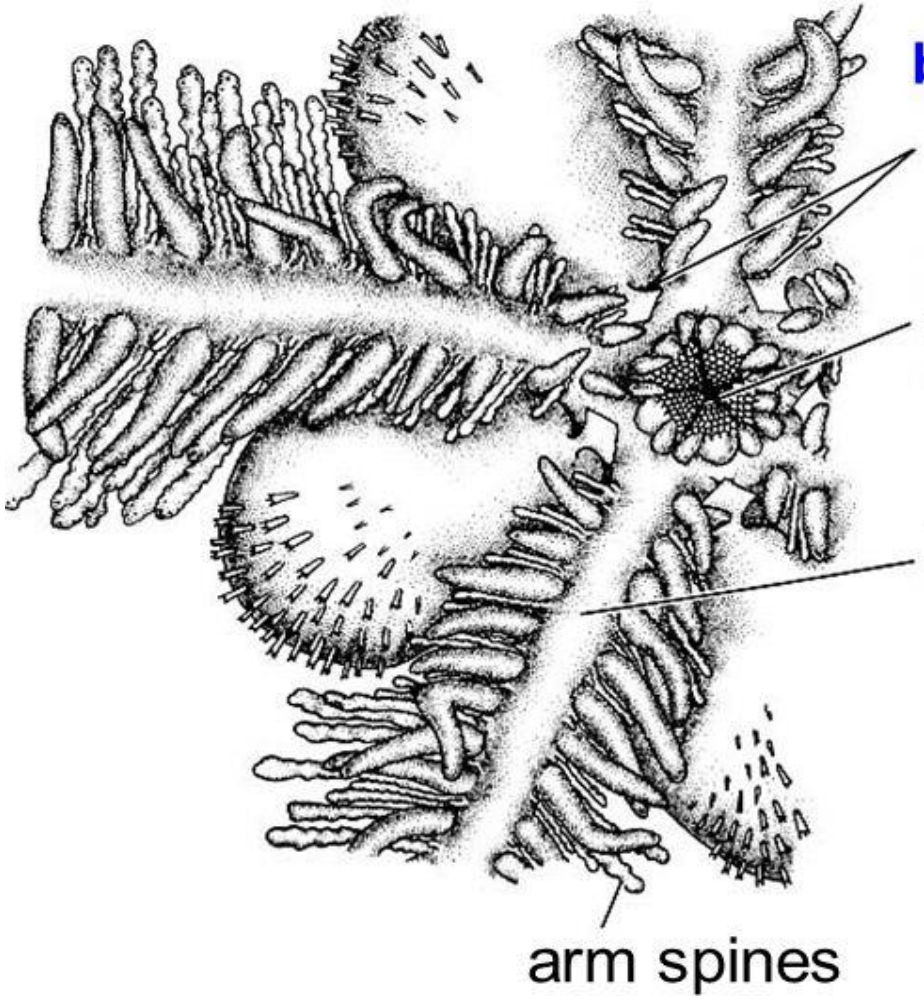
- Coelomocytes aid in the distribution of nutrients and the expulsion of particulate wastes.

# Gaseous Exchange in Ophiuroids

- Ammonia is the primary nitrogenous waste product, and it is lost by diffusion across tube feet and membranous sacs, called **bursae, that invaginate from the oral surface of the central disk.**
- Slits in the oral disk, near the base of each arm, allow cilia to move water into and out of the bursae.

# Gaseous Exchange in Ophiuroids

## Ophiuroid Oral Surface



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Gaseous Exchange in Ophiuroids

Lecture 241

# Gaseous Exchange in Ophiuroids

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# Gaseous Exchange in Ophiuroids

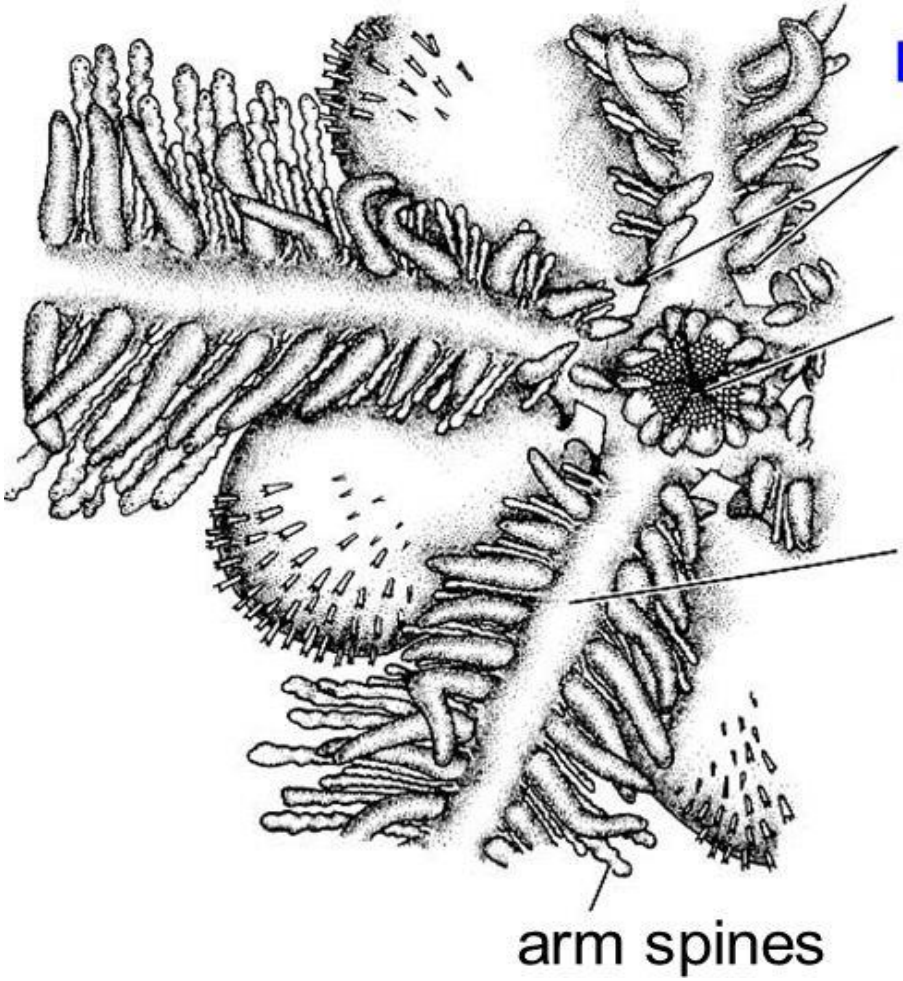
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# Gaseous Exchange in Ophiuroids

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# Gaseous Exchange in Ophiuroids

## Ophiuroid Oral Surface



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Regeneration in Ophiuroids

# Regeneration in Ophiuroids

- Like sea stars, ophiuroids can regenerate lost arms.
- If a brittle star is grasped by an arm, the contraction of certain muscles may sever and cast off the arm hence the common name brittle star.

# Regeneration in Ophiuroids

- This process, called **autotomy** (Gr. *autos,* self *tomos,* to cut), is used in escape reactions.
- The ophiuroid later regenerates the arm.
- Some species also have a fission line across their central disk.

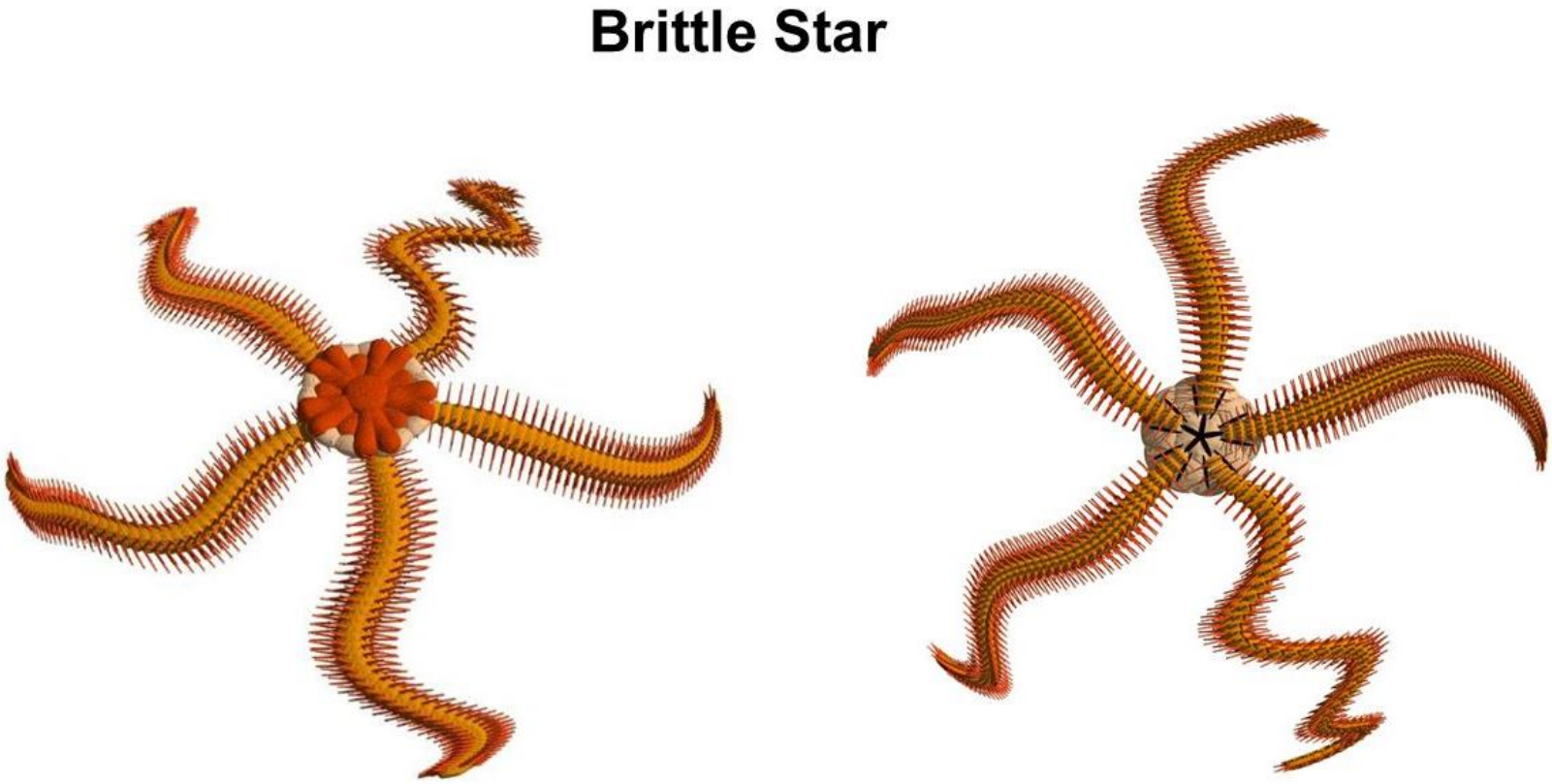
# Regeneration in Ophiuroids

- When an ophiuroid splits into halves along this line, two ophiuroids regenerate.
- Ophiuroids use this ability to escape predators, in a way similar to lizards which deliberately shed the distal part of their tails to confuse pursuers.

# Regeneration in Ophiuroids

- Moreover, the ophiuroids can regenerate gut and gonad fragments lost along with the arms.
- Discarded arms have not been shown to have the ability to regenerate.

# Regeneration in Ophiuroids



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Reproduction in Ophiuroids

Lecture 242

# Reproduction in Ophiuroids

- Ophiuroids are dioecious. Males are usually smaller than females, who often carry the males.
- The gonads are associated with each bursa, and gametes are released into the bursa.

# Reproduction in Ophiuroids

- Eggs may be shed to the outside or retained in the bursa, where they are fertilized and held through early development.
- Embryos are protected in the bursa and are sometimes nourished by the parent.

# Reproduction in Ophiuroids

- A larval stage, called an **ophiopluteus**, is planktonic.
- Its long arms bear ciliary bands used to feed on plankton, and it undergoes metamorphosis before sinking to the substrate.

# Reproduction in Ophiuroids

- In other species the fertilized eggs develop directly into young stars.
- Some species of brittle and basket stars **make** both eggs and sperm, and the eggs unite with sperm and develop in pouches at the bases of the female's arms.

# Reproduction in Ophiuroids



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Development in Ophiuroids

Lecture 243

# Development in Ophiuroids

- Ophiuroid ova undergo germinal vesicle breakdown (GVBD) prior to or soon after spawning in most species, irregardless of developmental mode.

# Development in Ophiuroids

- Species with pelagic, plankotrophic larvae tend to shed thousands of small (70–200 μm diameter), oligolecithal eggs and undergo external fertilization.

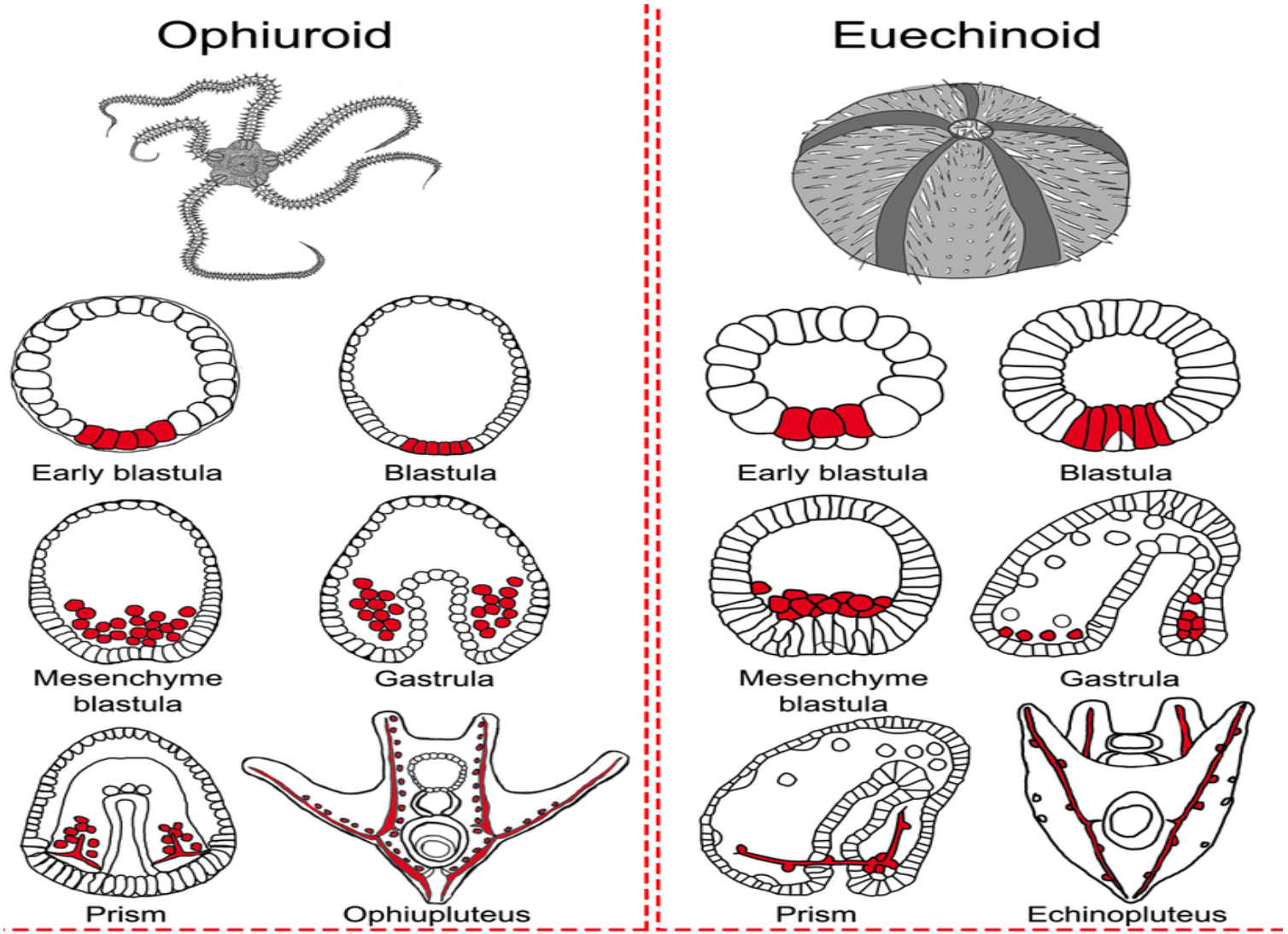
# Development in Ophiuroids

- They develop in the plankton into ophioplutei and have a pelagic period of about 20–90 days, depending on the water temperature.
- Other species with pelagic larval stages are lecithotrophic.

# Development in Ophiuroids

- These species tend to shed several thousand medium to large sized eggs (130–420 μm diameter) which are nutrient rich and opaque.
- Development into non-feeding larvae with pelagic periods of roughly 1 week or less occurs in the water column or near the bottom.

# Development in Ophiuroids



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Class Echinoidea: General features

Lecture 244

# Class Echinoidea: General Features

- The sea urchins, sand dollars, and heart urchins make up the class Echinoidea (Gr. *echinos,* spiny *oeides,* in the form of).
- The approximately one thousand species are widely distributed in nearly all marine environments.

# Class Echinoidea: General Features

- Sea urchins are specialized for living on hard substrates, often wedging themselves into crevices and holes in rock or coral.
- Sand dollars and heart urchins usually live in sand or mud, and burrow just below the surface.

# Class Echinoidea: General Features

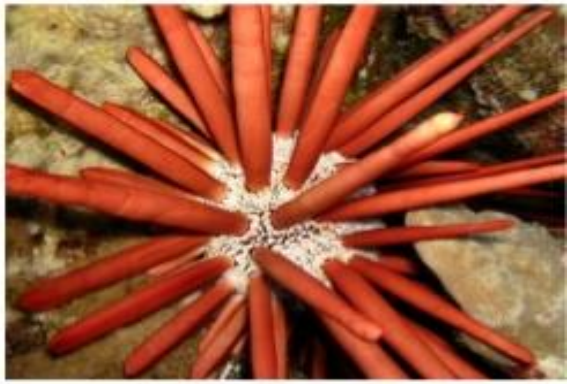
- They use tube feet to catch organic matter settling on them or passing over them.
- Sand dollars often live in dense beds, which favors efficient reproduction and feeding.

# Class Echinoidea: General Features

Class Echinoidea – **sea urchins** and sand dollars











# Animal Diversity: Invertebrates

## Phylum Echinodermata

Skeleton of Sea Urchins

Lecture 245

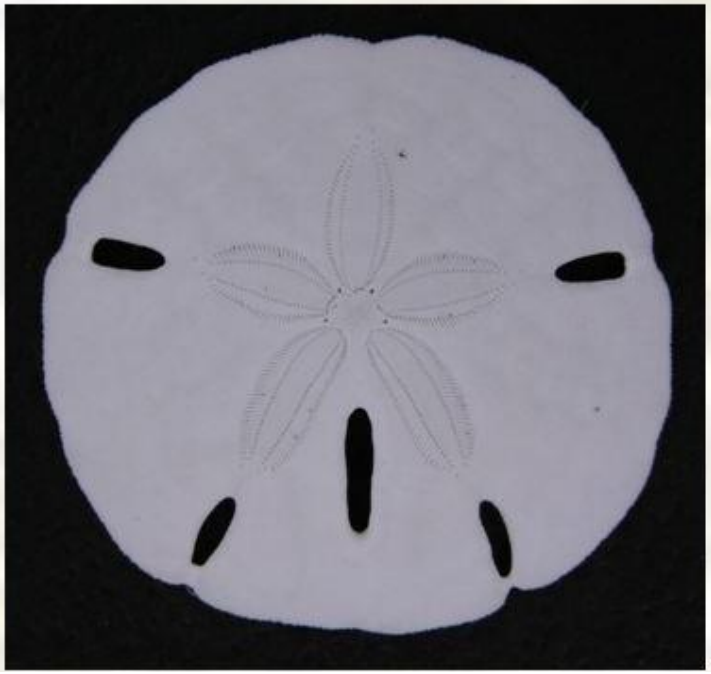
# Skeleton of Sea Urchins

- **Sea urchins** are rounded, and their oral end is oriented toward the substrate.
- Their skeleton, called a **test**, consists of 10 closely fitting plates that arch between oral and aboral ends.

# Skeleton of Sea Urchins

**Sea Urchin & Sand Dollar – skeleton plates fused into shell called "test"**





# Skeleton of Sea Urchins

- Five rows of ambulacral plates have openings for tube feet, and alternate with five inter ambulacral plates, which have tubercles for the articulation of spines.

# Skeleton of Sea Urchins

- The base of each spine is a concave socket, and muscles at its base move the spine.
- Spines are often sharp and sometimes hollow, and they may contain venom dangerous to swimmers.

END

# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Role of Pedicellariae in Sea Urchins

Lecture 246

# Role of Pedicellariae in Sea Urchins

- The pedicellariae of sea urchins have either two or three jaws and connect to the body wall by a relatively long stalk.

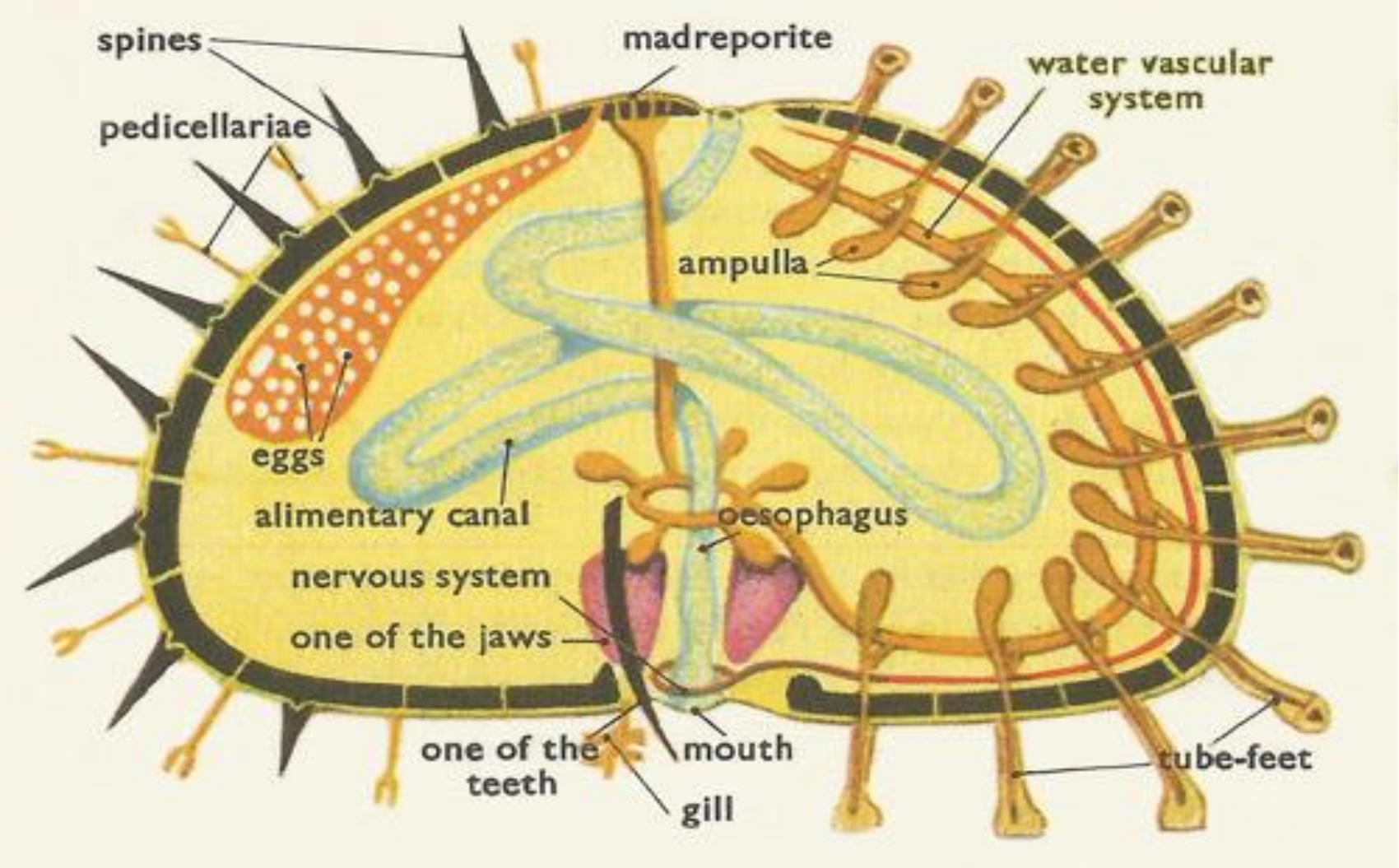
# Role of Pedicellariae in Sea Urchins

- They clean the body of debris and capture planktonic larvae, which provide an extra source of food.
- Pedicellariae of some sea urchins contain venom sacs and are grooved or hollow to inject venom into a predator, such as a sea star.

# Role of Pedicellariae in Sea Urchins

- Each pedicellaria is an effector organ with its own set of muscles, neuropils, and sensory receptors and is therefore capable of reflex responses to the environment.

# Role of Pedicellariae in Sea Urchins



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Water vascular System of Sea Urchins

Lecture 247

# Water Vascular System of Sea Urchins

- The water-vascular system is similar to that of other echinoderms.
- Radial canals run along the inner body wall between the oral and the aboral poles.

# Water Vascular System of Sea Urchins

- Tube feet possess ampullae and suction cups, and the water-vascular system opens to the outside through many pores in one aboral ossicle that serves as a madreporite.

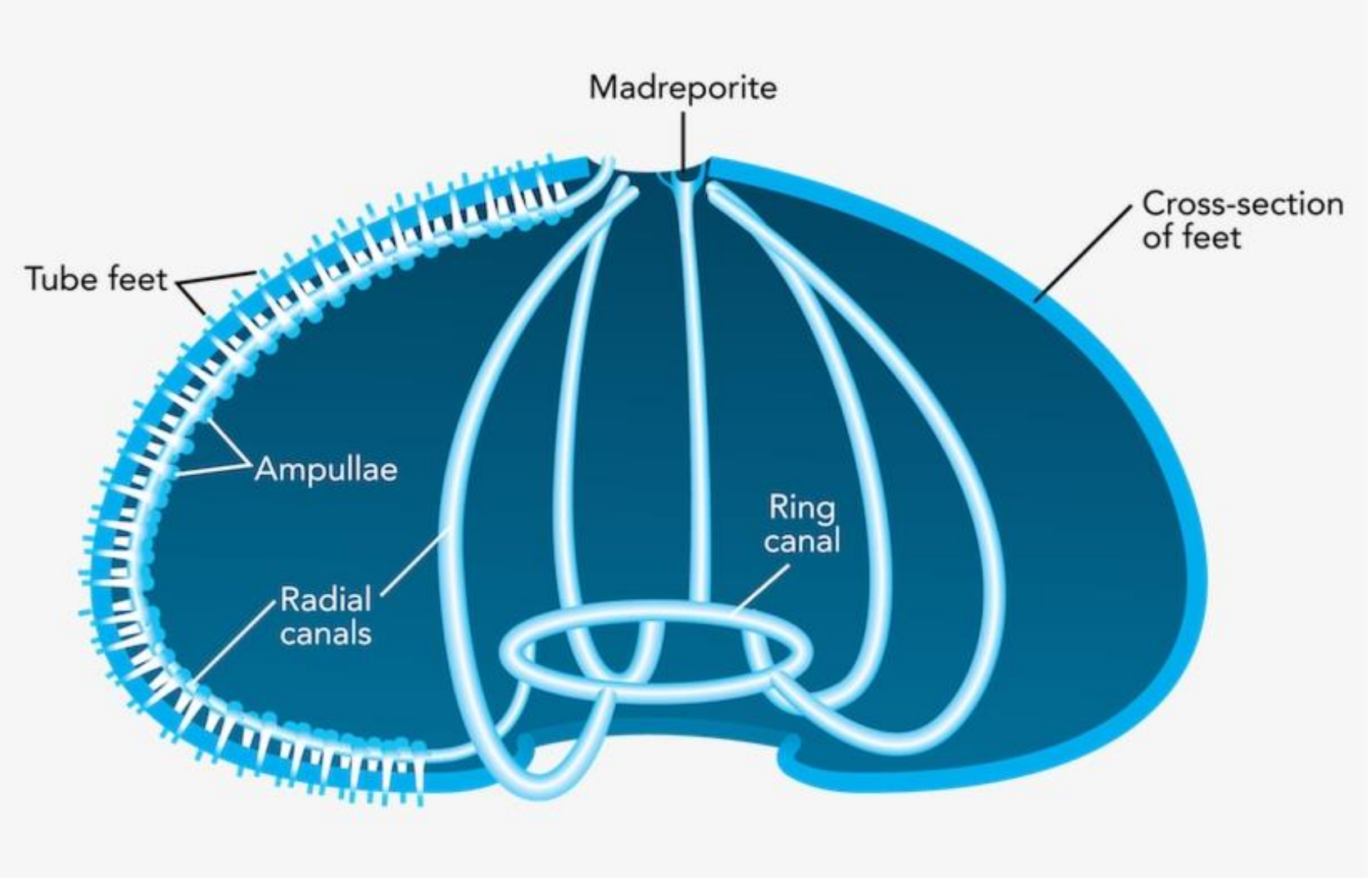
# Water Vascular System of Sea Urchins

- The madreporite of sea urchins is located within one of the plates surrounding the anus on the upper surface of the animal.
- The stone canal descends from the madreporite to the ring canal, which lies around the oesophagus, and includes a number of polian vesicles.

# Water Vascular System of Sea Urchins

- Because sea urchins have no arms, the five radial canals simply run along the inside of the solid skeletal "test", arching upwards towards the anus.
- The ampullae branching off from either side of the radial canals give rise to ten rows of tube feet, which penetrate through holes in the test to the outside.

# Water Vascular System of Sea Urchins



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Locomotion in Echinoids

Lecture 248

# Locomotion in Echinoids

- Echinoids move by using spines for pushing against the substrate and tube feet for pulling.
- Sand dollars and heart urchins use spines to help burrow in soft substrates.

# Locomotion in Echinoids

- Some sea urchins burrow into rock and coral to escape the action of waves and strong currents.
- They form cup-shaped depressions and deeper burrows, using the action of their chewing Aristotle's lantern.

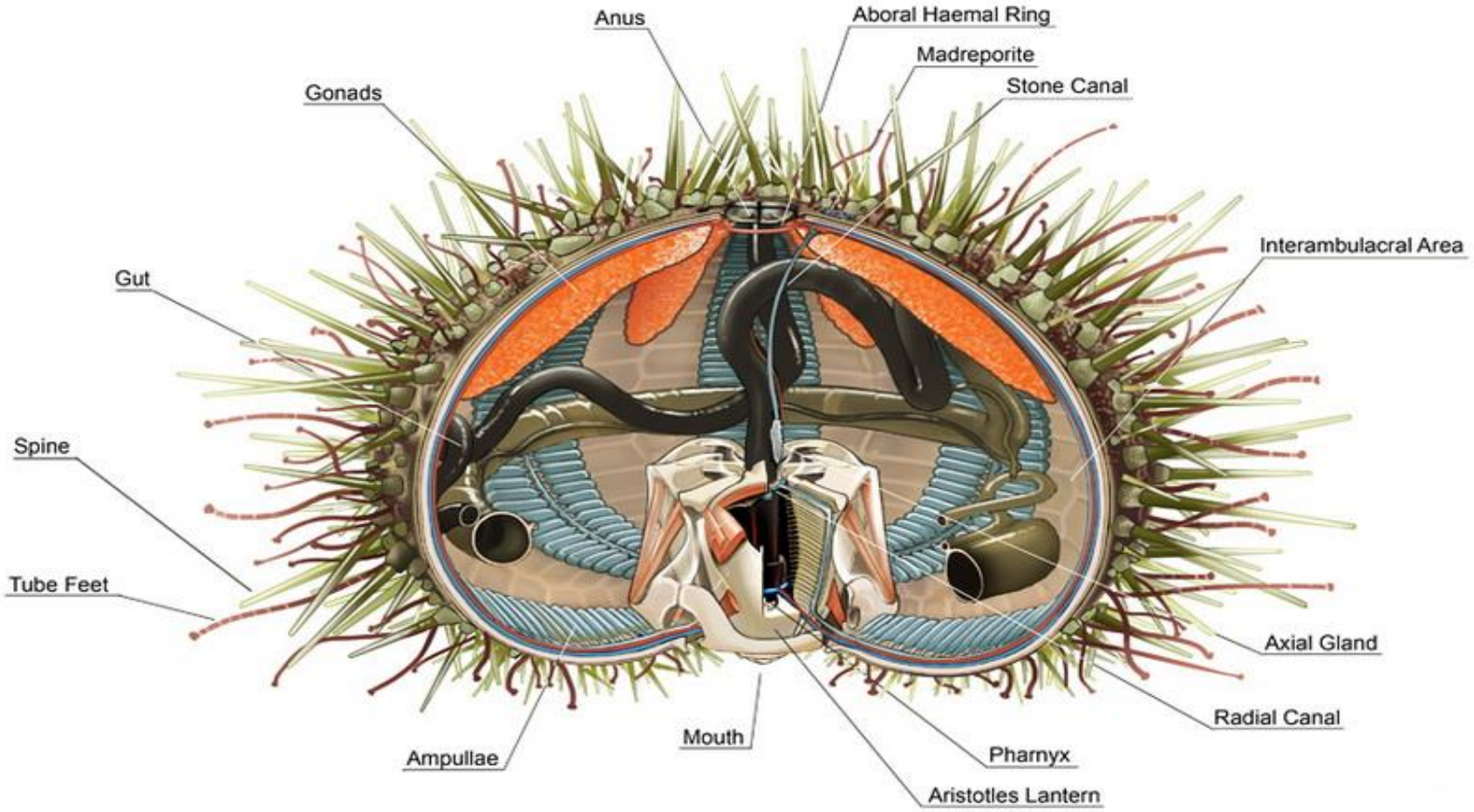
# Locomotion in Echinoids

- Muscles in the tube wall allow the sea urchins to bend or retract their feet, while increased fluid pressure is used to elongate the feet.
- Although the extremities of the feet look like suction naps, the actual mechanism used to hold on to the substrate is adhesion.

# Locomotion in Echinoids

- Sea urchins basically produce a sticky liquid at the end of their feet and glue themselves to their substrate.
- This adhesion is reversible and the animals can detach their feet from the substrate without damaging them.

# Locomotion in Echinoids



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Aristotle's Lantern in Echinoids

Lecture 249

# Aristotle's Lantern in Echinoids

- Echinoids feed on algae, bryozoans, coral polyps, and dead animal remains.
- Oral tube feet surrounding the mouth manipulate food.
- A chewing apparatus, called **Aristotle's lantern, can be projected** from the mouth.

# Aristotle's Lantern in Echinoids

- It consists of about 35 ossicles and attached muscles and cuts food into small pieces for ingestion.
- The mouth cavity leads to a pharynx, an esophagus, and a long, coiled intestine that ends aborally at the anus.

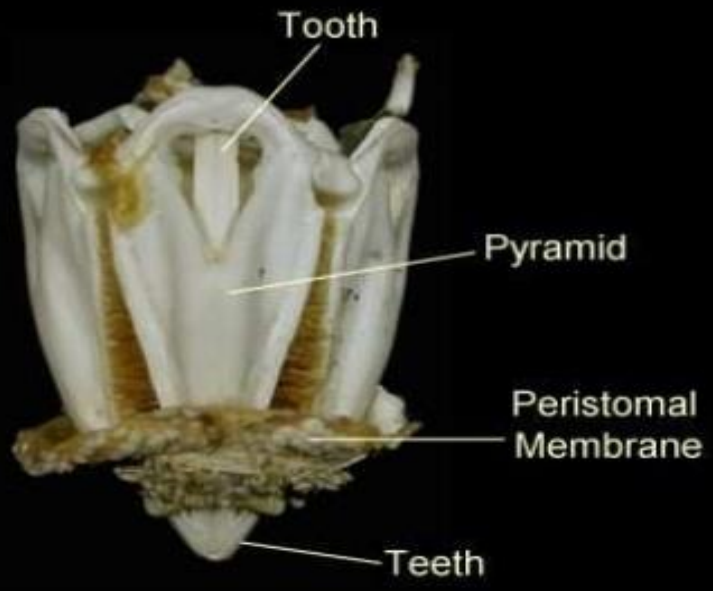
# Aristotle’s Lantern in Echinoids

- The teeth of Aristotle’s lantern are typically extruded to scrape algae and other food from rocks, and some urchins can excavate hiding places in coral or rock even in steel.

# Aristotle's Lantern in Echinoids

## Phylum Echinodermata
## Echinoidea

- Aristotle's lantern, equipped with five strong teeth, used for scraping food from hard substrates.
- 40 skeletal ossicles and 60 muscles that support and operate the teeth.



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Maintenance Functions of Echinoids: Gaseous Exchange, Excretion and Nervous System

Lecture 250

# Maintenance Functions of Echinoids

- Echinoids have a large coelom, and coelomic fluids are the primary circulatory medium.
- Small gills, found in a thin membrane surrounding the mouth, are outpockets of the body wall and are lined by ciliated epithelium.

# Maintenance Functions of Echinoids

- Gas exchange occurs by diffusion across this epithelium and across the tube feet.
- Ciliary currents, changes in coelomic pressure, and the contraction of muscles associated with Aristotle's lantern move coelomic fluids into and out of gills.

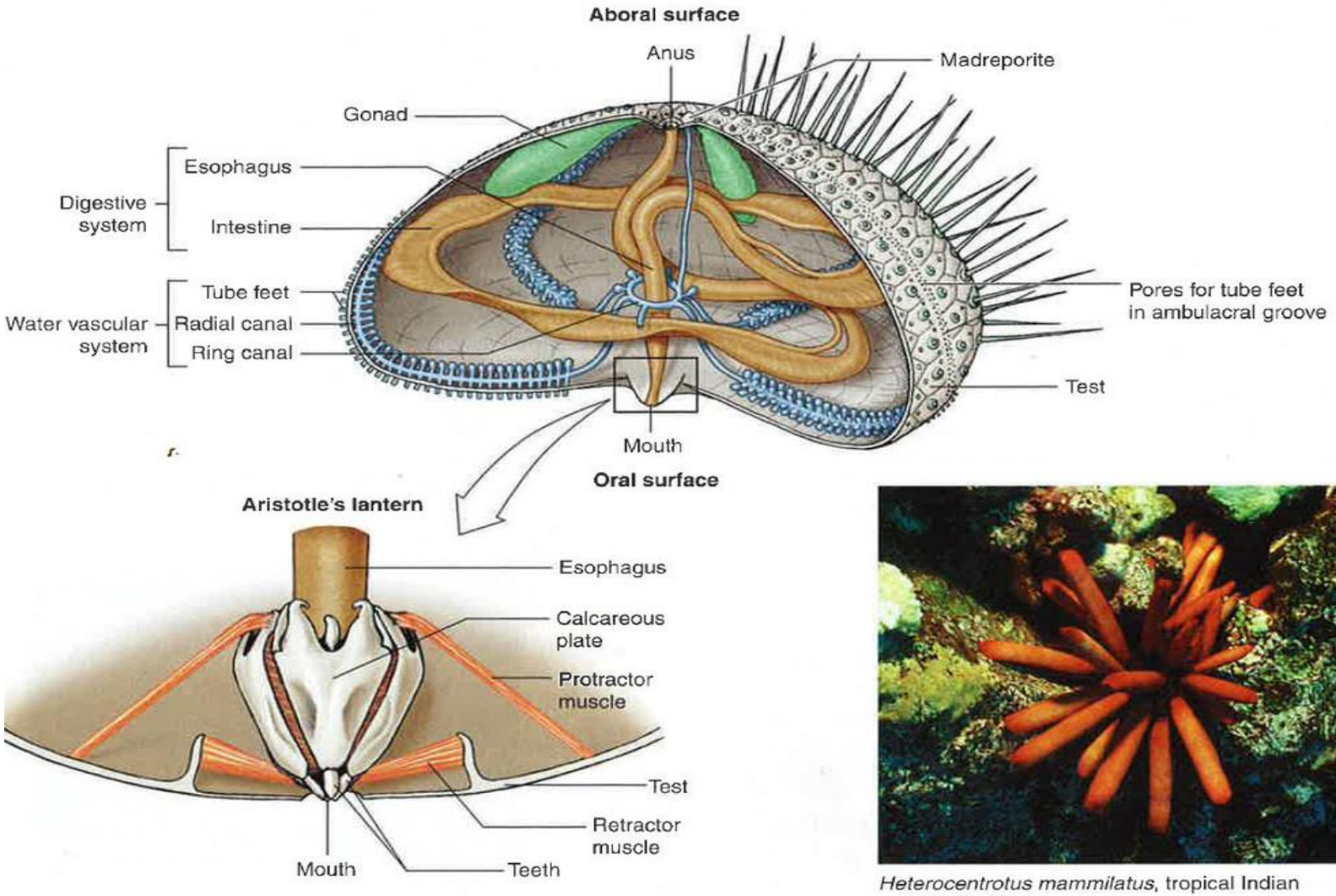
# Maintenance Functions of Echinoids

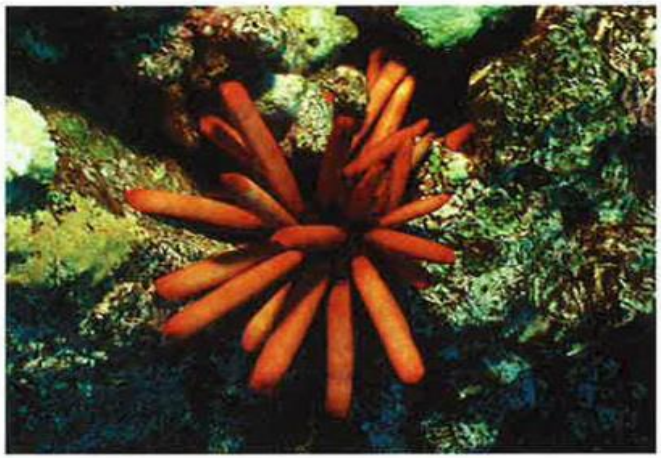
- The nervous system of sea urchins has a relatively simple layout. With no true brain, the neural center is a large nerve ring encircling the mouth just inside the lantern.

# Maintenance Functions of Echinoids

- From the nerve ring, five nerves radiate underneath the radial canals of the water vascular system, and branch into numerous finer nerves to innervate the tube feet, spines, and pedicellariae.

# Maintenance Functions of Echinoids



*Heterocentrotus mammilatus*, tropical Indian and Pacific Ocean.

# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Reproduction and Development in Echinoids

Lecture 251

# Reproduction and Development in Echinoids

- Echinoids are dioecious.
- Gonads are on the internal body wall of the inter ambulacral plates.
- During breeding season, they nearly fill the spacious coelom.

# Reproduction and Development in Echinoids

- One gonopore is in each of five ossicles, called genital plates, at the aboral end of the echinoid, although the sand dollars usually have only four gonads and gonopores.
- Gametes are shed into the water, and fertilization is external.

# Reproduction and Development in Echinoids

- The reproductive cycle has three main phases:

1 The growing phase (late autumn and winter) when gonads accumulate reserve material;

2 The maturation phase (spring and early summer) in which gametogenesis then spawning take place;

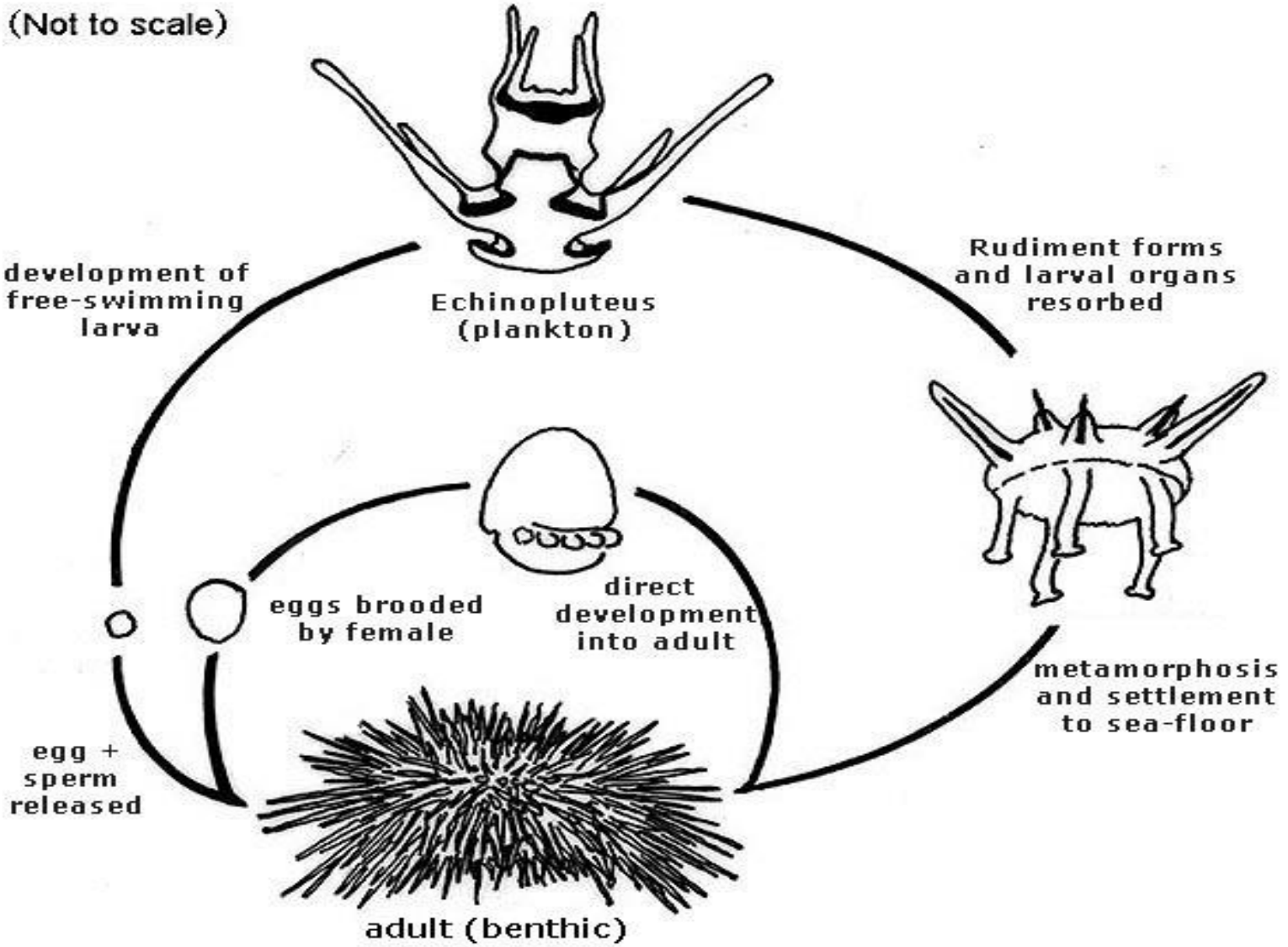
# Reproduction and Development in Echinoids

3 The spent/regenerating phase when relict gametes are resorbed by the nutritive phagocytes, the gonads being virtually devoid of sexual cells.

# Reproduction and Development in Echinoids

- Development eventually results in a pluteus larva that spends several months in the plankton and eventually undergoes metamorphosis to the adult.

# Reproduction and Development in Echinoids



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Class Holothuroidea: General Features

Lecture 252

# Class Holothuroidea: General Features

- The class Holothuroidea (Gr. *holothourion,* sea cucumber *oeides*, in the form of) has approximately 1,500 species, whose members are commonly called sea cucumbers.

# Class Holothuroidea: General Features

- Sea cucumbers are found at all depths in all oceans, where they crawl over hard substrates or burrow through soft substrates.
- Sea cucumbers have no arms, and they are elongate along the oral-aboral axis.

# Class Holothuroidea: General Features

- They lie on one side, which is usually flattened as a permanent ventral side, giving them a secondary bilateral symmetry.
- Tube feet surrounding the mouth are elongate and referred to as tentacles.
- Most adults range in length between 10 and 30 cm.

# Class Holothuroidea: General Features

- **Class Holothuroidea**
  - Sea cucumbers



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Body Wall of Sea Cucumber

Lecture 253

# Body Wall of Sea Cucumber

- Their body wall is thick and muscular, and it lacks protruding spines or pedicellariae.
- Beneath the epidermis is the dermis, a thick layer of connective tissue with embedded ossicles.

# Body Wall of Sea Cucumber

- Sea cucumber ossicles are microscopic and do not function in determining body shape.
- Larger ossicles form a calcareous ring that encircles the oral end of the digestive tract, serving as a point of attachment for body wall muscles.

# Body Wall of Sea Cucumber

- Beneath the dermis is a layer of circular muscles overlying longitudinal muscles.
- The body wall of sea cucumbers, when boiled and dried, is known as trepang in Asian countries.

# Body Wall of Sea Cucumber

- It may be eaten as a main-course item or added to soups as flavoring and a source of protein.

# Body Wall of Sea Cucumber

## Sea Cucumbers

plates over a soft muscular body wall



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Water Vascular System of Sea Cucumber

Lecture 254

# Water Vascular System of Sea Cucumber

- The madreporite of sea cucumbers is internal, and the water-vascular system is filled with coelomic fluid.
- The ring canal encircles the oral end of the digestive tract and gives rise to one to ten Polian vesicles.

# Water Vascular System of Sea Cucumber

- Five radial canals and the canals to the tentacles branch from the ring canal.
- Radial canals and tube feet, with suction cups and ampullae, run between the oral and aboral poles.

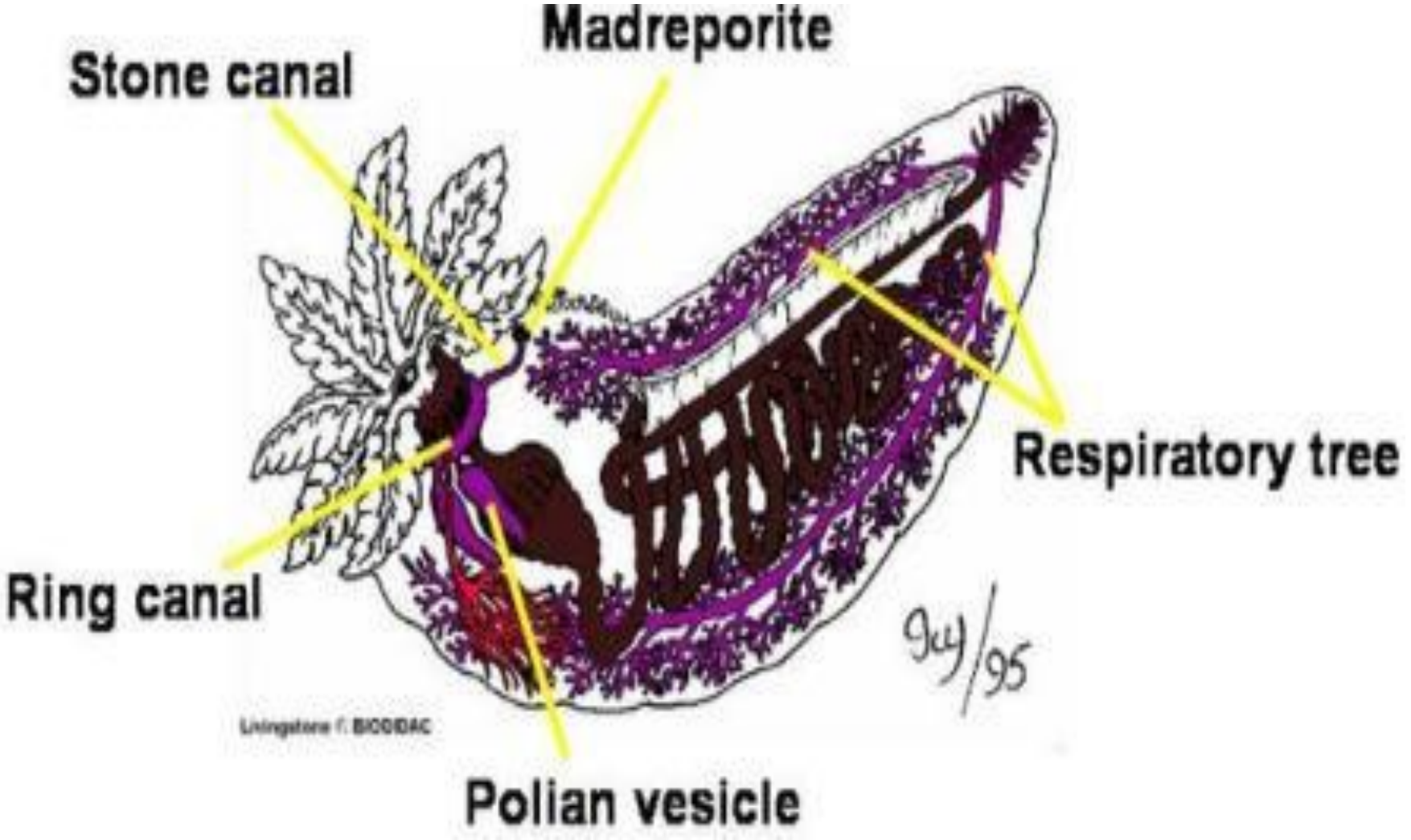
# Water Vascular System of Sea Cucumber

- The side of a sea cucumber resting on the substrate contains three of the five rows of tube feet, which are primarily used for attachment.
- The two rows of tube feet on the upper surface may be reduced in size or may be absent.

# Water Vascular System of Sea Cucumber

- The madreporite is present, but lies within the body cavity, just below the pharynx.
- The stone canal is relatively short.

# Water Vascular System of Sea Cucumber



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Locomotion in Sea Cucumber

Lecture 255

# Locomotion in Sea Cucumber

- Sea cucumbers are mostly sluggish burrowers and creepers, although some swim by undulating their bodies from side to side.
- Locomotion using tube feet is inefficient, because the tube feet are not anchored by body wall ossicles.

# Locomotion in Sea Cucumber

- Locomotion more commonly results from contractions of body-wall muscles that produce wormlike, locomotor waves that pass along the length of the body.

# Locomotion in Sea Cucumber

- One way that **sea cucumbers** can confuse or harm predators is by propelling their own toxic internal organs from their bodies in the direction of an attacker.
- The organs grow back, and it may save them from being eaten.

# Locomotion in Sea Cucumber

## Locomotion of Sea Cucumbers

- Tube feet



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Feeding and Digestion in Sea Cucumber

Lecture 256

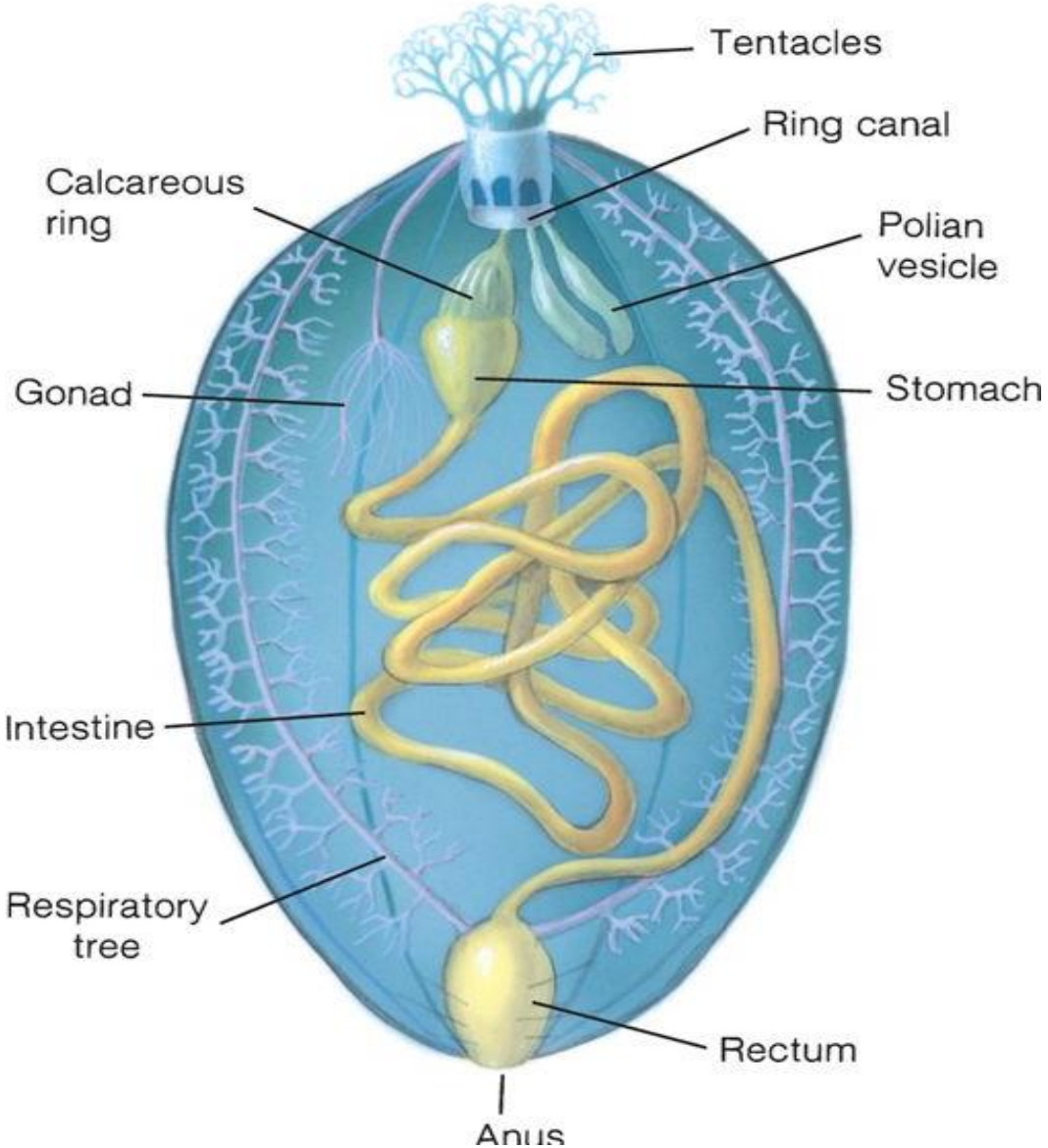
# Feeding and Digestion in Sea Cucumber

- Most sea cucumbers ingest particulate organic matter using their tentacles.
- Mucus covering the tentacles traps food as the tentacles sweep across the substrate or are held out in seawater.

# Feeding and Digestion in Sea Cucumber

- The digestive tract consists of a stomach; a long, looped intestine; a rectum; and an anus.
- Sea cucumbers thrust tentacles into the mouth to wipe off trapped food.

# Feeding and Digestion in Sea Cucumber



# Feeding and Digestion in Sea Cucumber

- During digestion, coelomocytes move across the intestinal wall, secrete enzymes to aid in digestion and engulf and distribute the products of digestion.

# Feeding and Digestion in Sea Cucumber

- The coelom of sea cucumbers is large, and the cilia of the coelomic lining circulate fluids throughout the body cavity, distributing respiratory gases, wastes, and nutrients.

# Feeding and Digestion in Sea Cucumber

- The hemal system of sea cucumbers is well developed, with relatively large sinuses and a network of channels containing coelomic fluids.
- Its primary role is food distribution.

END

# Animal Diversity: Invertebrates

## Phylum Echinodermata

Respiratory Tree

Lecture 257

# Respiratory Tree

- A pair of tubes called **respiratory trees attach at the rectum** and branch throughout the body cavity of sea cucumbers.
- The pumping action of the rectum circulates water into these tubes.

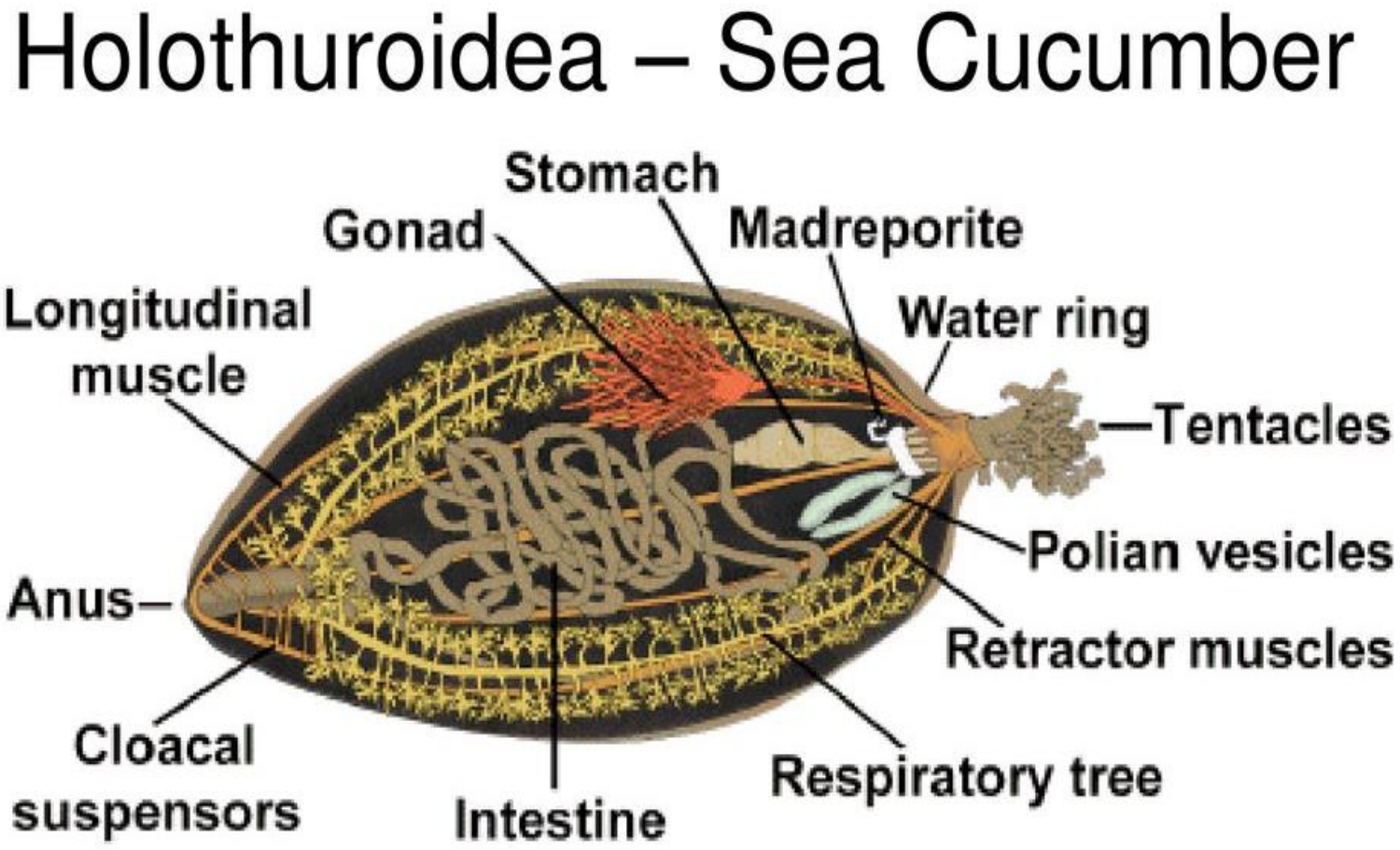
# Respiratory Tree

- When the rectum dilates, water moves through the anus into the rectum.
- Contraction of the rectum, along with contraction of an anal sphincter, forces water into the respiratory tree.
- Water exits the respiratory tree when tubules of the tree contract.

# Respiratory Tree

- Respiratory gases and nitrogenous wastes move between the coelom and seawater across these tubules.

# Respiratory Tree



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Nervous and Sensory Functions in Sea Cucumber

Lecture 258

# Nervous and Sensory Functions in Sea Cucumber

- The nervous system of sea cucumbers is similar to that of other echinoderms but has additional nerves supplying the tentacles and pharynx.
- Sea cucumbers have no true brain.

# Nervous and Sensory Functions in Sea Cucumber

- A ring of neural tissue surrounds the oral cavity, and sends nerves to the tentacles and the pharynx.
- The animal is, however, quite capable of functioning and moving about if the nerve ring is surgically removed, demonstrating that it does not have a central role in nervous coordination.

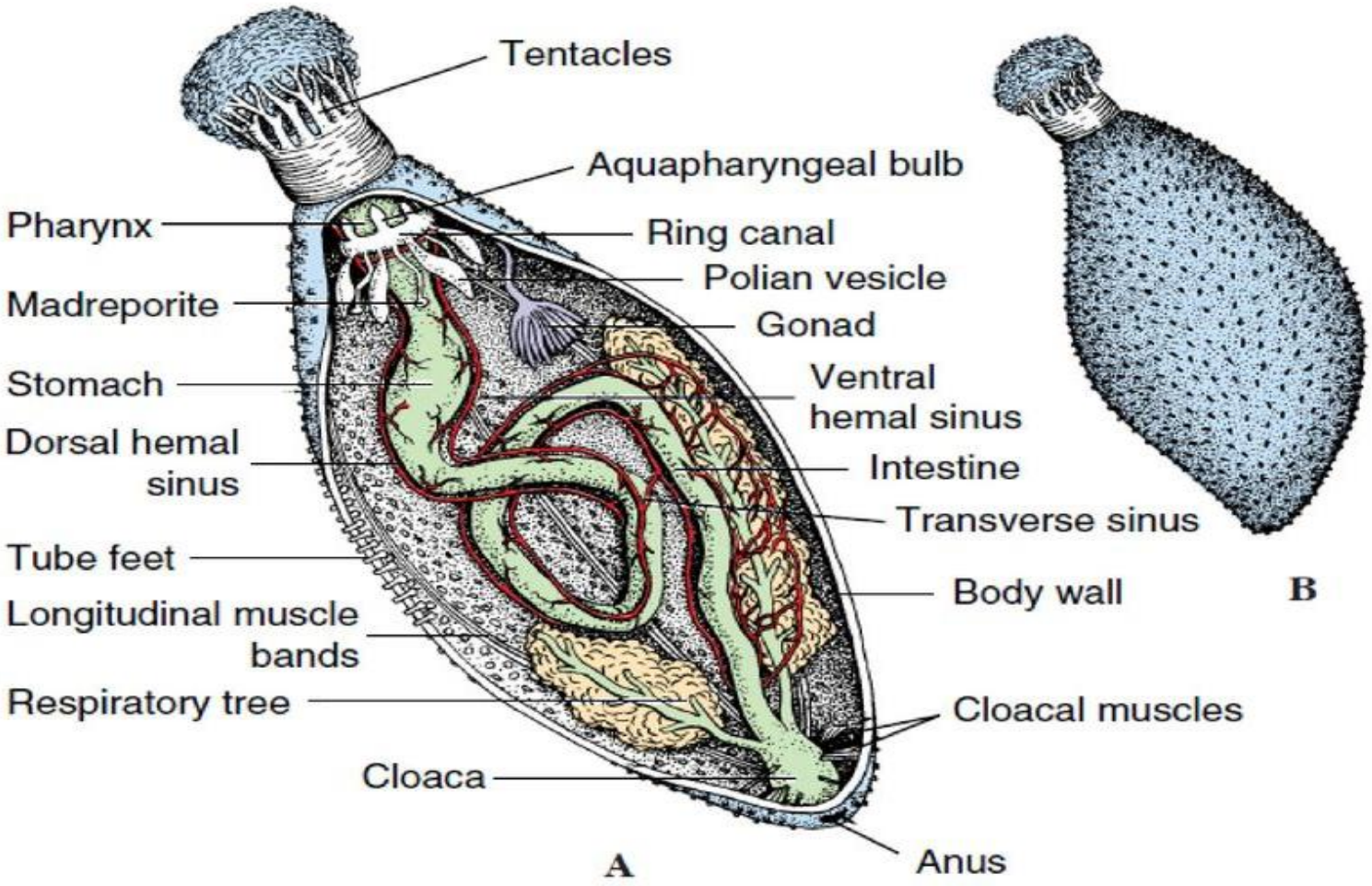
# Nervous and Sensory Functions in Sea Cucumber

- In addition, five major nerves run from the nerve ring down the length of the body beneath each of the ambulacral areas.
- Some sea cucumbers have statocysts, and others have relatively complex photoreceptors.

# Nervous and Sensory Functions in Sea Cucumber

- Most sea cucumbers have no distinct sensory organs, although there are various nerve endings scattered through the skin, giving the animal a sense of touch and a sensitivity to the presence of light.
- Some species possess small eye-spots near the bases of their tentacles.

# Nervous and Sensory Functions in Sea Cucumber



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Defense in Sea Cucumber

Lecture 259

# Defense in Sea Cucumber

- Casual examination suggests that sea cucumbers are defenseless against predators.
- Many sea cucumbers, however, produce toxins in their body walls that discourage predators.

# Defense in Sea Cucumber

- Other sea cucumbers can evert tubules of the respiratory tree, called **Cuverian tubules**, through the anus.
- These tubules contain sticky secretions and toxins capable of entangling and immobilizing predators.

# Defense in Sea Cucumber



# Defense in Sea Cucumber

- In addition, contractions of the body wall may result in expulsion of one or both respiratory trees, the digestive tract, and the gonads through the anus.

# Defense in Sea Cucumber

- This process, called evisceration, is a defensive adaptation that may discourage predators.
- Regeneration of lost parts follows.

END

# Animal Diversity: Invertebrates

## Phylum Echinodermata

Regeneration, Reproduction and Development in Sea Cucumber

Lecture 260

# Regeneration, Reproduction & Development in Sea Cucumber

- Sea cucumbers are dioecious.
- They possess a single gonad, located anteriorly in the coelom, and a single gonopore near the base of the tentacles.
- Fertilization is usually external, and embryos develop into planktonic larvae.

# Regeneration, Reproduction & Development in Sea Cucumber

- Metamorphosis precedes settling to the substrate.
- In some species, a female's tentacles trap eggs as the eggs are released.
- After fertilization, eggs are transferred to the body surface, where they are brooded.

# Regeneration, Reproduction & Development in Sea Cucumber

- Although rare, coelomic brooding also occurs.
- Eggs are released into the body cavity, where fertilization (by an unknown mechanism) and early development occur.

# Regeneration, Reproduction & Development in Sea Cucumber

- The young leave through a rupture in the body wall.
- Sea cucumbers can also reproduce by transverse fission, followed by regeneration of lost parts.

# Regeneration, Reproduction & Development in Sea Cucumber

## *Sea Cucumber Larva & Adult*





# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Class Crinoidea: General Features

Lecture 261

# Class Crinoidea: General Features

- Members of the class Crinoidea (Gr. *krinon,* lily *oeides,* in the form of) include the sea lilies and the feather stars.
- They are the most primitive of all living echinoderms and are very different from any covered thus far.

# Class Crinoidea: General Features

- Approximately 630 species are living today; however, an extensive fossil record indicates that many more were present during the Paleozoic era, 200 to 600 million years ago.

# Class Crinoidea: General Features

- **Crinoids** are pentamerous, stalked echinoderms with a cuplike body bearing five usually branched and commonly featherlike arms.
- Most of a crinoid's body consists of an endoskeleton composed of numerous calcareous pieces, called plates or ossicles.

# Class Crinoidea: General Features



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Difference Between Sea Lily and Feather Star

Lecture 262

# Difference Between Sea Lily and Feather Star

- Sea lilies attach permanently to their substrate by a stalk.
- The attached end of the stalk bears a flattened disk or root like extensions that are fixed to the substrate.

# Difference Between Sea Lily and Feather Star

- Disk like ossicles of the stalk appear to be stacked on top of one another and are held together by connective tissues, giving a jointed appearance.
- The stalk usually bears projections, or cirri, arranged in whorls.

# Difference Between Sea Lily and Feather Star

- The unattached end of a sea lily is called the crown.
- The aboral end of the crown attaches to the stalk and is supported by a set of ossicles, called the **calyx.**
- Five arms also attach at the calyx.

# Difference Between Sea Lily and Feather Star

- They are branched, supported by ossicles, and bear smaller branches (pinnules) giving them a featherlike appearance.
- Tube feet are in a double row along each arm.
- Ambulacral grooves on the arms lead toward the mouth.

# Difference Between Sea Lily and Feather Star

- The mouth and anus open onto the upper (oral) surface.
- Feather stars are similar to sea lilies, except they lack a stalk and are swimming and crawling animals.

# Difference Between Sea Lily and Feather Star

- The aboral end of the crown bears a ring of root like cirri, which cling when the animal is resting on a substrate.
- Feather stars swim by raising and lowering the arms, and they crawl over substrate by pulling with the tips of the arms.

# Difference Between Sea Lily and Feather Star



# Animal Diversity: Invertebrates

## Phylum Echinodermata

Class Crinoidea: Maintenance Functions, Regeneration, Reproduction and Development

Lecture 263

# Maintenance Functions

- Circulation, gas exchange, and excretion in crinoids are similar to these functions in other echinoderms.
- In feeding, however, crinoids use outstretched arms for suspension feeding.

# Maintenance Functions

- A planktonic organism that contacts a tube foot is trapped, and cilia in ambulacral grooves carry it to the mouth.
- Although this method of feeding is different from how other modern echinoderms feed, it probably reflects the original function of the water-vascular system.

# Maintenance Functions

- Crinoids lack the nerve ring found in most echinoderms.
- Instead, a cup-shaped nerve mass below the calyx gives rise to radial nerves that extend through each arm and control the tube feet and arm musculature.

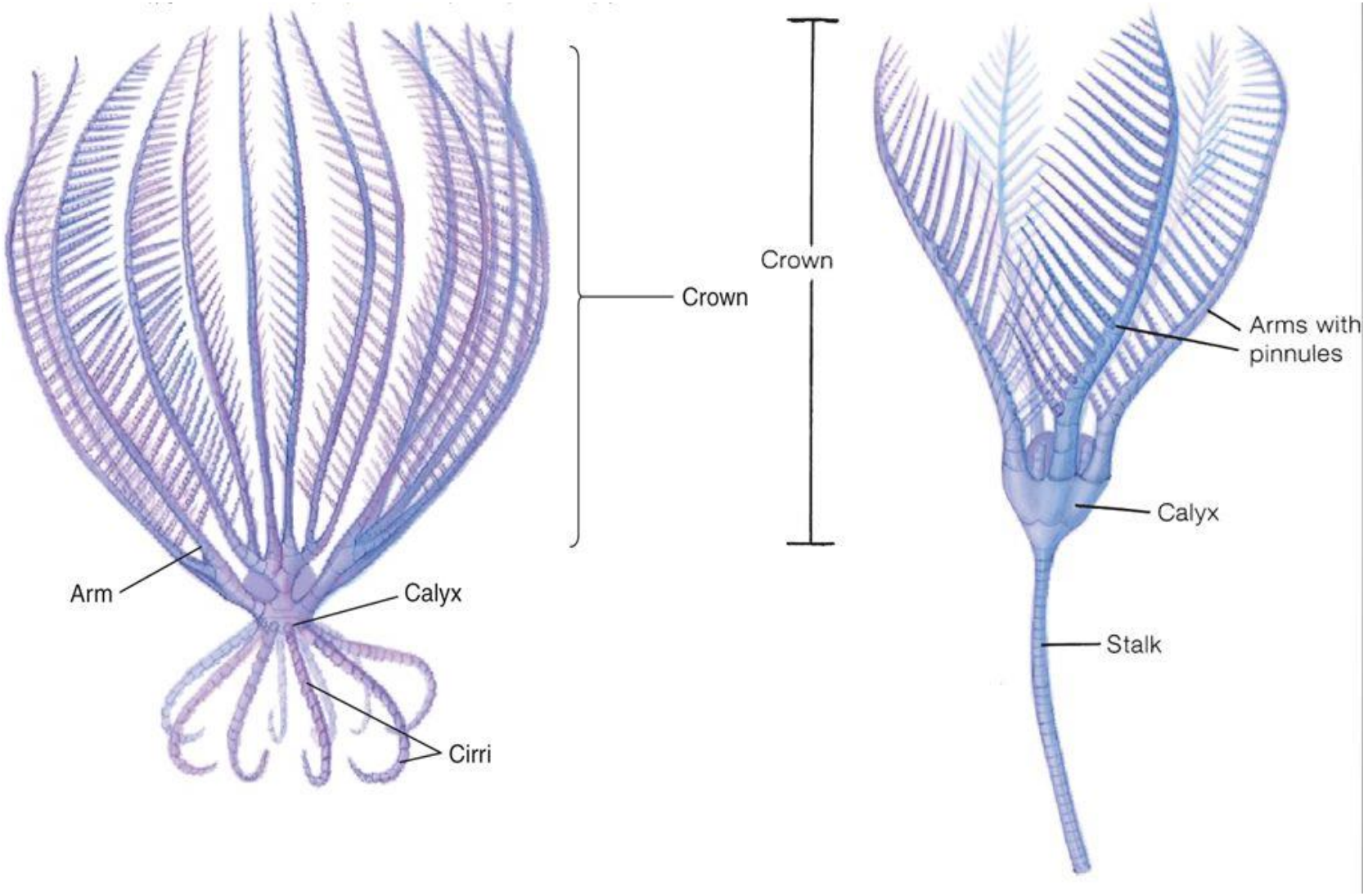
# Maintenance Functions

- Crinoids, like other echinoderms, are dioecious.
- Gametes form from germinal epithelium in the coelom and are released through ruptures in the walls of the arms.
- Some species spawn in seawater, where fertilization and development occur.

# Maintenance Functions

- Other species brood embryos on the outer surface of the arms.
- Metamorphosis occurs after larvae attach to the substrate.
- Like other echinoderms, crinoids can regenerate lost parts.

# Maintenance Functions



# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Class Concentricycloidea

Lecture 264

# Class Concentricycloidea

- The class Concentricycloidea (ME *consentrik,* having a common center and Gr. *kykloeides,* like a circle) contains a single described species, known as the sea daisy.
- Sea daisies have been recently discovered on debris in deep oceans.

# Class Concentricycloidea

- They lack arms and are less than 1 cm in diameter.
- The most distinctive features of this species are the two circular water-vascular rings that encircle the disk like body.

# Class Concentricycloidea

- The inner of the two rings probably corresponds to the ring canal of members of other classes, because it has Polian vesicles attached.
- The outer ring contains tube feet and ampullae and probably corresponds to the radial canals of members of other classes.

# Class Concentricycloidea

- Sea daisies lack an internal digestive system.
- Instead, a thin membrane, called a velum, covers the surface of the animal applied to the substrate (e.g., decomposing organic matter) and digests and absorbs nutrients.

# Class Concentricycloidea

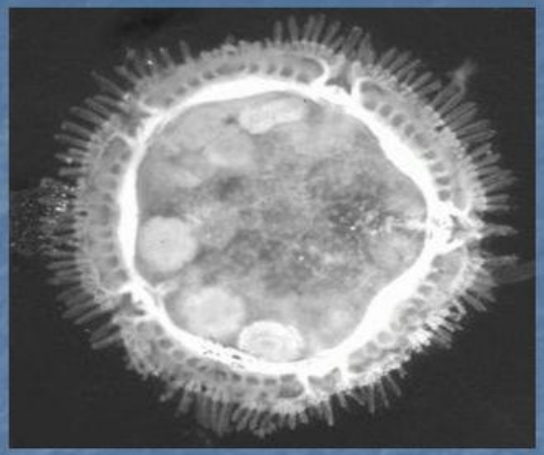
- Internally, five pairs of brood pouches hold embryos during development.
- No free-swimming larval stages are apparent.
- The mechanism for fertilization is unknown.

# Class Concentricycloidea

## Class Concentricycloidea

Example:
Sea Daisy









# Animal Diversity: Invertebrates

## Phylum Echinodermata

### Further Phylogenetic Considerations

Lecture 265

# Further Phylogenetic Considerations

- **As mentioned earlier, most zoologists believe that echinoderms** evolved from bilaterally symmetrical ancestors.
- Radial symmetry probably evolved during the transition from active to more sedentary lifestyles; however, the oldest echinoderm fossils, about 600 million years old, give little direct evidence of how this transition occurred.

# Further Phylogenetic Considerations

- Ancient fossils do give clues regarding the origin of the water-vascular system and the calcareous endoskeleton.
- Of all living echinoderms, the crinoids most closely resemble the oldest fossils.

# Further Phylogenetic Considerations

- Because crinoids use their water-vascular system for suspension feeding, filter feeding, rather than locomotion, was probably the original function of the water-vascular system.
- As do crinoids, early echinoderms probably assumed a mouth-up position and attached aborally.

# Further Phylogenetic Considerations

- They may have used arms and tube feet to capture food and move it to the mouth.
- The calcium carbonate endoskeleton may have evolved to support extended filtering arms and to protect these sessile animals.
- Many modern echinoderms are more mobile.

# Further Phylogenetic Considerations

- This free living lifestyle is probably secondarily derived, as is the mouth down orientation of most echinoderms.
- The mouth-down position is advantageous for predatory and scavenging lifestyles.

# Further Phylogenetic Considerations

- Similarly, changes in the water-vascular system, such as the evolution of ampullae, suction disks, and feeding tentacles, can be interpreted as adaptations for locomotion and feeding in a more mobile lifestyle.

# Further Phylogenetic Considerations

- The idea that the free-living lifestyle is secondary is reinforced by the observation that some echinoderms, such as the irregular echinoids and the holothuroids, have bilateral symmetry imposed upon a pentaradial body form.
- The evolutionary relationships among the echinoderms are not clear.

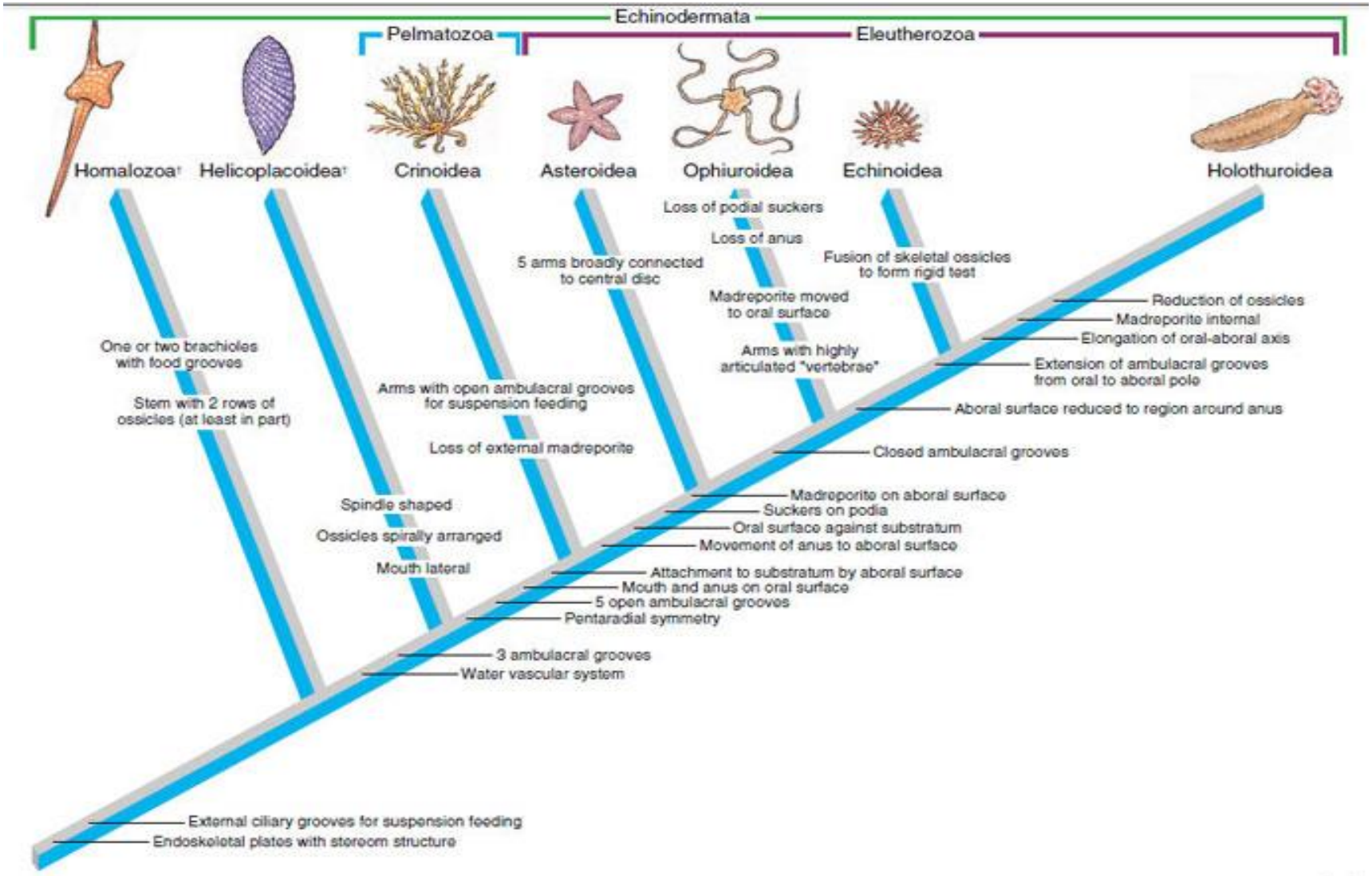
# Further Phylogenetic Considerations

- Numerous fossils date into the Cambrian period, but no interpretation of the evolutionary relationships among living and extinct echinoderms is definitive.

# Further Phylogenetic Considerations

- Most taxonomists agree that the echinoids and holothuroids are closely related.
- Whether the ophiuroids are more closely related to the echinoid/holothuroid lineage or the asteroid lineage is debated.
- The position of the Concentricycloidea in echinoderm phylogeny is highly speculative.

# Further Phylogenetic Considerations



# Animal Diversity: Invertebrates

## Lesser Invertebrates

### The Lophophorates

Lecture 266

# Lesser Invertebrates: The Laphophorates

- Brachiopoda, Ectoprocta, and Phoronida discussed in one major anatomical feature, the lophophore (Gr. *lophos*, crest or tuft and *phorein*, to bear) is a circumoral (around the mouth) body region characterized by a circular or U-shaped ridge, with either one or two rows of ciliated, hollow tentacles.

# Lesser Invertebrates: The Laphophorates

- An anus always open outside of the lophophore.
- The lophophore is a food collecting organ and a surface for gas exchange.
- Sensory cells receptive to chemicals and touch are concentrated on its tentacles.
- The lophophore can usually be extended for feeding or withdrawn for protection.

# Lesser Invertebrates: The Laphophorates

- All lophophorates are sessile or sedentary filter feeders that have a U-shaped digestive tract, a feature common in sedentary animals, and live in a secreted chitinous or calcareous tube.
- As in deuterostomes, these phyla have radial cleavage in embryonic stages and a coelom that is divided into compartments.

# Lesser Invertebrates: The Laphophorates

- As in protostomes, however, the embryonic mouth forms in the region of the blastopore.
- Recent molecular biological studies involving the sequencing of ribosomal DNA, however, suggest that the lophophorates are protostomes.

# Lesser Invertebrates: The Laphophorates

- **Phylum Brachiopoda**: The Brachiopods, or Lampshells (Gr. *brachion*, arm and *podos*, foot) bear a superficial resemblance to the bivalve molluscs because they have a bivalved, calcareous and/or chitinous shell that a mantle secretes and that encloses nearly all of the body.

# Lesser Invertebrates: The Laphophorates

- However, unlike the left and right valves in molluscs, the brachiopods have dorsal and ventral valves.
- In addition molluscs filter with their gills, whereas brachiopods use a lophophore.
- Brachiopods are commonly called **lampshells** because they resemble ancient Roman oil lamps.

# Lesser Invertebrates: The Laphophorates

- Brachiopods are exclusively marine; most species live from the tidal zone to the edge of the continental shelves (about 200 m deep).
- The phylum contains about three hundred living species.
- In the Articulata, the valves are composed primarily of calcium carbonate and have a hinge with interlocking teeth.

# Lesser Invertebrates: The Laphophorates

- The Inarticulata have unhinged valves that are composed primarily of calcium phosphate and held together only by muscles.
- Most members of both classes have a stalked pedicel that usually attaches to a hard surface.

# Lesser Invertebrates: The Laphophorates

- Some, such as *Lingula*, have a muscular pedicel used for burrowing and anchoring in mud or sand.
- The large, horseshoe-shaped lophophore in the anterior mantle cavity bears long, ciliated tentacles used in respiration and feeding.

# Lesser Invertebrates: The Laphophorates

- The cilia set up water currents that carry food particles (mainly organic detritus and algae) between the valves and over the lophophore into the mouth.
- A solitary, dioecious brachiopod reproduces sexually by releasing gametes from multiple gonads into the metacoel and discharging them into the water by the nephridia.

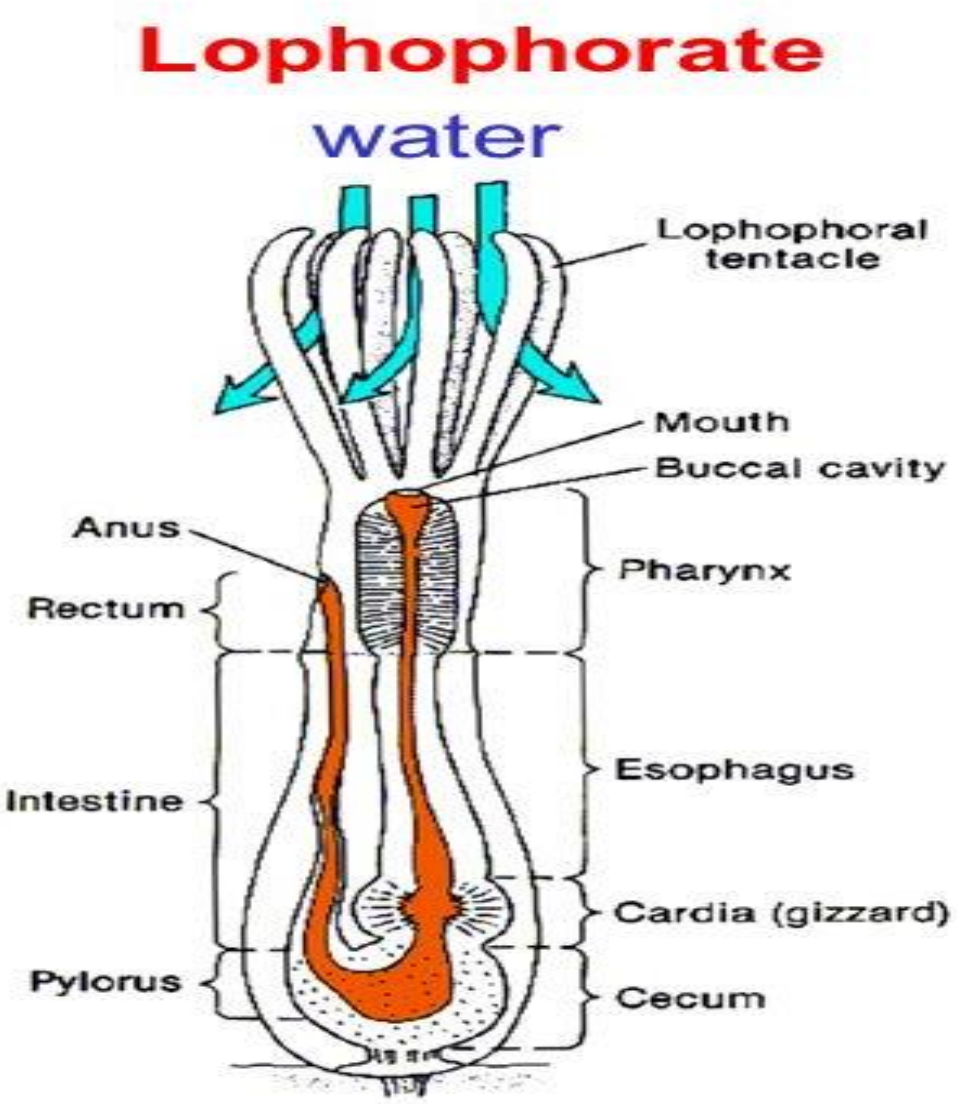
# Lesser Invertebrates: The Laphophorates

- Fertilization is usually external, and the nonfeeding, ciliated, free-swimming larva is planktonic before settling and developing into an adult.
- Development is similar to that of deuterostomes, with radial, mostly equal, holoblastic cleavage, and enterocoelous coelom formation.

# Lesser Invertebrates: The Laphophorates

- In the Inarticulata, the juvenile resembles a small brachiopod with a coiled pedicel in the mantle cavity.
- There is no metamorphosis; when the juvenile settles to the bottom, the pedicel attaches to a solid object, and adult existence begins.

# Lesser Invertebrates: The Lophophorates



# Animal Diversity: Invertebrates

## The Lophophorates

### Phylum Ectoprocta (Bryozoa): Moss Animals

Lecture 267

# Phylum Ectoprocta (Bryozoa): Moss Animals

- The ectoprocts (Gr. *ektos*, outside and *proktos*, anus) or bryozoans superficially resemble hydroids or corals.
- Bryozoa (Gr. *bryon*, moss and *zoon*, animal) means moss animals and refers to the moss like appearance of the colonies.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- The name ectoprocta refers to the position of the anus outside the ring of tentacles.
- The approximately four thousand living species of pseudocoelomates ectoprocts belong to three classes.
- All species are aquatic (both freshwater and marine) and less than 1.5 mm in length.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- Each body, or zooid, has a circular or horseshoe-shaped lophophore and is covered with a thin cuticle that encloses a calcified exoskeleton.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- The feeding body (lophophore, digestive tract, muscles, nerves) is called the **polypide**, the exoskeleton plus body wall (epidermis) is the cystid, and the secreted, nonliving part (exoskeleton) is the **zooecium** (Gr. *zoo*, animal and *oceus*, house).

# Phylum Ectoprocta (Bryozoa): Moss Animals

- Ectoprocts have an eversible lophophore that can be withdrawn into the body.
- Contraction of the retractor muscle rapidly withdraws the lophophore , whereas contraction of the muscles encircling the body wall exert pressure on the coelomic fluid, everting the lophophore.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- Ectoprocts grow by budding.
- Thin portion of the body wall grow out as small vesicles or tubes, and contain a complete zooid.
- The different budding patterns reflects the genetics of the individual animal and such factors as current flow and substrate.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- The factors determine the colony shape (e.g., thin sheets, convoluted folds, massive coral-like heads, upright tangles).
- Each colony can contain about 2 million zooids.
- Most ectoprocts are monoecious.
- In some species, heterozooids produce either eggs or sperm in different colonies.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- In others, simple gonads produce both sperm and eggs in the same autozooid.
- Sperm enter the coelomic cavity, exit through pores in the tips of the tentacles, and are caught by the tentacles of other colonies.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- Eggs are fertilized as they are released and are brooded in the coelom; some species have a modified ovicell in which the embryo develops.
- Marine species have radial cleavage and a free-swimming, ciliated larva.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- This larva swims for a variable period of time, depending on the species, and then sinks and attaches to a rock or other substrate and grows into a zooid.
- A colony forms by budding.
- Some freshwater ectoprocts produce a dormant stage called a **statoblast.**

# Phylum Ectoprocta (Bryozoa): Moss Animals

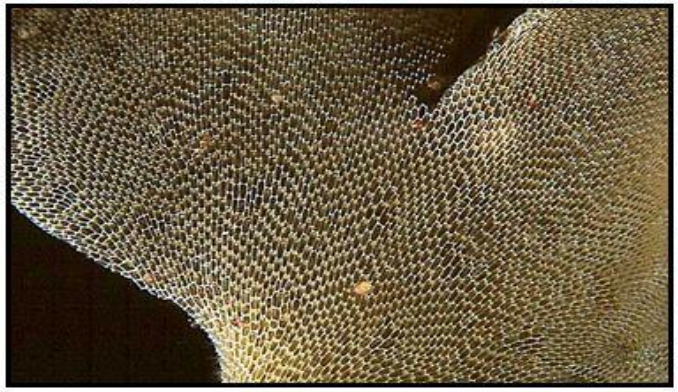
- A statoblast is a hard, resistant capsule containing a mass of germinative cells.
- Statoblasts are asexually produced and accumulate in the metacoel.
- They can survive long years in seasonally variable lakes and ponds.

# Phylum Ectoprocta (Bryozoa): Moss Animals

- Some float and are carried downstream, or are blown or carried from pond to pond, spreading ectoprocts over a large area.
- When environmental conditions become favorable, the statoblasts hatch and give rise to new polypides, and eventually, new colonies.

# Phylum Ectoprocta (Bryozoa): Moss Animals







**Phylum Bryozoa**

**"moss animals"**







# Animal Diversity: Invertebrates

## The Lophophorates

### Phylum Phoronida: The Phoronids

Lecture 268

# Phylum Phoronida: The Phoronids

- The Phoronids consist of about a dozen marine species divided between two genera: *Phoronis* and *Phoronopsis*.
- These animals live in permanent, chitinous tubes either buried in muddy or sandy sediments, or attached to solid surfaces.

# Phylum Phoronida: The Phoronids

- A few species bore into mollusc shells or calcareous rock.
- Generally, only the tentacles extend into the overlying water.
- Most phoronids are small, less than 20 cm long.
- The adult phoronid body consists of an anterior lophophore with two parallel rings of long tentacles.

# Phylum Phoronida: The Phoronids

- The tentacles of the lophophore are filled with coelomic fluid that serves as a hydrostatic skeleton to hold them upright.
- The cilia on the tentacles drive water into the ring of tentacles from the top of the lophophore and out through the narrow spaces between the tentacles.

# Phylum Phoronida: The Phoronids

- Suspended food particles are directed toward the mouth.
- A flap of tissue called the **epistome** (Gr. *epi*, around and soma, mouth) covers the mouth.
- Some phoronids reproduce asexually by budding and transverse fission; however, the majority are hermaphroditic.

# Phylum Phoronida: The Phoronids

- The gonads are in the coelom.
- Gametes pass from the coelom through the nephridiopore to the tentacles.
- Cross-fertilization is the rule, and zygote are either protected among the coils of the lophophore or released into the sea.

# Phylum Phoronida: The Phoronids

- Cleavage is radial, and a free-swimming larva called the actinotroch develops and feeds on plankton while drifting in the sea.
- It eventually settles to the bottom, metamorphoses, and begins to grow ventrally to form the body of the sedentary adult.

# Phylum Phoronida: The Phoronids

- As the animal grows, it burrows into the substrate.
- The body wall contains gland cells that eventually secrete the chitinous tube.

END

# Animal Diversity: Invertebrates

## Lesser Invertebrates

Cycliophores

Lecture 269

# Cycliophores

- Phylum cycliophores (Gr. *cyclion*, small wheel and *phora*, to carry) is the most recently described animal phylum.
- Members of its single species (*Symbionpandora*) are acoelomate, marine and bilaterally symmetrical.
- They live in association with the mouthparts of lobsters.

# Cycliophores

- The body of the feeding stage of *symbion* consists of a buccal funnel, a trunk, and a stalk that ends in a adhesive disk.
- The disk attaches to a seta on the mouthparts of the lobster host.
- The buccal funnel contains the mouth and is surrounded by a ring of compound cilia used in filter feeding.

# Cycliophores

- A cuticle lines the body on the outside.
- The mouth leads to a U-shaped gut tract.
- Protonephridia are used in excretion.
- The asexual feeding stage of *symbion* is about 0.3 mm tall.

# Cycliophores

- It can reproduce asexually by producing a larva stage, called the **pandora**, that is released and settles on the same host.
- Because of this asexual reproduction, large number of *symbion* can build up on one host.
- After settling and attachment, the new individual develops a buccal funnel and begins to feed.

# Cycliophores

- Sexual reproduction is apparently correlated to the molt cycle of the lobster.
- Some cue, possibly hormonal, from the lobster signals *symbion* that the lobster is about to molt.
- Male and female stages are produced within the bodies of feeding individual containing a maturing female.

# Cycliophores

- As the female escapes from its maternal feeding individual, fertilization occurs, and the female settles on the lobster's mouthparts.
- The female contains a single, large fertilized egg.
- The female dies, leaving a cuticular shell containing a **chordoid** larva.

# Cycliophores

- The chordoid larva is the dispersal stage.
- It is free-swimming and may settle on the mouth appendages of a new lobster host.

# Cycliophores

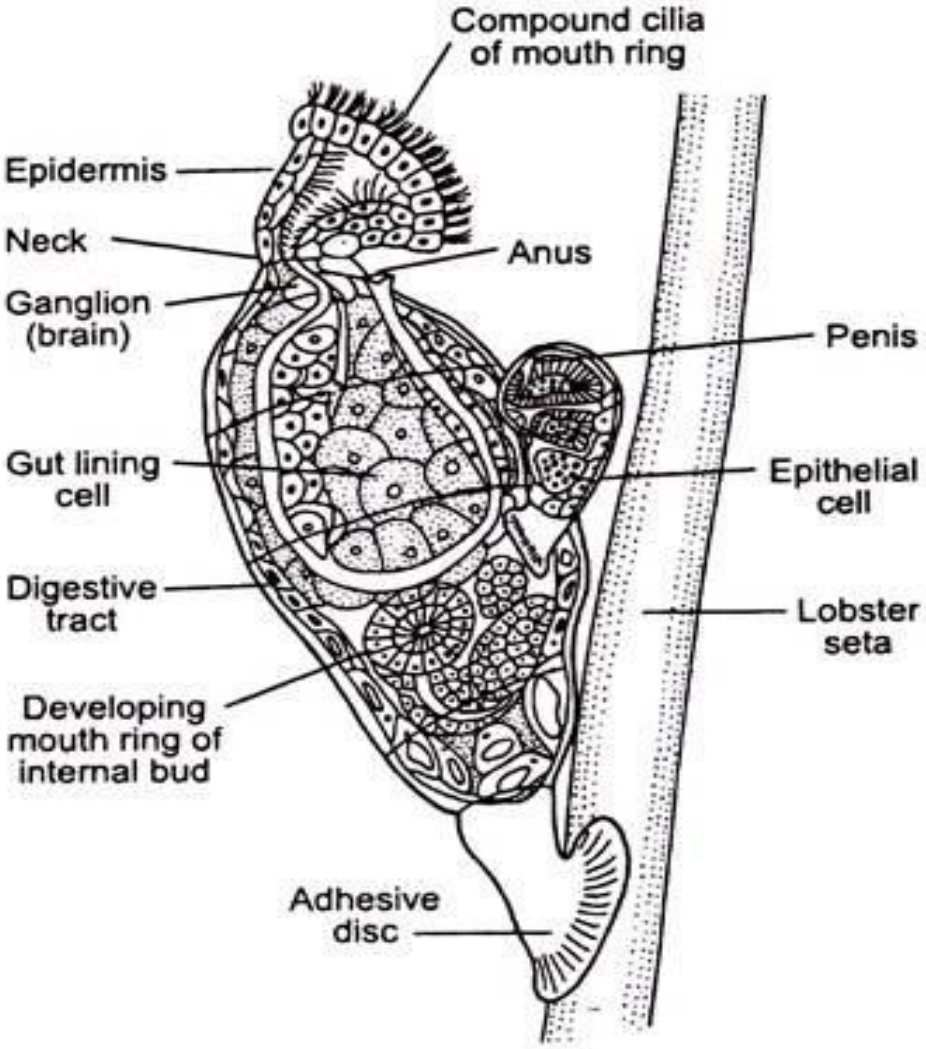



**Fig. 14.35:** A cycliophoran, *Symbion pandora*.

# Animal Diversity: Invertebrates

## Lesser Invertebrates

Entoprocts

Lecture 270

# Entoprocts

- The entoprocts (Gr. *entos*, within and *proktos*, anus) comprise a small phylum of about one hundred species of sedentary marine filter feeders.
- They are either solitary or colonial, and live in coastal waters.
- One group is commensalistic on the body surface of various invertebrates.

# Entoprocts

- Most entoprocts are microscopic.
- Entoprocts may form large, matlike colonies on rocks.
- An individual entoproct consists of a muscular stalk bearing a cup-shaped calyx with a crown of ciliated tentacles.

# Entoprocts

- The stalk is surrounded by a chitinous cuticle and may bear an attachment disk with adhesive glands.
- Entoprocts have a small body cavity that most zoologists consider a pseudocoelom.
- Loose connective tissue, however, fills this body cavity.

# Entoprocts

- How this cavity forms and its function are not well known.
- Entoprocts are filter feeders, and cilia on the tentacles convey food into the mouth.
- The filter-feeding apparatus is similar to the lophophore described for three previuos phyla.

# Entoprocts

- The entoproct feeding structure, however, is not a lophophore because the anus ends with the group of tentacles.
- Lophophores, by definition, have an anal opening outside their ring of tentacles.
- The digestive tract forms a U-shaped gut in the calyx.

# Entoprocts

- Also in the calyx is a pair of protonephridial tubules that open through a single pore in the mouth.
- The nervous system consists of a small, central ganglion and radiating nerves.
- Gas exchange occurs across the body surface.
- Entoprocts reproduce by asexual budding and also sexually.

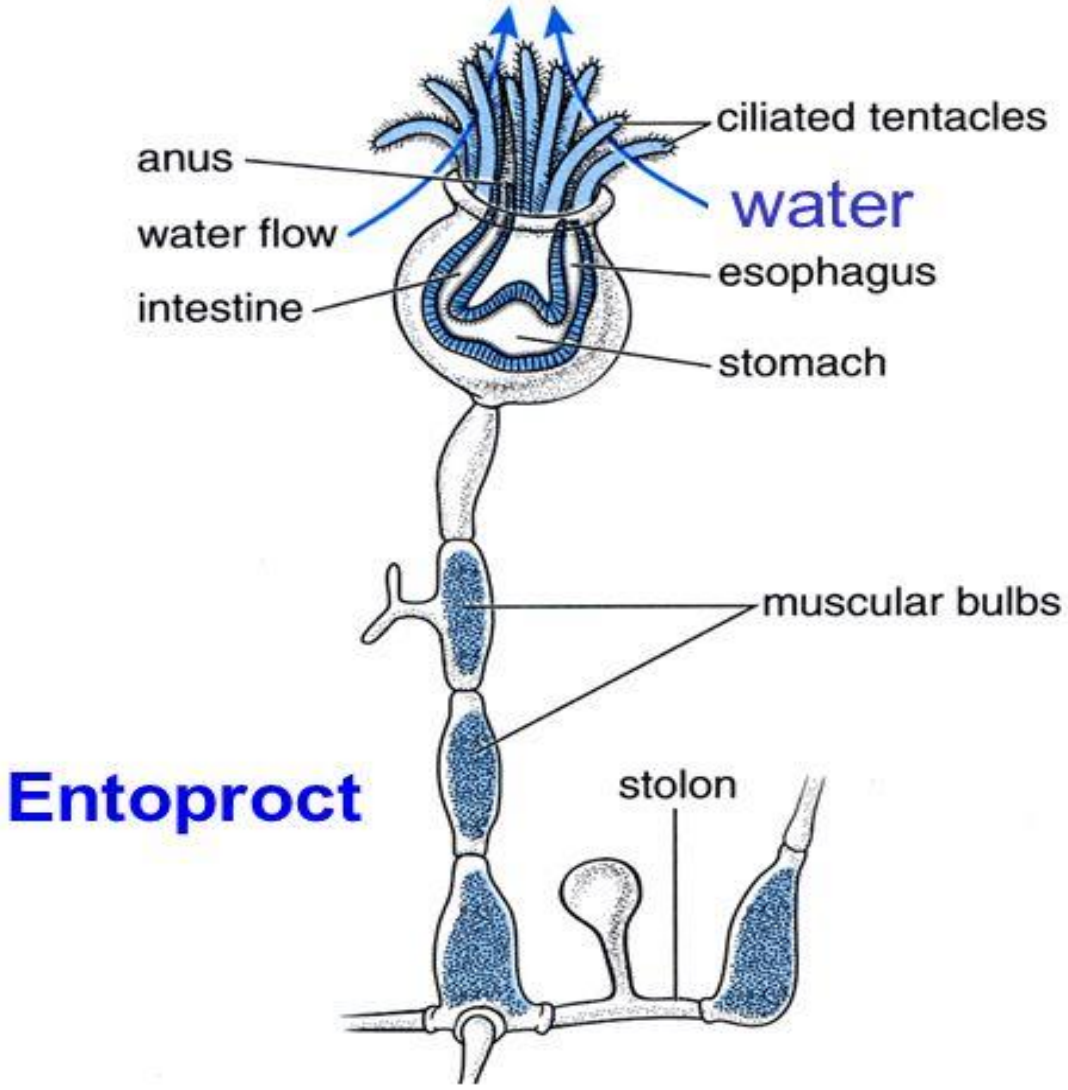
# Entoprocts

- Most entoprocts are monoecious, but eggs and sperm usually are produced at different times in one animal.
- Sperm are released freely into the water, and fertilization occurs internally.

# Entoprocts

- Embryos develop in a brood chamber, from which free-swimming larvae are released.
- Eventually, larvae settle to the substrate and develop into adults.

# Entoprocts


ciliated tentacles
anus
water
water flow
esophagus
intestine
stomach
muscular bulbs
Entoproct
stolon

# Animal Diversity: Invertebrates

## Lesser Invertebrates

Chaetognaths

Lecture 271

# Chaetognaths

- Members of the phylum Chaetognatha (Gr. *chaite*, hair or bristle and *gnathos*, jaw) are the arrow worms.
- The approximately one hundred species in this phylum are distributed throughout the world's oceans.
- Most are planktonic, 0.5 to 12 cm in length, streamlined, bilaterally symmetrical, and nearly transparent.

# Chaetognaths

- Their body consists of a head, trunk, and tail.
- The trunk has paired, lateral fins, and the tail has a single fin.
- The mouth is located ventrally on the head.
- Large grasping spines are lateral to the mouth, and smaller spines, called teeth are in front of the mouth.

# Chaetognaths

- Arrow worms are active swimmers.
- The near-perfect bilateral symmetry and streamlined shape of arrow worms permit a darting form of locomotion from laterally flexing the body.
- Their fins stabilize the body during swimming and help prevent sinking during moments of rest.

# Chaetognaths

- Arrow worms are active predators.
- They feed on planktonic crustaceans and small fish.
- An arrow worm darts forward quickly to grab its prey with mouth spines that inject a neurotoxic venom into the prey.

# Chaetognaths

- Interestingly, the neurotoxin is not produced by the arrow worm, but by commensalistic bacteria living in the head region.
- The gut of arrow worms is a simple, straight tube.
- Although much remains to be learned about arrow worm physiology, circulation, gas exchange, and excretion probably all occur by diffusion.

# Chaetognaths

- The nervous system consists of a cerebral ganglion in the head and other ganglia associated with sensory organs.
- Ciliary fans that cover the body probably detect water borne vibration.
- A single pair of eyes is below the epidermis on the dorsal surface of the haed.

# Chaetognaths

- Arrow worms probably use these photoreceptors to orient to the direction and intensity of light during vertical migrations within the water column.
- Chaetognaths are monoecious, with paired ovaries in the trunk and paired testes in the tail.

# Chaetognaths

- Arrow worms engage in a mating dance during which they deposit sperm in ball-like clusters on the mate's body.
- These sperms clusters rupture, and sperm migrate to the female gonopore.
- Fertilization is internal, and zygote in a jelly like coating float in the sea or sink to the substrate.

# Chaetognaths

- One genus has a marsupial pouch within which fertilized egg develops.
- There is no larval stage or metamorphosis.

END

# Chaetognaths

## Phylum Chaetognatha – arrow worms







